

Honeywell

Global User Station
Display Builder
User's Guide

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About This Document

This document describes how to use the Global User Station (GUS) Display Builder to create, open, and close the GUS Display Builder windows, manage displays, and manipulate and add objects on a display.

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Document ID
BasicScript 2.2 Language Reference	GU40
BasicScript 2.2 User's Guide	GU25
Display Authoring Tutorial	GU07
Display Scripting User's Guide	GU24

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Training Classes	Honeywell Automation College: http://www.automationcollege.com		

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

Symbol Definitions

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1. Display Builder Introduction

1.1 Overview

In this section, the Global User Station (GUS) Display Builder is introduced along with the tools needed to create, open, and close the GUS Display Builder windows.

Topics in this section are

- Display Builder Window
- Property Sheet Introduction
- Performing Actions on Objects

1.2 Display Builder Window

Components

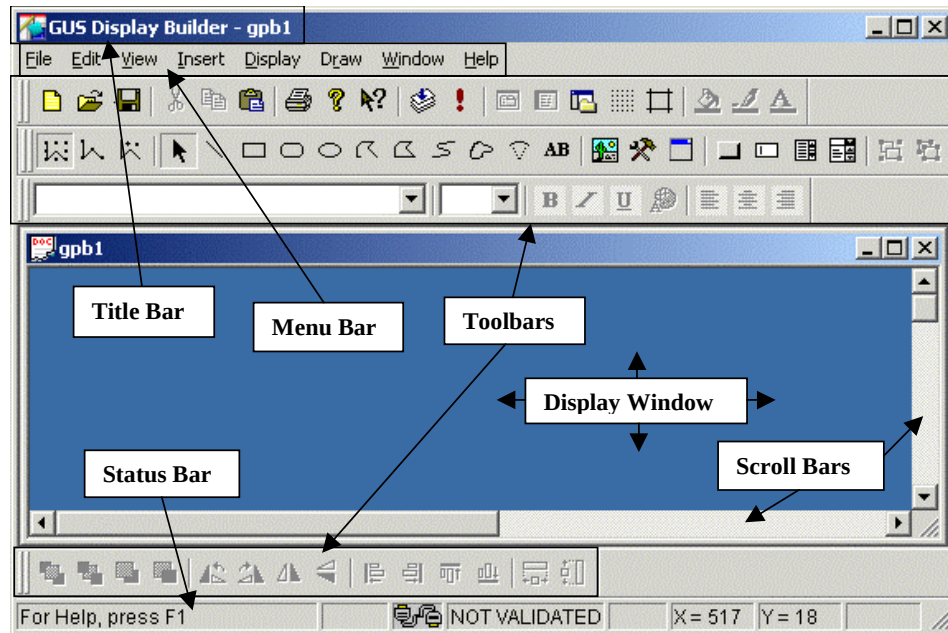
The Display Builder Window has the following components:

- Title Bar
- Menu Bar
- Toolbars
- Display Windows
- Scroll Bars
- Status Bar

See the illustration and descriptions that follow.

1. Display Builder Introduction

1.2. Display Builder Window



TITLE BAR—The title bar contains the Display Builder name and the name of the display being edited. The display being edited is the active display window.

DISPLAY WINDOW—The large area in the lower right is a display window, and several display windows can be opened at one time. When opened, each window displays a “GPBn” default name where “n” is a number.

MENU BAR—This menu bar contains the top-level menus. When Display Builder is running, you must click each menu name to see its menu items.

TOOLBARS—The toolbar areas can be made up of one to five toolbars: the Standard, Draw, Layout, and Text Format toolbars.

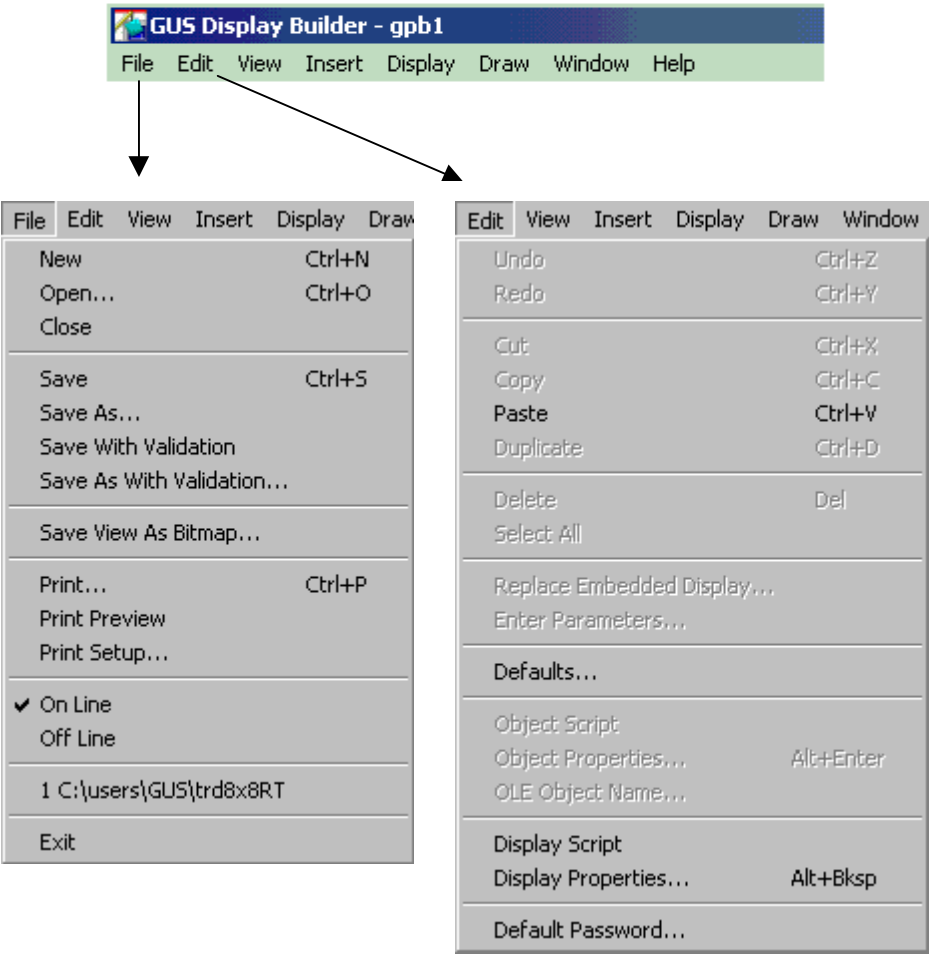
DISPLAY WINDOW—This is where you draw your displays. When first opened, the first new display window is named gpb1. Subsequent new display windows are named gpb2, and so on. Click any window to select it and make it the active window.

SCROLL BARS—Sometimes, the display being created is larger than the display window. The scroll bars allow you to move to areas of the display that are not currently displayed.

STATUS BAR—This bar can be found at the bottom of our display. It shows the status and help information that helps you use the Display Builder.

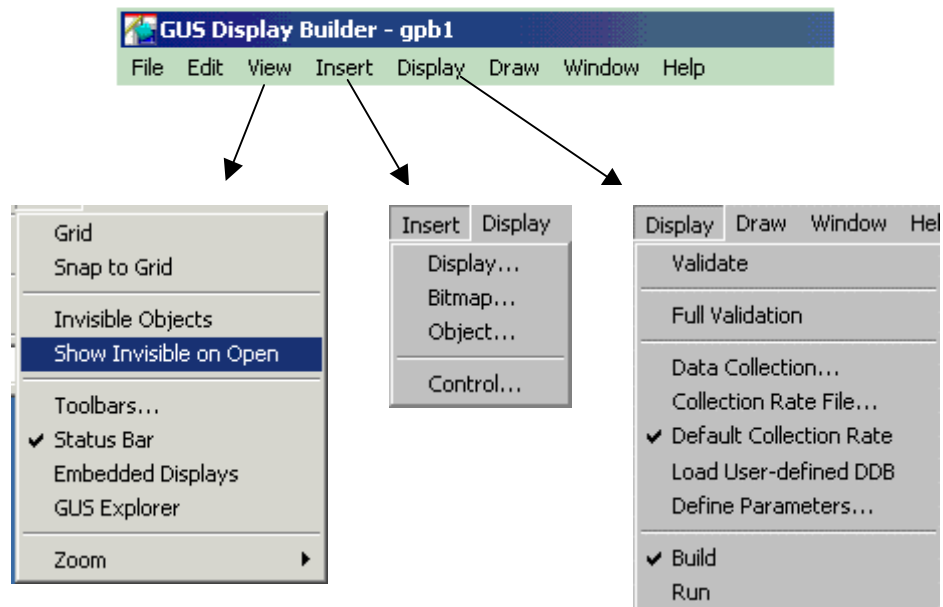
Menu Bar expanded

Here are the drop-down commands available under each menu item in the Menu Bar.

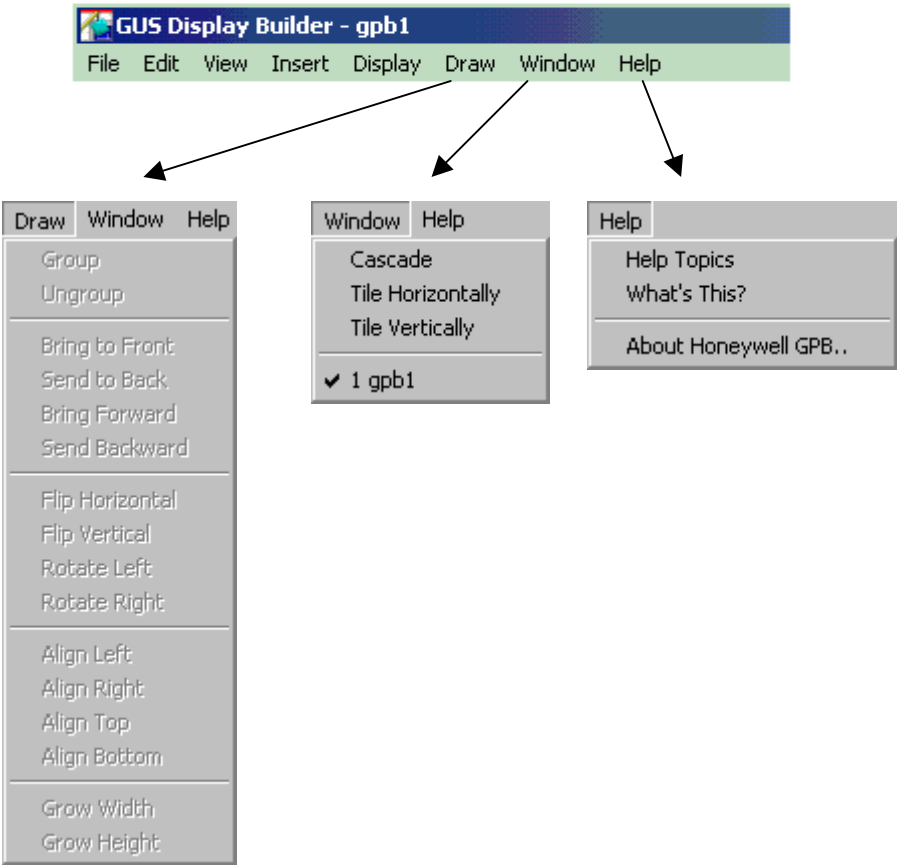


1. Display Builder Introduction

1.2. Display Builder Window



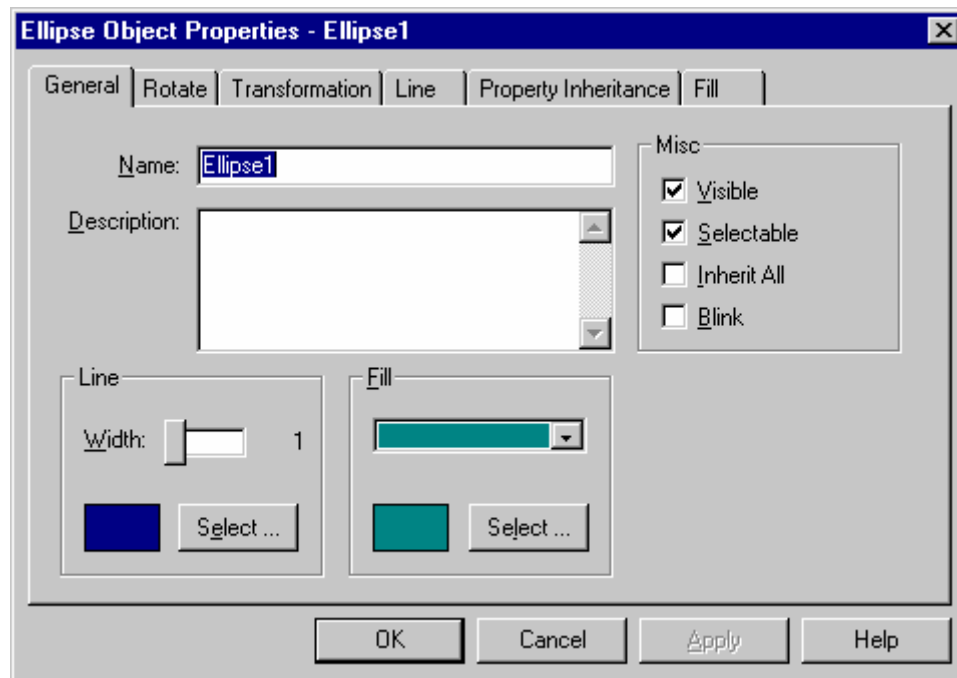
1. Display Builder Introduction
1.2. Display Builder Window



1.3 Property Sheet Introduction

Each object in a display window has a set of properties. You can view and change an object's properties by using a Property Sheet.

A typical Property Sheet, as shown below, is separated into Property Pages with each page showing different groups of information as indicated on the "tabs" (General, Rotate, etc.). When the tab for that page is clicked, the selected page moves to the front.



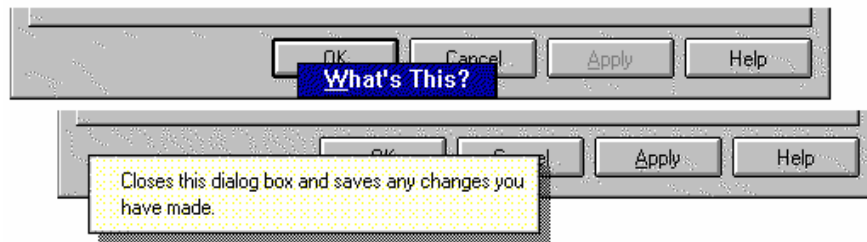
Property Sheet buttons (all property pages)

Unless specified otherwise, the following buttons apply to all object properties:

- **OK**—Syntax checks all variable expressions on the basic dynamics and saves all changes (also see **Apply**). Closes the Property Sheet.
- **Cancel**—Cancels all changes that have not been saved through the Apply button. Closes the Property Sheet.
- **Apply**—Syntax checks all variable expressions on the basic dynamics and saves all changes. Does not close the Property Sheet (also see **OK**).
- **Help**—Invokes help about the property page being displayed.

Property Sheet Buttons are also explained with a “What’s This?” menu:

- 1 Click any one of the Property Sheet Buttons with the Right-mouse Button and a “What’s This?” menu appears.



- 2 Click the “What’s This?” menu with the left mouse button to read a short explanation of the button’s function.

1.4 Performing Actions on Objects

The keyboard and the mouse are the tools you use with the GUS Display Builder.

With the Display Builder, you use both the left and right-mouse buttons to perform actions on objects. In this document, the **LEFT** mouse button is used for all such actions, unless specified otherwise.

Entering keyboard values

When you enter data using the Display Builder, you will often be required to use the keyboard. Note that all keystrokes you enter will be captured, even if you start typing before the actual data entry port appears on the screen.

Clicking the mouse

When you “click” (press once and release) the left mouse button on a displayed button or toolbar icon, the command indicated by the button is executed.

Selecting menu items with the mouse

Another common action you will perform is to “click and hold” (press once and hold) the left mouse button on an item from the menu bar, then drag your cursor down the menu until the required command is found. You then release the left mouse button. This is commonly called “selecting” a menu item.

Dragging the mouse

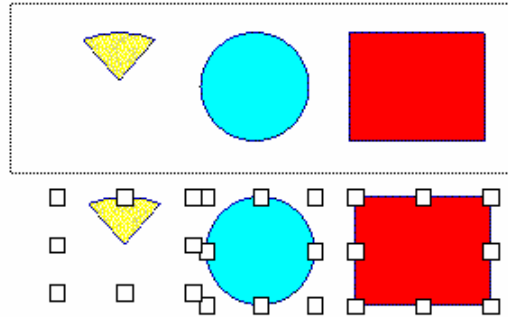
If you click an object and hold the left mouse button, then drag the object to another place and release it, you have performed a “drag and drop” action that is used to move or copy an object in the Display Builder.

Selecting objects with the mouse

Selecting a Single Object—If you click once on an object (a graphic object, for instance), small boxes appear on the corner and sides of the object called “handles.” When you do this, you are telling the computer that the next command (perhaps from the menu) is directed at that object.

Selecting Several Objects—If you want to select a group of objects with the mouse, proceed as follows:

- a) Using the selection tool, click-and-hold the mouse somewhere outside an object and drag a “bounding-box” around the required objects. An example of three objects is shown at the top right.
- b) Now release the mouse button. All objects will be selected (have handles). See example at the bottom right.



This process is also called **Net Selecting**.

Double-Clicking the Mouse

When you double-click the left mouse button on an object, the default action for that object occurs. Some examples are as follows:

- Double-click an object’s fill color or the line surrounding it to call up the Property Sheet for that object. This is the same as selecting the object, then selecting Edit/Object Properties from the menu.
- When drawing an object that can contain several segments (polygon, and Bezier curve), double-click at the point where you want the last segment to end.

1.5 Using the Right-mouse Button

Displaying menus

Use the right-mouse button to call up a popup menu of the frequently performed actions on objects. Use either of the following two methods:

Method 1:

- a) While the selection pointer is over either an Object or the Display Window background, click and hold the right-mouse button. A menu of frequently performed actions, as shown here, is displayed.
- b) Move the mouse to the required command and release the right-mouse button to select it.

Method 2:

- a) While the selection pointer is over either an Object or the Display Window background, click and release the right-mouse button. A menu of frequently performed actions, as shown here, is displayed.
- b) Move the mouse to the required command and click and release either mouse button to select it.

Object Menu



Display Menu



Runtime operation of the right-mouse button

When the Right-Mouse Button menu is visible on a runtime GUS display, the user-configured buttons and the dedicated buttons (such as FAST and ALM SUMM) on the operator keyboard are blocked from use, until you remove the Right-Mouse menu. The Right-Mouse menu affects buttons on either the Integrated Keyboard or the Operator Entry Panel.

Additional uses for the right-mouse button

The right-mouse button is also used to:

- Terminate drawing Polygon and Bezier Curve segments as described in Drawing an Open Polygon (Open Polygon Draw Tool) and Drawing an Open Bezier Curve (Open Bezier button on Draw Toolbar).
- Call “What’s This?” help descriptions of
 - Property Sheet buttons as described in the Property Sheet Introduction.
 - Standard dialog boxes as described in Managing Displays.

1. Display Builder Introduction

1.5. Using the Right-mouse Button

2. Managing Displays

2.1 Overview

In this section, you are introduced to the tools needed to manage Display Builder displays.

Topics in this section are

- Help on standard dialog boxes
- Creating a display (File/New menu and New on Toolbar)
- Opening a display
- Protecting a display
- Entering a default password
- Saving Displays
- Validating Displays (Display/Validate menu and Validate on Toolbar)
- Running Displays
- Printing
- Switching Between Displays (Window menu)

2. Managing Displays

2.2. Help on standard dialog boxes

2.2 Help on standard dialog boxes

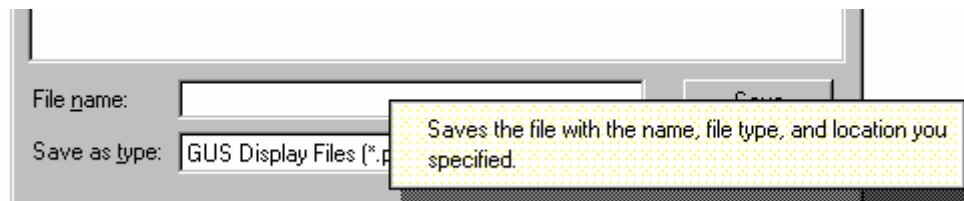
Several of these tools, such as Open and Save, use standard Microsoft Dialog boxes.

Use the Help icon  in the upper right corner of these boxes to obtain help.

- 1 Click the Help icon so it appears to be depressed.

RESULT: Your cursor changes to a question mark .

- 2 Click a button or list that you need help on (for example, the Save button as shown next).



RESULT: A pop-up box explains the function.

NOTE: As an alternate, you can use the Right-mouse Button to click directly on the button or list you need help on.

2.3 Creating a display

File/New menu and New on Toolbar

The New command creates a new display. After selecting New, a new display window opens. The builder defaults, located under the Edit/Defaults menu, are applied.

Creating a new display

To create a new display, select File/New (Ctrl + N) from the menu or click the New icon on the Standard Toolbar.

RESULT: A new Display Builder drawing window is created.

2.4 Protecting a display

The GUS Display Builder uses a password-based approach to prevent the unauthorized viewing and editing of the contents of a GUS Display. When you add a password to a GUS Display, it becomes "password-protected." This password protection mechanism applies to the opening of GUS displays with the Display Builder, and to the viewing of script text contained in a GUS Display file.

Password-protected displays can be migrated without knowing the display's password.

NT File System Security

This password protection feature is not intended as a replacement for NT file system security. It is not an impenetrable defense against unauthorized viewing of GUS Display contents. It protects against the unauthorized use of conventional access mechanisms to GUS Displays. It does not prevent people from running displays or from copying and deleting display files. Use NT file system security to control these types of operations on display files.

Using Password Protection

You can use the Display Builder protection mechanism to block the following actions:

- Opening a password-protected display using the Display Builder
- Viewing the contents of a password-protected embedded display by using <Alt-L> in the Display Builder
- Viewing the contents of the password-protected embedded display by using the Object Browser in the Display Builder
- Viewing the script text of a password-protected display by using a binary dump utility

When you open a password-protected display in the Display Builder, a dialog box will appear that prompts you for a password. If you enter a password that matches the assigned password, the display will be opened. If the password you enter doesn't match, the display will not be opened and an error message will be displayed.

2. Managing Displays

2.5. Entering a default password

Embedding Password-protected Displays

You can freely embed password-protected displays without knowing the display's password. Password protection does, however, restrict you from viewing the contents of embedded displays in the Display Builder when you select <Alt-L>. When you use <Alt-L>, the contents of password-protected embedded displays will not be listed. Only those properties visible to the containing display will be shown. For each password-protected embedded display, the display will indicate that viewing the contents of that embedded display is blocked.

Password Protection Functions

The Display Builder provides three basic password functions:

- Adding a Password
- Changing a Password
- Removing a Password

These functions are executed from the Display Object Property Sheet. Refer to General Page-Display Object Property Sheet for details.

2.5 Entering a default password

Edit/Default Password menu

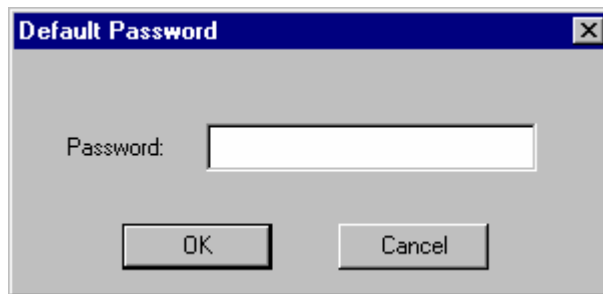
In order to manage a large number (or even all) of your GUS Displays effectively, the Display Builder provides a default password. Once a default password has been entered, every time the Display Builder tries to open a password-protected display, it will first compare the password of the display being opened to the default password. If there is a match, the display will be opened. If the passwords do not match, the user will be prompted for a password just as if no password existed.

The Display Builder also uses the default password when you select < Alt-L> to display the contents of an embedded display. The Display Builder attempts to match the default password to that of the embedded display. If it matches, the contents of the embedded display will be shown.

Enter a default password

The maximum length of a password is 255 characters. A password can consist of any sequence of characters. Passwords are case sensitive. Follow the normal security recommendations regarding password length (use 8 or more characters) and content for maximum security.

- 1 Select Edit > Default Password on the menu bar. The Default Password dialog box appears.



- a. Enter the required password.

CAUTION: You must remember your password. Consider writing it down and keeping it in a safe place. If you forget the password, there is no way to reconstruct it or remove the password protection. You will not be able to retrieve the display.

- b. Click Cancel to close the dialog box without entering a new default password.

- 2 Click OK to save the default password.
- 3 If there is no existing default password, the password you enter will be set up as the default password and the Default Password dialog box will disappear.
- 4 If a default password already exists, asterisks will appear in the Password data entry port.
- 5 To enter a new default password, type it in and click OK. The password you enter will be set up as the new default password and the Default Password dialog box will disappear.
- 6 To retain the current default password, click Cancel. The Default Password dialog box will disappear and the default password will be unchanged.
- 7 Save the display.

NOTE: The display is not password-protected until you save it.

2. Managing Displays

2.6. Opening a display

2.6 Opening a display

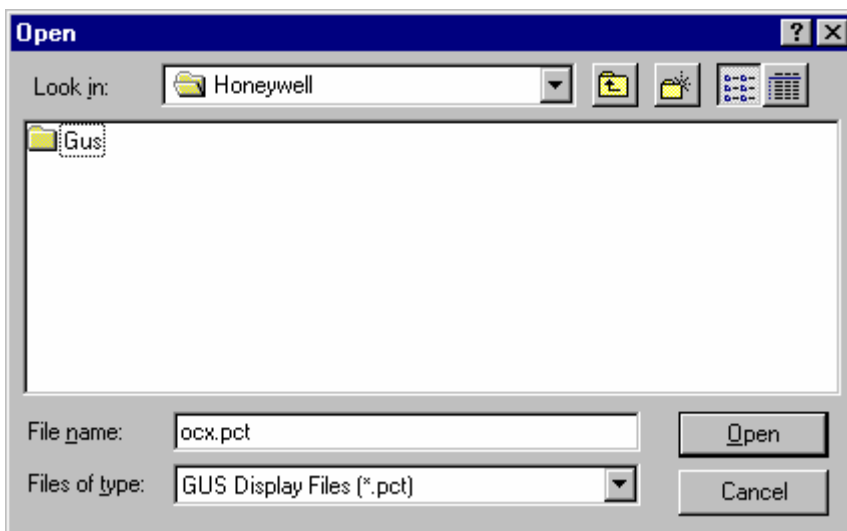
There are two ways to open a display:

- 1 By Opening an existing display (File/Open menu and Open on Toolbar)
- 2 By Opening a recently used display (File/Recent menu)

2.7 Opening an existing display

File/Open menu and Open on Toolbar

The File/Open command opens an existing display window file. After selecting it, the Open dialog box appears.



The options that can be changed include,

- Look in:—lists current folder (directory).
 - Down Arrow is used to search the hierarchy of the drives and folders.
 - Folder Icon with up arrow moves up one level in the hierarchy.
 - Folder Icon with a star creates a new folder.
 - List Icon with 6 lines lists all filenames in the current folder.
 - List Icon with many lines lists details of filenames in the current folder.
- File name:—lists files of the specified type in the selected directory.
- Files of type:—allows display of files of the following types:
 - GUS Display Files (*.pct)
 - All Files (*.*)

To learn about each feature of the Open dialog, click the  in the upper right corner of the box; then click the required feature for a short explanation.

How to open a display

- 1 Select File/Open (Ctrl + O) from the menu or click the Open icon on the Standard Toolbar.
- 2 The Open dialog box appears.
- 3 Select Look in, and Files of type for the requested file.
- 4 Select or type in the File name. Optionally, you can double-click the filename and skip step 5.
- 5 Click OK.

RESULT: A new window opens displaying the required file.

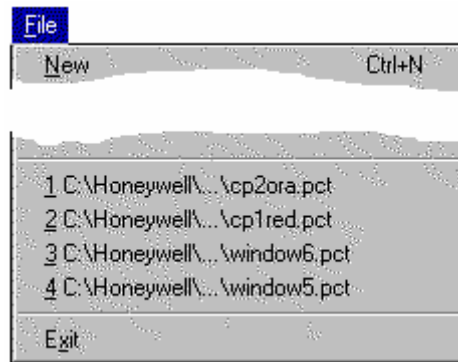
2. Managing Displays

2.8. Opening a recently used display

2.8 Opening a recently used display

File/Recent menu

As you use the GUS Display Builder, up to the last four (most recent) files you have opened are displayed near the bottom of the File menu.

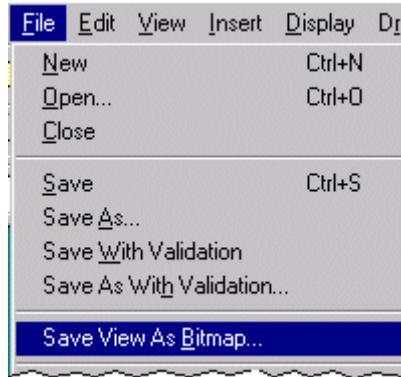


To quickly open one of these files, simply select it from the File menu.

2.9 Saving Displays

There are five ways to save a display.

- Saving an existing Display file (File/Save menu and Save on Toolbar)
- Saving a new, unnamed display (File/Save As menu)
- Saving and validating an existing display (File/Save with Validation menu)
- Saving and validating a new, unnamed display (File/Save As with Validation menu)
- Saving the current display view as a bitmap (.bmp) file (Save View As Bitmap)



2.10 Saving an existing Display file

File/Save menu and Save on Toolbar

This command saves a display that has been named and saved before.

- 1 If the display window to be saved is not active, select it by clicking once on the window's title bar.
- 2 Select **File/Save** (Ctrl + S) from the menu or click the Save icon on the Standard Toolbar.

RESULT: The file is saved, using the current name and current directory.

2. Managing Displays

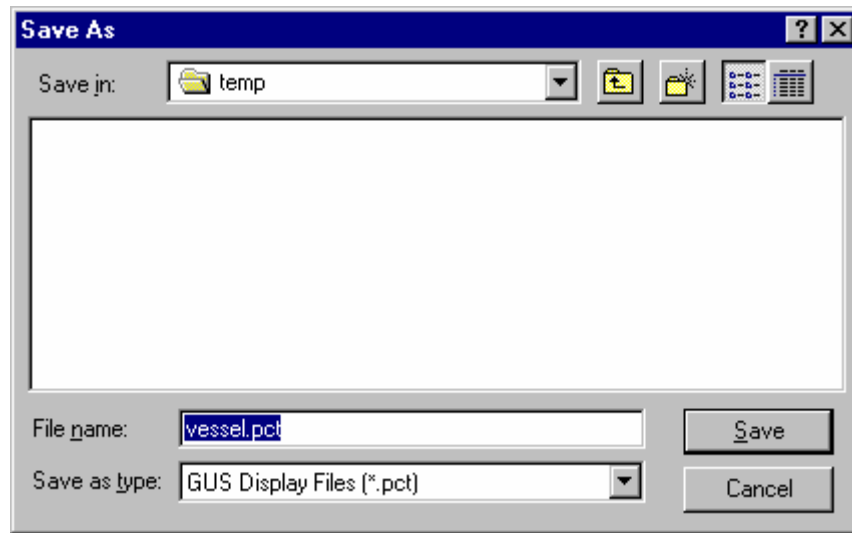
2.11. Saving a new, unnamed display

2.11 Saving a new, unnamed display

File/Save As menu

This command saves a new, unnamed display. You can also use this command to save a display with another filename, in another directory, or on another drive.

- 1 If the display window to be saved is not active, select it by clicking once on the window's title bar.
- 2 Select **File/Save As** from the menu. The Save As dialog box appears.



- 3 Select options as appropriate to save the display in the required folder (directory). To learn about each feature of the Save As dialog, click the  in the upper right corner of the box; then click the required feature for a short explanation.

NOTE: For Display Builder compatibility, save all displays with a **“.PCT”** extension.

- 4 Click **OK**.

RESULT: The file is saved using the specified filename, directory, and drive.

2.12 Saving and validating an existing display

File/Save with Validation menu

This command combines saving and validation into one step. Validation takes place first. If validation is successful, the file is saved.

- 1 If the display window to be saved is not active, select it by clicking once on the window's title bar.
- 2 Select **File/Save** and **Validate** from the menu.

RESULT: Validation occurs first. If an error is encountered, the Save is aborted.

If the save is aborted, you can then either use the Save command (without validating) or correct the errors shown in the Error List and retry. Refer to Validating Displays

Display/Validate menu and Validate on Toolbar and Validation Error List (Error List Help) for additional information.

With GUS R340, you can cancel a validation when it is in progress. This is accomplished by the addition of a Cancel button to the Validation dialog box. Refer to Validating Displays

Display/Validate menu and Validate on Toolbar for additional information about Canceling a Validation Command.



ATTENTION

If you are validating an existing .pct file, (which has already been validated) you can stop the validation using the Cancel button and use any one of the following options:

Cancel the validation and do not save the file. The status of the file goes back to the earlier validated stage

OR

Cancel the validation and save the file. The earlier validated information is canceled.

2. Managing Displays

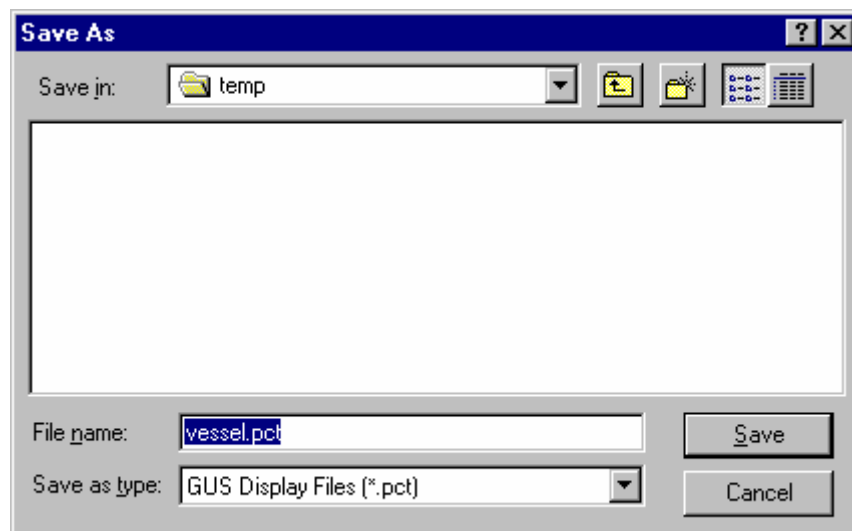
2.13. Saving and validating a new, unnamed display

2.13 Saving and validating a new, unnamed display

File/Save As with Validation menu

This command combines saving and validation into one step. Validation takes place first. If validation is successful, the Save As command is processed. You can also use the Save As command to save a display with another filename, in another directory, or on another drive.

- 1 If the display window to be saved is not active, select it by clicking once on the window's title bar.
- 2 Select **File/Save As with Validation** from the menu. Validation occurs first, then the Save As dialog box appears.



- 3 Select options as appropriate to save the display in the required folder (directory). To learn about each feature of the Save As dialog, click the  in the upper right corner of the box; then click the required feature for a short explanation.

NOTE: For Display Builder compatibility, save all displays with a “.PCT” extension.

- 4 Click **OK**.

RESULT: Validation occurs first. If an error is encountered, the Save As is aborted.

If the save is aborted, you can then either use the Save As command (without validating) or correct the errors shown in the Error List and retry. Refer to Validating Displays

Display/Validate menu and Validate on Toolbar and Validation Error List (Error List Help) for additional information.

With GUS R340, you can cancel a validation when it is in progress. This is accomplished by the addition of a Cancel button to the Validation dialog box. Refer to Validating Displays

Display/Validate menu and Validate on Toolbar for additional information about Canceling a Validation Command.

2.14 Saving the current display view as a bitmap

Save View As Bitmap overview

This command allows you to save the current GUS display view in the standard Windows bitmap (.bmp, 24-bit “TrueColor”) file format. The resulting bitmap file can then be used in most of the commercial graphical editing applications.

This command is available in both the GUS build-time and runtime operation modes. It creates a high-fidelity bitmap image of the associated GUS display in a WYSIWYG (what-you-see-is-what-you-get) form. All scrolling, zooming, and clipping are reflected in the resulting bitmap, including any occlusion by other windows that may cover up the GUS display view.

In conjunction with the “Save View As Bitmap” command, there is a configuration option in the standard TPSSConfig tool to configure a “Save-As-Bitmap Default Directory.” Refer to Save-As-Bitmap Default Directory in the *Configuration Utility User's Guide*, in the Configuring GUS Displays for Runtime section for further information.

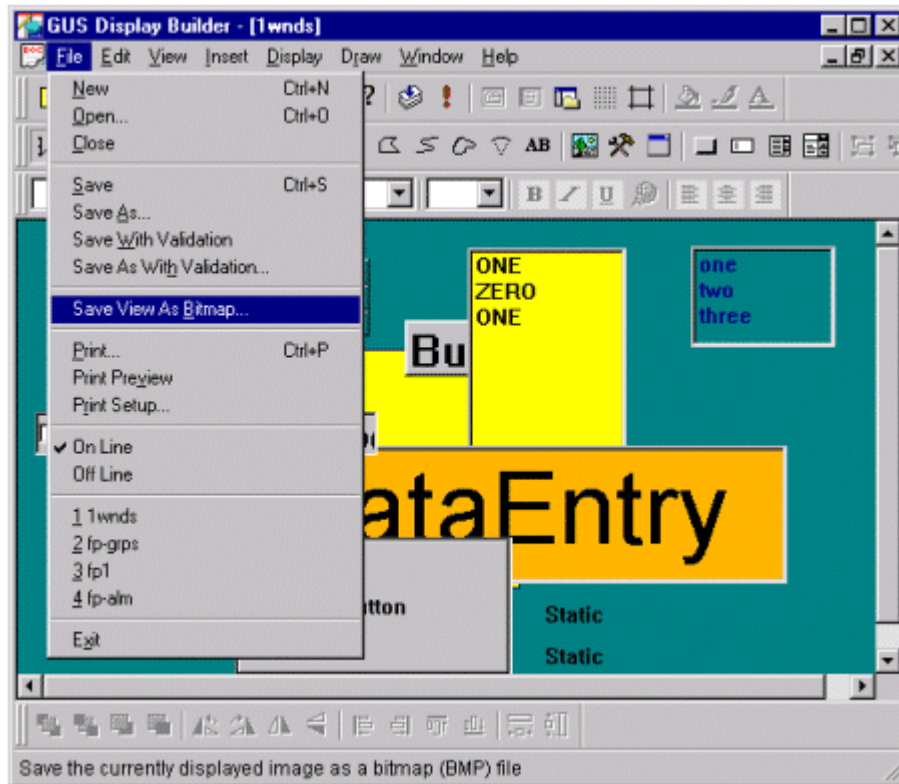
2. Managing Displays

2.14. Saving the current display view as a bitmap

Use in GUS build-time mode

The build-time “Save View As Bitmap” menu option is available whenever a GUS display is active (see the example that follows). This command applies to the currently *active* display. It allows you to capture the current “WYSIWYG” (what-you-see-is-what-you-get) rendering of the GUS display and save it with your specified bitmap file name.

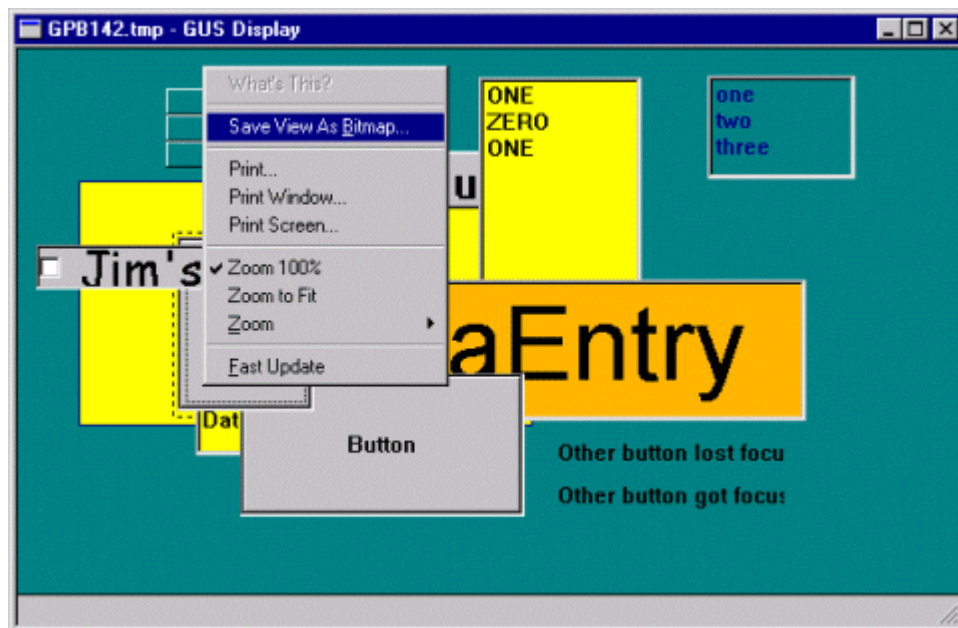
This command captures a high-fidelity image of the GUS display view as displayed “on screen.” Therefore, “always-on-top” overlapping windows should be removed from occluding the current GUS display, unless such occlusions are required in the resulting bitmap file.



Use in GUS runtime mode

At runtime, you can access the “Save View As Bitmap” option by right-clicking to obtain a pop-up menu (shown below). Selection of this menu item results in the same user interface dialog as provided for the build-time user.

Because the function is WYSIWYG, the runtime version of this function captures all of the “live” process information you see on the screen. As with the build-time version of this function, the existing node administration determines the user’s file and directory access rights and restrictions.

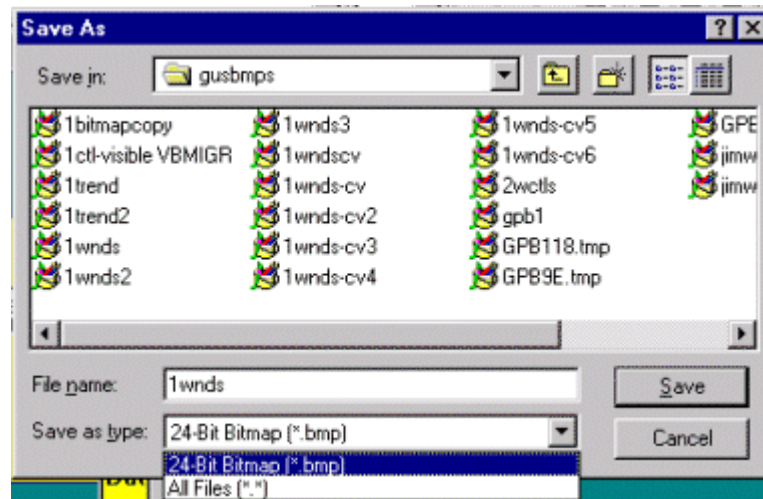


2. Managing Displays

2.14. Saving the current display view as a bitmap

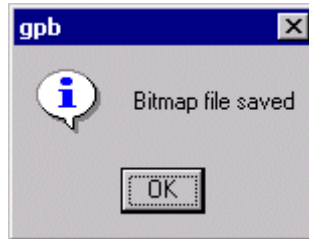
Save View As Bitmap dialog

Whether in the GUS build-time mode or the GUS runtime mode, upon selection of the “Save View As Bitmap” command you will see the standard Windows File/Save As dialog, as follows.



- The file name defaults to the name of the GUS display, with the “.bmp” extension.
- The directory defaults to the specified “GUS Bitmap Default Directory” as configured by the TPSCConfig utility.
- If the above GUS default bitmap directory is unspecified or empty, the directory defaults to that of the GUS display itself.
- By default, only bitmap files in the directory are shown.
- In addition to the “Bitmap Files (*.bmp)” option, you may also select the “All Files” option.
- The Network button is excluded. Therefore, you cannot use this function to mount additional network drives.
- Use of the function depends on the existing node administration for proper authorization and access rights.
- The function will not allow you to overwrite the current GUS display (.pct) file itself.

Once you have successfully completed the bitmap save, the following dialog appears.



2.15 Validating Displays

Display/Validate menu and Validate on Toolbar

The Display Validate function verifies that data references in scripts and objects' dynamics are correct and binds the references in the display to the data to allow efficient access during display execution.

The Validate function must be run before a display can run. Also, the validation will fail if there are any scripts that have not been successfully compiled, or if any embedded (sub) displays exist with parameters not entered or invalid parameters.

A display can run with validation errors, but invalid data references will result because that data is not accessed during execution.



ATTENTION

Display Validation should not be done on a node that is being used for critical control. Validation of a display may require many requests to the TPN nodes to validate data references. This may slow down data collection for other displays running on the same node.



ATTENTION

Do not use the #IF directive in GUS scripts. The Display Validate function ignores conditionally compiled code; therefore, Data Collection references are not resolved for code inside a #IF directive.

Offline Validation

If the Display Builder is in offline mode, the following occurs during display validation.

- 1** Compilation of all scripts (object scripts, embedded display scripts, display script, and so on.).
NOTE: The syntax checking done during the compilation is what is provided in the basic language.
- 2** Syntax checking of the variable expressions on objects' dynamics (For example, bar, rotate).

NOTES:

- The syntax checking done during the compilation is what is provided in the basic language.
- Any previously validated LCN references will remain unchanged. If new LCN references have been added since the display was validated on-line, then offline validation will report them as errors.

Online Validation

If the Display Builder is in online mode, the following occurs during display validation.

- 1** All actions in offline validation.
- 2** Validation of external data references in all scripts and the variable expressions on all objects' dynamics.

NOTES:

- Function calls used in scripting or for object dynamics are not validated. If improper function calls are programmed, errors will be detected and annunciated at runtime.
- Before moving a display to production, be sure to run the display to validate that all data references are valid.

Validation messages

When the display validation process has completed, one of these actions occurs.

Condition	Dialog box displayed	Status Bar Indication
Errors found during display validation	"Fatal error occurred during display validation"	Not Validated
No errors found during display validation	"Online [names resolved] validation"	Validated

How to invoke display validation

Invoke Display Validation in any of the following three ways:

- 1 From the Display menu, select the Validate command.
- 2 From the Display's Right-mouse Button, select the Validate command.
- 3 From the Standard toolbar, select the Validate tool  .

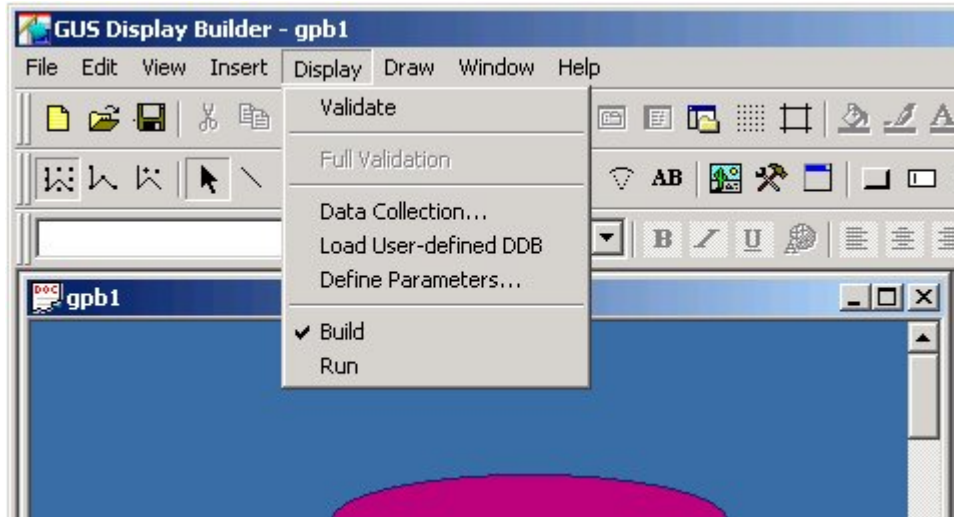
If errors occur, refer to Validation Error List (Error List Help) for help in resolving errors.

2. Managing Displays

2.15. Validating Displays

Full Validation (Display/Full Validation menu item)

Full Validation is found in the GUS Display Builder “Display” menu item. It allows you to set up the context for validation of TPN data access references.



If Full Validation is checked, the display validation command (such as Validate or Save With Validation) will recompile all object scripts and revalidate all TPN data access references.

If Full Validation is NOT checked, the display validation command will recompile all object scripts and validate only the unresolved TPN data access names.

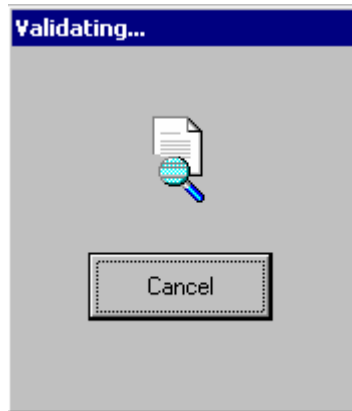
At GUS Display Builder startup, by default, Full Validation is not checked.

When performing batch validations using the validator tool, all TPN data access references are revalidated.

If the TPN is not available (off-line), then the revalidation of TPN data access names is not possible. When this is true, the Full Validation is not checked and is disabled.

Canceling a Validation Command

To cancel a validation while it is in process, select the Cancel button from the Validation dialog box.



The Cancel button will be disabled:

- When the Cancel button is clicked and the message Validation being Canceled appears in the dialog box, or
- When the validation reaches a stage where the action cannot be canceled.

NOTES:

- When the validation is cancelled, the validation status pane indicates a NOT VALIDATED status.
- If the validation of a display is cancelled and the file is saved, the previously validated information will be deleted and the display must be validated from the beginning. If you do not save the file, earlier validation status remains.

2. Managing Displays

2.16. Batch Validation

Validation Cancel Sequences

Canceling the validation applies to Offline Validation and Online Validation and is available from the Validate menu and the Build menu, or whenever a Validate operation occurs.

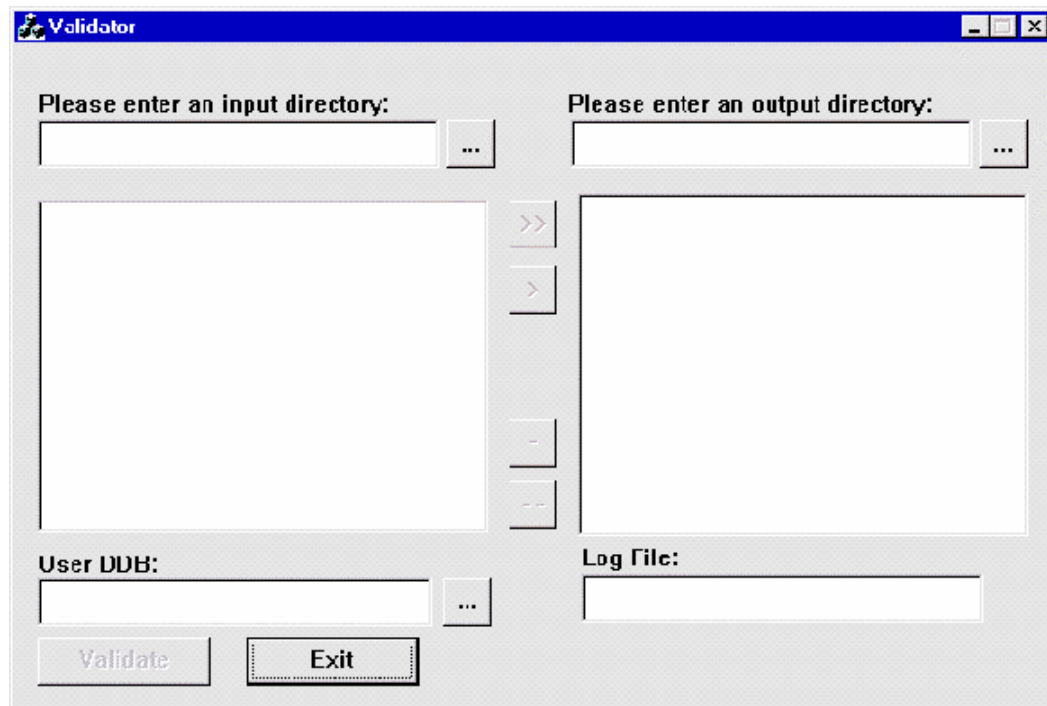
Validation Cancel is not available during validation when the following GPB tools are used:

- Bulk Replace
- Validator
- Migrator
- Translator

2.16 Batch Validation

The GUS Validator is an application that allows users to validate .PCT files outside of Display Builder. Users can validate multiple files in a single session (batch validation).

Validator uses Display Builder OLE Automation routines to perform batch validation on .PCT files, therefore the Validator will only work on a GUS node with the Display Builder application installed. The Validator will validate either on-line or off-line according to the current configuration for GPB on that node. For example, if GPB is currently configured to run in off-line mode, the Validator will validate pictures off-line.



To validate a directory of .PCT files

- 1 Enter the directory path in the Input Directory field in the Validator dialog box.
To locate the directory, click the browse button  to the right of the Input Directory field and navigate to the required directory.
When selected, the directory path appears in the Input Directory field.
- 2 Specify the output directory path in the Output Directory field.
- 3 Click .PCT filenames in the input directory window to select files to be validated.

Select  to add selected files to the validation list, or select {bmc30390.BMP} to add all input directory files to the output directory list.
The output list displays each file name with a grayed-out icon.

2. Managing Displays

2.17. Validation Error List (Error List Help)

NOTE: If you make a mistake, select  to remove a selected .PCT file from the output list, or select  to remove all .PCT files from the output list.

When the output list is complete, select the Validate button. Validation will run on each of the files in the output list.

When validation of a file is successful, the icon is no longer gray. If validation of a file does not succeed, the icon will have a red X through it.

Validation results are logged in a log file. If a full path is not specified, the default path for the log file is the output directory. If no validation log file is specified, a default log file, Validate.txt, is created in the output directory.

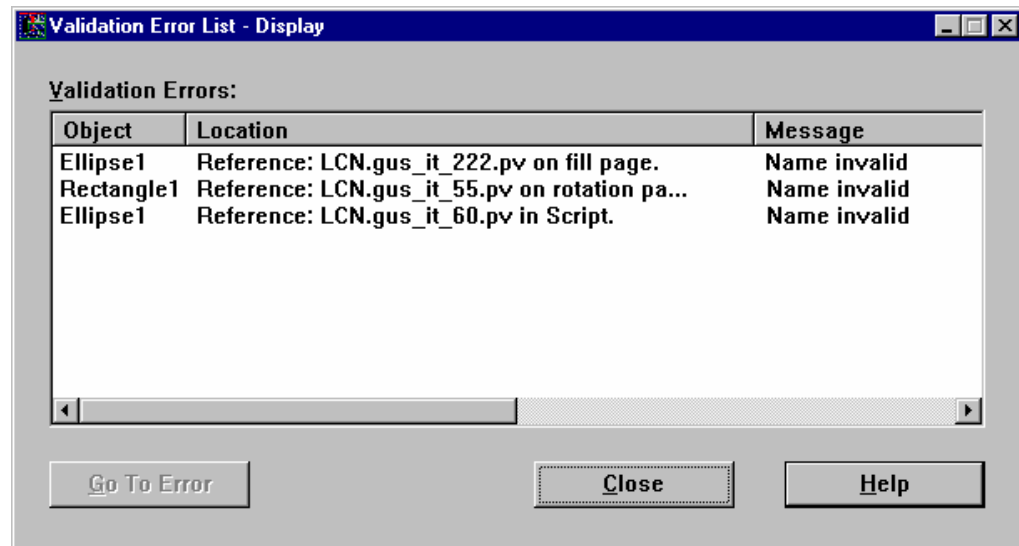
User-defined DDBs are optional for validation. When the User DDB browse button is selected, the .df files appear in a tree control along with drives and directory folders.

2.17 Validation Error List (Error List Help)

The Validation Error List displays a list of errors identified during the validation of a display. The errors identified for a display are:

- Syntax errors found in all scripts (graphic object scripts, embedded display scripts, display script itself, and so on.) in a display and variable expressions on objects' dynamics (both offline and online validation).
- Data reference errors found in all scripts (graphic object scripts, embedded display scripts, display script itself, and so on.) in a display and variable expressions on objects' dynamics (online validation only).

The Validation Error List dialog box is displayed automatically at the end of a display validation if errors are found. This dialog box will always appear on top of all other displays.



Description of data in each column of the Validation Error List

These are the descriptions of the data in each column.

- **Object** - the name of the object whose script and/or dynamic has the error.
- **Location** - identifies the data reference in error and the location of the error (script or which object dynamics).
- **Message** - contains text explaining the error.

Behavior of the Validation Error List

Dividers between columns can be dragged left-and-right to set column widths.

Headings on the columns are buttons. Click one of them to sort that column in ascending alphabetical order.

Select one list item at a time. Multiple selections of list items are not allowed.

Double-clicking a list item invokes the “Go to Error” behavior.

2. Managing Displays

2.17. Validation Error List (Error List Help)

Behavior of the “Go to Error” button

Use the “Go to Error” button to find the source of the error. Clicking it invokes the following actions.

- 1 If the error is in the object's script, the object's script is open. You do not go to the line in error nor is the line in error highlighted in the script.
- 2 If the error is in the object's dynamics, the object's Property Sheet is opened to the page containing the error. The error is not highlighted on the page.

Persistence of the Error List

When the Validation Error List dialog box is closed, it is not saved and cannot be invoked again. You must execute the Display/Validate command to generate a new error list.

If two displays are open in the Display Builder and a Validation Error List dialog is open for one of the displays, running Display Validation on the other display and getting errors causes the Validation Error List in the dialog to be replaced with the new Validation Error List for the second display.

Validation Error List

This is a list of Validation Errors ONLY. For other errors, refer to GUS Display Builder Error List.

Error message	When error check done	Explanation	Resolution
“A HOPC Server error occurred during display validation.\n”	During Display Validation	The HOPC Server compile method failed.	Try again, or restart the server or display.
“Cannot locate validation error.\nDisplay object may have been deleted.”	When invoking “Go to Error” function on Validation Error List dialog	The builder cannot find the object having the selected error.	Verify that the object has been deleted. You may want to validate the display again.
“Name index error”	During Display Validation	An expression used as the index of an array point or array parameter was not valid.	Using the Validation Error List dialog, “go to” the specific error to fix it.

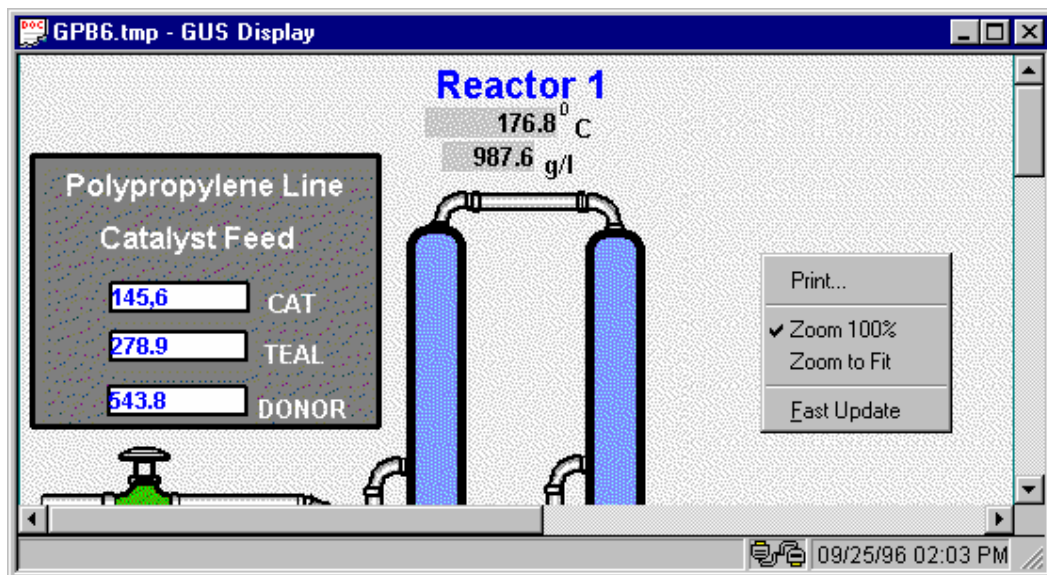
2. Managing Displays

2.17. Validation Error List (Error List Help)

Error message	When error check done	Explanation	Resolution
"Name not found"	During Display Validation	A point name used in a script or dynamic was not found on the system.	Using the Validation Error List dialog, "go to" the specific error to fix it.
"Name syntax error"	During Display Validation	A badly formed data object name was used.	Using the Validation Error List dialog, "go to" the specific error to fix it.
"This display contains more than 100 LCN scanned variables. If you are using the Display 100 runtime option, your display will be unusable. Select "Yes" to disable this warning for future display validations."	During Display Validation	Warning to user (message is explicit)	If you have Display Max or Multiple Displays installed, disable this message. If you have Display 100 installed, remove some points in the display.

2.18 Running Displays

Once a display is constructed in Build mode on the Display Builder, the display is placed in run mode to operate it. The display window looks like this when in run mode.



The **runtime** display window contains the following:

- **Title Bar**—at the top with standard minimize, maximize, and close icons.
- **Window**—(gray background area) containing the runtime display.
- **Menu**—accessed from the right-mouse button and shown opened here. Use this menu to:
 - **Save View As Bitmap** —saves a bitmap image of the area bounded by the display's window.
 - **Print**—sends a copy of the current runtime display to a printer. This includes portions of the display that are hidden, but may not include data displayed within ActiveX controls.
 - **Print Window** —sends a bitmap copy of the visible portion of the current runtime display to a printer.

- **Print Screen**—sends a bitmap of the entire screen (including the area covered by multiple monitors).
- **Zoom 100%**—sets the runtime display to full size.
- **Zoom to Fit**—shrinks the runtime display so it all fits in the window. The aspect ratio of the display is preserved while the full display is fit to the window.
- **Zoom >**—Offers a choice of Zooming contents of the display to 25%, 50%, 100%, 200%, or 400% of its configured size.
- **Fast Update**—sets the data from the Fast Update Group to be updated once per second. Only one runtime display at a time can be in fast update. Note: the LED on the Fast IKB key reflects the Native Window status and is not affected by placing a Gus Display in Fast Update mode.
- **Status Bar**—at the bottom containing:
 - Prompt Dialog, at the left (blank here), showing prompt information for the operator.
 - Connect/Disconnect icon, in the middle, showing if the display is connected to a process.
 - Date/Time, at the right, showing system date and time.

2. Managing Displays

2.19. Setting Build Mode

There are several ways to run a display:

- Running a Display from the Toolbar.
- Running a Display from the Builder using the Display Builder menu.
- Running a Display from the Desktop using RUNPIC.
- Running a Display from a Batch file.
- Running a Display from a Script using the **InvokeDisplay** function.

Refer to Built-in Functions and Statements in the *Display Scripting User's Guide*, in the GUS Scripting section for further information.

Unless you have the Multiple Displays software, you can ONLY run one display at a time, regardless of how it is invoked.

At runtime, up to four GUS displays may be active concurrently.

If runtime errors are encountered, refer to GUS Display Builder Error List for an explanation. You may also want to supply OnError handlers in your scripts to handle errors differently than in the default manner. For instance, entering an invalid point.parameter displays a default “1052 config error” error message. Refer to Bad Status Examples in the *Display Scripting User's Guide*, in the GUS Scripting Introduction section for further information.

When you finish running the display, you can simply close the runtime display by using the Close menu item in the upper left corner or the Close icon in the upper right corner.

2.19 Setting Build Mode

Display/Build

Use the Build mode to construct displays and write scripts. To set the build mode

- 1 Notice there is no check mark beside the Build menu item.
- 2 Simply select the Display/Build menu item.

RESULT: The mode changes and the menu closes. If the Display menu is opened again, you will see that the check mark in the menu now marks the Build mode.

2.20 Running a Display from the Builder

The Run command launches a Display runtime, using the active display. There are two ways to set the run mode: (1) from the Display/Run menu, and (2) from the Standard toolbar.

How to run a display from the Display/Run menu

- 1 Notice there is no check mark beside the Run menu item.
- 2 Simply select the Display/Run menu item.

RESULT: The active display is run. Also, the mode changes and the menu closes. If the Display menu is opened again, you will see that the check mark in the menu now marks the Run mode.

How to run a display from the Standard toolbar

To run a display from the Standard toolbar, select the Run tool .

RESULT: The active display will be run.

2.21 Running a Display from the Desktop using RUNPIC

- 1 Select Start > Run on your desktop. The Run dialog box appears.
- 2 Enter runpic <filename> into the Open text input port, where filename can be either
 - a full pathname specification, or
 - a filename with a .pct extension, or
 - a display name with no extension.

An example of a typical input string is "runpic d:\gus\displays\valve1.pct."

NOTE: If the filename string is not a full pathname, the display search path will be used to locate the display. The directory where your displays are stored must appear in the Pathname Catalog. If it isn't there, you will get an error message stating that the abstract cannot be found.

- 3 Select OK. The Run dialog box disappears and the requested display runs.

2. Managing Displays

2.22. Running a Display from a Batch File

2.22 Running a Display from a Batch File

Here is an example of how to run a display called "tank1" from a batch file:

```
operator.bat
start picmanc.exe
start runpic -w sss d:\gus\displays\tank1.pct
```

w is an optional command line "wait" switch that checks to see if PICMANC (the GUS Picture Manager) is initialized. Without this flag, requests to run GUS displays made before picmanc.exe has completed its startup will fail and an error message will be displayed.

sss is an optional parameter that can be used with **w**. It specifies the number of seconds to wait for PICMANC to get initialized. For example, an **sss** value of "600" would specify a wait time of 10 minutes (600 seconds). If no wait time parameter, or a value of 0, or a negative value is entered, a default value of "900" seconds (15 minutes) will be used.

If PICMANC initialization is in progress but doesn't get completed within the specified wait time, the error message "Picture Manager initialization is not complete - please try invocation later." will be displayed.

If PICMANC initialization has not yet begun, for example, PICMANC has not been invoked, after the specified wait time the error message "Picture Manager has not been initialized - run picmanc.exe first" will be displayed.

2.23 Printing

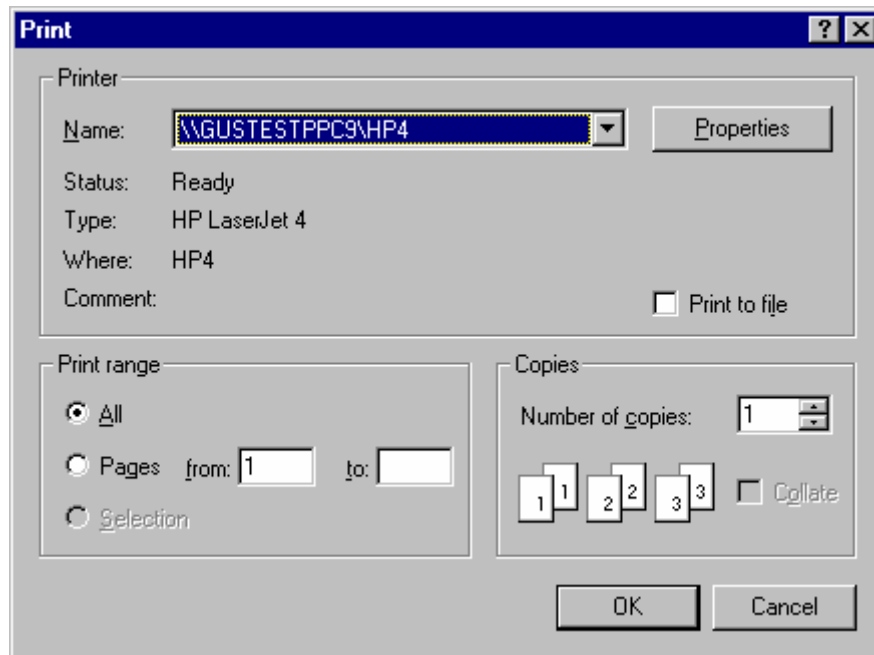
There are three Print commands:

- Printing a display or script (File/Print menu and Print on Toolbar)
- Previewing a display or script before printing (File/Print Preview menu)
- Setting printing options (File/Print Setup menu)

2.24 Printing a display or script

File/Print menu and Print on Toolbar

- 1 If the Display or Script window to be printed is not active, select it by clicking once on the window's title bar.
- 2 Select File/Print from the menu. The Print dialog box appears.



- 3 Select a printer Name.
- 4 Select dialog box options as appropriate to print the range of Pages and Number of Copies. To learn about each feature of the Print dialog, click the  in the upper right corner of the box; then click the required feature for a short explanation.
- 5 Click OK.

RESULT: The print job is submitted to the printer.

2. Managing Displays

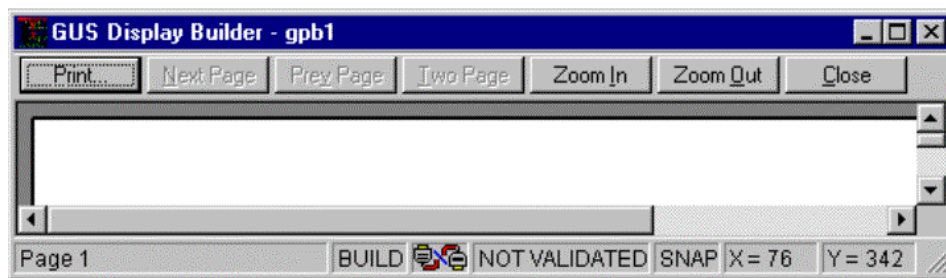
2.25. Previewing a display or script before printing

NOTE: To print a listing of all settings, scripts, and other display information for a specific display, position the cursor anywhere on the display window and select <Alt-L>. A text list appears and can be printed.

2.25 Previewing a display or script before printing

File/Print Preview menu

- 1 If the Display or Script window to be printed is not active, select it by clicking once on the window's title bar.
- 2 Select File/Print Preview from the menu. The Print Preview window appears showing a small view of the display window to be printed with control buttons along the top.



- 3 You can choose to
 - Print the file,
 - Zoom In and Zoom Out, or
 - Close the box without printing.

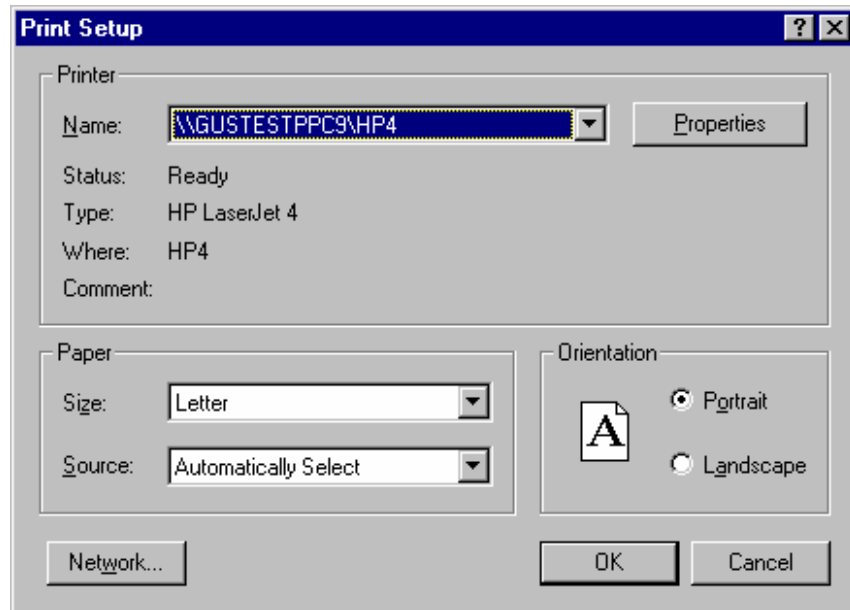
2.26 Setting printing options

File/Print Setup menu

Use the Print Setup command to select options such as which printer to choose and the required page orientation.

Changing the printer setup

- 1 Select File/Print Setup from the menu. The Print Setup dialog box appears.



- 2 Select dialog box options as appropriate. To learn about each feature of Print Setup, click the  in the upper right corner of the box; then click the required feature for a short explanation.
- 3 Click OK to set the options selected. Click Cancel to keep current settings.

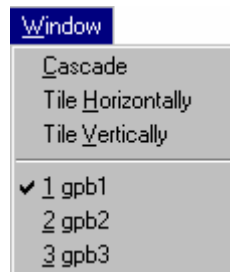
Invert Print Preview

The Invert Print Preview option reverses the light and dark background and foreground of displays, making displays with predominantly dark backgrounds much easier to view when printed.

2.27 Switching Between Displays

Window menu

The Window Menu contains the following selections:



At the bottom of this example Window menu, three display windows (GPB1, GPB2, and GPB3) have been opened and the check mark shows the display window named GPB1 is currently active.

To switch from one display to another, click the name of the display you want to open.

RESULT: The previously active display becomes inactive and the display you selected moves to the front and becomes active. The menu closes. If the Window menu is opened again, you will see that the check mark in the menu now marks the display you have selected as active.

2.28 Considerations when using Zoom to Fit displays at Runtime

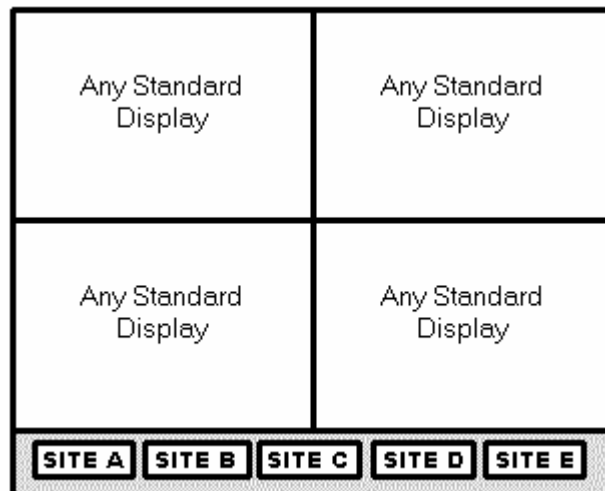
Display files have a “Zoom to Fit” option that can be accessed by either of these property pages:

- General Page—Display Builder Default Property Sheet used to set properties for ALL DISPLAYS you will build.
- General Page—Display Object Property Sheet used to set properties for ALL OBJECTS you build in the CURRENT DISPLAY.

When you check the Zoom to Fit option, the display is always sized to the window in which it is invoked, and its aspect ratio is retained regardless of the size of the window. However, in many cases it is better if the display’s “canvas” (background size) is the same size as the client area of the SafeView window in which you plan to use the display.

Consider the SafeView example labeled “Example 1.” The long display in the bottom window is a Navigator display that has object “buttons” scripted to allow a user to call up other standard displays that appear in the windows above it.

Let us also assume the Navigator Display was built so that its canvas size is the same as the client area at the bottom of the overall SafeView window.



Example 1

2. Managing Displays

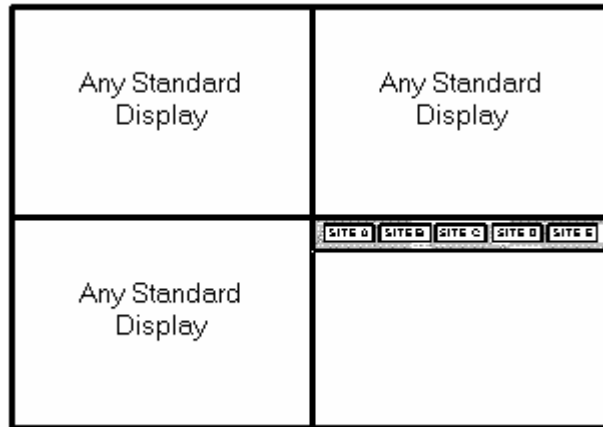
2.28. Considerations when using Zoom to Fit displays at Runtime

If the Zoom to Fit option was checked when the Navigator Display was built, it will be resized to fit when opened in a much smaller window, as shown in Example 2.

In this case, text on the buttons are unreadable because the Zoom to Fit option scaled the display so small.

This problem of scaling text in displays is also present when scaling OLE Objects like buttons, etc.

Consider the rules that follow when you let the Zoom to Fit option scale objects.



Example 2

Rules that apply when objects are allowed to scale

- 1 Text Bounding Box**—Do not let the bounding box that describes a Text Object form a “tight fit” around the text. Increase the bounding box size to allow for some “padding” because, when the display is scaled, the text may appear larger than the box enclosing it.
- 2 TrueType Fonts compared to others**—If non-TrueType fonts are used, the bounding box must be increased because they need more “padding” than TrueType fonts.
- 3 OLE Object text does not scale**—consider these workarounds when entering text in OLE objects.
 - Select a font that is readable at more “reasonable” sizes, like 10 or 12 point fonts.
 - Set the “Selectable property” of a Text Object on a button to false (check off) so the button sees the click and the display will scale to any size.

3. Managing Your Environment

3.1 Overview

Procedures used to view, change, and set modes of various display windows are presented in this section.

Topics in this section are:

- Setting Build Session Data Access Mode (File/Online, Offline)
- Changing the Mode to Online
- Changing the Mode to Offline
- Changing the View
- Zooming the Display to Different Magnifications
- Organizing Multiple Windows
- Closing a Window
- Working with Control Menus
- Exiting the Display Builder (File/Exit menu)
- Managing Pathnames

3.2 Setting Build Session Data Access Mode

File/Online, Offline

You can have only one data access mode (Online or Offline) active at a time. The mode that is active has a check mark beside it. These commands control the level of validation that takes place when validation occurs.

- When in the Offline mode, the validation process checks for syntax errors in all scripts and on all object dynamics.
- When in the Online mode, validation checks are also performed on runtime parameters such as process data references.

3.3 Changing the mode to Online

Toggling from offline to online

- 1 Notice that there is a check mark beside the offline menu item.
- 2 Select the File/Online menu item.

RESULT: The mode changes and the menu closes. If the File menu is opened again, you will see that the check mark in the menu now marks the online mode.

3.4 Changing the mode to Offline

Toggling from online to offline

- 1 Notice that there is a check mark beside the online menu item.
- 2 Simply select the File/Offline menu item.

RESULT: The mode changes and the menu closes. If the File menu is opened again, you will see that the check mark in the menu now marks the offline mode.

3.5 How Default Properties Propagate

Basic properties

The Display Builder can contain one or more display windows and each display window can in turn include display objects such as lines, rectangles, etc. Display objects have properties, such as fill color and line width, which govern the way the objects appear in the display. These display object properties are set to defaults when the objects are drawn and then can be altered.

Display Object Properties—The Display Builder allows you to set these default object properties for new display objects. Because these properties apply to all newly drawn objects for a given display these properties are included with the list of display properties for that display. These display object properties are accessed in several ways:

- by double-clicking the display background,
- by using the right-mouse button menu to choose Properties with the cursor over the display background, or
- by choosing Display Properties from the Edit menu.

Individual Object Properties—When you first draw an individual object (like a rectangle or an ellipse) it contains properties taken from the display object properties mentioned above. You can change an individual object's properties in several ways:

- by double-clicking the individual object,
- by using the Right-mouse Button menu to choose Properties with the cursor over the object, or
- by selecting the object and choosing Object Properties from the Edit menu.

Default Properties—When displays are first created, the display properties are defaulted to values set in the “Default Properties” dialog box. Properties that apply to the display window alone, such as background color, are also set in the Default properties dialog box.

They are set using the Defaults command from the Edit menu. Once these default properties are changed, they apply to NEW display windows.

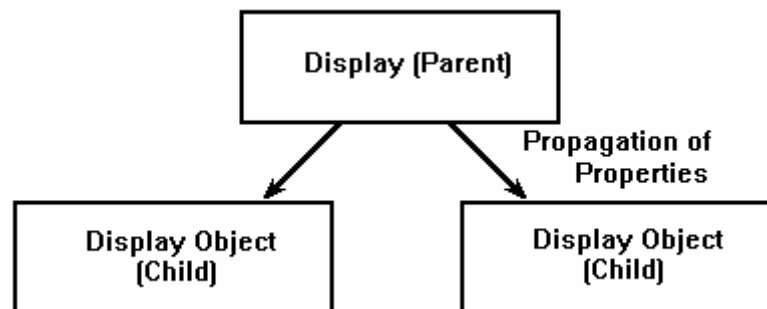
Inheritance—Because new objects added to a display get their default properties from the display window properties they are said to “inherit” these properties from the display.

Propagation—Because display properties are passed on as defaults for new objects they are said to propagate down to these objects as in a Parent/Child relationship.

- A child inherits properties from its parent.
- Properties are propagated from a parent to its children.

Propagation of properties

The concept of propagation can be best represented as a hierarchy of objects:



3. Managing Your Environment

3.6. Inheriting Properties

The properties that propagate to display objects include:

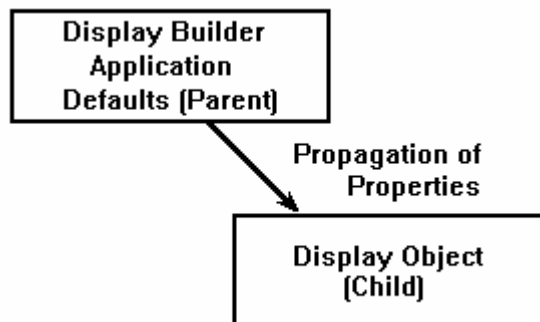
- Line Properties (Width, Color, and Style)
- Fill Properties (Color and Pattern)
- Text Properties for Text Objects (Color, Font, Alignment, and so on.)
- Blink Property

NOTE: The Blink property does not apply to embedded displays.

3.6 Inheriting Properties

Default properties are inherited by display object properties

The inheritance/propagation concept with respect to display objects is represented like this:



Note that default properties are assigned for displays when the display is created. Subsequent changes to the Display Builder defaults will have no effect on existing displays.

Inheritance of group object properties

A set of objects in a display can be grouped together into a new display object called a group. Once objects are grouped, the group becomes the new parent to the grouped objects and becomes a child of the display. The group has a set of properties just like a display that includes the same properties mentioned above.

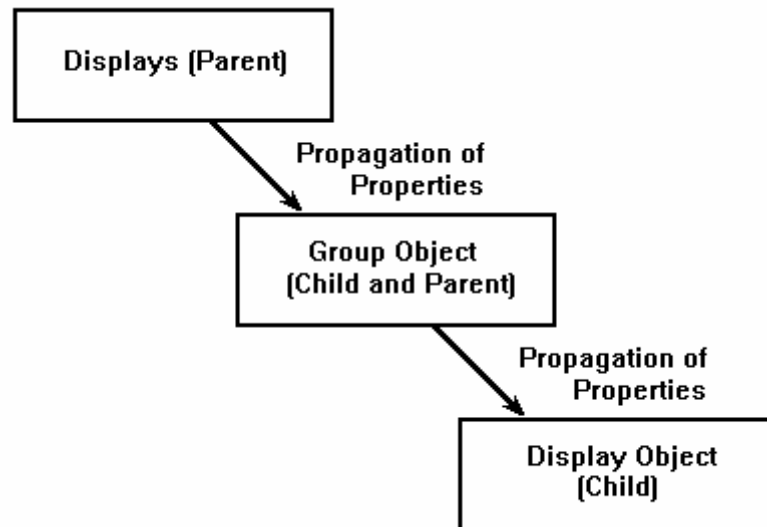
Group Object Properties—Group properties can be accessed in several ways:

- by double-clicking the group,
- by using the right-mouse button menu to choose Properties with the cursor over the group, or
- by selecting the group and choosing Object Properties from the Edit menu.

These properties are propagated to the objects that have been programmed to inherit from their parent. The Group itself can inherit properties from the display.

Following the creation of a group you can specify, for that group, whether subsequent changes to the Display properties will be propagated to the group. This is done by selecting the “Inherit Enabled” checkbox on the group object Property Sheet.

Group Inheritance—The inheritance/propagation concept with respect to groups is represented like this:



3. Managing Your Environment

3.6. Inheriting Properties

Procedure for Setting Group Properties—Groups provide one way to change the properties of a set of display objects without having to change the properties for each object individually. The steps to do this are:

- 1 Set the Inherit Enabled property on the individual objects.
- 2 Group the objects.
- 3 Change the properties for the group.
- 4 If the properties of the display are likely to change, ungroup the grouped objects and reset the Inherit Enabled property on the individual objects to keep them from changing.

These changed properties then propagate down to the individual objects.

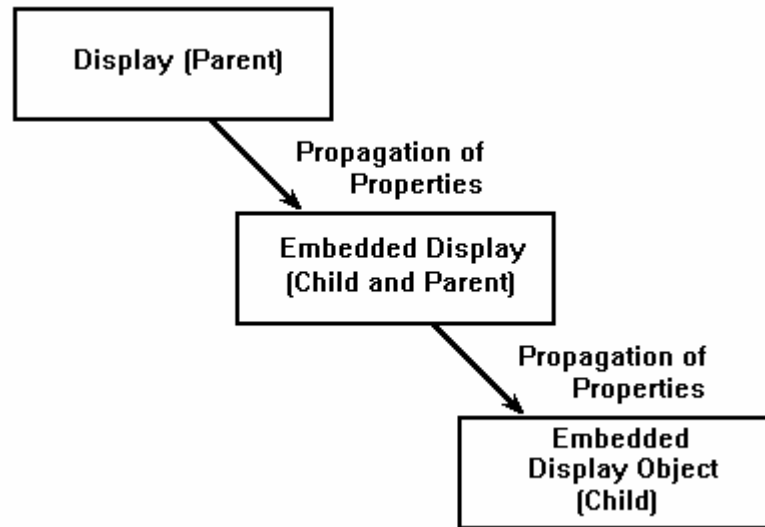
Inheritance of embedded display object properties

Pre-built displays can be inserted into larger displays. The inserted or “embedded” displays then become display objects in the containing display.

Inheritance—These embedded displays automatically inherit the properties of the containing display when they are inserted; however, only the objects within the display that have been set to inherit will take on these inherited properties.

NOTE: The “.fill” property is not available on the property sheets, but is available through scripting. Using the “.fill” property in scripting is conceptually the same as setting the fill pattern to none. This property is inherited and propagated along with FillPattern. (Refer to, Changing object properties through scripts)

Propagation—Following insertion, the behavior of embedded displays, with respect to inheritance and propagation of properties, is identical to that described for groups.



Procedure for Setting Embedded Display Properties—An embedded display has a set of properties just like any display object and includes the same properties that can be propagated. Embedded display properties can be accessed in several ways

- by double-clicking the embedded display,
- by using the right-mouse button menu to choose Properties with the cursor over the embedded display, or
- by selecting the embedded display and choosing Object Properties from the Edit menu.

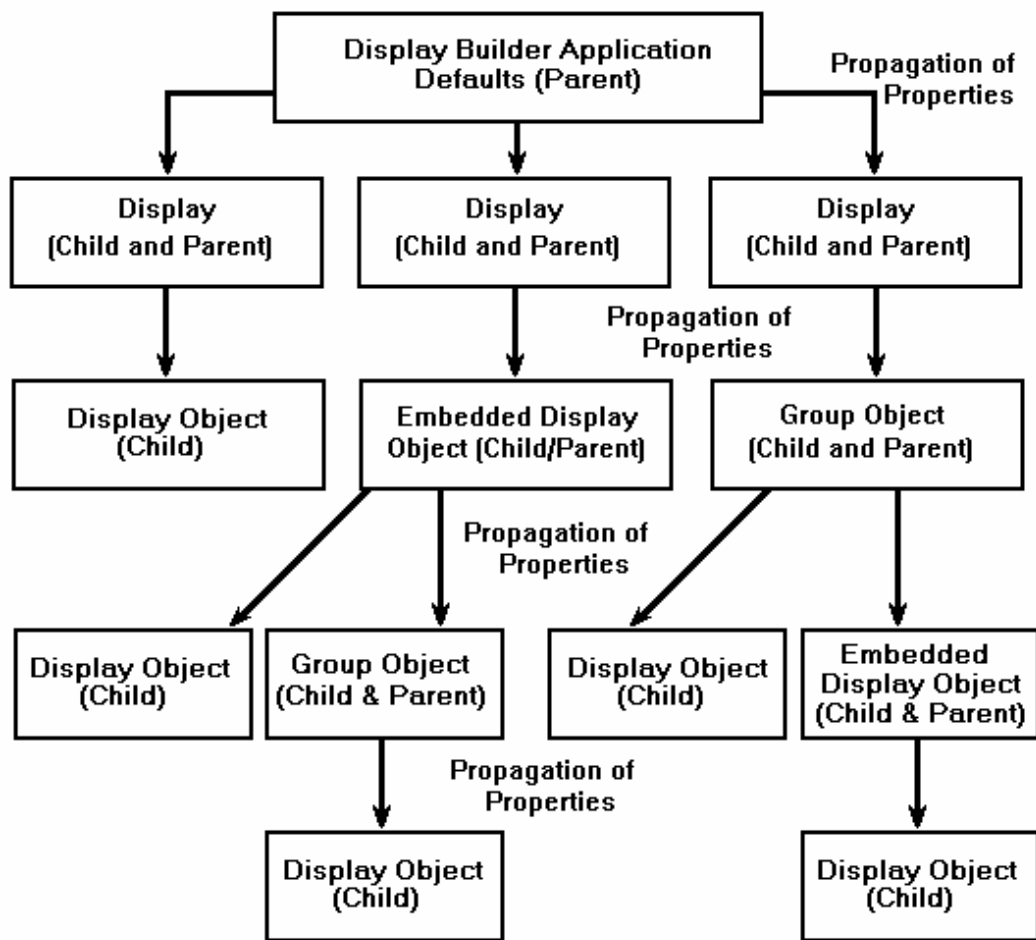
These properties are propagated to the objects that have been programmed to inherit properties. The embedded display itself can inherit properties from the display.

Allowing Inherited Properties—Following the insertion of an embedded display you can specify whether subsequent changes to the Display properties will be propagated to the embedded display. Do this by selecting the “Inherit Enabled” checkbox on the embedded display object Property Sheet.

Summary

This overview of propagation hierarchy for the Display Builder covers most combinations.

Notice that, after initial creation, properties of the parent propagate to a child ONLY if the child “Inherit Enabled” checkbox is true.



3.7 Changing the View

Several aids are built into the Display Builder. Those that change the view are:

- Viewing the Grid (View/Grid menu)
- Snapping to Grid (View/Snap to Grid menu)
- Viewing Invisible Objects (View/Invisible Objects menu)
- Showing Invisible Objects on Open (View menu)
- Viewing Toolbars (View/Toolbars menu)
- Viewing the Status Bar (View/Status Bar menu)

3.8 Viewing the Grid

View/Grid menu and Show Grid on Toolbar

Use this command to display a grid on the Display Builder window, behind the objects in your display, and to remove it. This grid can help you to align and size objects. There are two ways to display or remove the grid: (1) from the View menu, and (2) from the Standard toolbar.

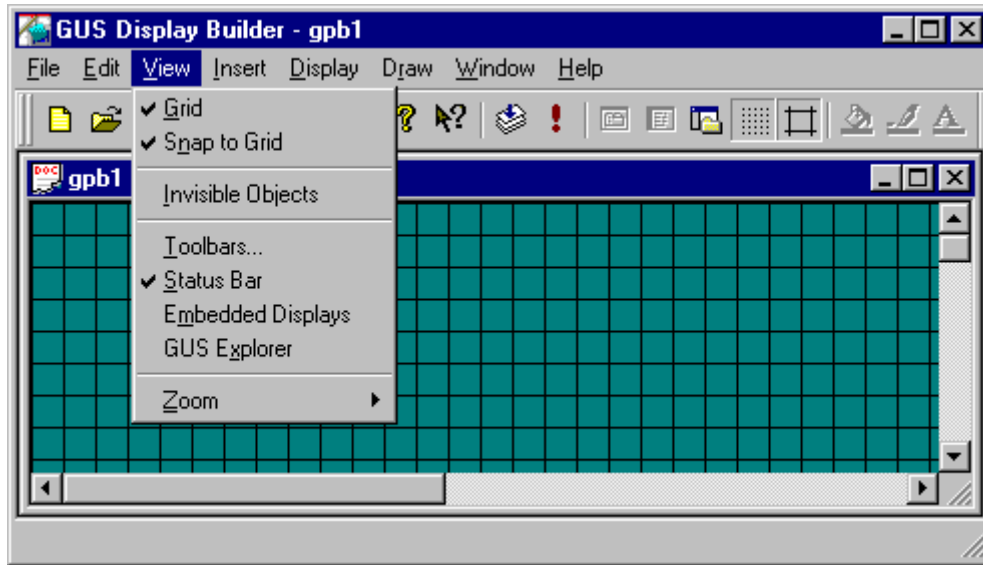
How to display or remove the grid from the View menu

To display or remove the grid, select View, then select the Grid command.

RESULT: The View/Grid command toggles between displaying the grid and removing it. A check mark marks the enabled state.

3. Managing Your Environment

3.9. Snapping to Grid



How to display or remove the grid from the Standard toolbar

To display or remove the grid using the Standard toolbar, select the display grid button . If the grid was not on display, it will be displayed. If it was on display it will be removed.

3.9 Snapping to Grid

View/Snap to Grid menu and Snap on Toolbar

Use this command to cause the outline of objects you are constructing to “snap” to the grid. In this way, similar objects can be set to the same size and aligned in a pleasing way.

This command is a toggle and only affects an object that you create or move after the command was invoked.

There are two ways to display or remove the grid: (1) from the View menu, and (2) from the Standard toolbar.

How to enable or disable Snap-to-Grid from the View menu

Click the View/Snap to Grid command.

RESULT: The command toggles between enabling Snap-to-Grid and disabling it. A check mark marks the enabled state.



How to enable or disable Snap-to-Grid from the Standard toolbar

Select the snap to grid button . The button toggles between enabling Snap-to-Grid functionality or disabling it.

3.10 Viewing Invisible Objects

View/Invisible Objects menu

Use this command to make invisible objects visible to the viewer. When a display is opened or a new display is invoked, Invisible Objects is always OFF.

This command is a toggle. It shows or hides all invisible objects in a display.

To toggle Invisible Objects

Click the View/Invisible Objects command.

RESULT: The command toggles between showing Invisible Objects or hiding them. A check mark marks the "show" state.

3. Managing Your Environment

3.11. Showing Invisible Objects on Open

3.11 Showing Invisible Objects on Open

Show Invisible Objects on Open menu

Use this command to make invisible objects appear in each newly opened or created display in the Builder session. When a display is opened or a new display is invoked, Invisible Objects is always ON.

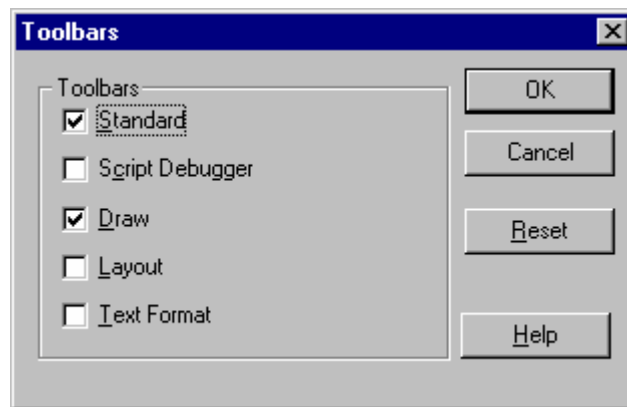
Click the View/Show Invisible on Open command.

RESULT: Invisible objects in each display that is opened or created will be visibly display within Display Builder.

3.12 Viewing Toolbars

View/Toolbars menu

All toolbars in the Display Builder can be selected for display with this command.



How to use the Toolbars dialog box

- 1 Select View/Toolbars from the menu or click the Help button on the Toolbar dialog box.
- 2 Click the checkboxes so that a "✓" is beside each toolbar you want displayed. Toolbars without check marks will not be displayed.
- 3 Click OK to accept the command and close the dialog box.

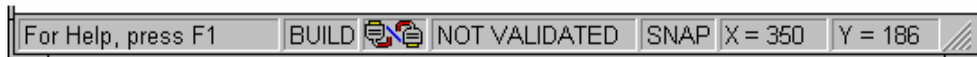
3.13 Viewing the Status Bar

View/Status Bar menu

The status bar shows information at the bottom of the Display Builder Window that helps you use the Display Builder.

The status bar changes, depending on whether you are building an object (Buildtime) or running GUS with the TPS Network (Runtime). The status bar cannot be hidden at Runtime.

Buildtime Status Bar



The buildtime status bar contains these areas:

- **Message Area**—Used for operator help prompts and various system messages.
- **Mode of Operation**—Displays the current Display Mode (Build or Run).
- **Online/Offline**—Displays whether you are building online (cable connected) or offline (cable broken as shown here).
- **Display Validation State**—Tells the operator if the display is Validated (has no errors) or Not Validated (contains errors).
- **Snap to Grid**—Indicates if Snap to Grid is turned on. If off, the field is blank.
- **X and Y Cursor Coordinates**—Shows the current cursor position (X, Y) in pixels from the upper left corner of the display window.

3. Managing Your Environment

3.14. Zooming the Display to Different Magnifications

Runtime Status Bar



The runtime status bar contains these areas

- **Message Area**—Used for error messages and various system messages.
- **Communication Status**—Tells the operator if communications with the TPS Network is currently connected (Online with cable connected) or disconnected (Offline with cable broken as shown).
- **Date Time**—Shows the current date and time.

How to Show or Hide the Status Bar

- Click the View/Status Bar.

RESULT: The Status Bar toggles ON or OFF. If the Status Bar is displayed, a check mark appears on the menu. The status bar cannot be hidden at Runtime.

3.14 Zooming the Display to Different Magnifications

Use the Zoom commands to show more or less detail in the current display. They are

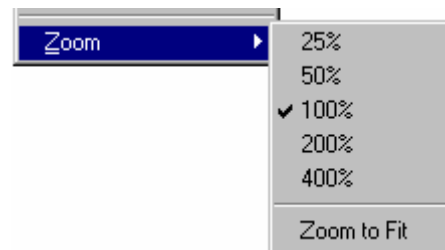
- How to zoom the display to another magnification (View/Zoom/ %)
- How to zoom the display to fit the window size (View/Zoom/Zoom to Fit menu)

3.15 How to zoom the display to another magnification

View/Zoom/%

There are five percentages of magnification on the menu. The current magnification is checked.

Select another percentage from the Zoom menu.



RESULT: The display is set to the required magnification. As the zoom occurs, Display Builder attempts to keep the selected object(s) in view.

3.16 How to zoom the display to fit window size

View/Zoom/Zoom to Fit menu

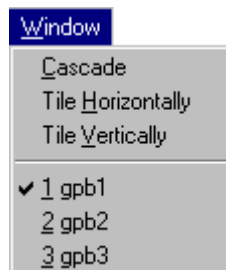
Zooms to show the full display in the display window.

Select View, then select Zoom to Fit from the Zoom menu. (See previous illustration)

RESULT: The entire display is shown in the display window.

3.17 Organizing Multiple Windows

The Window Menu contains the following selections:



3. Managing Your Environment

3.17. Organizing Multiple Windows

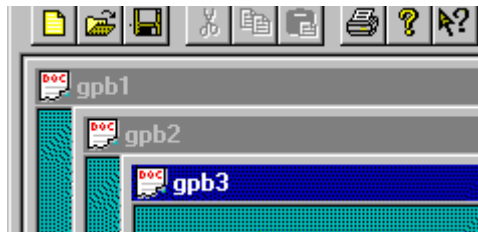
Use the first three window items (Cascade, Tile Horizontally, and Tile Vertically) to determine how you would like to view multiple windows.

Topics in this section are:

- Cascading Windows (Window/Cascade menu)
- Tiling Windows Horizontally (Window/Tile Horizontally menu)
- Tiling Windows Vertically (Window/Tile Vertically menu)

Cascading Windows (Window/Cascade menu)

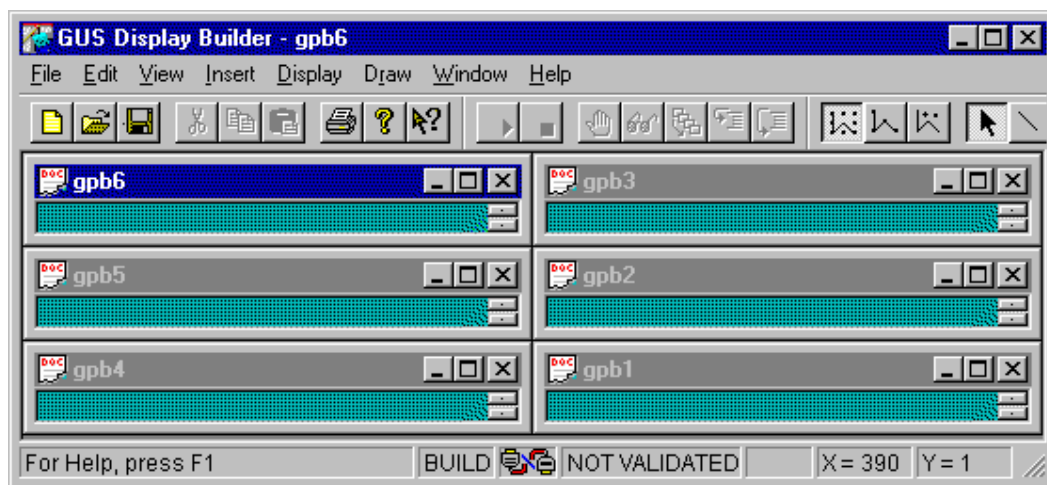
Select Cascade to display multiple windows that appear to overlap each other with only the title bar and left edge of each window, except the active (top) window, showing.



Tiling Windows Horizontally (Window/Tile Horizontally menu)

Select Tile Horizontally to display multiple windows that appear as if they were “tiles” in the Display Builder window, displayed in sequence, with the last active window at the top left.

Here is an example of six display windows tiled horizontally.



NOTE: Dialog Editor, Embedded Pictures, and Properties windows are not affected by tiling within the Display Builder.

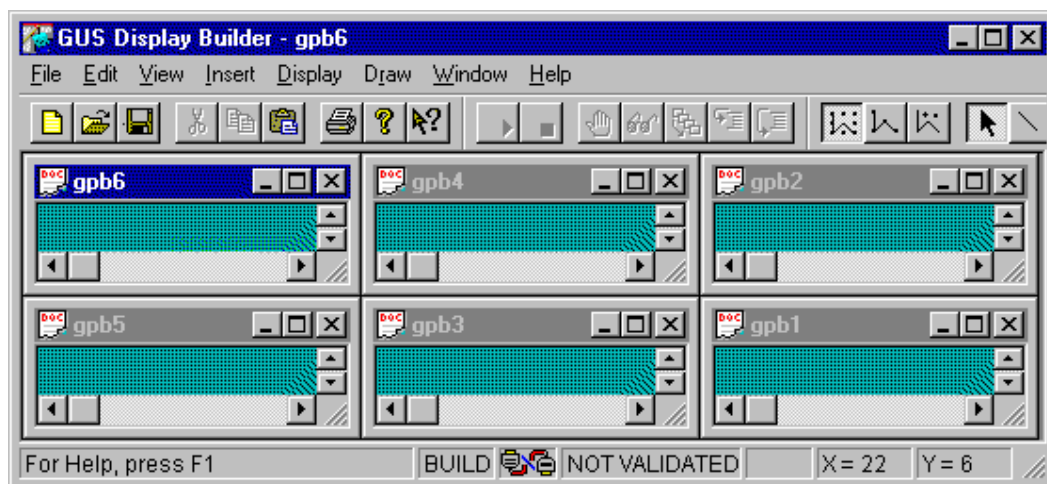
3. Managing Your Environment

3.17. Organizing Multiple Windows

Tiling Windows Vertically (Window/Tile Vertically menu)

Select Tile Vertically to display multiple windows that appear as if they were “tiles” in the Display Builder window, displayed in sequence, with the last active window at the top left.

Here is an example of six display windows tiled vertically.



3.18 Closing a Window

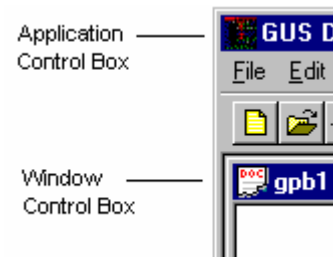
The Close command closes the selected Display Builder window. When a given window is closed, the data file associated with it is closed and, sometimes, the Display Builder application is closed as well.

The Application Control Box is located in the upper-left corner of the GUS Display Builder application. When the Close command is executed from this box, the application is closed and the Display Builder task is terminated. This function is the same as "Exit" from the File menu. Refer to Exiting the Display Builder for details.

A Window Control Box is located in the upper-left corner of each display window that is opened. When the Close command is issued from this box, the function is the same as "Close" from the File menu. The two ways to close a window are:

- from the File menu, and
- from the window control menu.

Refer to How to close a window from the File menu, and How to close a window from the window control menu.



How to close a window from the File menu

- 1 Select the display window to be closed.
- 2 Select File/Close from the menu.
- 3 If necessary, the Display Builder will prompt you to save changes.

RESULT: The selected display window is closed.

3. Managing Your Environment

3.19. Working with Control Menus

How to close a window from the window control menu

There is a Window Control Box located in the upper-left corner of each display window (it has a display window icon in it and is located just below the word "File" on the menu bar). When opened, it displays the Window Control Menu shown at right.

- If you select Close from this menu, the active display window closes.
- If you double-click the control box, the window also closes.

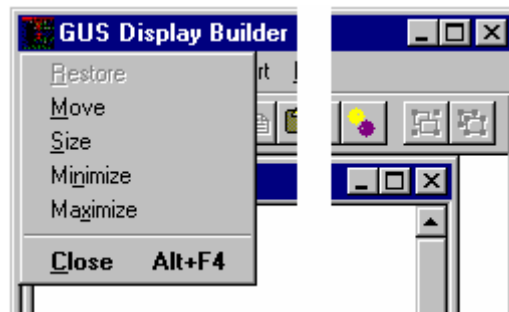
If necessary, the Display Builder will prompt you to save changes.



3.19 Working with Control Menus

Both the Application Control Box and the Window Control Box each open control menus that have a similar appearance to the ones shown at the right.

An explanation of how each of the commands in the drop-down menu works appears as follows.



- **Restore**—causes a minimized window to be opened again. You can do the same task by double-clicking the control box.
- **Move**—causes the cursor to change to a cross of double-ended arrows. Place the cursor over the title bar, then click and drag the title bar to move the window. You can do the same task by placing the arrow cursor over the title bar, then click and drag the title bar to move the window.

- **Size**—causes the cursor to change to a cross of double-ended arrows. Place the cursor over any window border or corner and it changes to a single double-arrow. Then move the cursor in the required direction.
You can do the same task by clicking on and clicking and dragging a window border or corner as you move the cursor in the required direction.
- **Minimize (also the icon with an underline in the upper right corner)**—causes the window to collapse to a button on the Taskbar at the bottom of the screen or Display Builder window.
You can do the same task by clicking the Underline icon in the upper-right corner of the window. To restore the display window, click the Taskbar button at the bottom of the Display Builder window.
- **Maximize (also the icon with a rectangle in the upper-right corner)**—causes the Display Builder window to fill the screen.
RESULT: The rectangular icon in the upper right corner changes to a double rectangle.
To return the window to its normal size, click the double rectangle.
- **Close (also the icon with an “X” in the upper right corner)**—causes the active Display Builder window (or the display window) to be closed. If changes have been made, Display Builder will prompt you to save changes.
Refer to Closing a Window for instructions pertaining to how the close function works with the Application Control Menu and the Window Control Menu.

3.20 Exiting the Display Builder

File/Exit menu

The Exit command closes the Display Builder application and ends all interfaces with Display Builder windows and their data files. Altered window data files must be saved or they will be lost.

How to exit the Display Builder

- 1 Select File, then select Exit.
- 2 If necessary, the Display Builder will prompt you to save changes.

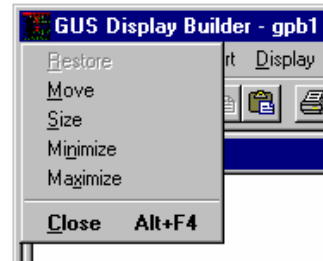
RESULT: The Display Builder window closes and the Display Builder task is ended.

3. Managing Your Environment

3.21. Managing Pathnames

CAUTION: The Application Control Box is in the upper-left corner of the GUS Display Builder application. As shown here, it has a Display Builder icon in it and is located above the word “File” on the menu bar.

The “Close” command on this menu is really an Exit command that will immediately close the GUS Display Builder application and end the task. If changes have been made, the Display Builder will prompt you to save changes.



3.21 Managing Pathnames

Custom displays (Schematics) can be invoked in a variety of ways on the Global User Station (GUS). This document details how to setup your system to take advantage of distributing classic schematics (.do files) and GUS schematics (.pct files) on local hard drives, servers or on removable media. This document will not cover the distribution of classic schematics on the TPS Network, but will focus on adding PC style pathnames in the pathname catalog in order to access new GUS Schematics.

Topics in this section are:

- Pathname Planning
- NT Registry Initialization
- Schematic Invocation

3.22 Pathname Planning

Pathname Catalog limitations

Pathnames are currently limited to a total of 14 characters. This limitation is a constraint set by the area database pathname catalog.



TIP

Use short folder names to achieve the required folder hierarchical structure.

To increase the pathname field requires extensive changes to TPS Network base software and also requires a translator for the area database.

How to change the Pathname Catalog

Before changing the Pathname Catalog, view current pathnames by following the instructions in View Current Pathnames in Catalog. Then, return here to change the pathname catalog by executing the following steps:

- 1** Select "AREA DATA BASE" on the "ENGINEERING MAIN MENU."
- 2** Select "PATHNAME CATALOG."
- 3** Push the CMND key on the keyboard to request the command display.
- 4** Select "SELECT AREA."
- 5** Select the box that corresponds to the area you wish to change.

 RESULT: The selected pathname is displayed in the box and the color is cyan.
- 6** Press CTRL+F1 to return to the Pathname Catalog window.

 RESULT: The area number is displayed at the top of the window (AREA:nn).
- 7** Press "ENTER" to clear the UNENTERED message at the top of the window.
- 8** Press CTRL+F7 to reconstitute the current pathname catalog

 NOTICE: If the warning "USING UNINSTALLED WORKFILE" appears, follow the instructions in Delete Area Work File (wa) then start again at step 1.
- 9** Update the pathnames.
- 10** Press "ENTER" to save data in memory.
- 11** Press CTRL+F12 to save data in a temporary work file.
- 12** Push the CMND key on the keyboard to request the command display.
- 13** Select "INSTALL AREA."

 RESULT: Pathname appears in white.
- 14** Press "ENTER" to validate pathname.

 RESULT: Pathname changes to cyan if valid.
- 15** Request the "Console Status Display" in the Native Window or on a standard US.
- 16** Select the node that you wish to load the new pathname catalog into.

3. Managing Your Environment

3.23. View Current Pathnames in Catalog

- 17 Select target box "AREA CHG."
- 18 Select target box, which corresponds to the area you wish to change to.
- 19 If the area database is on the HM, select target box "DEFAULT SOURCE."
- 20 Select target box "EXECUTE COMMAND" to execute the area change.

By repeating the steps in View Current Pathnames in Catalog you can validate that the pathname changes have been loaded.

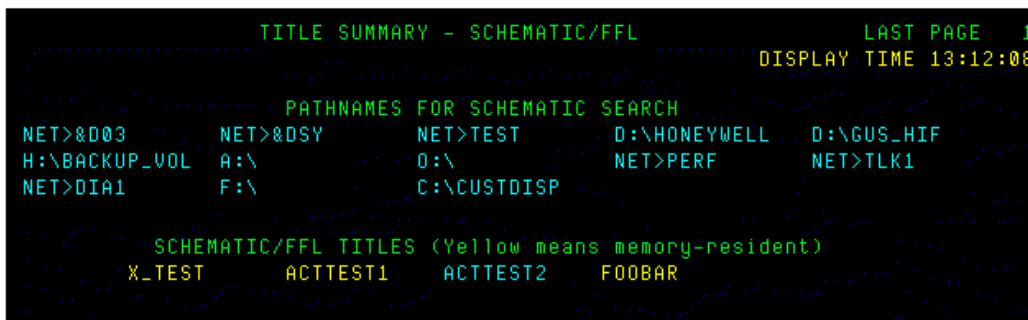
To continue reading the "Managing Pathnames" section, jump to NT Registry Initialization.

3.23 View Current Pathnames in Catalog

To view the current pathname settings, execute the following steps:

- 1 Request "SYSTEM MENU" in the Native Window or on a standard US.
- 2 Select the "ORGANIZATIONAL SUMMARY MENU" item on the menu.
- 3 Select the "SCHEMATIC/FFL TITLES" item on the menu.
- 4 Select "DISPLAY" target.

At this point, you will see a list of pathnames configured into the system for the station that the display is being viewed on (similar to this sample portion of the Native Window).



```
TITLE SUMMARY - SCHEMATIC/FFL                                LAST PAGE 1
                                                                DISPLAY TIME 13:12:08
PATHNAMES FOR SCHEMATIC SEARCH
NET>&DQ3      NET>&DSY      NET>TEST      D:\HONEYWELL  D:\GUS_HIF
H:\BACKUP_VOL A:\        O:\        NET>PERF      NET>TLK1
NET>DIA1     F:\        C:\CUSTDISP

SCHEMATIC/FFL TITLES (Yellow means memory-resident)
X_TEST      ACTTEST1    ACTTEST2    FOOBAR
```


3.24 Delete Area Work File

To delete the work file (.wa), execute the following steps:

- 1 Execute one of the following steps:
 - If you are currently in the pathname catalog Configurator display, press the “ESC” key on the keyboard. This will put you in the command processor.
 - If you are not in the pathname catalog Configurator display, go to the Main Engineering Menu and select the “COMMAND PROCESSOR” target.

RESULT: You are now in the command processor.
- 2 While in the command processor enter “DL NET>&Dnn>AREAnn.W.”

3.25 NT Registry Initialization

LCNFILELOAD.EXE is an executable file that is started when a user logs into GUS. The LcnFileLoad program will run in an idle mode until a GUS personality is loaded and running in the LCNP.

When the GUS personality is ready:

- LcnFileLoad will make a connection to GUS and maintain the link.
- After successfully connecting, LcnFileLoad will load the button file from LCNP memory into the NT memory, and copy the PC style pathnames into the NT Registry.

Communication link between NT and LCNP

If LcnFileLoad does not start at login, you can start it manually, following this procedure.

- 1 Locate LCNFILELOAD.EXE in the “Honeywell\TPS\bin” directory.
- 2 Double-click LCNFILELOAD.EXE.
- 3 Verify LcnFileLoad is running by using PVIEW.EXE, located in the same directory.

NOTE: If LcnFileLoad is started manually, make sure that all GPB processes have been terminated and restarted after you have manually started LcnFileLoad.

3. Managing Your Environment

3.25. NT Registry Initialization

Using the Shutdown Wait Limit to avoid infinite loops

When a GUS display is shutting down, pending scripts are allowed to run to completion. If the display script is in an infinite loop, the display will not be able to close. This will affect the ability to invoke new GUS displays, which can cause a number of different problems. An example of a problem that could occur is with a “Msg” object (used to provide progress dialog) in a GUS display under SafeView control that will not close if the code never breaks out of the loop.

To protect against errant scripts locking up GUS displays at display shutdown, a “Shutdown Wait Limit” configuration parameter is provided with a default value of 10 seconds, instead of 0 (infinite). With this default value, GUS displays are given up to 10 seconds to complete their shutdown scripts, after which further script processing is terminated and a new display can be invoked.

You can change the Shutdown Wait Limit for GUS displays, but if you do so you must be very cautious and consider the following before making any changes to the default Shutdown Wait Limit:

- The new value should allow sufficient time for scripts to complete, but shut down the display after a reasonable time if the script gets into a very long or infinite loop.
- The new value should not be set so high that it causes a delay in the invocation of new displays.
- A value set between 1 and 249 milliseconds will be automatically rounded up to 250 milliseconds (1/4 second).

Pathnames in the NT registry

Pathnames used by the NT applications can be viewed in the NT registry. To view the pathname settings, execute the following steps:

- 1 Start the REGEDT32.EXE application (located in D:\winnt35\system32).
- 2 Select “HKEY_LOCAL_MACHINE\SOFTWARE\HONEYWELL\TPS.”

At this point, the value in the right-hand pane named “PCT Search Path” will contain the NT pathnames that were configured into the pathname catalog.

To conclude reading the “Managing Pathnames” section, jump to Schematic Invocation.

3.26 Schematic Invocation

Schematics are invoked differently when the focus is on the Native Window or on the GUS Application.

Native Window focus search priorities

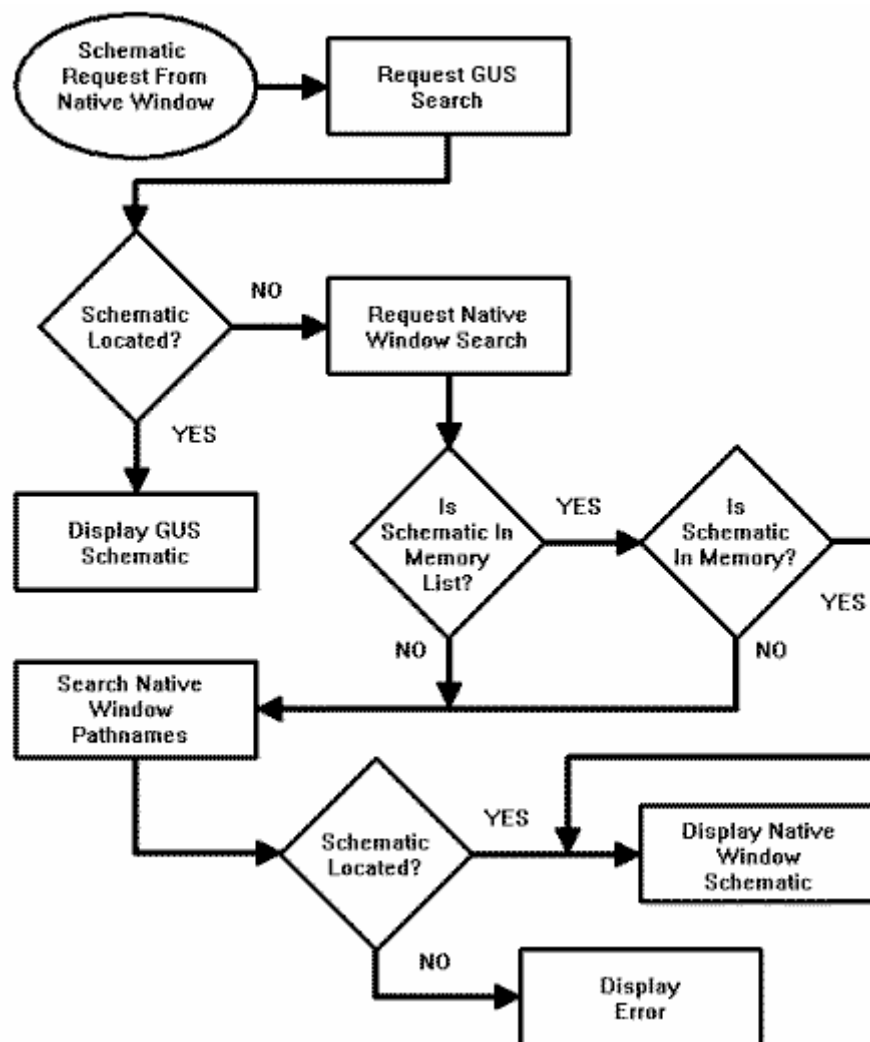
These search priorities are followed when focus is on the Native Window.

LcnFileLoad Not Running—If LcnFileLoad has not been started, or if it has failed to execute or link with the GUS personality, the search for a schematic will only include the TPS Network. This is obvious if you check the search paths displayed at the top right side of the Native Window under time and date. If this condition exists and a GUS schematic was requested, an error will be displayed in the Native Window.

LcnFileLoad Running—The search priorities followed when the LcnFileLoad is running are shown in this flowchart.

3. Managing Your Environment

3.26. Schematic Invocation



GUS application focus search priorities

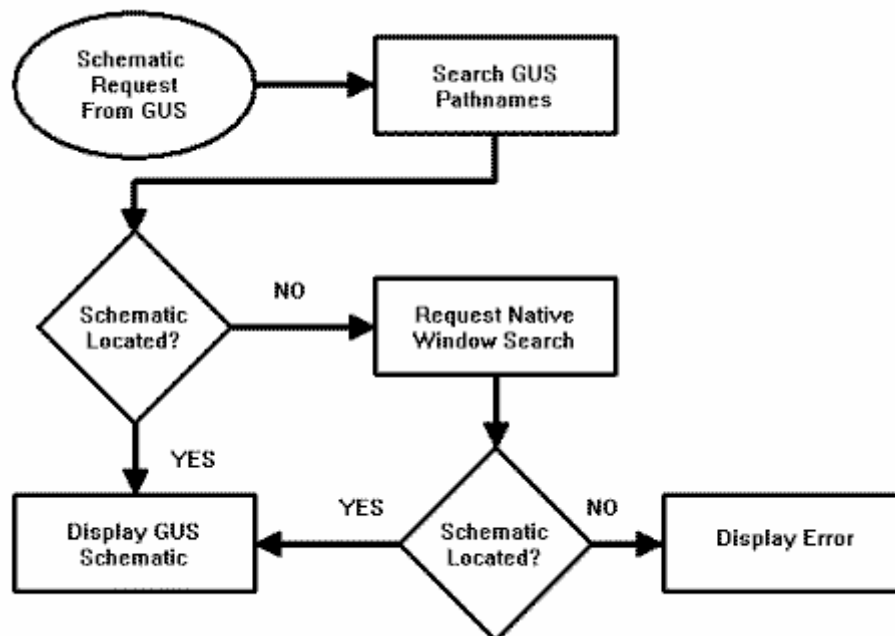
These search priorities are followed when focus is on the GUS Application.

LcnFileLoad Not Running—If LcnFileLoad has not been started, or if it has failed to execute, or if it has failed to link with the GUS personality, the search for a schematic will not occur. This is primarily due to the pathnames not being defined in the registry.

If the pathname is manually inserted as defined in the section named “Pathnames in the NT Registry” in NT Registry Initialization, a schematic search will occur.

If a Classic Schematic has been requested from the GUS side, an error message will be displayed indicating that the schematic does not exist.

LcnFileLoad Running—The search priorities followed when the LcnFileLoad is running and a schematic is requested from a GUS display follows this flowchart.



3. Managing Your Environment

3.26. Schematic Invocation

4. Manipulating Objects on the Screen

4.1 Overview

Objects in the Display Builder window can be manipulated in several ways. This section will familiarize you with the tools available and the methods used to manipulate graphic objects.

Topics in this section are:

- Draw Toolbar
- Selecting Objects
- Moving Objects
- Sizing Objects
- Reshaping Objects
- Grouping and Ungrouping Objects
- Rotating Objects
- Flipping Objects
- Scaling Objects
- Stacking Objects
- Deleting Objects (Edit/Delete menu)
- Copying Objects
- Changing Object Colors
- Changing Text Object Properties
- Correcting your mistakes

4.2 Draw Toolbar

The Draw toolbar is the main drawing tool in the Display Builder. Following is a display that shows the Draw toolbar.



4.3 Selecting Objects

You can select (or deselect) a single object, multiple objects, or all objects in a display window.

Selection tool (Arrow button on Draw toolbar)

Use the Selection tool  to select, move, and resize objects.

How to select an object

- 1 Move the mouse until the Selection tool points to the object to be selected.
- 2 Click the mouse button to select the object. Handles appear on the selected object, and are in different places on the object, depending upon which mode tool (Size, Shape, or Rotate) was selected. In the Size mode, the handles describe the width and height of the object.

How to deselect an object

- 1 Move the mouse until the Selection tool points to a part of the display window where there are no objects.
- 2 Click the mouse button.

RESULT: The handles disappear from the object and it is deselected.

How to select multiple objects

There are two ways you can select multiple objects.

Method 1

- 1 Select the first object. Handles appear on the selected object.
- 2 Move the Selection tool to the second object.
- 3 While holding the shift key down, select the second object. Handles appear on both objects.
- 4 Select more objects as required.
- 5 All selected objects will be treated as a single object for actions such as cut, copy, and drag.

Method 2

With the mouse, drag a “bounding box” around the required objects. This procedure is called Net Selecting, and is fully described in the “Selecting objects with the mouse” procedure in Section 1.4.

4.4 Selecting all objects in a display

Edit/Select All

The Select All command allows you to select all the items in the window.

How to select all objects in a display

- 1** Select Edit/Select All.
- 2** All objects in the active display window are selected. These objects can be cut, copied, deleted, duplicated, grouped, or moved to another location in the display window.

4.5 Moving Objects

The Drag-and-Drop method is used to move graphic objects in a Display Builder window or between Display Builder windows.

How to move an object by using drag-and-drop editing

All objects can be moved using the same procedure:

- 1** If not selected, click the object to select it.
- 2** Without clicking on one of the handles on the object, click and hold the mouse either on the body (fill) of the object or on a line of the object.
- 3** Move the mouse to drag the object to its new location.
- 4** Release the mouse button.

RESULT: The object has been moved to its new location.

NOTE: If the Snap-to-Grid command is active, the object aligns itself to the grid.

4. Manipulating Objects on the Screen

4.6. Sizing Objects

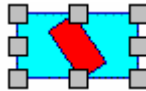
4.6 Sizing Objects

You can make an object taller, shorter, wider, or narrower by sizing. You can make an object appear larger or smaller by using proportional sizing.

NOTE: These rules apply to sizing objects.

You cannot size a grouped object that contains one or more rotated objects.

Gray handles on a grouped object that is in sizing mode indicate that it can't be resized.



How to resize objects

All objects can be sized by using the same procedure:

- 1 If not selected, make sure the SIZE tool on the draw toolbar is depressed then click the object to select it.

RESULT: Handles appear on all four sides and all four corners of the object.
- 2 Perform one or more of these tasks
 - a) Move a centered top or bottom handle to make the object taller or shorter.
 - b) Move a centered left or right handle to make the object wider or narrower.
 - c) Move a corner handle in or out to simultaneously change the width and height of the object.
- 3 Position the pointer in another part of the drawing area and click the mouse button. The object is deselected and the handles disappear.

Proportional sizing

If you want to keep the same proportions when sizing an object, press and hold the Shift key, then change the object size with a corner handle.

RESULT: The object shrinks or gets larger but keeps its proportional dimensions.

NOTE: If View/Snap to Grid is on, the object may distort slightly as it snaps from grid to grid. Refer to Scaling Objects to size an object by a known ratio.

4.7 Reshaping Objects

Other than sizing (width and height), some graphic objects can be reshaped using the SHAPE tool  on the draw toolbar.

Topics covered in this section are

- How to reshape the corners of a Rounded Rectangle.
- How to reshape the length of an Arc (or Pie).
- How to reshape the Open and Closed Polygon.
- Reshaping Closed and Open Bezier Curves.

4.8 How to reshape the corners of a Rounded Rectangle

Follow this procedure to reshape the corners of a Rounded Rectangle.

- 1 If not selected, make sure the SHAPE tool  on the draw toolbar is depressed then click the Rounded Rectangle to select it.

RESULT: A special “rounding handle” appears near the lower right corner but inside the rounded rectangle.



The square handle is the rounding handle.

- 2 To change the radius of the corners, click and hold the mouse button on the rounding handle and drag the mouse toward the center of the rectangle. All four corners become more rounded. Release the mouse when you reach the required radius.
- 3 When finished, position the pointer in another part of the drawing area and click the mouse button. The Rounded Rectangle is deselected and the rounding handle disappears.

NOTE: The corner has an equal radius only when you move the mouse at a 45-degree angle to the corner. Each of the 4 corners reshape to an arc when the mouse is moved at other angles.

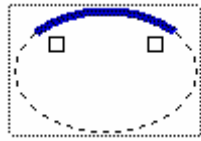
4. Manipulating Objects on the Screen

4.9. How to reshape the length of an Arc (or Pie)

4.9 How to reshape the length of an Arc (or Pie)

When the arc/pie was first drawn, its length was described as a portion of an ellipse (or circle) with endpoints at 30 degrees to the horizontal. To reshape the length of the arc, follow this procedure:

- 1 If not selected, make sure the SHAPE tool  on the draw toolbar is depressed then click the arc/pie to see two “endpoint handles” that mark the endpoints of the arc. In the following illustration, each endpoint handle is in the default 30-degree position.

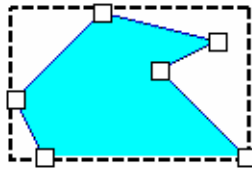


- 2 Using the mouse button, click and drag one endpoint handle to change the length of the arc as required. Release the mouse button. If required, click and drag the other endpoint handle to change that length also.
- 3 When finished, position the pointer in another part of the drawing area and click the mouse button. The arc/pie is deselected and the endpoint handles disappear.

4.10 How to reshape the Open and Closed Polygon

Follow this procedure to reshape a Open or Closed Polygon:

- 1 If not selected, make sure the SHAPE tool  on the draw toolbar is depressed, then click the Open or Closed Polygon to see “endpoint handles” at the end of each segment, as shown in this illustration.



- 2 Click and hold the mouse on an endpoint handle that you want to move. Drag the mouse to its new position and release. If required, repeat this drag-and-drop procedure for other endpoints.
- 3 When finished, position the pointer in another part of the drawing area and click the mouse button. The polygon object is deselected and the endpoint handles disappear.

4.11 Reshaping Closed and Open Bezier Curves

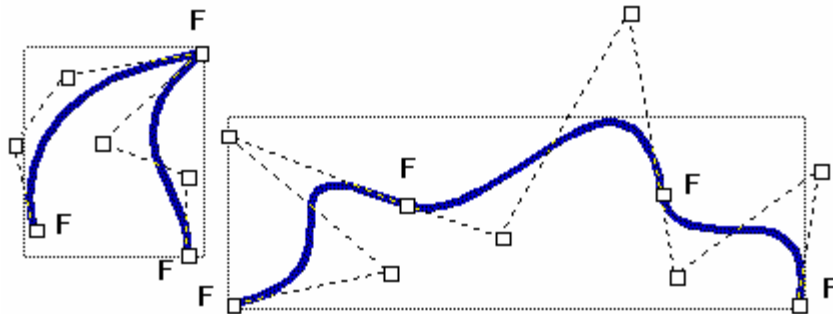
An Open Bezier Curve is open on one side, as in the diagrams below.

A Closed Bezier Curve is similar but its line is completely closed. If curved segments do not completely encircle a closed curve, the line across the closed Bezier curve will be straight.

Follow this procedure to reshape either a closed or open Bezier Curve:

1

If not selected, make sure the SHAPE tool  on the draw toolbar is depressed, then click the Bezier curve object to show fixed endpoint handles that mark the ends of each Bezier segment (each "F" in these diagrams). The end of each segment also contains control handles on dotted lines as shown.



2

Move the fixed endpoints and/or control handles as required to reshape the Bezier curve. Notice you can make a smooth transition between Bezier line segments when the second control handle of one line segment is in line with the first control handle of the next line segment, as shown in the second drawing.

3

When finished, position the pointer in another part of the drawing area and click the mouse button. The object is deselected and the endpoint handles disappear.

4. Manipulating Objects on the Screen

4.12. Grouping and Ungrouping Objects

4.12 Grouping and Ungrouping Objects

Several objects can be grouped together so that their positions in relation to each other are constant.

In a similar way, a set of grouped objects can be ungrouped (separated) into individual objects.

Topics in this section are

- Grouping Objects (Draw/Group menu and Group on Toolbar)
- Ungrouping Objects (Draw/Ungroup menu and Ungroup on Toolbar)

4.13 Grouping Objects

Draw/Group menu and Group on Toolbar

Several objects can be selected and “grouped” with this command. When grouped, all objects together behave like a single object.

How to group objects

- 1 Select the objects either by
using the selection tool to drag a box around the objects, or
by selecting each object using the shift key and selection tool.
- 2 Select Draw/Group from the menu or select the GROUP tool from the toolbar.
- 3 Notice the handles on the individual objects disappear and a single set of handles appears around the entire group.

RESULT: The group is treated as a single object for actions such as cut, copy, and move.

4.14 Ungrouping Objects

Draw/Ungroup menu and Ungroup on Toolbar

Objects that have been grouped can be “ungrouped” into separate objects with this command.

How to ungroup objects

- 1 Select the grouped object. A single set of handles appears.
- 2 Select Draw/Ungroup from the menu or select the UNGROUP tool  from the toolbar.
- 3 Notice the handles around the group of objects disappear and handles appear on the individual objects.

RESULT: The objects are no longer treated as a group but as individual objects for actions such as copy, cut, and move.

4.15 Rotating Objects

Objects can be rotated using four tools in the Display Builder.

- 1 Rotate 90 Degrees Using the Draw Menu—Commands allow objects to be rotated by 90 degrees to the left or to the right. Refer to
 - Rotating an object to the left
 - Draw/Rotate Left menu and Rotating an object to the right
 - Draw/Rotate Right menu.



- 2 Rotate 90 Degrees Using the Layout Toolbar—Commands allow objects to be rotated by 90 degrees to the left or to the right. Refer to Rotating an Object (Layout toolbar).
- 3 Rotate Freely—An object can be rotated freely using the ROTATE tool on the draw toolbar. Refer to How to freely rotate an object.
- 4 Rotate a Number of Degrees—You can also rotate an object a precise number of degrees using the object's Rotate Property Page. Refer to Setting an arbitrary rotation angle of an object.

4. Manipulating Objects on the Screen

4.16. Rotating an object to the left

4.16 Rotating an object to the left

Draw/Rotate Left menu

An object can be rotated 90 degrees to the left with this command.

How to rotate an object 90 degrees to the left

- 1 Select the object. Handles appear on the object.
- 2 Select the Rotate Left command from the Draw menu.

RESULT: The object rotates 90 degrees counter-clockwise. This command can be repeated any number of times to make successive rotations by 90 degrees.

NOTE: There are many ways to rotate objects, some more precise than others. Refer to Rotating Objects for the full list of functions.

4.17 Rotating an object to the right

Draw/Rotate Right menu

An object can be rotated 90 degrees to the right with this command.

How to rotate an object 90 degrees to the right

- 1 Select the object. Handles appear on the object.
- 2 Select the Rotate Right command from the Draw menu.

RESULT: The object rotates 90 degrees clockwise. This command can be repeated any number of times to make successive rotations by 90 degrees.

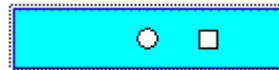
NOTE: There are many ways to rotate objects, some more precise than others. Refer to Rotating Objects for the full list of functions.

4.18 How to freely rotate an object

An object can be freely rotated in either direction by using the ROTATE tool on the draw toolbar. Follow this procedure:

1

If not selected, make sure the ROTATE tool  on the draw toolbar is depressed, and then click the object to be rotated. Note the round "center of rotation" handle and the square "rotator handle" that appear on the object.



2

Click and drag the rotator handle in a circular motion to rotate the object around the center of rotation, then release when finished.

3

Optionally, notice you can move the center of rotation handle so it is anywhere within the object or outside of the object.

NOTE: Use care if the center of rotation is outside the object because you may lose it. If you should accidentally lose it, first make sure the ROTATE tool is selected, Minimize the object's display window and then Maximize it again. To learn to use Minimize and Maximize, refer to Working with Control Menus

4

When finished, position the pointer in another part of the drawing area and click the mouse button. The object is deselected and the rotation handles disappear.

NOTE: There are many ways to rotate objects, some more precise than others. Refer to Rotating Objects for the full list of procedures.

4.19 Setting an arbitrary rotation angle of an object

An object can be rotated an arbitrary number of degrees by integers.

To rotate an object by degree integers, use the Rotate Property Page for that object. For more information, refer to Rotate Properties.

4.20 Flipping Objects

Objects can be flipped using two tools in the Display Builder.

- 1 Flipping Objects Using the Draw Menu—Commands allow objects to be flipped from left-to-right horizontally or from top-to-bottom vertically. Refer to Flipping an object horizontally

Draw/Flip Horizontal menu and Flipping an object vertically

Draw/Flip Vertical menu



- 2 Flipping Objects Using the Layout Toolbar—Commands allow objects to be flipped from left-to-right horizontally or from top-to-bottom vertically. Refer to

Flipping an Object (Layout toolbar).

The following rules apply to flipping objects:

- You cannot flip a grouped object that contains one or more rotated objects.
- You cannot flip a Text object unless you go to Transformation Properties and change Scale X and/or Scale Y to -1. Scale X = -1 flips the text horizontally and Scale Y = -1 flips the text vertically.
- Flipping is based on the original orientation. If an object has been rotated 90 degrees, flipping “horizontally” will vertically flip the rotated object.

4.21 Flipping an object horizontally

Draw/Flip Horizontal menu

An object can be flipped from left-to-right with this command. A mirror image of the object is created.

The following rules apply to flipping objects:

- You cannot flip a grouped object that contains one or more rotated objects.
- You cannot flip a Text object with this command. Refer to Transformation Properties for instruction in using Scale X = -1 to flip a Text object horizontally.

How to flip an object horizontally

- 1 Select the object. Handles appear on the object.
- 2 Select the Flip Horizontal command from the Draw menu.



RESULT: The object flips from left-to-right. This command can be repeated any number of times to cause successive flips.

4. Manipulating Objects on the Screen

4.22. Flipping an object vertically

4.22 Flipping an object vertically

Draw/Flip Vertical menu

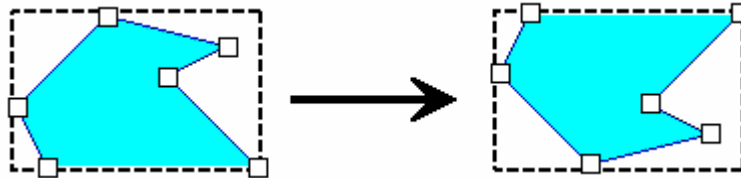
An object can be flipped from top-to-bottom with this command. An inverted-image of the object is created.

The following rules apply to flipping objects:

- You cannot flip a grouped object that contains one or more rotated objects.
- You cannot flip a Text object with this command. Refer to Transformation Properties for instruction in using Scale Y = -1 to flip a Text object vertically.

How to flip an object vertically

- 1 Select the object. Handles appear on the object.
- 2 Select the Flip Vertical command from the Draw menu.



RESULT: The object flips from top-to-bottom. This command can be repeated any number of times to produce successive flips.

4.23 Scaling Objects

Objects can be scaled in size; that is, an object can be made larger or smaller by a decimal ratio of the original or can be changed to a fixed size.

How to scale an object

To scale an object, use the Transformation Property Page for that object. For more information, refer to Transformation Properties.

4.24 Stacking Objects

Objects can be stacked using two tools in the Display Builder.

- 1 Stacking Objects Using the Draw Menu—Commands allow objects to be moved to the front or the back of the stack. Refer to Bringing objects to the front of the stack

 Draw/Bring to Front menu and Sending objects to the back of the stack

 Draw/Send to Back menu. Other commands allow objects to be moved forward or back one level in the stack. Refer to Bringing an object forward one level in the stack

 Draw/Bring Forward menu and Sending an object back one level in the stack

 Draw/Send Backward menu.
- 2 Stacking Objects Using the Layout Toolbar—Commands allow objects to be moved to the front or the back of the stack, and forward or back one level in the stack. Refer to Stacking an Object (Layout toolbar).

Each object in a display window is placed on a different “level” as if all the objects were drawn on transparent sheets and you were viewing the objects through the “stack” of sheets.

If objects are drawn so they overlap, those in front may need to be moved up or down in their levels so the overall display is correct. There are also special “rules” that must be observed for OLE objects.

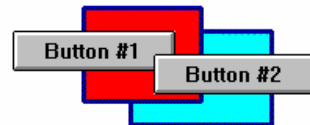
4. Manipulating Objects on the Screen

4.24. Stacking Objects

OLE Object Rules—OLE Objects may seem to overrule the stacking order of other objects. Here are some OLE Object stacking rules:

- OLE Objects have their own “stack” and are always in front of the “stack” of all other objects in the window.
- Like other objects, an OLE Object can be moved through the “stack” of all OLE objects.

This illustration may help to clarify these rules. The colored rectangles are objects that can be moved in front of, or behind each other.



However, the buttons are OLE objects. In the Display Builder, they can be moved in relation to each other in the same way as the rectangles, but will ALWAYS be stacked in front of all other objects in the display window.

Order of position—All objects in a display have an order of position, not just overlapping objects. When sending an object backward one level at a time it may appear that nothing happened if the next object in position order does not overlap the one being sent backward.

For example, the red object above (the darker gray rectangle) may be in level 1 and the blue object (the lighter gray rectangle) may be in level 4 (because some unnoticed objects in the display window occupy levels 2 and 3). In this case, you should simply continue sending the selected red object backward one command at a time until the required position is reached.

Stacking order of duplicated (copied and pasted) objects—The stacking order of a group of objects relative to each other may be lost when the objects are duplicated (copied and pasted). Use net selection (refer to Selecting objects with the mouse) to select the group of objects whose stacking order you want to maintain, and copy and paste the entire selection. This will maintain the stacking order.

How to move an object through its stack

Use these commands to move objects through the stack in which it is located.

- Bringing objects to the front of the tack (Draw/Bring to Front menu)
- Sending objects to the back of the stack (Draw/Send to Back menu)
- Bringing an object forward one level in the stack (Draw/Bring Forward menu)
- Sending an object back one level in the stack (Draw/Send Backward menu)

4.25 Bringing objects to the front of the stack

Draw/Bring to Front menu

Use this command when an object needs to be located in front of all other objects on the screen.

NOTE: OLE Objects may seem to overrule the stacking order of other objects. For further information, refer to Stacking Objects.

How to bring an object to the front

- 1** Select the object. Handles appear on the object.
- 2** Select Draw/Bring to Front from the menu.

RESULT: The selected object is placed in front of all other objects on the screen.

NOTE: If the object's position prevents the command from being executed within that object's "stack," that menu command may be shown in gray on the menu.

4.26 Sending objects to the back of the stack

Draw/Send to Back menu

Use this command when an object needs to be located behind all other objects on the screen.

NOTE: OLE Objects may seem to overrule the stacking order of other objects. For further information, refer to Stacking Objects.

How to send an object to the back

- 1** Select the object. Handles appear on the object.
- 2** Select Draw/Send to Back from the menu.

RESULT: The selected object is placed in back of all other objects on the screen.

NOTE: If the object's position prevents the command from being executed within that object's "stack," that menu command may be shown in gray on the menu.

4. Manipulating Objects on the Screen

4.27. Bringing an object forward one level in the stack

4.27 Bringing an object forward one level in the stack

Draw/Bring Forward menu

Use this command when an object needs to be located between other objects. This command moves the selected object forward, one position at a time.

NOTE: OLE Objects may seem to overrule the stacking order of other objects. For further information, refer to Stacking Objects.

How to bring an object forward

- 1 Select the object. Handles appear on the object.
- 2 Select Draw/Bring Forward from the menu.

RESULT: The selected object is brought forward one position order.

NOTE: If the object's position prevents the command from being executed within that object's "stack," that menu command may be shown in gray on the menu.

4.28 Sending an object back one level in the stack

Draw/Send Backward menu

Use this command when an object needs to be located between other objects. This command moves the selected object backwards, one position at a time.

NOTE: OLE Objects may seem to overrule the stacking order of other objects. For further information, refer to Stacking Objects.

How to send an object backward

- 1 Select the object. Handles appear on the object.
- 2 Select Draw/Send Backward from the menu.

RESULT: The selected object is sent backward one position order.

NOTE: If the object's position prevents the command from being executed within that object's "stack," that menu command may be shown in gray on the menu.

4.29 Deleting Objects

Edit/Delete menu

CAUTION: When an object is deleted, it CANNOT be recovered from the clipboard.

If you DON'T ISSUE FURTHER COMMANDS after a delete command, it may be recovered by using the Undo command (Edit/Undo or Ctrl + Z). Refer to Undoing your last action.

Objects can be deleted individually or in sets. To delete multiple objects, select them by

- holding the shift key and selecting the required objects or by
- using the selection tool to drag a boundary around the required objects.

Then delete as if the objects were a single object.

How to Delete an object

- 1 Select the object. Handles appear on the object.
- 2 Press the Delete key on the keyboard (or select Edit/Delete from the menu).

RESULT: The selected object is PERMANENTLY removed from the display window.

RECOVERY: To get the object back, you must IMMEDIATELY undo this procedure by pressing the Ctrl + Z key combination or by selecting Edit/Undo from the menu.

4.30 Copying Objects

Overview

Objects can be cut, copied, pasted, and duplicated. When you cut, copy, and paste objects, the clipboard is used to temporarily store the object. These procedures use both the menu and the keyboard.

How to copy a single Display Object using the menu

- 1 In a display window, select the object by clicking on it with the selection tool.
- 2 Select Edit/Copy (Ctrl + C) from the menu.

RESULT: The selected object remains, but a copy is placed on the Windows Clipboard.
- 3 If you want the copied object to be placed in another display window, select that window by clicking on its Title Bar (otherwise, it will be placed in the same display window).
- 4 Select Edit/Paste (Ctrl + V) from the menu.
Also refer to Pasting Objects

Edit/Paste menu and Paste on Toolbar.

RESULT: The selected object is placed on the required display window from the clipboard.

How to copy a single Display Object using drag-and-drop editing

- 1 Press and hold the Ctrl key, then click and hold the required object (handles will appear).
- 2 Drag the object to a new location (either on the same display or on another display) and release the mouse button.

RESULT: The selected object remains, but a copy is placed in the new location.

How to copy multiple Display Objects using the menu

- 1 In a display window, select more than one object by either
 - holding the shift key and individually selecting the required objects or
 - using the selection tool to drag a boundary around the required objects.
- 2 Select Edit/Copy (Ctrl + C) from the menu.

RESULT: The selected objects remain, but copies of them are placed on the clipboard.
- 3 If you want the copied objects to be placed in another display window, select that window by clicking on its Title Bar (otherwise, they will be placed in the same display window).
- 4 Select Edit/Paste (Ctrl + V) from the menu.
Also refer to Pasting Objects

Edit/Paste menu and Paste on Toolbar.

RESULT: The selected objects are placed on the required display window from the clipboard.

How to copy multiple Display Objects using drag-and-drop editing

- 1 In a display window, select more than one object by either
 - holding the shift key and individually selecting the required objects or
 - using the selection tool to drag a boundary around the required objects.
- 2 Drag the objects to a new location (either on the same display or on another display) and release the mouse button.

RESULT: The selected objects remain, but copies of them are placed in the new location.

4.31 Cutting Objects

Edit/Cut menu and Cut on Toolbar

Objects can be cut individually or in sets. To cut a set of objects, select multiple objects by

- holding the shift key and selecting the required objects or by
- using the selection tool to drag a boundary around the required objects.

Then, cut as if the set were a single object.

4. Manipulating Objects on the Screen

4.32. Pasting Objects

How to cut Display Objects

- 1 Select the object(s). Handles appear on the object(s).
- 2 Select Edit/Cut (Ctrl + X) from the menu.

RESULT: The selected objects are removed and placed on the Windows Clipboard.

NOTE: You must use the Paste command to recover objects placed on the clipboard. Refer to Pasting Objects.

4.32 Pasting Objects

Edit/Paste menu and Paste on Toolbar

Objects or groups that have been Cut or Copied from a display window can be pasted

- inside the current display window,
- into another display window within the same Display Builder application, or
- into another display window in a different Display Builder application.

How to paste an object

- 1 Before using this command, you must have selected an object or objects in a display window and have used the Copy or Cut commands to place the object(s) on the Windows Clipboard.
- 2 If you have not already done so, make the destination display window active by clicking the title bar. This may be the current display window, a second display window within the same Display Builder application, or a display window in another Display Builder application.
- 3 Select Edit/Paste (Ctrl + V) from the menu.
- 4
 - a) If you are in the same display window where the clipboard object was cut or copied, the object is pasted **on top of** where the cut or copied object was, but **shifted to the right and down one grid line** from the area from which the object was copied.
 - b) If you are pasting into another display window or into a window in another Display Builder application, the object is pasted in the same relative location as in the original display window but **shifted to the right and down one grid line**.
- 5 If required, select and drag the pasted object to the required location in the display window.

4.33 Duplicating Objects

Edit/Duplicate menu

The Duplicate command allows you to duplicate one or more objects in the display window.

How to duplicate an object

- 1 Select the object(s) to be duplicated.
- 2 Select Edit/Duplicate (Ctrl + D).

RESULT: The object is duplicated. The duplicated object is in the same relative location as the original object **but shifted to the right and down one grid line.**

4.34 Changing Object Colors

GUS Display Builder uses the RGB (Red, Green, Blue) color system to support the 256-color (8-bit) palette or the 65,536-color palette.

- When you request an object color change, the high-color palette is displayed.
- You can choose colors from the palette (at the top) or select a new color from the RGB or the Hue-Saturation-Luminance selection boxes.

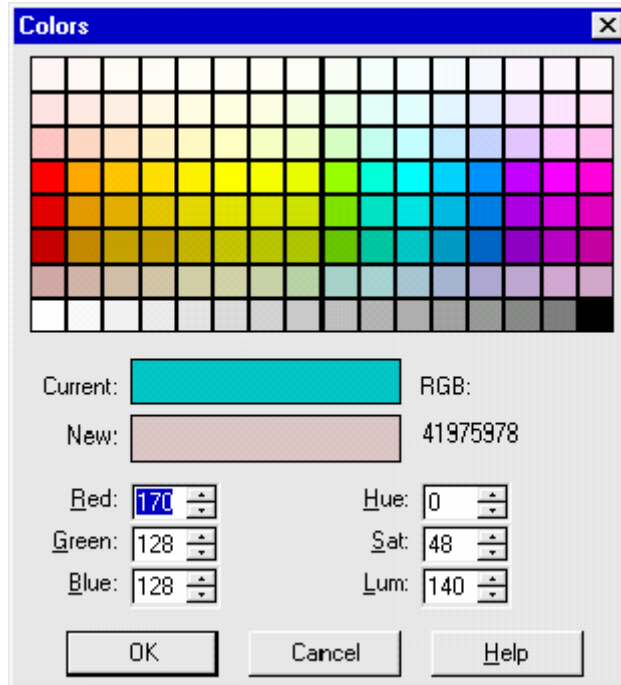
4. Manipulating Objects on the Screen

4.34. Changing Object Colors

High-Color Palette

The high-color palette enables mixing of colors and adjustment of hue (color), saturation (concentration), and luminance (brightness).

If a custom color is created with the high-color palette, then displayed on a 256-color GUS, the closest standard color from the 256-color palette will replace the custom color.



You can change the fill, line, and text color.

Topics in this section are:

- Changing Object Fill Colors (Fill color button on Toolbar)
- Changing Object Line Colors (Line color button on Toolbar)
- Changing Text Color (Text color button on Toolbar)

4.35 Changing Object Fill Colors

Fill color button on Toolbar

There are two ways to change the fill color of objects: (1) using the Color Select button on the property page of the object, and (2) using the fill color button on the Standard toolbar.

The Standard toolbar procedure is described below.

How to change the fill color of objects

- 1 Select the required object(s).
- 2 Select the fill color button  on the Standard toolbar. The high-color palate is displayed. Refer to Changing Object Colors for details.
- 3 Select the required color on the palate if it is on display or else create the required color by selecting the required Red-Green-Blue or Hue-Saturation-Luminance values.
- 4 Click OK. The fill color of the selected object(s) is changed.

4.36 Changing Object Line Colors

Line color button on Toolbar

There are two ways to change the line color of objects: (1) using the Color Select button on the property page of the object, and (2) using the line color button on the Standard toolbar.

The Standard toolbar procedure is described below.

How to change the line color of objects

- 1 Select the required object(s).
- 2 Select the line color button  on the Standard toolbar. The high-color palate is displayed. Refer to Changing Object Colors for details.
- 3 Select the required color on the palate if it is on display or else create the required color by selecting the required Red-Green-Blue or Hue-Saturation-Luminance values.
- 4 Click OK. The line color of the selected object(s) is changed.

4.37 Changing Text Color

Text color button on Toolbar

There are two ways to change the color of text: (1) using the text button on the Text Format toolbar, and (2) using the text color button on the Standard toolbar.

The Standard toolbar procedure is described below.

How to change the text color

- 1** Select the required text object(s).
- 2** Select the text color button  on the Standard toolbar. The high-color palate is displayed. Refer to Changing Object Colors for details.
- 3** Select the required color on the palate if it is on display or else create the required color by selecting the required Red-Green-Blue or Hue-Saturation-Luminance values.
- 4** Click OK. The color of the selected text object(s) is changed.

4.38 Changing Text Object Properties

The properties of text objects in a Display Builder window can be changed in several ways. This section will familiarize you with the main tool, the Text Format toolbar, and the methods used to change text object properties.

Topics in this section are

- Using the Text Format Toolbar
- Changing the Font (Text Format Toolbar)
- Changing the Font Size (Text Format Toolbar)
- Changing the Text Properties (Text Format Toolbar)
- Changing the Text Color (Text Format Toolbar)
- Changing the Text Alignment (Text Format Toolbar)

4.39 Using the Text Format Toolbar

The Text Format toolbar can be docked or floating. It can be docked only to the top or bottom edge of the main window. Following is a display that shows the Text Format toolbar as a floating window:



The Text Format toolbar works only when a display edit window is active. It does not apply when a script editor window is active. To use the Text Format toolbar functions, you must first select one or more text objects. The toolbar will be updated to show the current settings for the selected objects. When multiple objects are selected, and not all of the selected objects have the same value for a text property, that property will be grayed out on the toolbar.

4.40 Changing the Font

Text Format toolbar

Use the font name pull-down list.



How to change the font

There are two procedures for changing the font.

Procedure 1

- 1 Select the required text object(s).
- 2 Type the name of the required font into the font name pull-down list.
- 3 Press Enter.

Procedure 2

- 1 Select the required text object(s).
- 2 Scroll down the font name pull-down list until you find the required font.
- 3 Select it.

4. Manipulating Objects on the Screen

4.41. Changing the Font Size

4.41 Changing the Font Size

Text Format toolbar

Use the font size pull-down list.



How to change the font size

There are two procedures for changing the font size.

Procedure 1

- 1 Select the required text object(s).
- 2 Type the required font size into the font size pull-down list.
- 3 Press Enter. The selected text object(s) will be updated with the new font.

Procedure 2

- 1 Select the required text object(s).
- 2 Scroll down the font size pull-down list until you find the required size.
- 3 Select it. The selected text object(s) will be updated to the new size.

4.42 Changing the Text Properties

Text Format toolbar

You can make text bold, indent it, and underline it.

How to make text bold

- 1 Select the required text object(s).
- 2 Click the bold button . The text will be made bold.

NOTE: If the selected text was initially bold, clicking the bold button will change it to normal.

How to italicize text

- 1 Select the required text object(s).
- 2 Click the italic button . The text will be italicized.

NOTE: If the selected text was initially italicized, clicking the bold button will change it to normal.

How to underline text

- 1 Select the required text object(s).
- 2 Click the underline button . The text will be underlined.

NOTE: If the selected text was initially underlined, clicking the bold button will change it to normal.

4. Manipulating Objects on the Screen

4.43. Changing the Text Color

4.43 Changing the Text Color

Text Format toolbar

GUS displays can be built using the RGB (Red Green Blue) color system to support the 256 color (8 bit) or the 65,536 color palette. Refer to Changing Object Colors for more details.

How to change text color

- 1 Select the required text object(s).
- 2 Click the text color button . The high-color palate is displayed.
- 3 Select the required color on the palate if it is on display or else create the required color by selecting the required Red-Green-Blue or Hue-Saturation-Luminance values.
- 4 Click OK. The color of the selected text is changed.

4.44 Changing the Text Alignment

Text Format toolbar

You can align text to the left, to the center, and to the right.

How to left align text

- 1 Select the required text object(s).
- 2 Click the Align Left button . The text will be aligned against the left margin.

How to center text

- 1 Select the required text object(s).
- 2 Click the Align Center button . The text will be centered between the left and right margins.

How to right align text

- 1 Select the required text object(s).
- 2 Click the Align Right button . The text will be aligned against the right margin.

4.45 Changing Object Layout

The layout of objects in a Display Builder window can be changed in several ways. This section will familiarize you with the main tool, the Layout toolbar, and the methods used to change the layout of objects.

Topics in this section are:

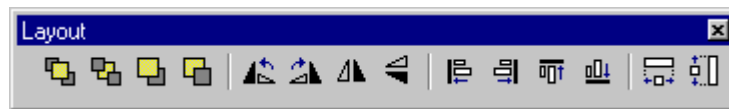
- Using the Layout Toolbar
- Stacking an Object (Layout Toolbar)
- Rotating an Object (Layout Toolbar)
- Flipping an Object (Layout Toolbar)
- Positioning Objects Relatively (Layout Toolbar)
- Sizing Objects Relatively (Layout Toolbar)

4. Manipulating Objects on the Screen

4.46. Using the Layout Toolbar

4.46 Using the Layout Toolbar

The Layout toolbar can be docked or floating. It can be docked to any edge (top, bottom, left, or right) of the main window. Following is a display that shows the Layout toolbar as a floating window:



The Layout toolbar works only when a display edit window is active. It does not apply when a script editor window is active. To use the Layout toolbar functions, you must first select one or more objects.

4.47 Stacking an Object (Layout toolbar)

There are two tools you can use to stack an object. Refer to Stacking Objects for details on the other tool and order of position, and stacking order.

NOTE: OLE Objects may appear to overrule the stacking order of other objects.

Using the Layout toolbar, you can move an object as follows:

- to the front of the stack
- to the back of the stack
- one level forward in the stack
- one level back in the stack

NOTE: If an object's position prevents the command from being executed within that object's "stack," that command may be grayed out on the toolbar.

How to bring an object to the front of the stack

- 1 Select the object. Handles appear on the object.
- 2 Select the Bring to the Front button . The selected object is placed in front of all other objects on the screen.

How to send an object to the back of the stack

- 1 Select the object. Handles appear on the object.
- 2 Select the Send to the Back button . The selected object is placed in back of all other objects on the screen.

How to bring an object one level forward in the stack

- 1 Select the object. Handles appear on the object.
- 2 Select the Bring Forward button . The selected object is moved one level forward in the stack.

How to send an object one level back in the stack

- 1 Select the object. Handles appear on the object.
- 2 Select the Send Back button . The selected object is moved one level back in the stack.

4.48 Rotating an Object (Layout toolbar)

There are four tools you can use to rotate an object. Refer to Rotating Objects for details on the other three tools.

Using the Layout toolbar, you can rotate an object as follows:

- 90 degrees to the left
- 90 degrees to the right

How to rotate an object to the left

- 1 Select the object. Handles appear on the object.
- 2 Select the Rotate Left button . The object rotates 90 degrees counter-clockwise.

NOTE: This command can be repeated any number of times to make successive 90-degree rotations.

4. Manipulating Objects on the Screen

4.49. Flipping an Object (Layout toolbar)

How to rotate an object to the right

- 1 Select the object. Handles appear on the object.
- 2 Select the Rotate Right button . The object rotates 90 degrees clockwise.

NOTE: This command can be repeated any number of times to make successive 90-degree rotations.

4.49 Flipping an Object (Layout toolbar)

There are two tools you can use to flip an object. Refer to Flipping Objects for details on the other tool.

The following rules apply to flipping objects:

- You cannot flip a grouped object that contains one or more rotated objects.
- You cannot flip a Text object unless you go to Transformation Properties and change Scale X and/or Scale Y to -1. Scale X = -1 flips the text horizontally and Scale Y = -1 flips the text vertically.

Using the Layout toolbar, you can flip an object as follows:

- from left-to-right horizontally
- from top-to-bottom vertically

How to flip an object horizontally

- 1 Select the object. Handles appear on the object.
- 2 Select the Flip Horizontal button . The object flips from left to right.

NOTE: This command can be repeated any number of times to make successive flips.

How to flip an object vertically

- 1 Select the object. Handles appear on the object.
- 2 Select the Flip Vertical button . The object flips from top to bottom.

NOTE: This command can be repeated any number of times to make successive flips.

4.50 Positioning Objects Relatively (Layout toolbar)

Using the Layout toolbar, you can align one or more objects relative to a reference object so that they are aligned as follows:

- horizontally at the left edge
- horizontally at right edge
- vertically at the top edge
- vertically at the bottom edge

How to align objects at the left edge

- 1 Select the reference object. Handles appear on the object.
- 2 Select the first object to be aligned. Handles appear on the object.
- 3 While holding down the <Shift> key, select any other objects to be aligned.
- 4 Select the Align Left button . All objects selected for alignment will be moved horizontally so that their left edges are aligned with the left edge of the reference object.

How to align objects at the right edge

- 1 Select the reference object. Handles appear on the object.
- 2 Select the first object to be aligned. Handles appear on the object.
- 3 While holding down the <Shift> key, select any other objects to be aligned.
- 4 Select the Align Right Button . All objects selected for alignment will be moved horizontally so that their right edges are aligned with the right edge of the reference object.

4. Manipulating Objects on the Screen

4.51. Sizing Objects Relatively (Layout toolbar)

How to align objects at the top edge

- 1 Select the reference object. Handles appear on the object.
- 2 Select the first object to be aligned. Handles appear on the object.
- 3 While holding down the <Shift> key, select any other objects to be aligned.
- 4 Select the Align Top button . All objects selected for alignment will be moved vertically so that their top edges are aligned with the top edge of the reference object.

How to align objects at the bottom edge

- 1 Select the reference object. Handles appear on the object.
- 2 Select the first object to be aligned. Handles appear on the object.
- 3 While holding down the <Shift> key, select any other objects to be aligned.
- 4 Select the Align Bottom button . All objects selected for alignment will be moved vertically so that their bottom edges are aligned with the bottom edge of the reference object.

4.51 Sizing Objects Relatively (Layout toolbar)

Using the Layout toolbar, you can size one or more objects relative to a reference object so that they have:

- the same width
- the same height

How to make objects the same width

- 1 Select the reference object. Handles appear on the object.
- 2 Select the first object to be sized. Handles appear on the object.
- 3 While holding down the <Shift> key, select any other objects to be sized.
- 4 Select the Size Width button . The width of all the objects selected for sizing will be changed so that it is the same as the width of the reference object.

How to make objects the same height

- 1 Select the reference object. Handles appear on the object.
- 2 Select the first object to be sized. Handles appear on the object.
- 3 While holding down the <Shift> key, select any other objects to be sized.
- 4 Select the Size Height button . The height of all the objects selected for sizing will be changed so that it is the same as the height of the reference object.

4.52 Correcting your mistakes

You can correct mistakes if you act immediately because the Undo and Redo commands act on **ONLY** the last action you performed.

Topics in this section are:

- Undoing your last action (Edit/Undo menu)
- Redoing your last action (Edit/Redo menu)

4.53 Undoing your last action

Edit/Undo menu

Use the Undo command to remove the effect of the last item you changed, as follows:

IMMEDIATELY select Edit/Undo (Ctrl + Z) from the menu.

RESULT: The effect of the last command you entered is recovered.

4.54 Redoing your last action

Edit/Redo menu

Use the Redo command to remove the effect of the Undo command, as follows:

IMMEDIATELY select Edit/Redo (Ctrl + Y) from the menu.

RESULT: The effect of the Undo command you entered is recovered.

4. Manipulating Objects on the Screen

4.54. Redoing your last action

5. Adding Objects to Your Display

5.1 Overview

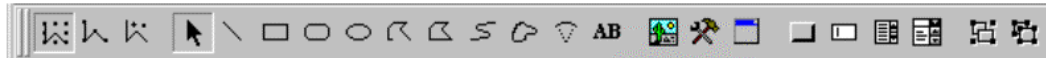
You can add many different types of objects to the Display Builder. This section will familiarize you with the tools available and the methods used to create and insert graphic objects.

Topics in this section are:

- Using the Draw Tools
- Adding Basic Graphic Objects

5.2 Using the Draw Tools

Basic objects are created and manipulated with tools on the draw toolbar.



The Draw Toolbar is divided into six sections:

- **The Object modifier tools** (the 3 leftmost buttons)—These tools are used to Resize, Reshape, and Rotate drawn objects. Each of these tools “latch” when in use. That is, once a tool is depressed it will remain down until another of the three tools is clicked. However, if the selection tool is used to click the display background, Display Builder defaults to the Resize tool.
To Resize an object, refer to Sizing Objects.
To Reshape an object, refer to Reshaping Objects.
To Rotate an object, refer to How to freely rotate an object.
- **The Selection tool** (the next button)—This tool is used to select objects in the display window. Refer to Adding Basic Graphic Object.

5. Adding Objects to Your Display

5.2. Using the Draw Tools

- **The Object drawing tools** (the next 10 buttons)—These tools are used to construct basic graphics. Moving from left to right, they are the Selection Tool, Line, Rectangle (also Square), Rounded Rectangle (also Rounded Square), Ellipse (also Circle), Open Polygon, Closed Polygon, Open Bezier, Closed Bezier, Arc (also Pie), and Text tools. Refer to Adding Basic Graphic Object.
NOTE 1: Certain objects can be drawn with a 1-1 aspect ratio. This is accomplished by holding down the Shift key while you drag the mouse to create the object. Using this method, you can use
 - the Rectangle tool to draw a Square
 - the Rounded Rectangle tool to draw a Rounded Square
 - the Ellipse tool to draw a Circle
 - the Text tool to insert a square Text object.NOTE 2: If you are using the Line tool, holding down the Shift key while you drag the mouse will create the line at a 45-degree angle.
NOTE 3: None of the drawing tools can be “latched.” That is, you cannot “latch” a drawing tool so that successive objects of the same type can be drawn without selecting the tool again.
- **The Insert tools** (the next 3 buttons)—These tools are used to insert separate, previously built objects into the display window. Moving from left to right, these tools are the Insert Bitmap, Insert Display, and Insert OLE Control tools.
To insert a bitmap, refer to Inserting a Bitmap.
To insert a display, refer to Inserting an Embedded Display.
To insert an OLE control, refer to Inserting Controls.
- **The Insert Primitives tools** (the next 4 buttons)—These tools are used to insert pre-built primitive objects into the display window. They will enable you to insert fairly complex objects into a display with relative ease. Moving from left to right, these tools are the Insert Button, Insert Data Entry, Insert Listbox, and Insert ComboBox tools.
- **The Group/Ungroup tools** (the two rightmost buttons)—These tools are used to group and ungroup objects.
To group objects, refer to Grouping Objects.
To ungroup objects, refer to Ungrouping Objects.

Re-installing the Toolbar

If the draw toolbar is accidentally removed from the Display Builder, use the Toolbar dialog box to place it back in the toolbar area. Refer to Viewing Toolbars for details.

5.3 Adding Basic Graphic Object

The following topics identify the basic Draw Toolbar icons and show how to use them.

- Making a Selection (Selection tool on Draw Toolbar)
- Drawing a Line (Line button on Draw Toolbar)
- Drawing a Rectangle (Rectangle button on Draw Toolbar)
- Drawing a Rounded Rectangle (Rounded Rectangle button on Draw Toolbar)
- Drawing an Ellipse (Ellipse button on Draw Toolbar)
- Drawing an Open Polygon (Open Polygon Draw Tool)
- Drawing a Closed Polygon (Closed Polygon Draw Tool)
- Drawing an Open Bezier Curve (Open Bezier button on Draw Toolbar)
- Drawing a Closed Bezier Curve (Closed Bezier button on Draw Toolbar)
- Drawing an Arc (Arc button on Draw Toolbar)
- Drawing a Text Object (Text button on Draw Toolbar)

These topics cover how to insert other graphic objects from the menu.

- Inserting a Bitmap (Insert Bitmap menu and Draw toolbar)
- Inserting an Embedded Display (Insert/Display menu and Draw toolbar)
- Inserting OLE Controls (Insert/OLE Controls menu)
- Inserting a Button Object (Draw toolbar)
- Inserting a Data Entry Object (Draw toolbar)
- Inserting a Listbox Object (Draw toolbar)
- Inserting a ComboBox Object (Draw toolbar)

5. Adding Objects to Your Display

5.4. Making a Selection

5.4 Making a Selection

Selection tool on Draw Toolbar

Use the Selection tool  to select, move, and resize objects.

How to select an object or objects

To learn the various methods used to make a selection, refer to [Selecting Objects](#) and [Selecting all objects in a display](#).

How to deselect an object or objects

- 1 Move the mouse until the Selection tool points to a part of the display window where there are no objects.
- 2 Press the mouse button. The handles disappear from the object(s).

5.5 Drawing a Line

Line button on Draw Toolbar

Use the Line tool  to draw lines. Lines can have properties that include width, color, and style, such as dots and dashes.

How to draw a line

- 1 Click the Line tool to select it.
- 2 Position the pointer in the drawing area. The cursor changes to cross hairs.
- 3 Click and hold the mouse button. Drag the mouse. The line appears with a handle at each end. A circle appears in the center of the line to define the center of rotation.

NOTE: To draw a line at a 45-degree angle, hold down the Shift key while you drag the mouse.
- 4 Continue to drag the mouse until the line is the required length.
- 5 Release the mouse button when the line is the required length. Because the line is still selected, the handles are displayed.

- 6 Position the pointer in another part of the drawing area and click the mouse button. The line is deselected and the handles disappear

Other functions

The section titled *Manipulating Objects on the Screen* provides details on how to move, resize, rotate, flip horizontally or vertically, delete, and copy a line.

Line properties

If required, modify the line using Property Pages:

- Line Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)

5.6 Drawing a Rectangle

Rectangle button on Draw Toolbar

Use the Rectangle tool  to draw rectangles and squares.

How to draw a rectangle

- 1 Click the Rectangle tool to select it.
- 2 Position the pointer in the drawing area. The cursor changes to cross hairs.
NOTE: To draw a square, hold down the Shift key while performing the next three steps
- 3 Click and hold the mouse button. Drag the mouse. Lines appear that define the sides of the rectangle. Handles appear at each corner of the rectangle and in the center of each side. A circle appears in the center of the rectangle to define the center of rotation.
- 4 Continue to drag the mouse until the rectangle is the required size.
- 5 Release the mouse button when the rectangle is the required size. Because the rectangle is still selected, the handles are displayed.
- 6 Position the pointer in another part of the drawing area and click the mouse button. The rectangle is deselected and the handles disappear.

5. Adding Objects to Your Display

5.6. Drawing a Rectangle

Other functions

The section titled Manipulating Objects on the Screen provides details on how to move, resize, rotate, flip horizontally or vertically, delete, and copy a rectangle.

Rectangle properties

If required, modify the rectangle using Property Pages:

- Rectangle Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

5.7 Drawing a Rounded Rectangle

Rounded Rectangle button on Draw Toolbar

Use the Rounded Rectangle tool  to draw rounded rectangles and rounded squares.

How to draw a rounded rectangle

- 1 Click the Rounded Rectangle tool to select it.
- 2 Position the pointer in the drawing area. The cursor changes to cross hairs.

NOTE: To draw a rounded square, hold down the Shift key while performing the next three steps.
- 3 Click and hold the mouse button. Drag the mouse. Lines appear that define the sides of the rounded rectangle. Handles appear at the corners of the rounded rectangle and in the center of each side. The lower-right corner has two handles. The corners appear square. A circle appears in the center of the rounded rectangle to define the center of rotation
- 4 Continue to drag the mouse until the outside dimensions of the rounded rectangle are the required size.
- 5 Release the mouse button when the rounded rectangle is the required size. Because the rounded rectangle is still selected, the handles are displayed.
- 6 To round the corners, select the handle that is in the lower-right corner but inside the rectangle. This is the rounding handle. Hold the mouse button and drag the mouse toward the center of the rectangle. The corners become rounded.
- 7 Position the pointer in another part of the drawing area and click the mouse button. The rounded rectangle is deselected and the handles disappear.

5. Adding Objects to Your Display

5.8. Drawing an Ellipse

Other functions

The section titled Manipulating Objects on the Screen provides details on how to reshape, move, resize, rotate, flip horizontally or vertically, delete, and copy a rounded rectangle.

Rounded rectangle properties

If required, modify the rounded rectangle using Property Pages:

- Rounded-Rectangle Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

5.8 Drawing an Ellipse

Ellipse button on Draw Toolbar

Use the Ellipse tool  to draw an ellipse and a circle.

How to draw an ellipse

- 1 Click the Ellipse tool to select it.
- 2 Position the pointer in the drawing area. The cursor changes to cross hairs.

NOTE: To draw a circle, hold down the Shift key while performing the next three steps
- 3 Click and hold the mouse button. Drag the mouse. Lines appear that define the edge of the ellipse. Handles appear around the ellipse. A circle appears in the center of the ellipse to define the center of rotation.
- 4 Continue to drag the mouse until the ellipse is the required size.
- 5 Release the mouse button when the ellipse is the required size. Because the ellipse is still selected, the handles are displayed.
- 6 Position the pointer in another part of the drawing area and click the mouse button. The ellipse is deselected and the handles disappear.

Other functions

The section titled Manipulating Objects on the Screen provides details on how to move, resize, rotate, flip horizontally or vertically, delete, and copy an ellipse.

Ellipse properties

If required, modify the ellipse using Property Pages:

- Ellipse Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

5.9 Drawing an Open Polygon

Open Polygon Draw Tool

Use the Open Polygon tool  to draw open polygons.

How to draw a open polygon

- 1** Click the Open Polygon tool to select it.
- 2** Position the pointer in the drawing area. The cursor changes to cross hairs.
- 3** Click the mouse button where you want to start the first segment of the open polygon. This will be one of the “open” ends of the completed open polygon. Drag the mouse. A line appears that defines the first edge of the open polygon and your mouse is “attached” to the end of that line.
- 4** Move to the point where you want the first segment to end and single-click the mouse button to establish the first segment of the open polygon. Continue to add segments in this manner until the open polygon is the required shape, using a single-click to establish each segment.

5. Adding Objects to Your Display

5.9. Drawing an Open Polygon

- 5 Double-click the mouse button (or single-click the right-mouse button) when the last segment is in place at the other “open” end of the open polygon.

RESULT: The open polygon is completed and handles define the line segments.
- 6 Position the pointer in another part of the drawing area and click the mouse button. The open polygon is deselected and the handles disappear

Other functions

The section titled Manipulating Objects on the Screen provides details on how to reshape, move, resize, rotate, flip horizontally or vertically, delete, and copy an open polygon.

Open polygon properties

If required, modify the open polygon using Property Pages:

- Open Polygon Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

5.10 Drawing a Closed Polygon

Closed Polygon Draw Tool

Use the Closed Polygon tool  to draw a closed polygon.

How to draw a closed polygon

- 1 Click the Closed Polygon tool to select it.
- 2 Position the pointer in the drawing area. The cursor changes to cross hairs.
- 3 Click the mouse button where you want to start the first segment of the closed polygon. Drag the mouse. A line appears that defines the first edge of the closed polygon and your mouse is "attached" to the end of that line.
- 4 Move to the point where you want the first segment to end and single-click the mouse button to establish the first segment of the closed polygon. Continue to add segments in this manner until the closed polygon is the required shape, using a single-click to establish each segment.
- 5 Double-click the mouse button (or single-click the right-mouse button) when the last segment is in place.

RESULT: The closed polygon is completed and handles define the line segments.
- 6 Position the pointer in another part of the drawing area and click the mouse button. The closed polygon is deselected and the handles disappear.

Other functions

The section titled Manipulating Objects on the Screen provides details on how to reshape, move, resize, rotate, flip horizontally or vertically, delete, and copy a closed polygon.

5. Adding Objects to Your Display

5.11. Drawing an Open Bezier Curve

Closed polygon properties

If required, modify the closed polygon using Property Pages:

- Closed Polygon Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

5.11 Drawing an Open Bezier Curve

Open Bezier button on Draw Toolbar

Use the Open Bezier Line tool  to draw a Bezier curve.

A SINGLE Bezier curve segment is defined as an irregular curve between two fixed endpoints that is described by the position of two (and only two) additional points. If several single Bezier curve segments are continuously attached end-to-end, they produce a complex Bezier curve.

Both the Open and Closed Bezier Line tools are capable of drawing several connected Bezier curve segments, each connected to the next by fixed endpoints.

How to draw a SINGLE open Bezier line segment

Let us start by drawing a single Bezier line between two fixed endpoints.

- 1 Click the Open Bezier Line tool to select it.
- 2 On the display window, click and release the mouse button to produce the first fixed endpoint. Notice the cursor changes to crosshairs and the drawing SHAPE tool is selected as shown here.



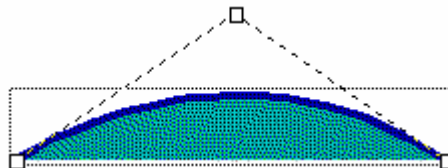
- 3 Now move the cursor to the right to a point about 2 inches (50 cm) away, to where you want the second fixed endpoint. Click the mouse to produce a single Bezier line that looks something like this:



- 4 Now click the **right-mouse** button to stop Display Builder from drawing more than the single line segment.

NOTICE: You can also stop Display Builder from drawing more than the single line segment by double-clicking the **left** mouse button.

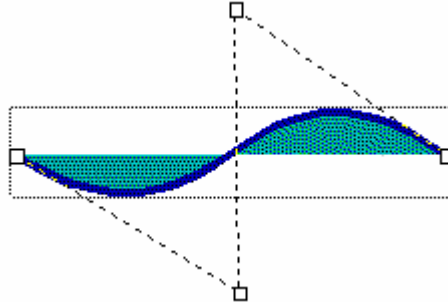
- 5 Now, with the SHAPE tool still selected, click and hold the mouse button on the first if required endpoint and drag out a control handle. Position this control handle about an inch (25 mm) **above** the line and halfway between the endpoints, then release the button. Your Bezier line segment now looks like this:



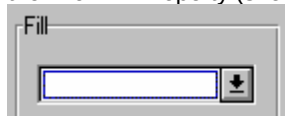
5. Adding Objects to Your Display

5.11. Drawing an Open Bezier Curve

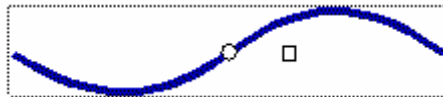
- 6 Again, with the SHAPE tool still selected, click and hold the mouse button on the first endpoint to drag out a second control handle. Position this control handle about an inch (25 mm) below the line and halfway between the endpoints, then release the button. The Bezier line segment now looks like this:



- 7 To get a “feel” for the shapes you can draw, try experimenting by moving the control handles around. Move both handles to the same side of the line and move them closer to and farther away from the fixed endpoints. Notice that you can move the handles beyond the right and bottom borders of the display window.
- 8 A default fill color was applied when this sample Bezier curve was drawn. You may want to change the fill color so it matches the background by double-clicking on the filled area and selecting a suitable fill color (white, in this case) from the Bezier Curve General Property Page. A better way is to change the fill color to the “No Fill” Property (shown below).



- 9 You can also change the line width and color. The finished curve is shown below (shown in the ROTATE mode).

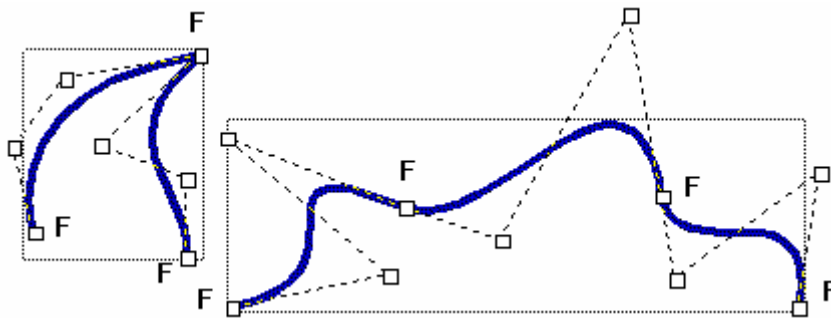


How to draw a COMPLEX Bezier line

You can easily construct a complex Bezier line by continuously joining several single Bezier line segments together.

- 1 When constructing a complex Bezier line, follow the instructions above but single-click at the endpoint of the first segment. Then continue on with the next joined Bezier line segment, then on-and-on with more line segments as required.
- 2 To finish the complex Bezier curve, double-click (or single-click the right-mouse button) at the endpoint of the last segment.

Here are samples of complex Bezier lines composed of two and three single Bezier line segments.



Each of the segment's fixed endpoints is marked by an "F" in these drawings.

Notice you can make a smooth transition between Bezier line segments when the second control handle of one line segment is in line with the first control handle of the next line segment, as shown in the second drawing.

Other functions

The section titled Manipulating Objects on the Screen provides details on how to reshape, move, resize, rotate, flip horizontally or vertically, delete, and copy an open Bezier curve.

5. Adding Objects to Your Display

5.12. Drawing a Closed Bezier Curve

Open Bezier properties

If required, modify the open Bezier curve using Property Pages:

- Open Bezier Curve Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

5.12 Drawing a Closed Bezier Curve

Closed Bezier button on Draw Toolbar

Use the Closed Bezier Line tool  to draw a closed Bezier curve.

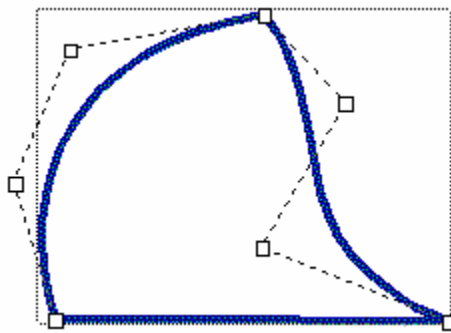
- 1 Select Closed Bezier Curve.
- 2 On the display grid, successively click 4 points to form a square, and then click the right mouse button to terminate the curve.

The only difference between closed and normal Bezier curves is, when the group of segments is completed, a straight line joins the first segment to the last as shown.

How to draw a closed Bezier line

Follow the basic instructions for drawing normal Bezier curves.

The only difference between closed and normal Bezier curves is that when the group of segments is completed, a straight line joins the first segment to the last, as in this illustration.



Other functions

The section titled *Manipulating Objects on the Screen* provides details on how to reshape, move, resize, rotate, flip horizontally or vertically, delete, and copy a closed Bezier curve.

Closed Bezier properties

If required, modify the closed Bezier curve using Property Pages:

- Closed Bezier Curve Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

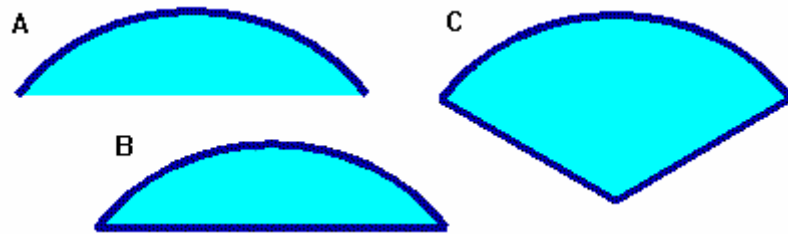
5. Adding Objects to Your Display

5.13. Drawing an Arc

5.13 Drawing an Arc

Arc button on Draw Toolbar

Use the Arc tool  to draw an arc (A), a Closed Arc (B), or a Pie (C).



The arc you draw can either be filled (as shown here) or empty (as in the examples that follow).

How to draw a “default” arc

- 1 Click the Arc tool to select it.
- 2 Position the pointer in the drawing area. The cursor changes to cross hairs. To draw an arc that is a portion of a circle, hold down the Shift key while performing the next three steps.
- 3 Click and hold the mouse button at the point where you want the upper-left-hand corner of the object. Drag the mouse at an approximate 45-degree angle to the right and down. Lines appear that define a “default” arc and a “ghost” outline of the ellipse it is part of (see left figure in the example that follows). Two handles show where you start and end the drag.

NOTE: If you had dragged the mouse from the upper-right-hand corner of the object and at an angle to the left and down, the arc appears upside-down from the example shown here.

- 4 Continue to drag the mouse until the ellipse/arc is the required shape (roundness) and size. You may want the finished arc to have a longer or shorter length, so use the ghost image to estimate where you want the final endpoints to fall—you will lengthen or shorten the “default” arc in the procedures that follow.



- 5 Release the mouse button. Because the ellipse/arc is still selected, “size” handles are displayed as shown to the right in the example above. At this time, a “default” arc is drawn with both endpoints at 30 degrees from a horizontal line through the center of the ellipse.

How to lengthen or shorten (reshape) the arc

- To reshape an arc, use the SHAPE tool  as described in How to reshape the length of an Arc (or Pie).

How to rotate the arc

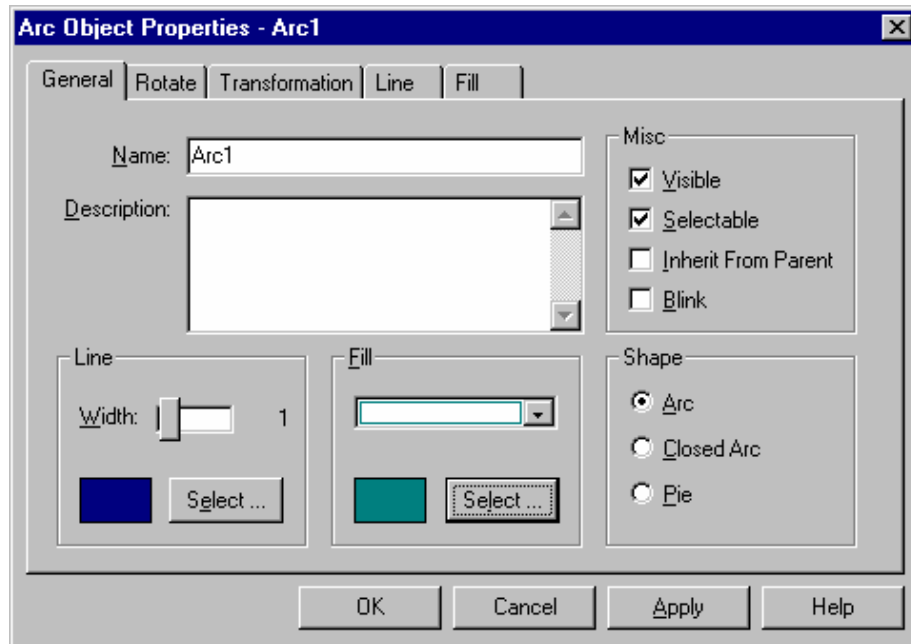
- To rotate an arc, use the ROTATE tool  as described in How to freely rotate an object or one of the other methods described in Rotating Objects.

5. Adding Objects to Your Display

5.13. Drawing an Arc

How to finish the arc

- 1 Either double-click the arc line or select Edit/Properties (Alt + Enter) to open the Arc General Properties Page, a portion of which is shown below.



- 2 Change the Line width, the Fill property ("no fill" is shown), and the Shape and Color as required. Click OK on the Arc General Properties Page to accept your changes
- 3 Position the pointer in another part of the drawing area and click the mouse button. The object is deselected and the handles disappear.

Other functions

The section titled Manipulating Objects on the Screen provides details on how to move, resize, flip horizontally or vertically, delete, and copy an arc.

Arc properties

If required, modify the arc using Property Pages:

- Arc Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)

5.14 Drawing a Text Object

Text button on Draw Toolbar

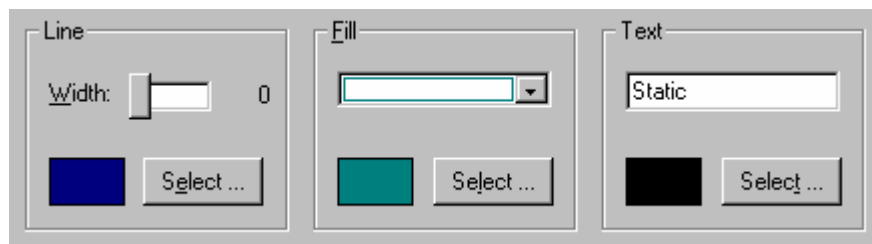
Use the Text tool  to add text to a drawing.

How to add text to a display

- 1 Click the Text tool to select it.
- 2 Position the pointer in the drawing area where text is required.
- 3 Click and hold the mouse button. Drag the mouse. A rectangle appears containing the word "Static" and "size" handles appear around the text.

NOTE: To draw a square containing the word "Static" instead of a rectangle, hold down the Shift key while you drag the mouse.

- 4 When the required size and length of text is reached, release the mouse button.
- 5 Select Edit/Properties (Alt + Enter) to open the Text General Properties Page, a portion of which is shown below.



5. Adding Objects to Your Display

5.14. Drawing a Text Object

- 6 As required, change the Line width and color of a box that can optionally surround the text (by default, the line doesn't show because it is of "0" width).
- 7 As required, change, the Fill property ("no fill" is shown) and color.
- 8 Double-click the word "Static" in the Text box and change it as required. Also, if required, change the color of the text.
- 9 Click OK on the Text General Properties Page to accept your changes.
- 10 Examine the finished object. If necessary, change the size of the bounding box to display your text properly.
- 11 Position the pointer in another part of the drawing area and click the mouse button. The object is deselected and the handles disappear.

Other functions

The section titled Manipulating Objects on the Screen provides details on how to move, resize, rotate, flip horizontally or vertically, delete, and copy a text object.

Text properties

If required, modify the text object using Property Pages:

- Text Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)
- Text Properties (Text Page)
- Value Properties (Value Page)

5.15 Inserting a Bitmap

Insert/Bitmap menu and Draw toolbar

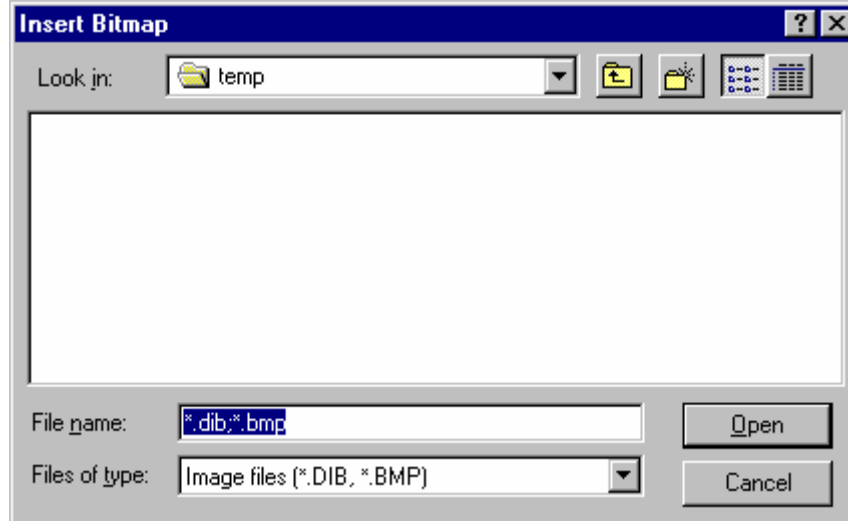
Use this command to insert a bitmap file (a file with a .bmp extension) into a display.

NOTE: You can only insert bitmap files that were created in 16 or 256 color format. An attempt to insert a 2-color (black and white) or 24-color bitmap will fail, and you will get an error message. These other bitmap formats can be inserted as OLE objects into a GUS Display by choosing Insert/Object on the menu bar.

There are two ways to insert a bitmap: (1) from the menu bar, and (2) from the Draw toolbar.

How to add a bitmap drawing

- 1 Select the display window where you want to place the bitmap drawing.
- 2 Select Insert/Bitmap on the menu bar, or
select the Insert Bitmap button  on the Standard toolbar.
The Insert Bitmap dialog box appears (it's similar to an Open dialog box).



- 3 Modify entries as required to locate the bitmap file. To learn about each feature of the Insert Bitmap dialog, click the  in the upper right corner of the box; then click the required feature for a short explanation.

5. Adding Objects to Your Display

5.15. Inserting a Bitmap

- 4 When the required bitmap file is in the File Name listbox, click the Open button.

RESULT: The dialog box closes and the cursor changes to crosshairs indicating that you are in the Insert mode.
- 5 Position the pointer in the display window and click the mouse button to select the location for the upper left corner of the bitmap.
- 6 Holding the mouse button down, drag the mouse to create a box for the bitmap. Release the mouse button.
- 7 The bitmap drawing is embedded into the display window. If the bitmap drawing is smaller than the box, the bitmap is positioned in the top-left corner of the box.
- 8 If required, modify the bitmap using its Property Pages:
 - Bitmap Object Properties
 - Rotate Properties (Rotate Page)
 - Transformation Properties (Transformation Page)

Other functions

The section titled Manipulating Objects on the Screen provides details on how to move, resize, rotate, flip horizontally or vertically, delete, and copy an open Bezier curve.

Bitmap properties

If required, modify the bitmap using Property Pages:

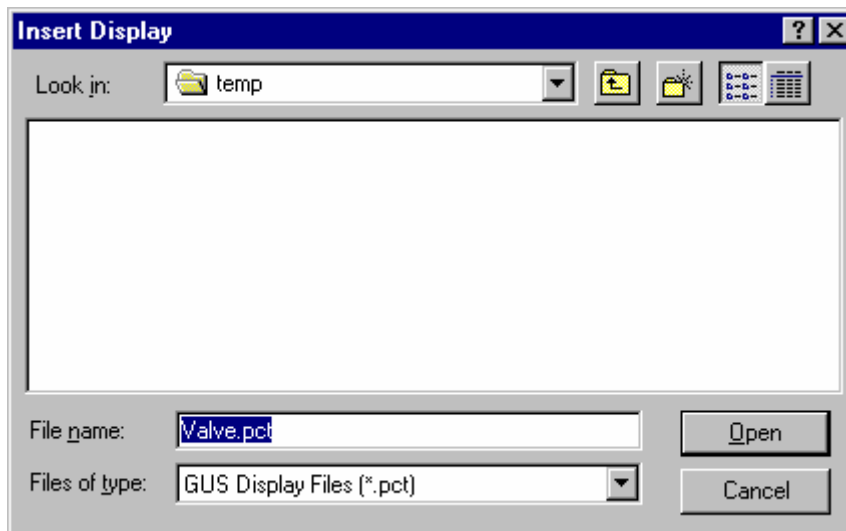
- Bitmap Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)

5.16 Inserting an Embedded Display

Insert/Display menu and Draw toolbar

Use this command to insert (embed) a display into a Display Builder display. There are two ways to insert a display: (1) from the menu bar, and (2) from the Draw toolbar.

- 1 Open the display where you want to embed another display
- 2 Select Insert\Display on the menu bar, or select the Insert Display button  on the Draw toolbar. The Insert Display dialog box is displayed



- 3 Modify entries as required to locate the Display file. To learn about each feature of the Insert Display dialog box, click the  in the upper right corner of the box; then click the required feature for a short explanation.

5. Adding Objects to Your Display

5.16. Inserting an Embedded Display

- 4 Select an object to embed (VALVE.PCT is selected in this example). When the filename of the required object is in the File Name listbox, click the Open button.

RESULT: The dialog box closes and the cursor changes to crosshairs indicating that you are in the Insert mode.

NOTE: If "Auto Param Prompts" has been set, the Entering Display Parameters (Edit/Enter Parameters) dialog box opens so you can insert parameters before the crosshairs are displayed.

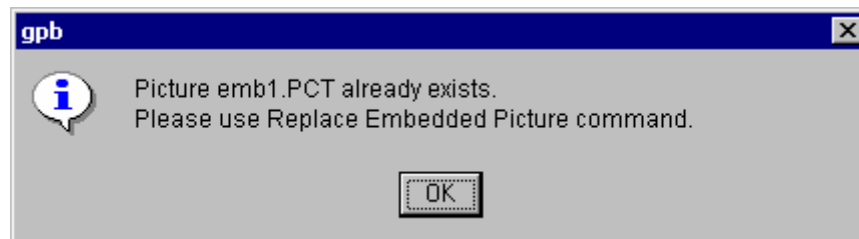
- 5 Move the crosshairs to where you want the upper-left-hand corner of the display to be located, then click once.

RESULT: The new embedded display is placed in the main display.

You can insert more than one new embedded display by:

- repeating steps 1, 2, and 3 and then selecting a new display in step 4, or by
- copying one display and pasting it several times.

NOTE: An embedded display is uniquely identified by its FILENAME. The version of the embedded display is identified by the MODIFICATION DATE and TIME of the embedded display's .PCT file. If you attempt to embed a second display with the same filename, a "Display already exists" error will occur.



NOTE: Embedded displays do not support the blinking property.

Copying an Embedded Display from one Display Window to Another

To copy an embedded display from one display window to another display window, execute the following steps:

- 1** Make sure there are at least two display windows open. The first window contains the embedded display that is to be copied to the second window.
- 2** Hold down the **Ctrl key** while you click and hold on the embedded display in the first window. Now, without releasing the mouse, drag the mouse so that it is over the second window in a position where the new embedded display is to be located.
- 3** Release the mouse button.

RESULT: A new embedded display is placed in the second window. If a script has been written for the first (parent) embedded display, the script will be associated with the second (child) display.

Copying an Embedded Display from one Location to Another in Same Display Window

To copy an embedded display to another location in the same display window, execute the following steps:

- 1** Hold down the **Ctrl key** while you click and hold on the embedded display you want to copy. Now, without releasing the mouse, drag the mouse so that it is over the area where the new embedded display is to be located.
- 2** Release the mouse button.

RESULT: A new embedded display is placed in the same window. If a script has been written for the first (parent) embedded display, the script will be associated with the second (child) display.

NOTE: You can insert the original embedded display any number of times into the same window.

5. Adding Objects to Your Display

5.17. Shape Library

5.17 Shape Library

The shape library contains a number of sample displays, shapes, and symbols that you can use in developing custom displays. These objects are provided on an AS IS basis, and are not certified as complete applications. Depending upon complexity, the objects are documented through a combination of embedded description/comment lines and text files. Text files have the same name as the .pct files they describe. The objects are contained in topical folders within C:\Program Files\Honeywell\TPS\RAC\Shapes. Each topical folder includes a Component Explorer Library (.cel) file and a set of embeddable components.

5.18 Inserting Displays

Insert/Display menu

The Insert/Display command allows you to insert displays into a display window.

To insert display objects

- 1 Select Insert\Display from the Menu Bar.
- 2 Browse to the required display object and then double-click it. The selected display is inserted in the display window.

To copy display objects from the Shape Library

- 1 Open the required sample .pct in the RAC\Shapes or RAC\symbols folder.
- 2 Select the required shape or symbol.
- 3 Press and hold the Ctrl key while dragging the object from the sample window to the required display window (or you can copy the object and paste it into the display window).
The object appears, as a group or embedded display (depending on the sample used), in the display window.

To use display objects in the Component Explorer

- 1 Register the .cel files with the Component Explorer. Refer to Component Explorer Management Functions for details.

NOTE: this operation only needs to be done once on each node.
- 2 Select the required display object from the Components tab of the GUS Explorer.
- 3 Select Insert from the right-mouse menu. The selected display object is inserted.

5.19 Inserting Controls

Insert/Control menu

The Insert/Control command allows control items to be inserted into a display window.

Available control items

The following Honeywell OLE controls are available:

- Honeywell GUS Alarm Window Control
- Honeywell GUS Faceplate Control
- Honeywell GUS Trend Control
- Honeywell xPM Logic Display Control (with GUS R360 and later, and with Experion Station-TPS (ES-T) nodes)
- 3rd-Party OLE Controls (not supplied with GUS)

The following Honeywell OLE controls are obsolete and have been replaced by GUS primitives that provide these functions more efficiently. They are supplied only to provide compatibility with earlier releases. They should not be inserted into new displays.

- Honeywell GUS Button Control
- Honeywell GUS Button Plus Control
- Honeywell GUS Checkbox Control
- Honeywell GUS Data Entry Control
- Honeywell GUS Listbox Control

5. Adding Objects to Your Display

5.19. Inserting Controls

Inserting an OLE Control

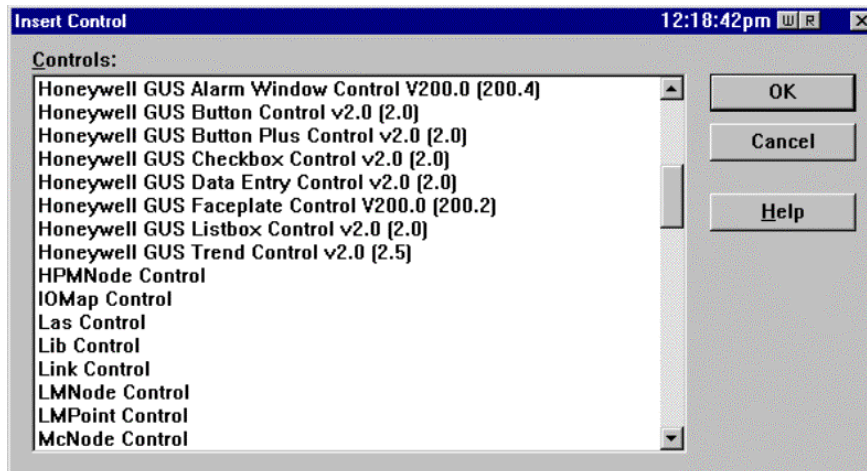


ATTENTION

ActiveX controls can appear in the Insert/Control dialog that are not appropriate for use in a GUS display. For example, some ActiveX controls may be automatically installed on a machine when an internet browser views a web page with such controls. Those controls may support the web page but should not be included in GUS displays. Example: the Microsoft Car-Point Auto-Pricer.

To avoid this problem, recognize that inserting unknown or unfamiliar controls into GUS displays, just because they appear in the Insert/Control dialog, may cause problems with the displays or the builder/runtime. Only use controls that you know are intended for persistent document applications such as GUS displays.

- 1 Select the display window in which you want the OLE control to appear.
- 2 Select Insert/Control from the menu.
- 3 The Insert Control dialog box appears listing the registered control items that are available.



- 4 Select the required control item.
- 5 Click OK. The control item is inserted in the **upper-left corner** of the display window.
IMPORTANT: If your display window is positioned so the **upper-left area** of your display doesn't show, you may not see that the control item has been inserted. Use the scroll bars to navigate to the upper-left corner before doing the next step.
- 6 Select the control item and drag it to the required point in your display.

5.20 Inserting a Button Object

Draw toolbar

- 1 Open the display where you want to insert the Button object.
 - 2 Select the Insert Button Object button  on the Draw toolbar. A pre-built Button object appears in the display
- 
- 3 With the Button object selected, you can either select Edit/Object Properties on the menu bar, or double-click the object, or press <Alt-Enter> on the keyboard, or click the right-mouse button. The Button Object Properties sheet appears, with the General page on display. Refer to Button Object General Properties for additional details.
 - 4 Enter the required parameters and select the required options.
 - 5 Display the remaining five Property pages one at a time, and make the required entries and selections.
 - 6 Click OK. The Button object is inserted.

5. Adding Objects to Your Display

5.21. Inserting a Data Entry Object

5.21 Inserting a Data Entry Object

Draw toolbar

- 1 Open the display where you want to insert the Display Entry Object.
- 2 Select the Insert Data Entry Object button  on the Draw toolbar. A pre-built Data Entry object appears in the display.



- 3 With the Data Entry object selected, you can either select Edit/Object Properties on the menu bar, or double-click the object, or press <Alt-Enter> on the keyboard, or click the right-mouse button. The Data Entry Object Properties sheet appears, with the General page on display. Refer to Data Entry Object General Properties for additional details
- 4 Enter the required parameters and select the required options.
- 5 Display the remaining two Property pages one at a time, and make the required entries and selections.
- 6 Click OK. The Data Entry object is inserted.

5.22 Inserting a Listbox Object

Draw toolbar

- 1 Open the display where you want to insert the Listbox object.
- 2 Select the Insert Listbox Object button  on the Draw toolbar. A pre-built Listbox object appears in the display.



- 3 With the Listbox object selected, you can either select Edit/Object Properties on the menu bar, or double-click the object, or press <Alt-Enter> on the keyboard, or click the right-mouse button. The Listbox Object Properties sheet appears, with the General page on display. Refer to Listbox Object General Properties for additional details.

- 4 Enter the required parameters and select the required options.
- 5 Display the remaining three Property pages one at a time, and make the required entries and selections.
- 6 Click OK. The Listbox object is inserted.

5.23 Inserting a ComboBox Object

Draw toolbar

- 1 Open the display where you want to insert the ComboBox object.
- 2 Select the Insert ComboBox Object button  on the Draw toolbar. A pre-built ComboBox object appears in the display.



- 3 With the ComboBox object selected, you can either select Edit/Object Properties on the menu bar, or double-click the object, or press <Alt-Enter> on the keyboard, or click the right-mouse button. The ComboBox Object Properties sheet appears, with the General page on display. Refer to ComboBox Object General Properties for additional details.



TIP

When resizing the ComboBox object, it is vertically limited in size that will accommodate font sizes of 5 to 76 points.

- 4 Enter the required parameters and select the required options.
 - 5 Display the remaining three Property pages one at a time, and make the required entries and selections.
- Click OK. The ComboBox object is inserted.

5. Adding Objects to Your Display

5.23. Inserting a ComboBox Object

6. Object Browser

6.1 Overview

There are two tools you can use to view and catalog display objects:

- the Object Browser to view display objects, and
- the Component Explorer to catalog display objects.

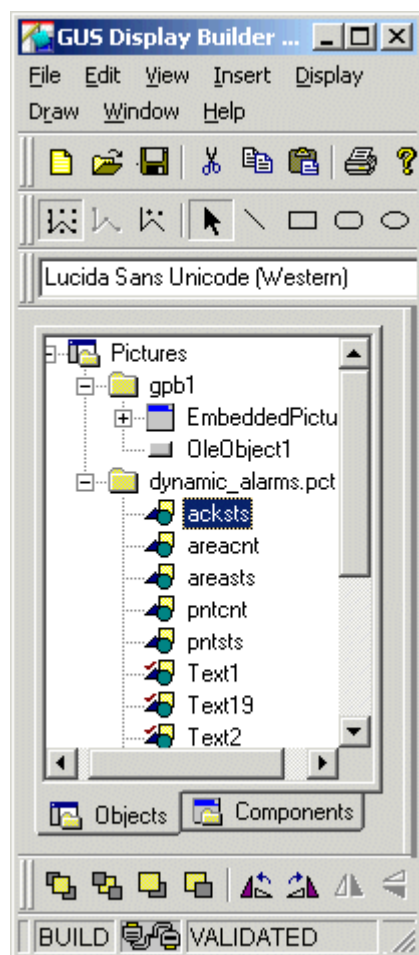
The Object Browser provides the GUS Display Builder with a hierarchical tree view of objects found in all open displays. A checkmark next to an object indicates script is available for that object.

6.2 Turn on the Object Browser

- 1 Turn on the GUS Display Builder.
- 2 Select GUS Explorer on the View menu (or click the GUS Explorer button  on the Standard toolbar). Either the Component Explorer or the Object Browser window is displayed.
- 3 If the Component Explorer is displayed, select the Objects tab. A view of the Object Browser is displayed.

6. Object Browser

6.2. Turn on the Object Browser



Each open picture in the GUS Display Builder has an associated tree node with the same name as the display title. Picture nodes are located under the “Pictures” root node:

- If a picture is closed in the Display Builder, its associated picture node disappears.
- If a new or an existing picture is opened in the Display Builder, a new picture node is added to the Object Browser’s root node.

Under each picture node is a listing of all the objects found in its associated picture. There are currently four classifications of objects found in the Object Browser: Primitives, Groups, Embedded Displays and ActiveX controls. OLE embedded objects such as a Word document will not appear in the Object Browser.

6.3 Docking

The Object Browser can be docked onto the Display Builder's top, bottom, left and right hand side, or can be used as an undocked floating window:

- To dock the Object Browser when it is a floating window simply drag the title bar to the region in the Display Builder where you want to dock it.
- To undock the Object Browser when it is docked, simply click the surrounding area of the tree control or drag the Object Browser off the side of the Display Builder.

When the Display Builder is first started, the Object Browser is docked on the left-hand side. Users can hide or show the Object Browser using the View / GUS Explorer menu item or the GUS Explorer button  on the Standard toolbar.

6.4 Updating

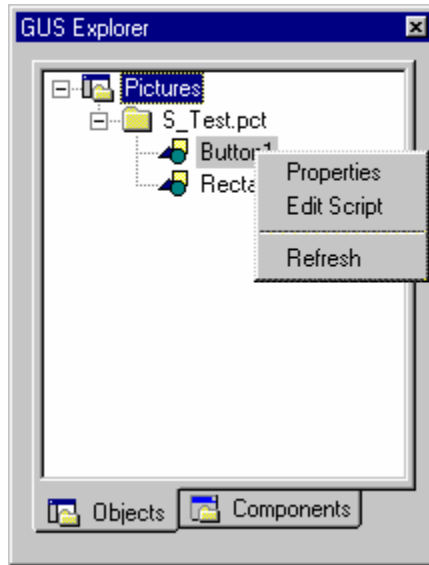
The Object Browser updates itself when changes are made to pictures in the Display Builder. When an object is inserted or removed from a display, an object is inserted or removed from the Object Browser. When an object is renamed, the Object Browser refreshes itself to pick up the new name.

6.5 Viewing Object Properties

Double-clicking a primitive node item in the Object Browser will display the object's Property Sheet in a modal state. Pressing the right-mouse button will display a popup menu that allows editing of properties and scripts for the current selected item in the Object Browser as shown below:

6. Object Browser

6.6. Editing Groups and Embedded Displays

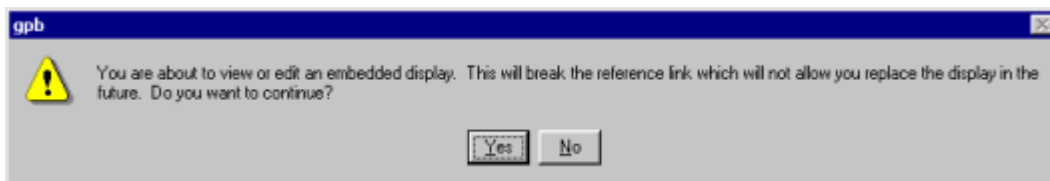


The Refresh menu item rebuilds the tree structure for the currently selected picture in the Object Browser and places the objects in alphabetical order.

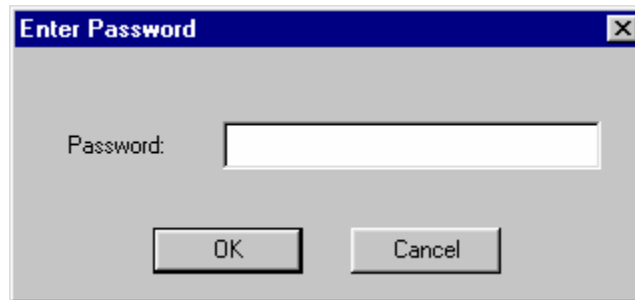
6.6 Editing Groups and Embedded Displays

It is possible to edit items internal to groups but users can not edit items in embedded displays. By expanding a group node in the Object Browser, users can see all the objects contained in the group and edit or view each group object's properties or scripts.

Viewing or editing an embedded display's internal objects will break the display's link with its original reference file. Once this link has been removed, the embedded display will be provided with a unique picture name as its reference. Prior to expanding an embedded display's tree item, the following dialog will prompt the user if they want to break the display's reference link:



If the embedded display has password protection, the user will be prompted for the password with the following dialog box:



If the correct password is not provided, the display tree item will not be expanded and the reference link will not be removed.

Embedded displays have a concept known as an internal script. This is the script attached to a display before it has been embedded. An embedded displays internal script can only be edited when the display has been unlocked. To access an internal script, highlight the unlocked embedded display in the object browser and click the right-mouse button.

6. Object Browser

6.6. Editing Groups and Embedded Displays

7. Component Explorer

7.1 Overview

There are two tools you can use to view and catalog display objects:

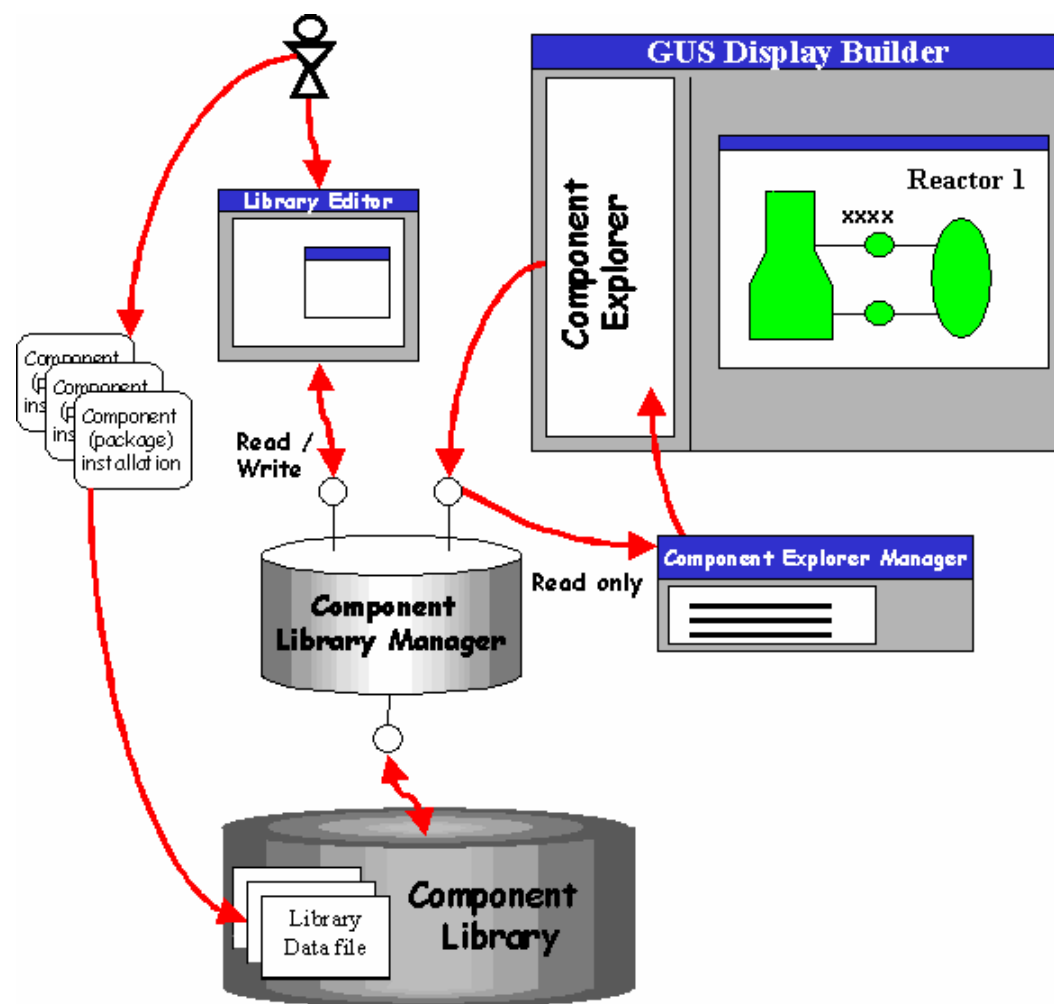
- the Object Browser to view display objects, and
- the Component Explorer to catalog display objects.

Component Explorer Introduction

The Component Explorer consists of a set of Display Builder tools that you can use to store, control, and access embedded displays easily, and to get an overview of all the Embedded Displays that can be accessed from your GUS. The functionality provided by the Component Explorer helps you to build displays quickly and accurately. The following functional block diagram shows the various Component Explorer tools and their interfaces.

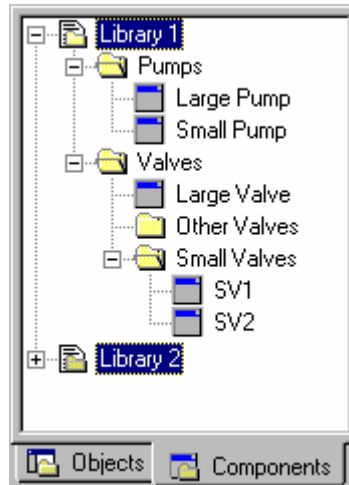
7. Component Explorer

7.1. Overview



Component Explorer Description

The Component Explorer stores embedded displays as “components” in a Component Library. Libraries are structured hierarchically, similar to the way Microsoft Windows folders and sub-folders are structured. An example of the configuration a typical Component Library is shown below.



In the above display, note that SV1 and SV2 represent two different embedded displays in the Library1\Valves\Small Valves folder structure. Note also that multiple libraries (Library 1 and Library 2) are supported.

The Component Explorer uses the following icons:

Icon	Representation
	Selected Folder
	Unselected Folder
	Embedded Display
	Non-existent Embedded Display
	Component Explorer Page
	Library

7.2 Component Explorer Tools

The Component Explorer consists of the following three tools:

- Basic Component Explorer
- Component Explorer Manager
- Component Library Editor

Basic Component Explorer

The basic Component Explorer tool is a part of the Display Builder. It provides a view to the embedded displays in the Component Library and access to a broad range of component, folder, library, and general application functions.

Component Explorer Manager

The Component Explorer Manager tool is also part of the Display Builder. It is used to configure the view that the Component Explorer presents to the user.

Library Editor

The Library Editor is an optional tool. It is not part of the Display Builder. It runs as a separate executable that can be installed along with the Display Builder. It is used to create new libraries and modify existing libraries. The Library Editor is especially useful during the installation of new software.

7.3 Component Explorer Functions

The functions provided by Component Explorer can be divided into the following three groups:

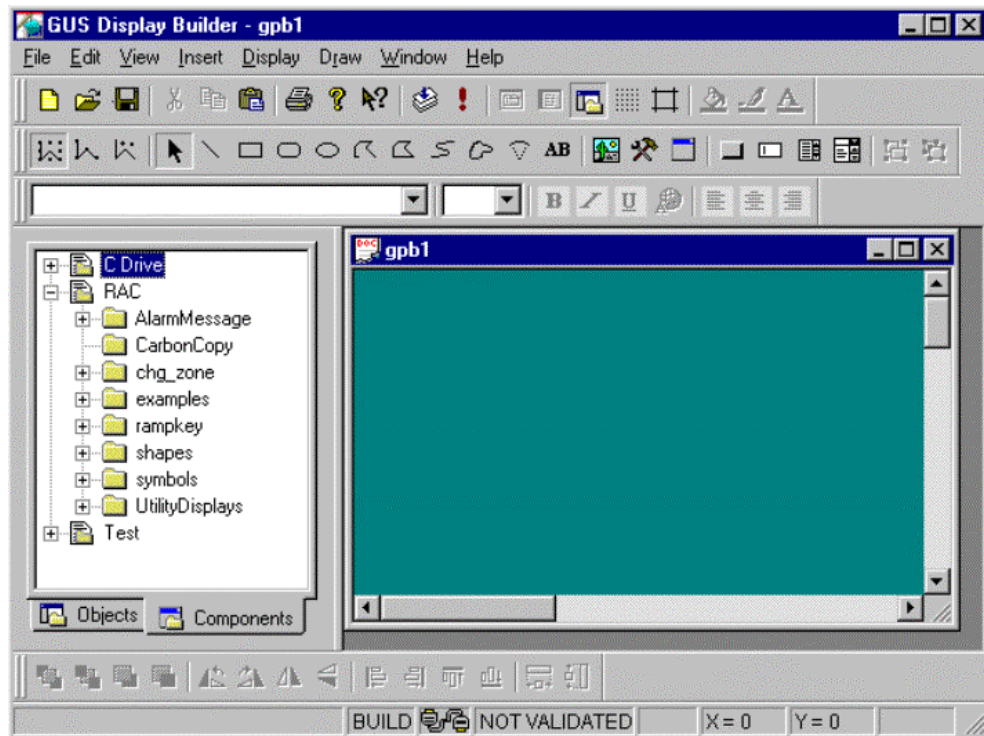
- Basic Component Explorer Functions
- Component Explorer Management Functions
- Component Library Editing and Display Functions, using the Library Editor

7.4 Basic Component Explorer Functions

All basic Component Explorer functions are initiated through the Component Explorer page.

Turn on the Component Explorer

- 1 Turn on the GUS Display Builder.
- 2 Select GUS Explorer on the View menu (or select the GUS Explorer button  on the Standard toolbar). Either the Object Browser or the Component Explorer window is displayed.
- 3 If the Object Browser is displayed, select the Components tab. A view of the Components Library is displayed.



The Components Library is displayed in the standard Microsoft hierarchical tree format.

7. Component Explorer

7.5. Component Functions

There are four different categories of basic Component Explorer functions:

- 1 Component Functions
- 2 Folder Functions
- 3 Library Functions
- 4 General Functions

7.5 Component Functions

Component functions can be initiated by double-clicking a component and also by using the Component Functions pop-up menu.

Display the Component Functions Popup Menu

To display the component Functions Popup Menu:

- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Open the library and folder that contains the component you want to access after you display it.
- 3 Right-click a component in the Component Library. The Component Functions pop-up menu is displayed.



The following Component functions are available on the Component Functions pop-up menu:

- Open a Component
- Insert a Component into an Existing Display
- Replace a Component
- View the Properties of a Component
- Refresh a Component
- Turn On the Component Explorer Manager

Error Handling

When a function is initiated that requires reading a file, the following checks are made:

- Is the file reference correct?
- Is the user authorized to access the file?
- Is the file the component type that goes with the file extension?

If an error is detected, a message explaining the problem will be displayed.

Component functions that can be affected are as follows:

- Opening a Component
- Inserting a Component
- Replacing a Component

Open a Component

To open a Component in Component Explorer:

- 1** Turn on the Component Explorer. The Component Library is displayed.
- 2** Right-click the component you want to open. The Component Functions pop-up menu appears.
- 3** Select Open.

A new Display Builder window containing the selected component opens. If a Display Builder window was already open, the new window will overlay it. This embedded display is now available for any normal Display Builder editing you want to perform.

Insert a Component into an Existing Display

- 1 Call up the Display Builder display into which you want to insert a component.
- 2 Turn on the Component Explorer. The Component Library is displayed.
- 3 Perform one or the other of these steps:

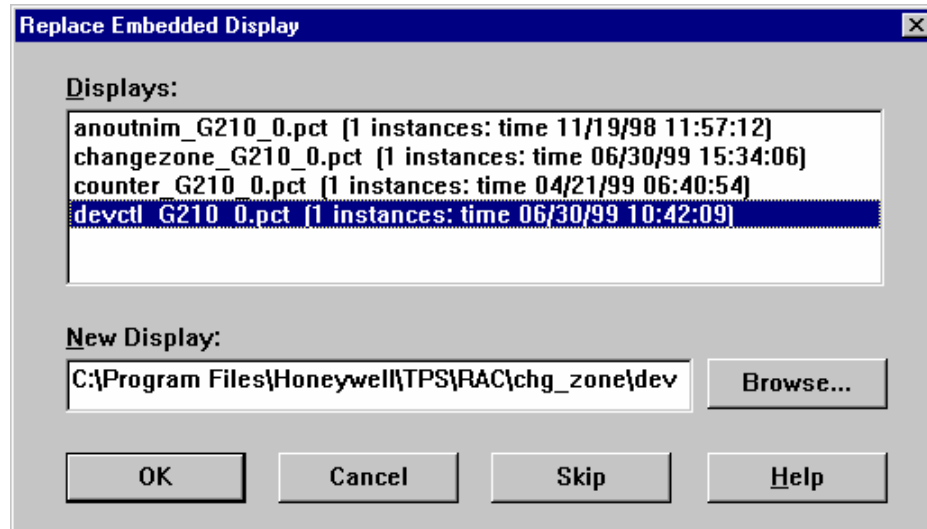
Right-click the component that you want to insert. The Component Functions pop-up menu appears. Then select Insert. The pop-up menu disappears and cross hairs appear in the Display Builder window.

OR

Double-click the component you want to insert. Cross hairs appear in the Display Builder window. NOTE: The Component Explorer "Opens it" option must be active before this function will work. For details, refer to Component Explorer Management Functions.
- 4 Click the mouse. The embedded display is inserted into the Display Builder window where the cross hairs appear.

Replace a Component

- 1 Call up the Display Builder display into which you want to insert a component.
- 2 Turn on the Component Explorer. The Component Library is displayed.
- 3 Right-click the new replacement component. The Component Functions pop-up menu appears.
- 4 Select Replace. The standard Display Builder Replace Embedded Display dialog box is displayed.



The Displays window lists all of the embedded displays in the Display Builder display. The New Display port displays the pathname of the new component in the Component Library.

- 5 In the Displays list, select (highlight) the embedded display you want to replace.
- 6 Click OK. The Replace Embedded Display dialog box disappears and the new component replaces the embedded display in the Display Builder display.

NOTE 1: You must save the Display Builder display before the replacement is actually made.

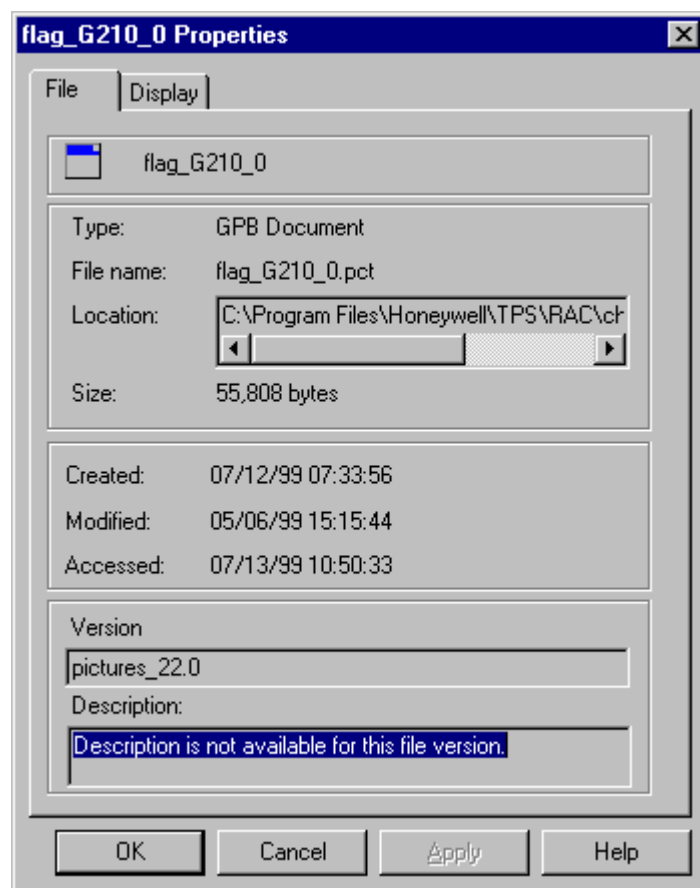
NOTE 2: The Skip button is not active. If you press it, it will function like the Cancel button.

View the Properties of a Component

- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click the component whose properties you want to view. The Component Functions pop-up menu appears.
- 3 Select Properties. The Properties window with two tabs is displayed.
- 4 Select the first tab. The name, file path, and description of the selected component are displayed.

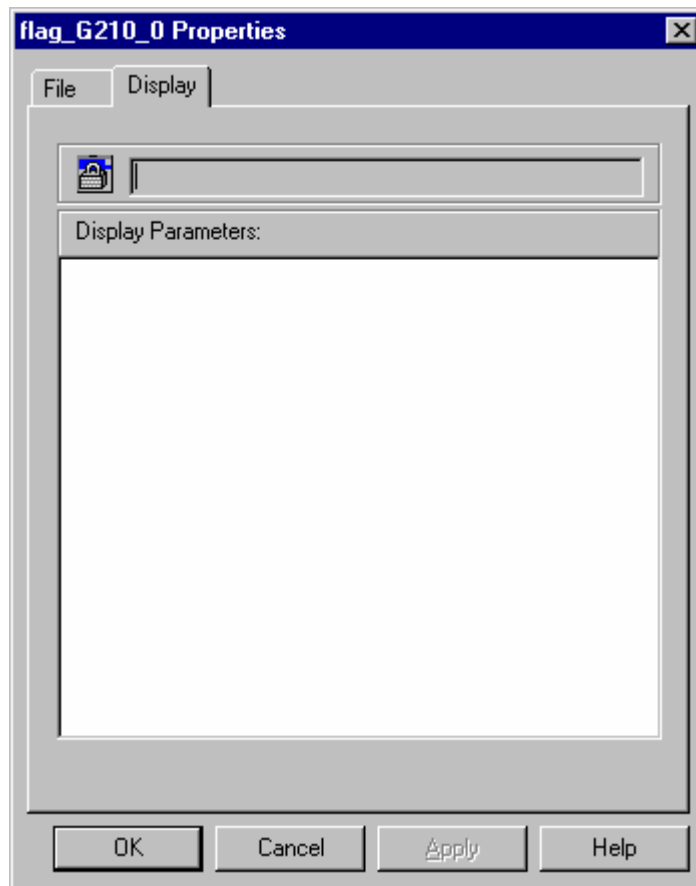
7. Component Explorer

7.5. Component Functions



If the file reference is validated, the size and dates of the file are also displayed.
If the file reference is not correct, the component is changed to a non-existent component.

- 5 Select the second tab. If the component is a display, the parameters and password protection indicator are displayed.



Refresh a Component

The Component Explorer refreshes a component by checking to see if the .pct file specified for the component, as listed in the Component Library still exists. If it does, the Component Explorer updates the version and description of the component in the Component Library from the values listed in the .pct file.

7. Component Explorer

7.6. Folder Functions

- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click the component whose properties you want to refresh. The Component Functions pop-up menu appears
- 3 Select Refresh on the Component Functions pop-up menu. The file reference for the selected component is checked.
- 4 If no problems were detected, the description and version number of the selected component are updated.
If the file reference for the component was incorrect, for example the .pct file couldn't be found, the component is changed into a non-existent component.

Turn On the Component Explorer Manager

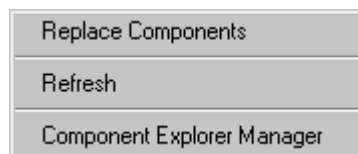
- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click any component in the Component Library. The Component Functions pop-up menu appears.
- 3 Select Component Explorer Manager. The Component Explorer Manager is turned on.

7.6 Folder Functions

Folder functions are initiated through the Folder Functions pop-up menu.

Display the Folder Functions Popup Menu

To display the Folder Functions Popup Menu, right-click a folder in the Component Library. The Folder Functions pop-up menu is displayed (See below).

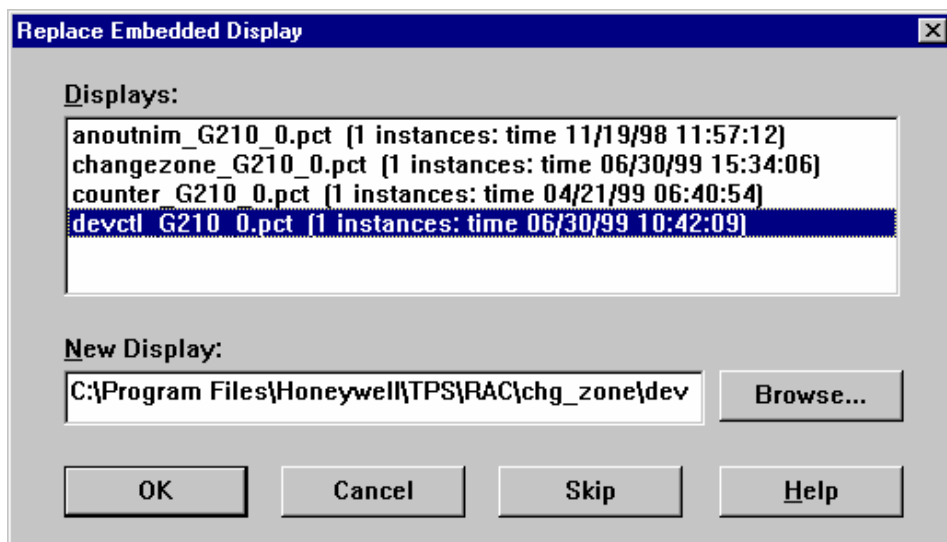


The following Folder functions are available on the Folder Functions pop-up menu:

- Replace Components
- Refresh Components
- Turn On the Component Explorer Manager 2

Replace Components

- 1 Call up the Display Builder display containing the components (embedded displays) you want to replace.
- 2 Turn on the Component Explorer. The Component Library is displayed.
- 3 Right-click the folder containing the new replacement components. The Folder Functions popup menu appears.
- 4 Select Replace Components. The standard Display Builder Replace Embedded Display dialog box is displayed.



The Displays window lists all of the embedded displays in the Display Builder display. The New Display port displays the pathname of the first component in the folder in the Component Library.

- 5 In the Displays list, select (highlight) the embedded display you want to replace.
- 6 Click OK. The new component replaces the embedded display selected in the Display Builder window and in the Displays list. The New Display port is updated to show the pathname of the next new component.

7. Component Explorer

7.6. Folder Functions

- 7 If required, click the Skip button until the pathname of required new component appears in the New Display.
- 8 Repeat steps 5, 6, and 7 until you have replaced all required Display Builder embedded displays. When the last display is replaced, the Replace Embedded Display dialog box disappears.

Refresh Components

The Component Explorer refreshes a component by checking to see if the .pct file specified for the component, as listed in the Component Library, still exists. If it does, the Component Explorer updates the version and description of the component in the Component Library from the values listed in the .pct file.

- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click the folder whose components you want to refresh. The Folder Functions pop-up menu appears.
- 3 Select Refresh. Each of the components in the folder is checked.

If no problems were detected, the description and version number of all the components in the selected folder are updated.

If the file reference for a component was incorrect, for example the .pct file couldn't be found, the component is changed into a non-existent component.

Turn On the Component Explorer Manager 2

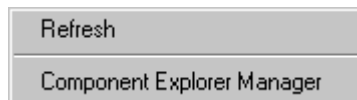
- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click any folder in the Component Library. The Folder Functions pop-up menu appears.
- 3 Select Component Explorer Manager. Component Explorer Manager is turned on.

7.7 Library Functions

Library functions are initiated through the Library Functions pop-up menu.

Display the Library Functions Popup Menu

To display the Library Functions Popup Menu, right-click a library in the Component Library. The Library Functions pop-up menu is displayed.



The following Library functions are available on the Library Functions pop-up menu:

- Refresh a Library's Components
- Turn On the Component Explorer Manager 3

Refresh a Library's Components

The Component Explorer refreshes a component by checking to see if the .pct file specified for the component, as listed in the Component Library, still exists. If it does, the Component Explorer updates the version and description of the component in the Component Library from the values listed in the .pct file.

- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click the library whose components you want to refresh. The Library Functions pop-up menu appears.
- 3 Select Refresh. Each component in all of the folders in the selected library is checked.

If no problems were detected, the description and version number of all the components in all the folders in the selected library are updated.

If the file reference for a component was incorrect, for example the .pct file could not be found, the component is changed into a non-existent.

7. Component Explorer

7.8. General Functions

Turn On the Component Explorer Manager 3

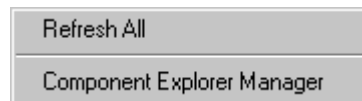
- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click any library in the Component Library. The Library Functions pop-up menu appears.
- 3 Select Component Explorer Manager. The Component Explorer Manager is turned on.

7.8 General Functions

General functions are initiated through the General Functions pop-up menu.

Display the General Functions Popup Menu

- 1 Right-click in any blank area in the Component Library. The General Functions pop-up menu is displayed.



There are two General functions available from the General Functions pop-up menu.

- Refresh All Components in the Component Library
- Turn On the Component Explorer Manager 4

Refresh All Components in the Component Library

- 1 Turn on the Component Explorer. The Component Library is displayed.
- 2 Right-click in any blank area in the Component Library. The General Functions pop-up menu appears.
- 3 Select Refresh. Each component, in each of the folders, in each library is checked.

If no problems were detected, the description and version number of all the components in all the folders in all the Component Explorer libraries are refreshed.

If the file reference for a component is incorrect, the component is changed into a non-existent.

Turn On the Component Explorer Manager 4

- 1** Turn on the Component Explorer. The Component Library is displayed.
- 2** Right-click in any blank area in the Component Library. The General Functions pop-up menu appears.
- 3** Select Component Explorer Manager. The Component Explorer Manager is turned on.

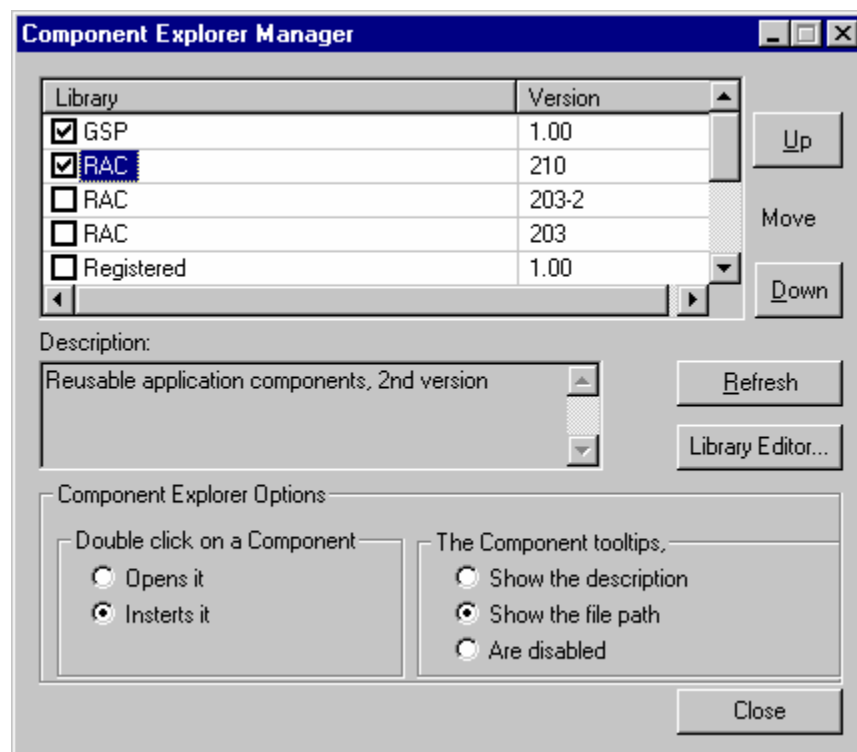
7.9 Component Explorer Management Functions**Component Explorer Manager**

Component Explorer Management functions are executed by the Component Explorer Manager (CEM) through the Component Explorer Manager window, which is displayed when the Component Explorer Manager is turned on. For details, refer to Turn On the Component Explorer Manager.

An example of a typical Component Explorer Manager window is shown below.

7. Component Explorer

7.9. Component Explorer Management Functions



The following basic Component Explorer Manager functions are available on the Component Explorer Manager window:

- Place a Library in the Component Explorer Library
- Refresh All Components in the Component Explorer Library
- Turn On the Library Editor
- Select the Component Explorer Options
- Turn Off the Component Explorer Manager

Place a Library in the Component Explorer Library

- 1** Turn on the Component Explorer Manager. The Component Explorer Manager window is displayed.
- 2** Select the library with the version you want to be controlled. The Library name is highlighted, the Version listed, and information about this library is displayed in the Description box.
- 3** If you want to relocate the selected library, click the Up or Down buttons to move it up and down the list.
- 4** Select the associated check box. A check mark is displayed. The selected library is immediately added to the Component Explorer Library and will be subject to Component Explorer control.
- 5** Turn off the Component Explorer Manager (select Close).

NOTE: Only one version of the same library can be controlled at any one time.

Refresh All Components in the Component Explorer Library

- 1** Turn on the Component Explorer Manager. The Component Explorer Manager window is displayed.
- 2** Select Refresh. Each component in each folder in each library in the Component Explorer Library is checked.

If no problems were detected, the description and version number of all the components in all the folders in all the Component Explorer libraries are refreshed.

If the file reference for a component is incorrect, the component will be flagged as "non-existent."

The library and version data in the Component Explorer Manager window will be updated to reflect any changes that may have been made to existing libraries, and to show any new libraries that may have been added since the last refresh.

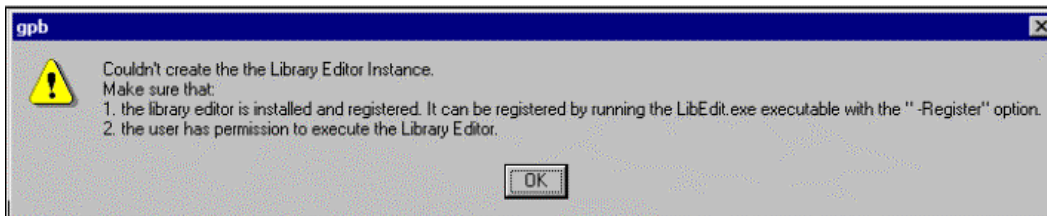
7. Component Explorer

7.9. Component Explorer Management Functions

Turn On the Library Editor

Prerequisites

- The Library Editor (an optional package) must be installed and executed with the “register” option before you can perform this function. If it is not, you will get the error message that follows.
- You must have permission (normal NT permission) to access the Library Editor. If you do not have permission and attempt to turn on the Library Editor, you will get the error message that follows.



Two ways to turn on the Library Editor

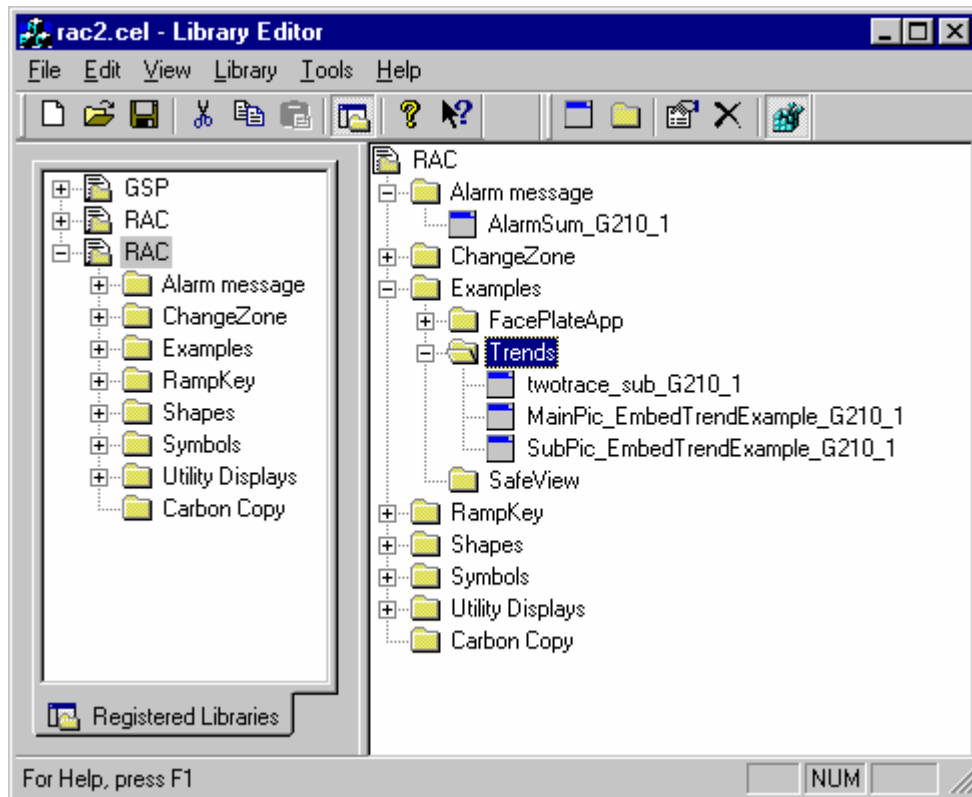
EITHER

- 1 Turn on the Component Explorer Manager. The Component Explorer Manager window is displayed..
- 2 Select the Library Editor... button in the Component Explorer Manager window.

OR

In the task bar at the bottom of the screen, select Start > Programs > Honeywell TPS > Component Library Editor.

With either method, the Library Editor window is displayed.



7. Component Explorer

7.9. Component Explorer Management Functions

Select the Component Explorer Options

- 1** Turn on the Component Explorer Manager. The Component Explorer Manager window is displayed.

- 2** Specify what you want to happen when you double-click a component by selecting either the Opens it or Inserts it radio button.

THEN

- 3** Specify what you want the Component tooltips to display by selecting either the description or file path radio button.

OR

Select the disabled radio button to disable the Component tooltips

NOTE: Opens it and Show the description are the default values of the options. These values will be used if a read or write error occurs.

Turn Off the Component Explorer Manager

- 1** Turn on the Component Explorer Manager. The Component Explorer Manager window is displayed.
- 2** Perform the required functions.
- 3** Select the Close button. The Component Explorer Manager window is turned off and the window disappears.

7.10 Component Library Editing and Display Functions

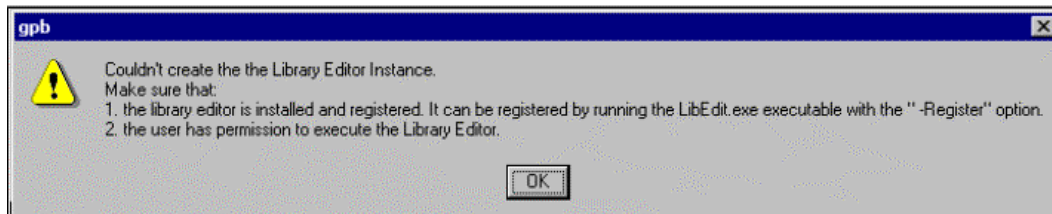
Library Editor

The Component Explorer library consists of two main parts, a registry segment and a file segment. Component Library editing and display functions are executed by the Library Editor, through the Library Editor window that displays both segments, the registry segment on the left side of the screen and the file segment on the right.

Turn On the Library Editor

Prerequisites

- The Library Editor (an optional package) must be installed and executed with the “register” option before you can perform this function. If it is not, you will get the error message shown below.
- You must have permission (normal NT permission) to access the Library Editor. If you do not have permission and attempt to turn on the Library Editor, you will get the error message shown below.



Two ways to turn on the Library Editor:

EITHER

- Turn on the Component Explorer Manager. The Component Explorer Manager window is displayed.
- Select the Library Editor... button in the Component Explorer Manager window.

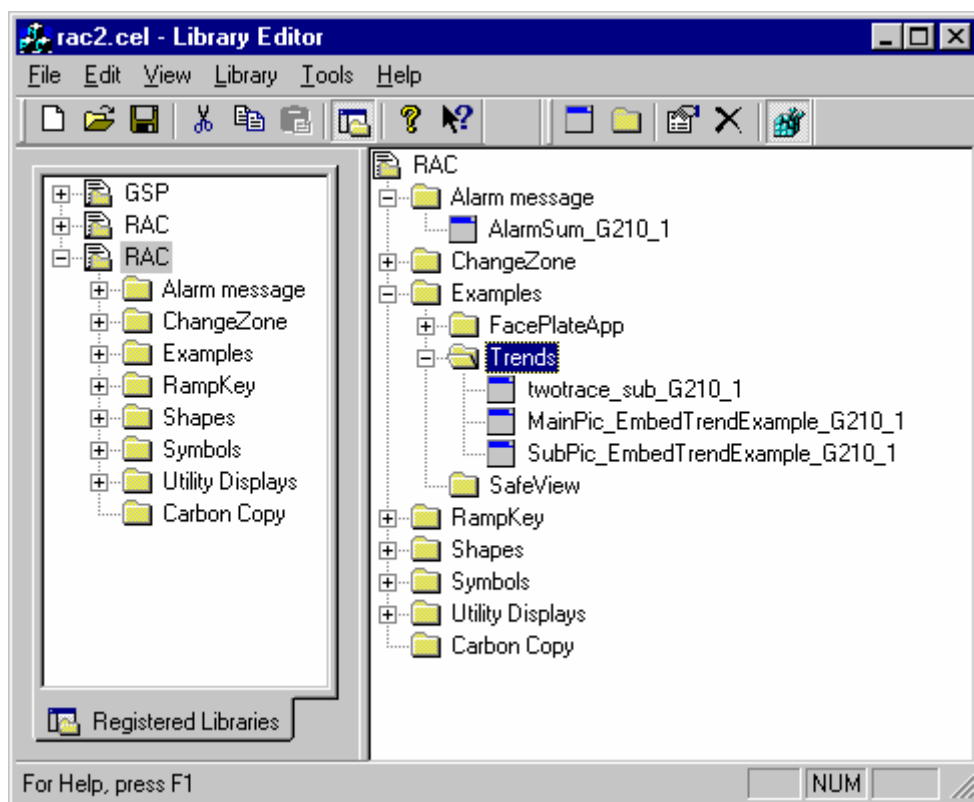
7. Component Explorer

7.10. Component Library Editing and Display Functions

OR

In the task bar at the bottom of the screen, select Start > Programs > Honeywell TPS > Component Library Editor.

With either method, the Library Editor window is displayed.



The Library Editor uses the following icons to depict the library structure:

Icon	Representation
	Selected Folder
	Unselected Folder
	Embedded Display
	Non-existent Embedded Display
	Library Data File (a main library)

Library editing and display functions are executed from the Library Editor window by right-clicking libraries and components and by using the following four menus:

Library Editor menu bar	
Library Editor toolbar	
Library popup menu	
Folder-Component popup menu	

7. Component Explorer

7.10. Component Library Editing and Display Functions

Two types of library functions

There are two types of library functions:

- Basic Library Editing and Display
- Registered Libraries Editing and Display

Basic Library Editing and Display

The following basic editing and display functions are available:

Library Editor File Functions

- Display the File Menu
- Create a Library
- Open a Library
- Open a Recently Opened Library
- Save a Library
- Save a Library with Changes
- Exit the Library Editor

Library Editor Edit Functions

- Display the Edit Menu
- Cut a Folder or Component
- Copy a Folder or Component
- Paste a Folder or Component

Library Editor View Functions

- Display the View Menu
- Display (or Hide) the Standard Toolbar
- Display (or Hide) the Library Editor Toolbar
- Display (or Hide) the Status Bar
- Display (or Hide) the Registered Libraries Window

Library Editor Library Functions

- Display the Library Menu
- Create a Folder
- Create a Component
- Register (or Unregister) a Library
- Delete a Folder or Component
- Edit Library Properties
- Edit Folder Properties
- Edit Component Properties

Library Editor Tool Functions

- Display the Tools Menu
- Clean a Library
- Create a Library from a Folder

Library Editor Help Functions

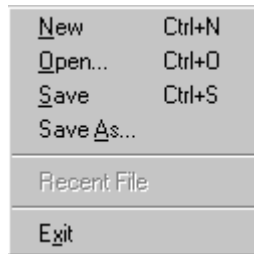
- Display the Library Editor Help Topics
- Display Library Editor Information
- Display Context-sensitive Help

7.11 Library Editor File Functions

Library Editor File functions are executed from the File menu.

Display the File Menu

- 1 Select File on the Library Editor menu bar. The File drop-down menu appears.



The File functions available are as follows:

- Create a Library
- Open a Library
- Open a Recently Opened Library
- Save a Library
- Save a Library with Changes
- Exit the Library Editor

Create a Library

- 1 Select New on the File menu (or click the New button  on the Standard toolbar).
A new, empty library called "New Library" is created and displayed in the Library Editor window, if it didn't already exist. If "New Library" already existed, an index is added and "New Library #1" is created, if it didn't already exist. This is repeated (the index # is incremented) until a unique library name is found.

Open a Library

- 1 Select Open on the File menu (or click the Open  button on the Standard toolbar).
The standard Microsoft Open dialog box is displayed.
- 2 Locate and open the required library. It is now available for editing.

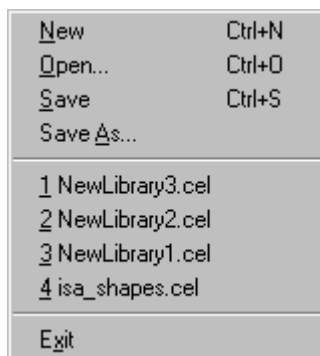
Open a Recently Opened Library

Prerequisite

To execute this function, you must have already opened and closed at least one configuration file. As you continue to open and close files, the last four files will always be listed between Save As... and Exit in the File drop-down menu.

Procedure

Select the required Library from the list of the last four files opened. The selected library is opened.



Save a Library

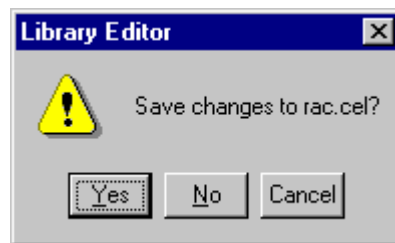
Select Save on the File menu (or click the Save  button on the Standard toolbar).
The Library Data file for the currently open library is saved.

Save a Library With Changes

- 1 Select **S**ave **A**s on the File menu. The standard Microsoft Save As dialog box is displayed.
- 2 Make the required changes to the file name or type of the currently open library and then click OK. The Library Data file for the currently open library is saved.

Exit the Library Editor

- 1 Select **E**xit on the File menu. The Library Editor is turned off and the Library Editor window disappears.
- 2 If you made changes to the open Library Data file and didn't save them, a Library Editor dialog box will appear asking if you want to save your changes. An example follows.



- 3 If you want to save the changes, click **Y**es. Click **N**o to ignore the changes.

The dialog box disappears and the Library Editor is turned off.

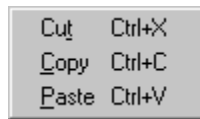
NOTE: to negate this procedure and return to the Library Editor, click Cancel.

7.12 Library Editor Edit Functions

Library Editor Edit functions are executed from the Edit menu.

Display the Edit Menu

Select Edit on the Library Editor menu bar. The Edit drop-down menu appears.



The Edit functions available are as follows:

- Cut a Folder or Component
- Copy a Folder or Component
- Paste a Folder or Component

Cut a Folder or Component

This function is not available for folder or components in a registered library.

- 1 Select the folder or component in the currently open library you want to cut.
- 2 **THEN EITHER**
Select Edit on the Library Editor menu bar. The Edit drop-down menu appears.
Select Cut on the Edit menu.
OR
Click the Cut button  on the Standard toolbar.

The selected item and all its sub-items are removed and placed in the Clipboard.

Copy a Folder or Component

- 1** Select the folder or component you want to copy.
- 2** **THEN EITHER**

Select Edit on the Library Editor menu bar. The Edit drop-down menu appears. Select Copy on the Edit menu.

OR

Click the Copy  button on the Standard toolbar

The selected item and all its sub-items are copied to the Clipboard.

Paste a Folder or Component

Prerequisite

The folder or component you want to paste must be in the Clipboard. There are three ways you can paste a folder or component.

Three Procedures to choose from:

- 1** Select the folder or component in the library below which you want the new folder or component to appear.
- 2** Select Edit on the Library Editor menu bar. The Edit drop-down menu appears.
- 3** Select Paste on the Edit menu.

OR

- 1** Select the folder or component in the library below which you want the new folder or component to appear.
- 2** Click the Paste  button on the Standard toolbar.

OR

- 1** Right-click the folder or component in the library below which you want the new folder or component to appear. The Library popup menu appears.
- 2** Select Paste on the Library popup menu.

The new folder or component and all its subordinate folders and components in the Clipboard are inserted into the library.

NOTE 1: If the Clipboard is empty, the Paste command and the Paste button will both be inoperable (grayed out).

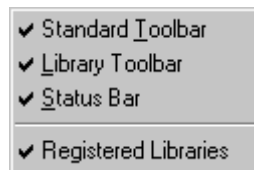
NOTE 2: If the items in the Clipboard already exist, the Paste command will do nothing.

7.13 Library Editor View Functions

Library Editor View functions are executed from the View menu.

Display the View Menu

- 1 Select View on the Library Editor menu bar. The View drop-down menu appears.



In the View menu, a check mark indicates that the item is currently being displayed. The absence of a check mark indicates that the item is currently hidden.

The View functions available are as follows:

- Display (or Hide) the Standard Toolbar
- Display (or Hide) the Library Editor Toolbar
- Display (or Hide) the Status Bar
- Display (or Hide) the Registered Libraries Window

7. Component Explorer

7.14. Library Editor Library Functions

Display (or Hide) the Standard Toolbar

Select Standard Toolbar on the View menu and click it. The Standard Toolbar is displayed (or hidden) and a check mark is added (or deleted) next to the name on the View menu.

Display (or Hide) the Library Editor Toolbar

Select Library Toolbar on the View menu and click it. The Library Editor toolbar is displayed (or hidden) and a check mark is added (or deleted) next to the name on the View menu.

Display (or Hide) the Status Bar

Select Library Toolbar on the View menu and click it. The Library Editor toolbar is displayed (or hidden) and a check mark is added (or deleted) next to the name on the View menu.

Display (or Hide) the Registered Libraries Window

Select Registered Libraries on the View menu and click it. The Registered Libraries window is displayed (or hidden) and a check mark is added (or deleted) next to the name on the View menu.

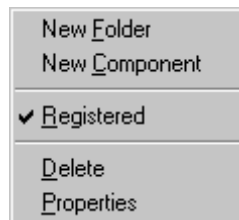
For a description of this window and details on the functionality provided, refer to Registered Libraries Editing and Display.

7.14 Library Editor Library Functions

Library Editor Library functions are executed from the Library menu.

Display the Library Menu

Select Library on the Library Editor menu bar. The Library drop-down menu appears.



The Library functions available are as follows:

- Create a Folder
- Create a Component
- Register (or Unregister) a Library
- Delete a Folder or Component
- Edit Library, Component, and Folder Properties

Create a Folder

- 1 Select the folder to which you want add a subordinate folder.
- 2 Display the Library menu.
- 3 Select New Folder on the Library menu.

OR

- 1 Select the folder to which you want add a subordinate folder.
- 2 Select the New Folder button  on the Library Editor toolbar.

OR

- 1 Right-click the folder to which you want to add a subordinate folder. The Folder-Component popup menu appears.
- 2 Select New Sub Folder on the Folder-Component popup menu.

A new, empty, subordinate folder called “New Folder” is created if it didn’t already exist. If “New Folder” already existed, an index is added and “New Folder #1” is created and displayed in the Library editor window, if it didn’t already exist. This is repeated (the index # is incremented) until a unique folder name is found.

7. Component Explorer

7.14. Library Editor Library Functions

Create a Component

1 Select the folder to which you want to add a component.

2 Select New Component on the Library menu.

OR

1 Select the folder to which you want to add a component.

2 Select New Component button  on the Library Editor toolbar.

OR

1 Right-click the folder to which you want to add a component. The Library popup menu appears.

2 Select New Component on the Library popup menu.

THEN

3 Locate and then select (highlight) the component(s) (.pct file(s)) you want to create.

4 Select Open. The Open dialog box disappears and the selected components are added to the selected folder.

Register (or Unregister) a Library

Prerequisite

The library must be saved before this function will work.

Procedure

1 Select the required library.

2 Click Registered on the Library menu.

OR

1 Open the required library.

2 Select the Register/Unregister button  on the Library Editor toolbar.

OR

Right-click the required library. The Library popup menu appears.

Select Registered on the Library popup menu.

The Library is registered (or unregistered) and a check mark is added next to (or deleted from) the Library name. Note that the Library File for an unregistered library is not deleted, it is just no longer part of the Component Library.

Delete a Folder or Component

1 Select the Folder or Component you want to delete.

2 Select Delete on the Library menu.

OR

2 Select the Delete button  on the Library Editor toolbar.

The selected Folder or Component, and all of its subordinate items are deleted.

NOTE: You cannot delete an entire library using this procedure. If you select a library, the Delete button is grayed out and becomes inoperable.

7.15 Edit Library, Component, and Folder Properties**Edit Library Properties**

There are property pages defined for all three data levels employed by the Component Explorer: Library, Folder, and Component. You can access all of these property pages through the Property command on the Library menu.

Prerequisite

The library must be saved before this function will work.

Four Possible Procedures

1 Select the required library.
Select Properties on the Library menu.

OR

Select the required library.
Select the Properties button  on the Library Editor toolbar.

OR

Right-click the required library. The Library popup menu appears.
Select Properties on the Library popup menu.

OR

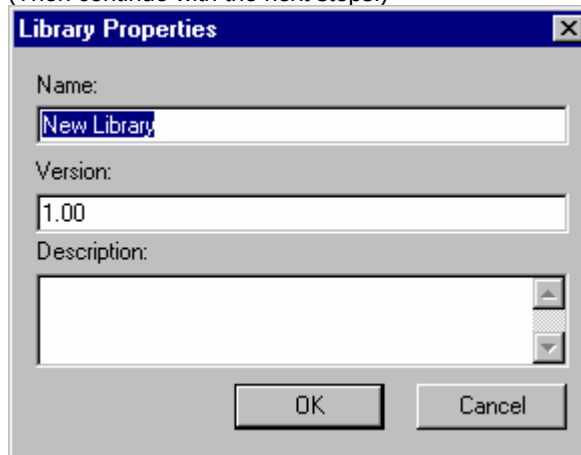
7. Component Explorer

7.15. Edit Library, Component, and Folder Properties

Double-click the required library.

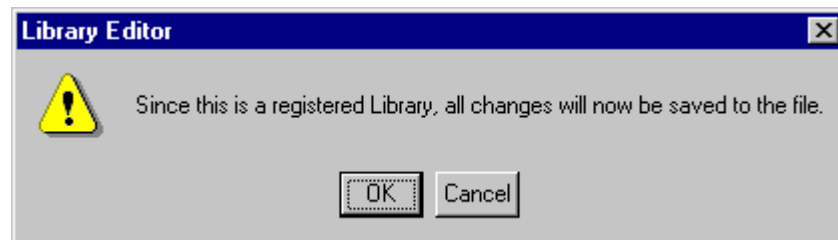
The Library Properties page is displayed.

(Then continue with the next steps.)



THEN

- 2 Edit the Library Name, Version and Description as required.
- 3 Click OK. If the library is registered, the following Library Editor dialog box appears.



- 4 Click OK to proceed.

The Library Properties page disappears.

If the Library is registered, name, version, and description changes are saved immediately.

NOTE: Description changes to a registered library are save only if there was an accompanying name or version change.

Edit Folder Properties

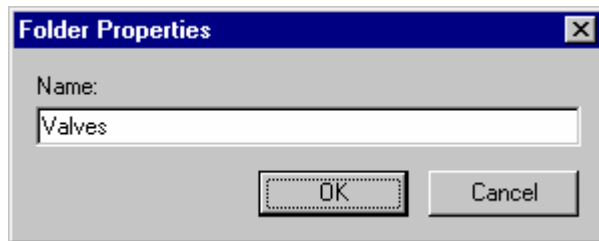
- 1 Select the required folder.
Select Properties on the Library menu.

OR

Right-click the required folder. The Folder-Component popup menu appears.
Select Properties on the Library popup menu.

OR

Double-click the required folder.
The Folder Properties page is displayed.



THEN

- 2 Edit the Folder Name, as required.

NOTE: All sub-folder names within a folder must be unique.
- 3 Click OK. The Folder Properties page disappears and the new folder name is saved.

Edit Component Properties

- 1 Select the required component.
Select Properties on the Library menu. The Component Properties page is displayed.

OR

Right-click the required component. The Folder-Component popup menu appears.
Select Properties on the Library popup menu.

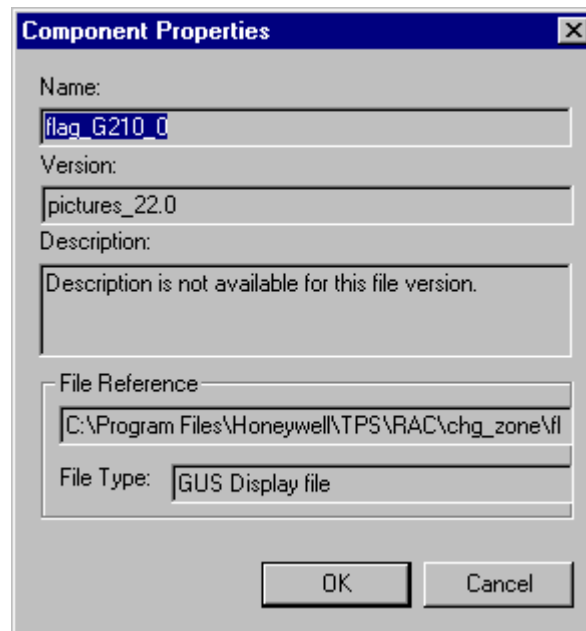
OR

Double-click the required component.

7. Component Explorer

7.15. Edit Library, Component, and Folder Properties

The Component Properties page is displayed.



THEN

- 2 Edit the component Name as required.

NOTE: The Version and Description cannot be edited.

- 3 To designate a different file for this component, select Browse, locate, and then select the new file.
- 4 Click OK. The Component Properties page disappears and the changes are saved.

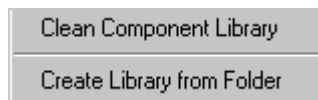
NOTE: If File Reference file has a .pct extension, File Type will be "GUS Display file." For any other extension, the File Type will be "Unknown."

7.16 Library Editor Tool Functions

Library Editor Tool functions are executed from the Tools menu.

Display the Tools Menu

Select Tools on the Library Editor menu bar. The Tools drop-down menu appears.



The Tools functions available are as follows:

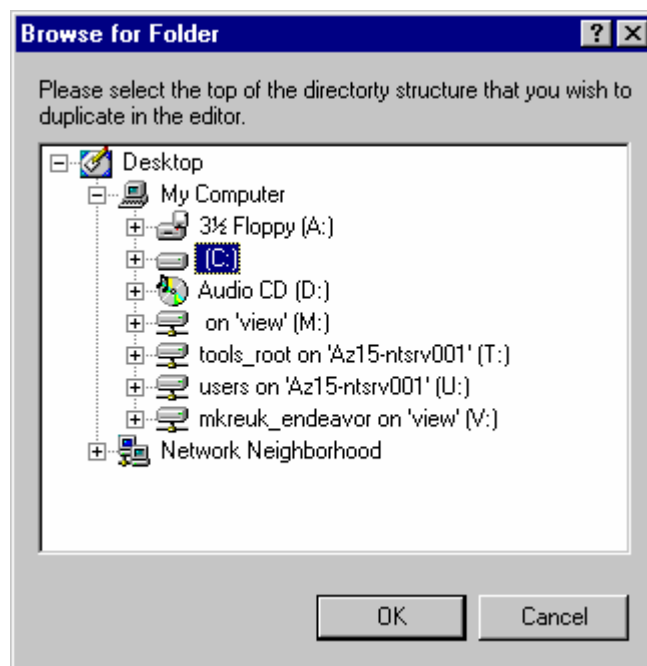
- Clean a Library
- Create a Library from a Folder

Clean a Library

- Select Clean Component Library on the Tools drop-down menu. The file references for each library are checked. If any reference in a library is found to be incorrect (for example, a file was deleted), the library will be removed.

Create a Library from a Folder

- 1 Select Create Library from Folder on the Tools drop-down menu. The Browse for Folder dialog box appears. An example follows.



- 2 Locate and select the required folder.
- 3 Click OK. A new library will be created from the selected folder that includes all subfolders and all files that qualify as components (files with the .pct extension).

7.17 Library Editor Help Functions

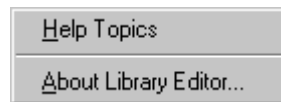
Library Editor Help functions are executed from both the Standard toolbar and the Help Menu.

The Help functions available are as follows:

- Display the Library Editor Help Topics
- Display Library Editor Information
- Display Context-sensitive Help

Display the Library Editor Help Topics

- 1 Select Help on the Library Editor menu bar. The Help drop-down menu appears.



- 2 Select Help Topics on the Help drop-down menu.

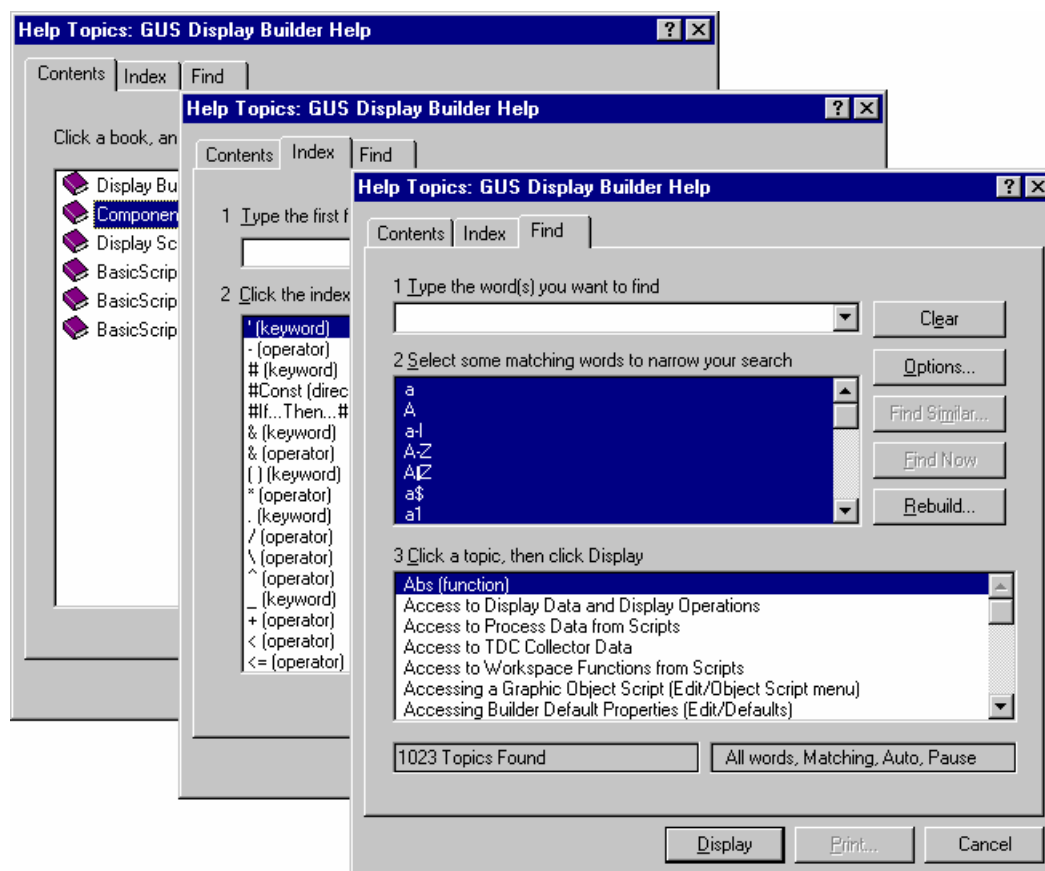
OR

- 1 Select the Help button  on the Standard toolbar

7. Component Explorer

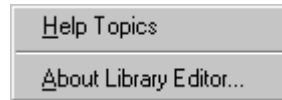
7.17. Library Editor Help Functions

The Library Editor Help Topics window appears, with the Contents, Index, or Find Page on display.

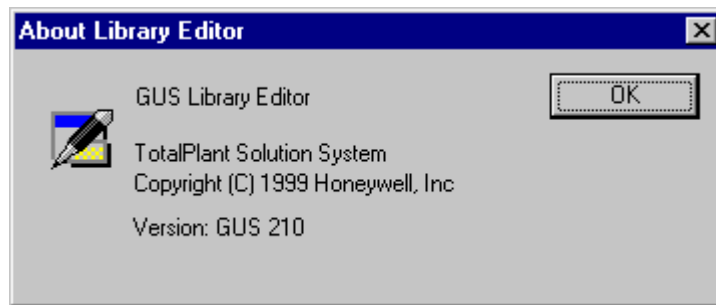


Display Library Editor Information

- 1 Select Help on the Library Editor menu bar. The Help drop-down menu appears.



- 2 Select About Library Editor on the Help drop-down menu. A dialog box containing release information about the Library Editor is displayed. A sample dialog box is shown below.



NOTE: The revision number shown will identify your particular system.

Display Context-sensitive Help

- 1 Select the What's This Help button  on the Standard toolbar.
The normal cursor is changed to a Help cursor by adding a question mark

- 2 Move the new Help cursor over a menu command or toolbar icon you want help information on and right-click. Help information for the item selected will be displayed.

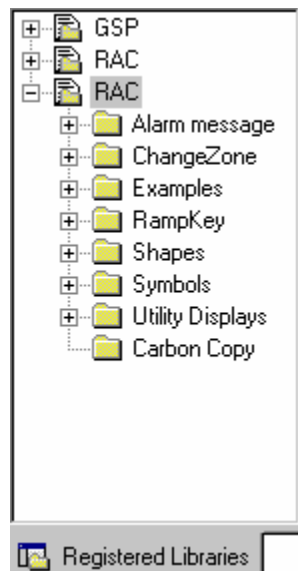
7. Component Explorer

7.18. Registered Libraries Editing and Display

7.18 Registered Libraries Editing and Display

The registry segment of the Component Library is displayed in the Registered Libraries page, on the left side of the Library Editor window. Internally, registered libraries are identified using a number of sub-keys that contain a description of the library, its location, name, and version number, and a flag that indicates whether or not it is visible in the Component Explorer. A library must be registered before it can be used by the Component Explorer.

Registered libraries are listed in the Registered Libraries window. An example of a Registered Libraries window for a GUS node with 2 Libraries (2 RACs) is shown below.



Registered libraries display functions are executed from the Registered Libraries window using the Registered Libraries and Registered Libraries Folder-Component popup menus.

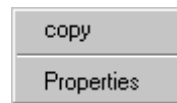
Display the Registered Libraries Popup Menu

Right-click the required library. The Registered Libraries popup menu is displayed.



Display the Registered Libraries Folder-Component Popup Menu

Right-click the required folder or component. The folder or component is selected (highlighted) and the Registered Libraries Folder-Component popup menu is displayed.



The following editing and display functions available on the Registered Libraries popup menus:

- Open a Registered Library
- Unregister a Library
- Display Library Properties
- Copy a Folder or Component
- Display Folder or Component Properties

Open a Registered Library

- 1 Right-click the required library. The Registered Libraries popup menu is displayed.
- 2 Select Open. The selected library is opened in the Library Editor window.

Unregister a Library

- 1 Right-click the required library. The Registered Libraries popup menu is displayed.
- 2 Select Unregister. The selected library is unregistered and removed from the Registered Libraries window.

Display Library Properties

- 1 Right-click the required library. The Registered Libraries popup menu is displayed.
- 2 Select Properties. The Library Properties page (read-only version) for the selected library is displayed.

Copy a Folder or Component

- 1 Right-click the required folder or component. The folder or component is selected (highlighted) and the Registered Libraries Folder-Component popup menu is displayed.
- 2 Select Copy. The selected item and all its sub-items are copied to the Clipboard.

Display Folder or Component Properties

- 1 Right-click the required folder or component. The folder or component is selected (highlighted) and the Registered Libraries popup menu is displayed.
- 2 Select Properties. The Folder or Component Properties page (read-only) for the selected item is displayed.

7.19 General Procedures

Using the functions described earlier, there are a number of general Component Explorer procedures that users will typically execute. They are as follows:

- Create a Library From a New Display Package
- Create a New Version of a Library Without Replacing Any Files
- Create a New Version of a Library by Overwriting Existing Files

The library creation procedures described below all require setting up a complete package installation, not just copying files.

Create a Library From a New Display Package

The new display package can be either Honeywell display packages, such as RAC or packages from non-Honeywell sources. It should be noted that the file and directory structure of the library would be identical to the file and directory structure of the display package.

- 1 Turn on the Library Editor.
- 2 Enter the name, version, and a description of the new library into a data file.
- 3 Save the new data file.
- 4 Include copying the data file to the installation procedure.
- 5 Include registering the data file to the installation procedure.
- 6 Start the installation procedure.
- 7 If no errors occur, the new library will have been created.
- 8 If a registered library with the same name and version number already exists on the node, a dialog box appears listing four options Replace, Rename, Ignore, and Cancel. Select Replace to replace the existing library with the new one. Select Rename to enter a different name or version number and try again. Select Ignore to skip the installation of this library. Select Cancel to terminate the procedure.

Create a New Version of a Library Without Replacing Any Files

- 1 Turn on the Library Editor.
- 2 Enter the same name, a new version, and any description of the new version of the library into a data file.
- 3 Save the new data file.
- 4 Include copying the data file to the installation procedure.
- 5 Include registering the data file to the installation procedure.
- 6 Set the installation procedure not to save the files of this new version in the same location as the files of the existing version.
- 7 Start the installation procedure.
- 8 If no errors occur, two versions of the same library are now in existence.
- 9 If a registered library with the same name and version number already exists on the node, a dialog box appears listing four options Replace, Rename, Ignore, and Cancel. Select Replace to replace the existing library with the new one. Select Rename to enter a different name or version number and try again. Select Ignore to skip the installation of this library. Select Cancel to terminate the procedure.

Create a New Version of a Library by Overwriting Existing Files

- 1** Turn on the Library Editor.
- 2** Enter the same name, a new version, and any description of the new version of the library into a data file.
- 3** Save the new data file.
- 4** Include copying the data file to the installation procedure.
- 5** Include registering the data file to the installation procedure.
- 6** Set the installation procedure to save the files of this new version in the same location as the files of the existing version.
- 7** Start the installation procedure.
- 8** If no errors occur, a new version of a former library will have been created.
- 9** If a registered library with the same name and version number already exists on the node, a dialog box appears listing four options Replace, Rename, Ignore, and Cancel. Select Replace to replace the existing library with the new one. Select Rename to enter a different name or version number and try again. Select Ignore to skip the installation of this library. Select Cancel to terminate the procedure.

8. Properties of Graphic Objects

8.1 Overview

All graphic objects have properties such as color, line width, fill, etc.

In the sections listed below, the properties for each graphic object are described.

Topics in this section are

- General Properties (General Pages)
- Properties Common to Objects
- Properties Unique to Object Types
- Line Object Properties
- Rectangle Object Properties
- Rounded Rectangle Object Properties
- Ellipse Object Properties
- Open Polygon Object Properties
- Closed Polygon Object Properties
- Open Bezier Curve Object Properties
- Closed Bezier Curve Object Properties
- Arc Object Properties
- Text Object Properties
- Bitmap Object Properties
- Group Object Properties
- Accessing Properties of Graphic Objects
- Accessing Properties of Graphic Object Scripts

8.2 General Properties (General Pages)

All Graphic Objects use the same General Properties Help button.

Select the page for the object you are working with from the list below.

- Line Object General Properties
- Rectangle Object General Properties
- Rounded-Rectangle Object General Properties
- Ellipse Object General Properties
- Open Polygon Object General Properties
- Closed Polygon Object General Properties
- Open Bezier Curve Object General Properties
- Closed Bezier Curve Object General Properties
- Arc Object General Properties
- Text Object General Properties
- Bitmap Object General Properties
- Button Object General Properties
- Data Entry Object General Properties
- ListBox Object General Properties
- ComboBox Object General Properties
- Group Object General Properties

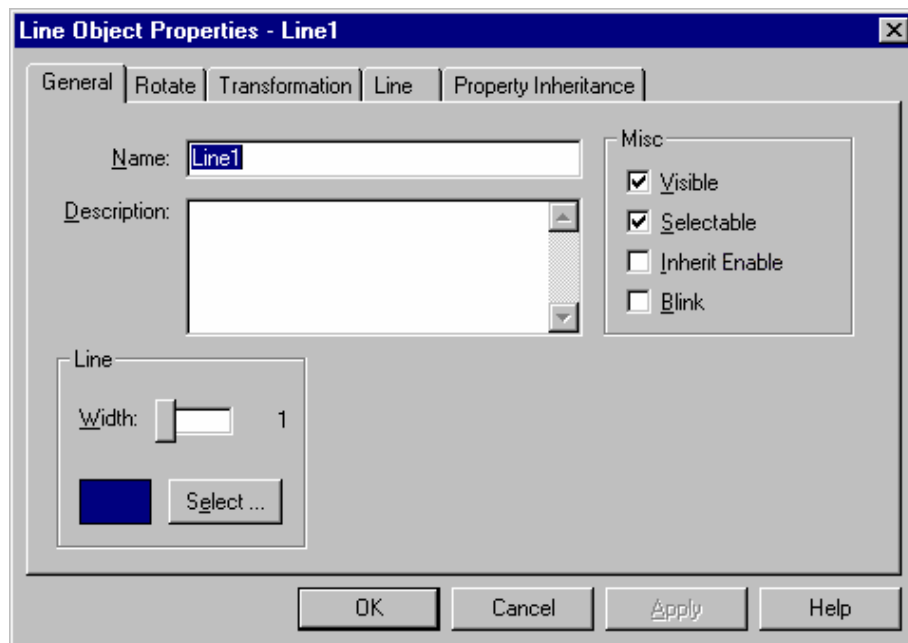
8.3 Line Object General Properties

Defines properties of a Line. Standard primitive pages are Rotate, Transformation, and Line.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting line object general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Line Object Properties Sheet appears with the General tab displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the line.
- 4 Check Selectable to allow line to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Accept default Line Color and Width or change as required.
- 8 Click OK or Apply to accept command.

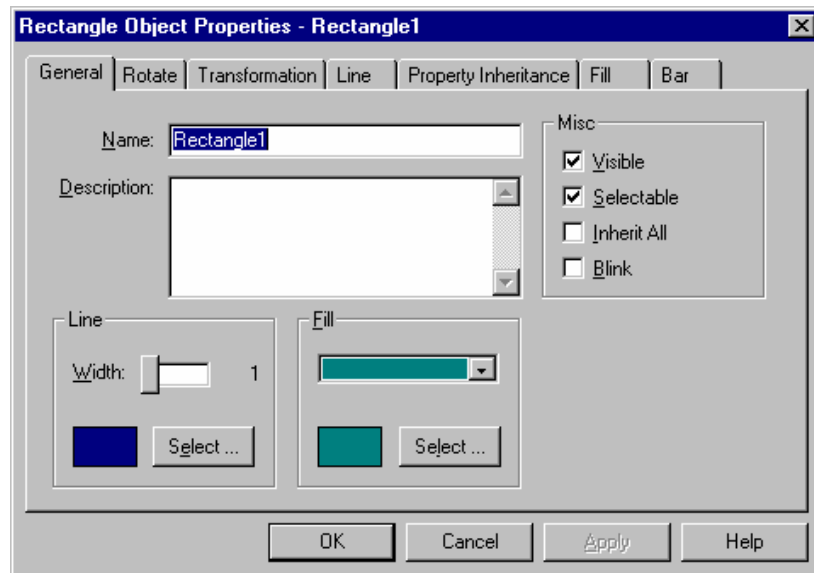
8.4 Rectangle Object General Properties

Defines properties of a Rectangle. Standard primitive pages are Rotate, Transformation, Line, Fill, and Bar.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Rectangle Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.

- 7** Accept default Line Color and Width or change as required.
- 8** Accept default Fill Color or change as required.
- 9** Click OK or Apply to accept command.

8.5 Rounded-Rectangle Object General Properties

Defines properties of a rectangular object with rounded corners. Standard primitive pages are Rotate, Transformation, Line, and Fill.

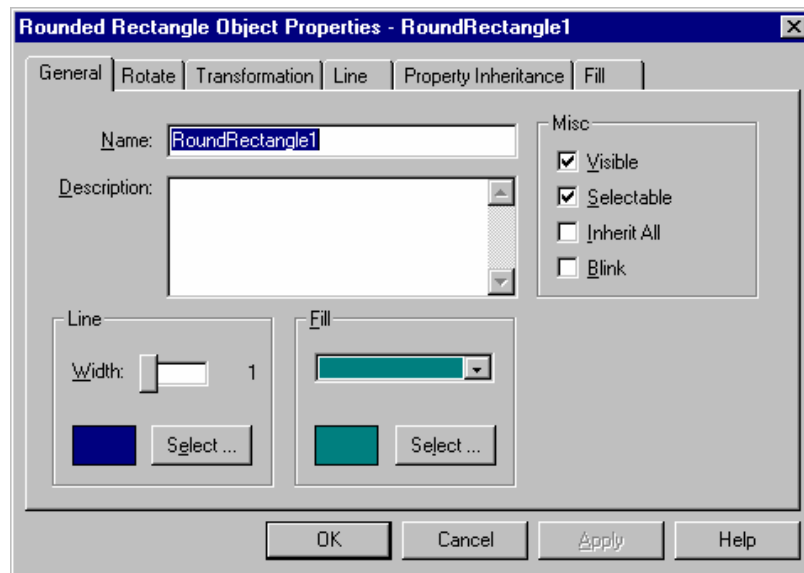
Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

8. Properties of Graphic Objects

8.5. Rounded-Rectangle Object General Properties

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Rounded Rectangle Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Accept default Line Color and Width or change as required.
- 8 Accept default Fill Color or change as required.
- 9 Click OK or Apply to accept command.

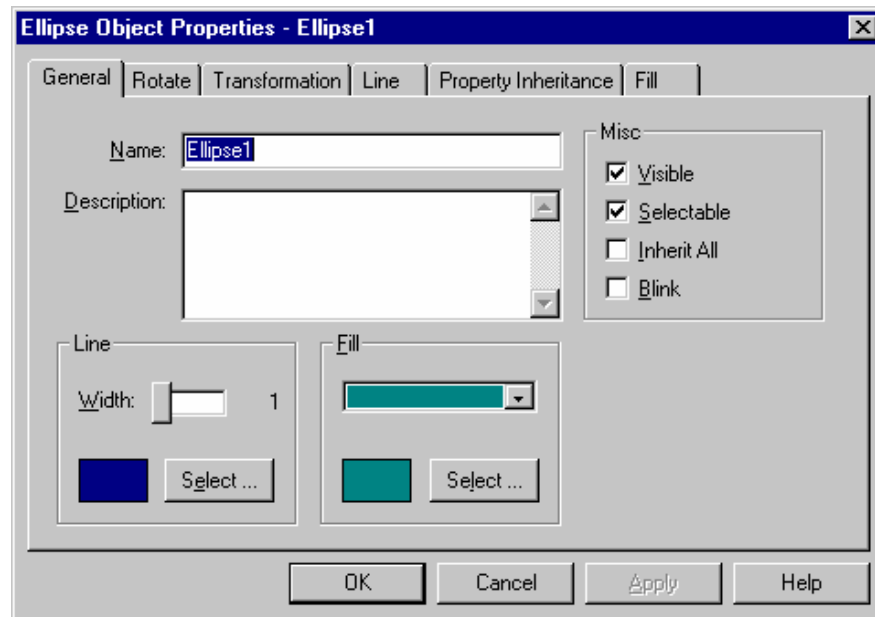
8.6 Ellipse Object General Properties

Defines properties of an Ellipse. Standard primitive pages are Rotate, Transformation, Line, and Fill.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Ellipse Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.

8. Properties of Graphic Objects

8.7. Open Polygon Object General Properties

- 6** Check Blink to cause the object to blink on and off.
- 7** Accept default Line Color and Width or change as required.
- 8** Accept default Fill Color or change as required.
- 9** Click OK or Apply to accept command.

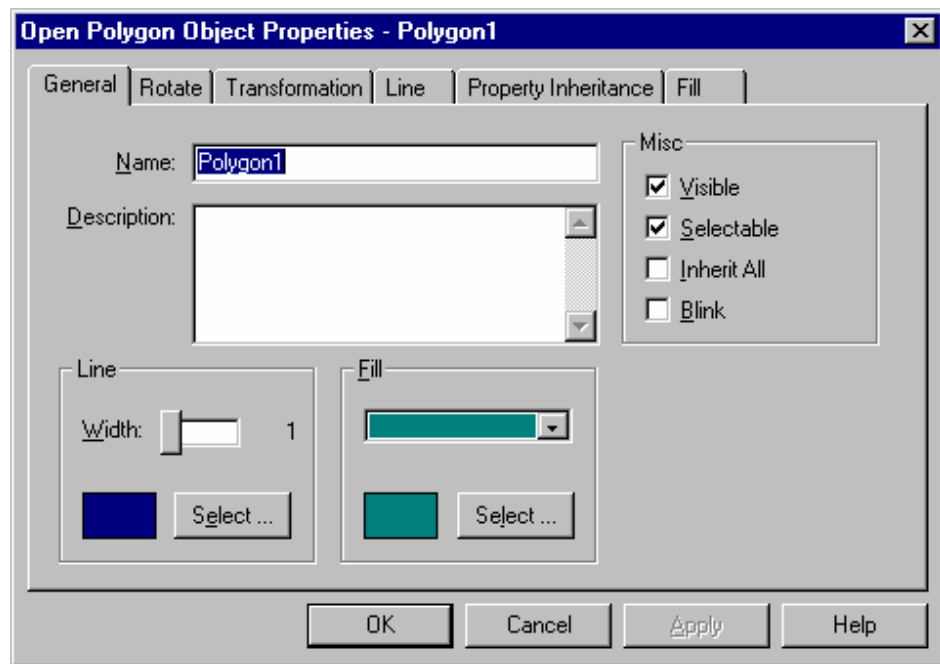
8.7 Open Polygon Object General Properties

The properties of a multisided, open object are determined. Standard primitive pages are Rotate, Transformation, Line, and Fill.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Open Polygon Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Accept default Line Color and Width or change as required.
- 8 Accept default Fill Color or change as required.
- 9 Click OK or Apply to accept command.

8. Properties of Graphic Objects

8.8. Closed Polygon Object General Properties

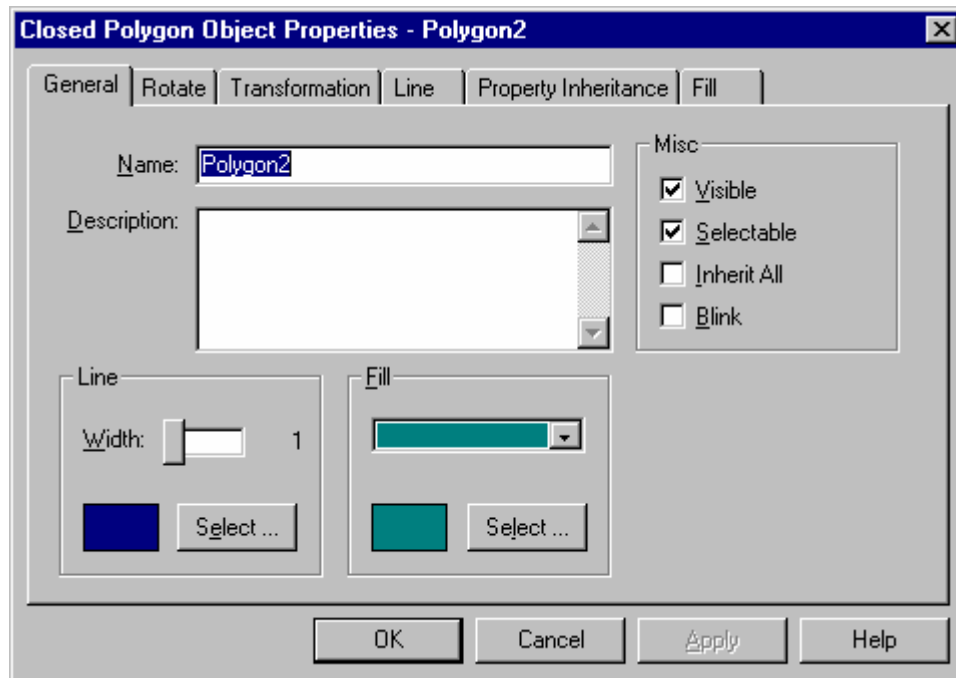
8.8 Closed Polygon Object General Properties

The properties of a multisided, closed object are determined. Standard primitive pages are Rotate, Transformation, Line, and Fill.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Closed Polygon Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.

- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Accept default Line Color and Width or change as required.
- 8 Accept default Fill Color or change as required.
- 9 Click OK or Apply to accept command.

8.9 Open Bezier Curve Object General Properties

The properties of a multi-curved line object are determined. Standard primitive pages are Rotate, Transformation, Line, and Fill.

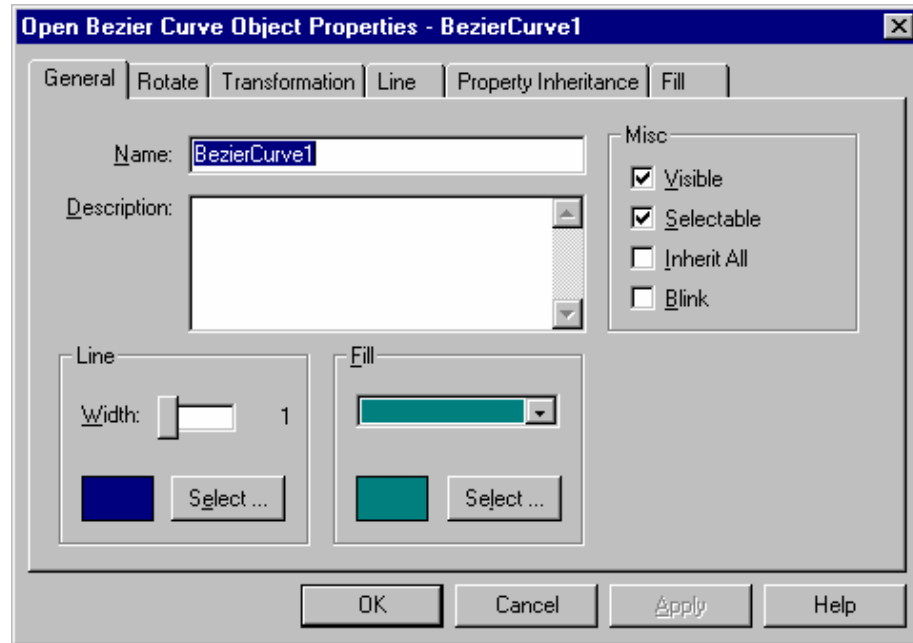
Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Open Bezier Curve Object General Property page displayed.

8. Properties of Graphic Objects

8.9. Open Bezier Curve Object General Properties



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Accept default Line Color and Width or change as required.
- 8 Accept default Fill Color or change as required.
- 9 Click OK or Apply to accept command.

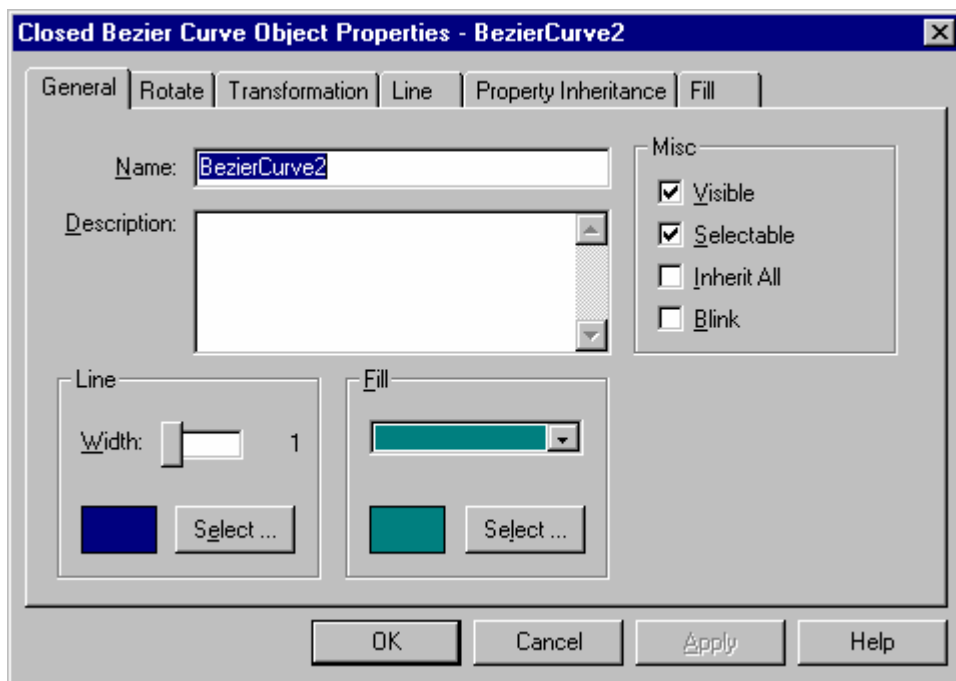
8.10 Closed Bezier Curve Object General Properties

The properties of a closed multi-curved line object are determined. Standard primitive pages are Rotate, Transformation, Line, and Fill.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Closed Bezier Curve Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.

8. Properties of Graphic Objects

8.11. Arc Object General Properties

- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Accept default Line Color and Width or change as required.
- 8 Accept default Fill Color or change as required.
- 9 Click OK or Apply to accept command.

8.11 Arc Object General Properties

The properties of an arc are determined. An arc is a portion of an ellipse. Standard primitive pages are Rotate, Transformation, Line, and Fill.

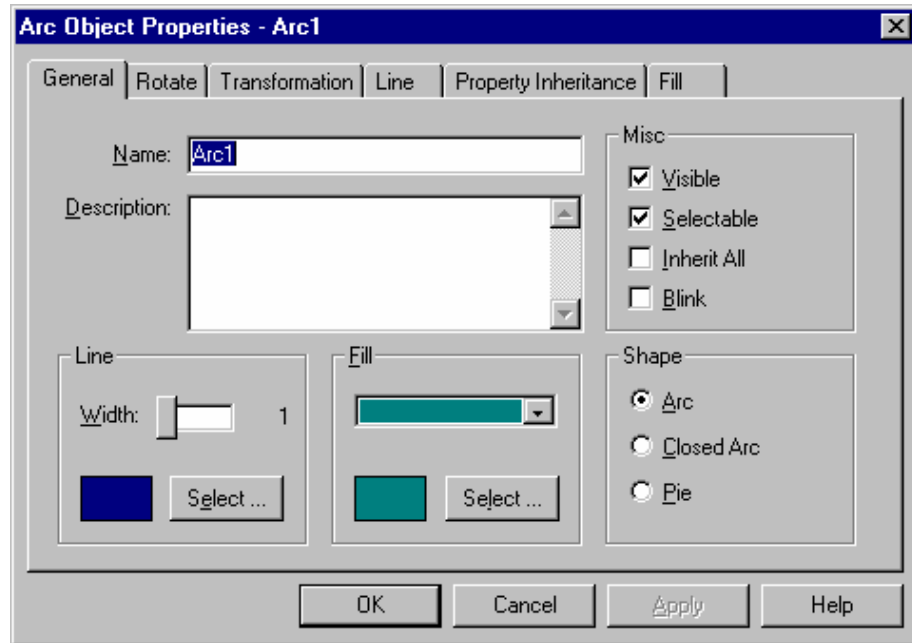
Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Arc Object General Property page displayed.

8. Properties of Graphic Objects

8.11. Arc Object General Properties



- 2 Enter Name and enter Description if required.
 - 3 Check Visible to show the object.
 - 4 Check Selectable to allow object to be selected while the display is running.
 - 5 Check Inherit Enabled to inherit properties from defaults or parent.
 - 6 Check Blink to cause the object to blink on and off.
 - 7 Accept default Line Color and Width or change as required.
 - 8 Accept default Fill Color or change as required.
 - 9 Chose Shape of open Arc, Closed Arc, or Pie.
- Click OK or Apply to accept command.

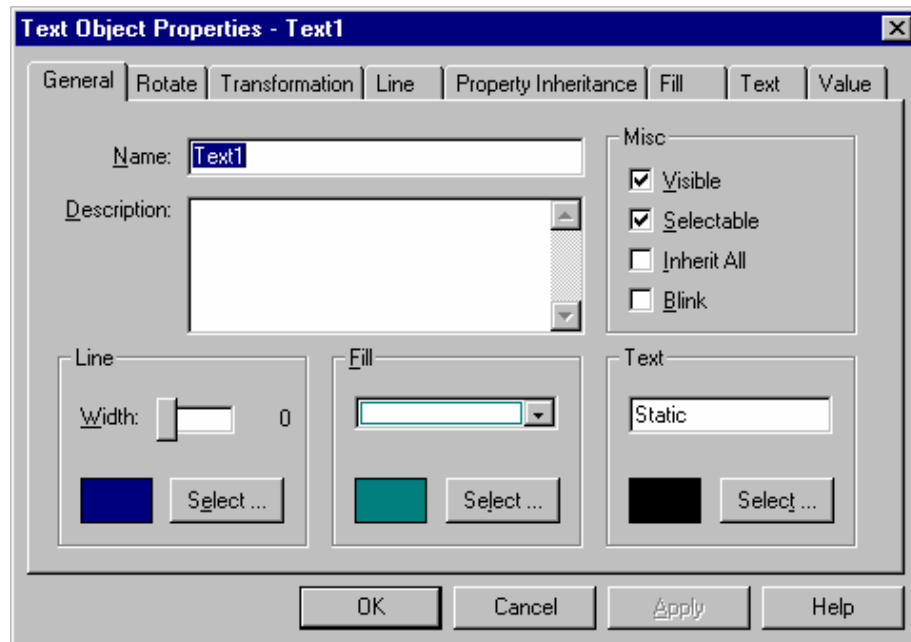
8.12 Text Object General Properties

The properties of a text object are determined. Standard primitive pages are Rotate, Transformation, Line, Fill, Text, and Value.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Text Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.

- 5** Check Inherit Enabled to inherit properties from defaults or parent.
- 6** Check Blink to cause the object to blink on and off.
- 7** Accept default Line Color and Width or change as required.
- 8** Accept default Fill Color or change as required.
- 9** Replace the Text word "Static" with the required Text and accept or change color as required.
- 10** Click OK or Apply to accept command.

8.13 Bitmap Object General Properties

The properties of Bitmap objects are determined. Standard primitive pages are Rotate and Transformation.

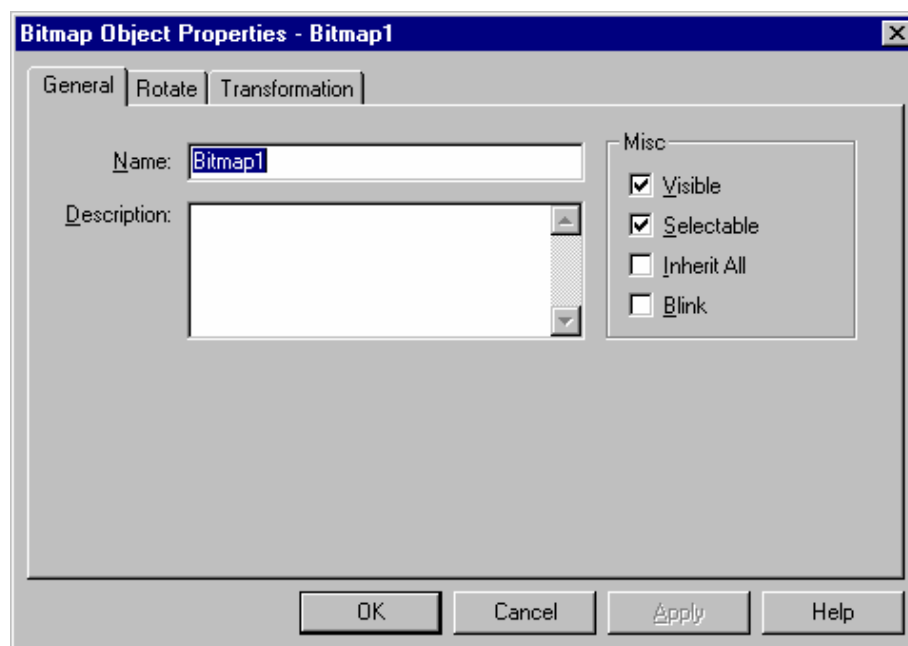
Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1** Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Bitmap Object General Property page displayed.

8. Properties of Graphic Objects

8.13. Bitmap Object General Properties



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Click OK or Apply to accept command.

8.14 Button Object General Properties

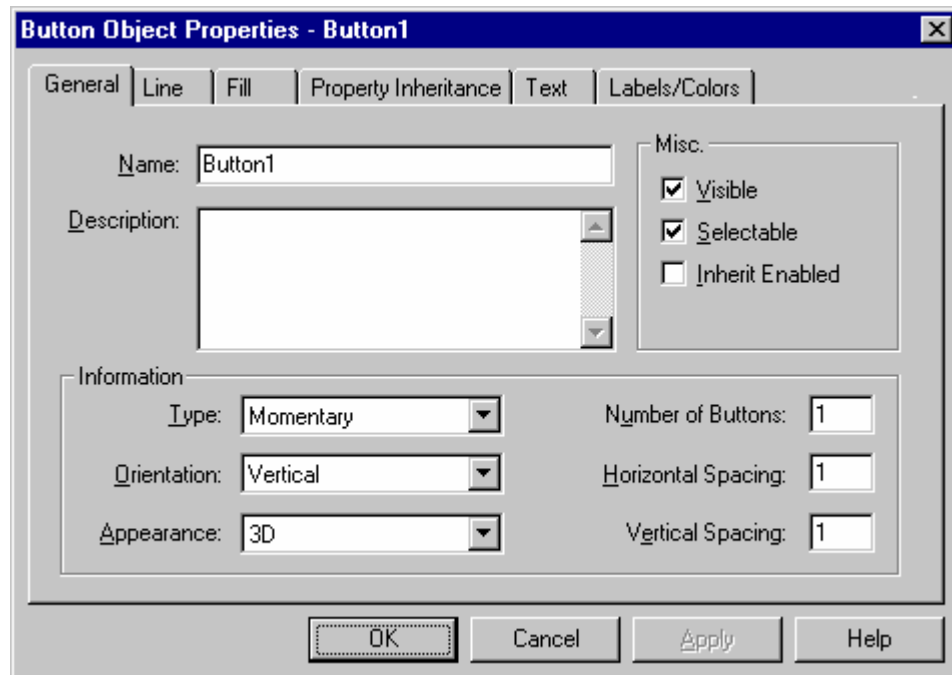
The properties of a button object can be determined by selecting the object and right-button clicking with the cursor positioned over the object. A drop-down menu appears with the **Properties...** dialog selection.

Standard primitive pages are Line, Fill, Property Inheritance, and Text. The Labels/Colors page is configured specifically for a button object.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Button Object Property Sheet appears with the Button Object General Property page displayed.



8. Properties of Graphic Objects

8.14. Button Object General Properties

- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Select Type of button from the pull-down list in the Information area.
Momentary = Immediate ON/OFF action; does not maintain state
Latching = Multiple selection list
Latching/Checkbox = Multiple selection list; Checkbox displays in front of text
Interlocking = Single selection list
Interlocking/Radio = Single selection list; Radio button displays in front of text

NOTE: The Interlocking options are available only if number of buttons is greater than one.
- 7 Select Orientation from the pull-down list in the Information area.
- 8 Select Appearance from the pull-down list in the Information area.
- 9 Enter the Number of Buttons required (1-12 buttons) and the Horizontal and Vertical Spacing (0-50 logical pixels).
- 10 Click OK or Apply to accept command.

Line properties define border of the rectangular backplate. The **Line** properties do not affect appearance of the pushbutton(s).

Fill properties affect appearance of the rectangular backplate. The **Fill** properties do not affect appearance of the pushbutton(s).

The **Inherited Fill** and **Line** properties apply to the rectangular backplate. The **Inherited Text** properties apply to the text on the pushbutton(s).

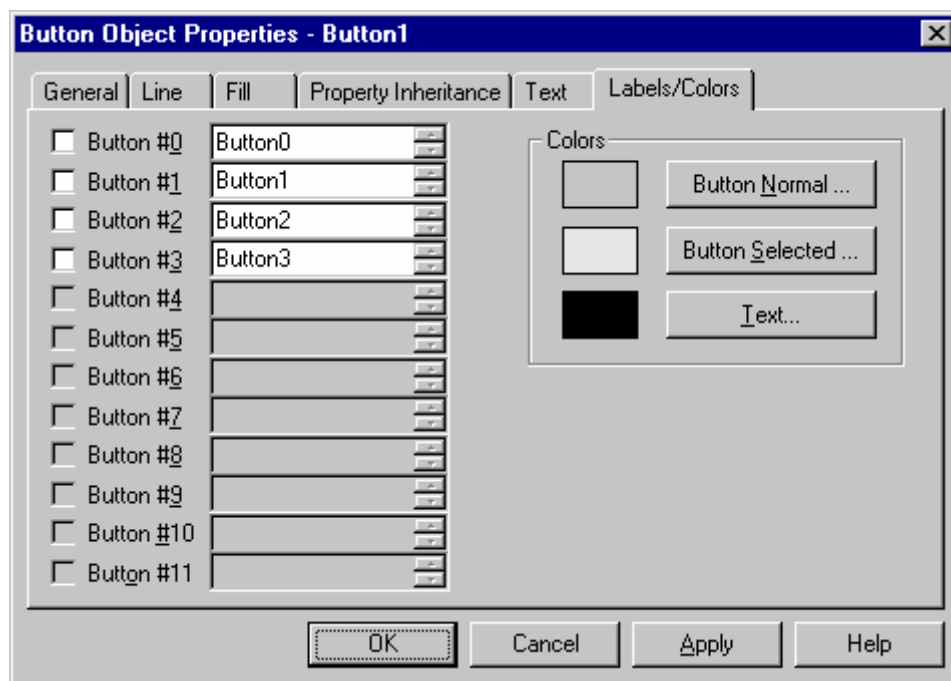
Text properties apply to the text on the pushbutton(s). The **Select...** pushbutton invokes a standard font selection dialog box.

For **Label** and **Color** properties, refer to Button Label/Color Properties.

8.15 Button Label/Color Properties

Labels/Colors Page

The **Labels/Colors** property page defines texts on buttons, button/text colors and, for some button types, initial selection states. As shown below, this property page applies to a 4-item latching-type button.



Button #0-11

Edit fields: Text labels on buttons.

If button type is Latching or Latching/Checkbox

Checkbox buttons: Initial selection state for each pushbutton.

If button type is Interlocking or Interlocking/radio

Radio buttons: Indicate initially selected pushbutton. A single pushbutton must be selected. In the pushbutton numbering scheme, the first pushbutton is denoted by the index of 0.

8. Properties of Graphic Objects

8.16. Data Entry Object General Properties

Colors

Button Normal invokes GUS color selection dialog box; defines color of the pushbutton when it is in the OFF state (deselected).

Button Selected invokes GUS color selection dialog box; defines color of the pushbutton when it is in the ON state (selected).

Text invokes GUS color selection dialog box; defines text color. NOTE: text color is independent of the current state of the pushbutton.

Other colors (3D edges, pushbutton disabled) are fixed in software and may not be changed:

- disabled state is indicated by text color set to half-intensity of current color of the pushbutton
- 3D edge colors are based on the systems colors: 3DHILIGHT and 3DDKSHADOW.

NOTE: The Button object is the suggested recommendation for this functionality. If you have old OCXs, it is recommended you replace them with this Button primitive.

8.16 Data Entry Object General Properties

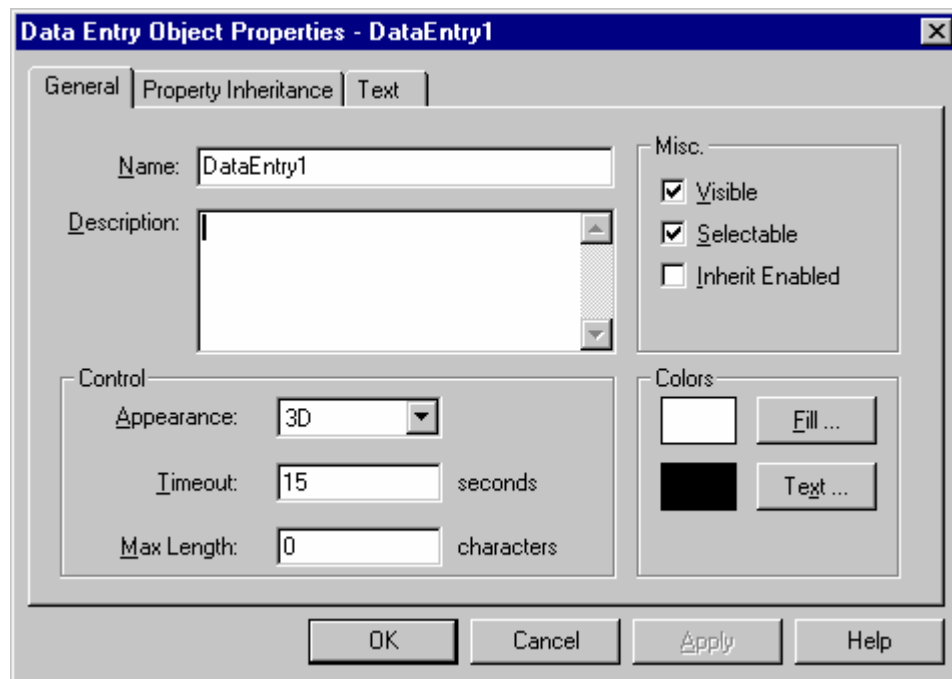
The properties of a data entry object can be determined by selecting the object and right-button clicking with the cursor positioned over the object. A drop-down menu appears with the **Properties...** dialog selection.

Standard primitive pages are Property Inheritance and Text.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Data Entry Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.

NOTE: At runtime, if the Selectable property is not set, the object will not accept operator input and the control window will be grayed out.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.

NOTE: Property Inheritance is NOT supported for the following properties: Blink, Fill, FillPattern, LineStyle, LineColor, LineWidth, TextAlignment, TextReverse, and HalfIntensity.

8. Properties of Graphic Objects

8.17. ListBox Object General Properties

- 6 Select Appearance from the pull-down list in the Control area.
- 7 Enter the Timeout value (in seconds) in the Control area field.

NOTE: At runtime, the timer must be activated by the script using the TimeoutActive property. Timeout value = 0 disables activation of the timer.
Refer to the OnTimeout() scriptable event in the *Display Scripting User's Guide*, in the GUS , Data Entry Primitive section for more information.
- 8 Enter the maximum number of allowable data entry characters in the Max Length field in the Control area. Value = 0 disables maximum length checking.
Refer to the OnMaxLength() scriptable event in the *Display Scripting User's Guide*, in the GUS , Data Entry Primitive section for more information.
- 9 Select Fill and Text values from the Colors area of the dialog box.
Multiline: In the multiline operation, text automatically wraps around at the border and the vertical scrollbar is visible. In single-line operation, text automatically scrolls horizontally if it does not fit into the edit window.
- 10 Click OK or Apply to accept command.

NOTE: The Data Entry Object is the suggested recommendation for this functionality. If you have old OCXs, it is recommended you replace them with this Data Entry primitive.

8.17 ListBox Object General Properties

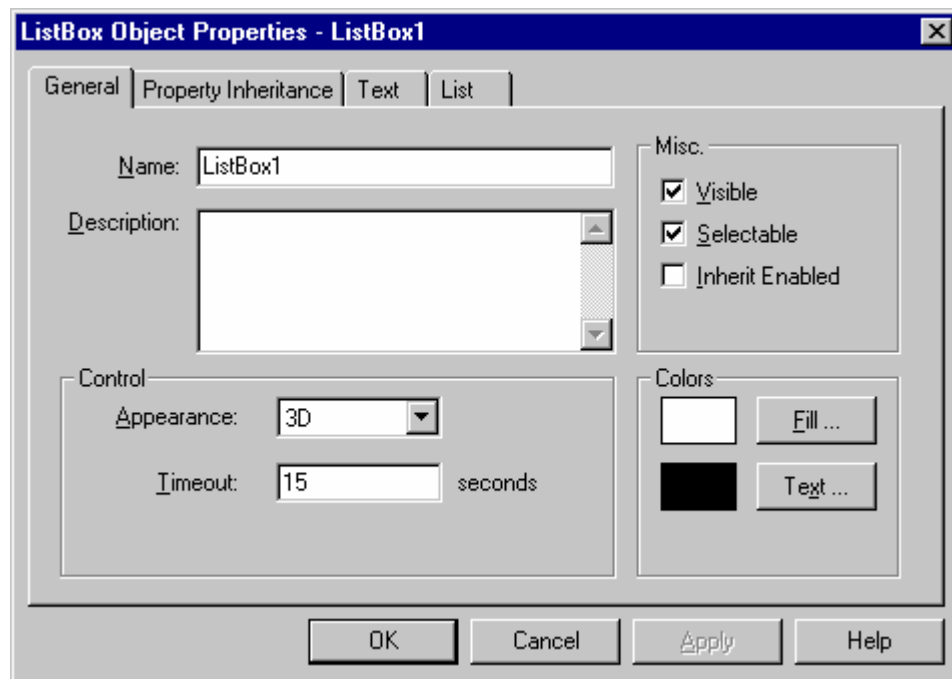
The properties of a listbox object can be determined by selecting the object and right-button clicking with the cursor positioned over the object. A drop-down menu appears with the **Properties...** dialog selection.

Standard primitive pages are Property Inheritance, Text, and List.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The ListBox Object Property Sheet appears with the Listbox Object General Property page displayed.



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.

NOTE: At runtime, if the Selectable property is not set, the object will not accept operator input and the control window will gray.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.

NOTE: Property Inheritance is not supported for the following properties: Blink, Fill, FillPattern, LineStyle, LineColor, LineWidth, TextAlignment, TextMultiline, TextReverse, and HalfIntensity.

8. Properties of Graphic Objects

8.18. ComboBox Object General Properties

6 Select Appearance from the pull-down list in the Information area.

7 Enter the Timeout value (in seconds) in the Control area field.

NOTE: At runtime, the timer must be activated by the script (through the property TimeoutActive). Timeout value = 0 disables activation of the timer. Refer to the OnTimeout() scriptable event in the *Display Scripting User's Guide*, in the GUS , Data Entry Primitive section for more information.

If timeout has been activated on a control, while the control has input focus any keyboard entry and mouse event restarts the timeout timer.

8 Select Fill and Text values from the Colors area of the dialog.

NOTE: The Alignment and Multiline text primitive properties are not supported.

9 Click OK or Apply to accept command.

Refer to ListBox Properties for more information.

NOTE: The ListBox Object is recommended for this functionality. If you have old OCXs, it is recommended you replace them with this ListBox primitive.

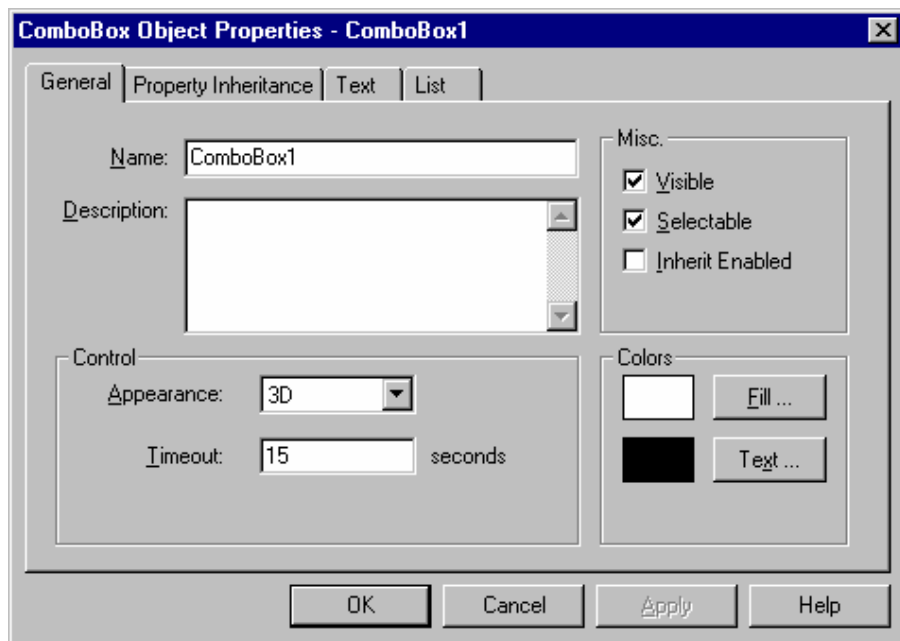
8.18 ComboBox Object General Properties

There are four ComboBox Property pages: the General, Property Inheritance, Text, and List pages.

Default properties are determined from an object's parent. To determine the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The ComboBox Object Property Sheet appears with the General Property page displayed.



- 2 Enter a Name and, if required, a Description.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.

NOTE: At runtime, if the Selectable property is not selected, the object will not accept operator input and the control window will be grayed out.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.

NOTE: Property Inheritance is not supported for the following properties: Blink, Fill, FillPattern, LineStyle, LineColor, LineWidth, TextAlignment, TextMultiline, TextReverse, and HalfIntensity.
- 6 Select the required Appearance value from the pull-down list in the Control area.

8. Properties of Graphic Objects

8.19. Group Object General Properties

- 7 Enter the required Timeout value (in seconds) in the Control area field.

NOTE: At runtime, the timer must be activated by the script (through the property TimeoutActive). Timeout value = 0 disables activation of the timer.
Refer to the OnTimeout() scriptable event in the *Display Scripting User's Guide*, in the GUS , Data Entry Primitive section for more information.
If timeout has been activated on a control, while the control has input focus, any keyboard entry and mouse event restarts the timeout timer.
- 8 Select Fill and Text values from the Colors area of the dialog.

NOTE: The Alignment and Multiline text primitive properties are not supported.
- 9 Click OK or Apply to set up the properties selected and information entered.

Setting other properties

Refer to the ComboBox Properties for more information. The Property Inheritance and Text pages are standard primitive pages.

NOTE: The ComboBox object is the suggested recommendation for this functionality. If you have old OCXs, it is recommended you replace them with this ComboBox primitive.

8.19 Group Object General Properties

The properties of grouped objects are determined. Standard primitive pages are Rotate and Transformation.

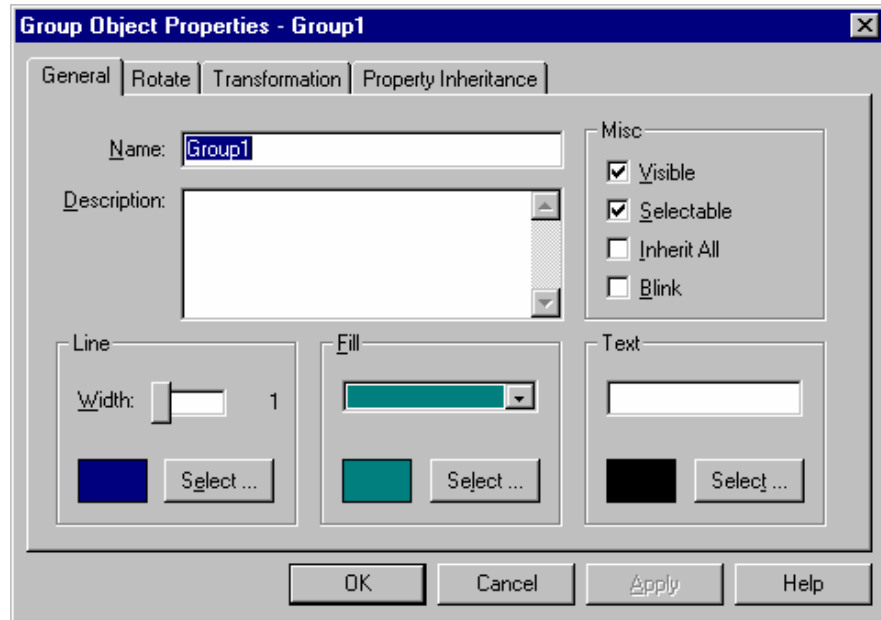
Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting general properties

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears with the Text Object General Property page displayed.

8. Properties of Graphic Objects

8.19. Group Object General Properties



- 2 Enter Name and enter Description if required.
- 3 Check Visible to show the object.
- 4 Check Selectable to allow object to be selected while the display is running.

NOTE: If a Group has an OnLButton_ or OnRButton_ script, that script will be activated only if the click is made on a member object whose selectable attribute is TRUE.
- 5 Check Inherit Enabled to inherit properties from defaults or parent.
- 6 Check Blink to cause the object to blink on and off.
- 7 Accept default Line Color and Width or change as required.
- 8 Accept default Fill Color or change as required.
- 9 Replace the Text word "Static" with the required Text and accept or change color as required.
- 10 Click OK or Apply to accept command.

8. Properties of Graphic Objects

8.20. Properties Common to Objects

8.20 Properties Common to Objects

This section contains properties common to all objects. They are

- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)
- Text Properties (Text Page)

8.21 Rotate Properties

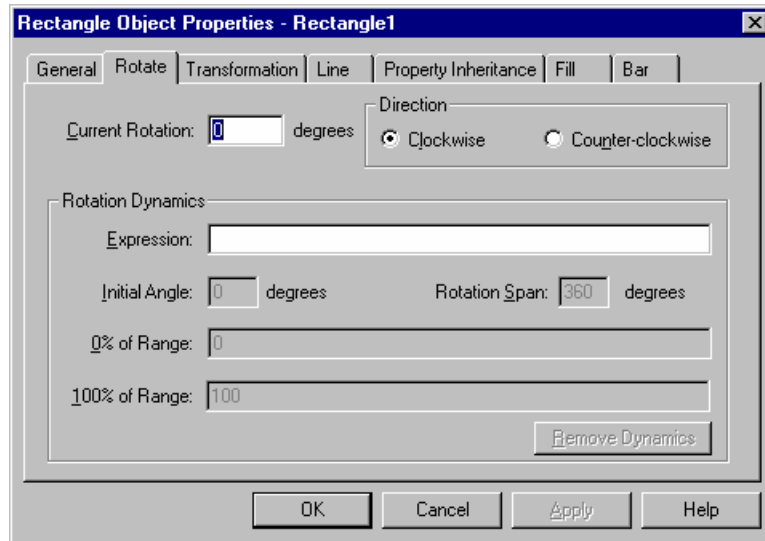
Rotate Page

This property page defines how an object rotates around the Center of Rotation mark. It may rotate either clockwise or counterclockwise from +360 to -360 by integer degrees.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Rotating an object with its rotate property page

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears.
- 2 Click the Rotate tab. The Rotate Property Page is displayed.



- 3 Accept Current Rotation and Direction or change as required.
 - a) Current Rotation reflects the static position of the object in respect to zero (0).
 - b) Enter a value into Current Rotation field for a static position and click the required Direction.
- 4 If Rotation Dynamics are required, the entries must be complete. Be sure to include an LCN or DDB reference as specified in the note below. Enter
 - a) Any valid Expression or data reference. Scrolling is supported.
 - b) Initial Angle in degrees. This is the angle of the object upon initial invocation of the display.
 - c) Rotation Span in degrees from -360 to 360.
 - d) 0% and 100% of Range accept any valid expression or data reference. Scrolling is supported.
 - e) Click Remove Dynamics to erase dynamics entries.
- 5 Click OK or Apply to accept attributes. OK also closes the Property Sheet.

NOTE: The controlling variable or expression for the basic object dynamics (for example, fill, rotate, value, and bar) may not contain a public scripting variable. It must include an LCN or DDB reference. Literals and constants should never be used alone in these fields.

8.22 Transformation Properties

Transformation Page

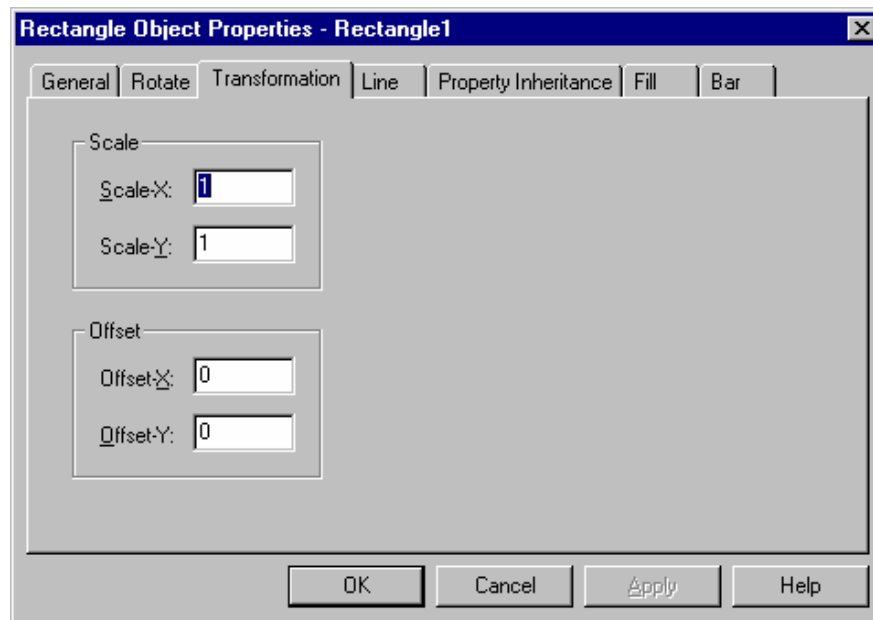
This property page defines how an object can be scaled. You can choose

- Object to be scaled in the X and Y direction by a decimal value.
- Object to be offset from its origin.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Transforming an object with its transformation property page

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears.



- 2 Click the Transformation tab to open the Transformation Property page.

- 3** Choose to scale the object in the X and Y direction as described by a decimal number (the value of "1" is the "100%" value). For example, changing the values of "1" in each direction to "1.25" in each direction has the effect of increasing the width and height of the object by 25%.
- RESULT: The object grows or shrinks in each direction about the center of the object.
- NOTE: In the special case of a Text Object, Scale X = -1 flips text horizontally and Scale Y = -1 flips text vertically.
- 4** Set the Offset of the upper left corner of the object in pixels.
- X direction from left to right.
 - Y direction from the top down.
- RESULT: The object is offset by the number of pixels specified.
- 5** Click OK or Apply to accept attributes. OK also closes the Property Sheet.

8.23 Line Properties

Line Page

This property page defines the line in an object. Each line is described by

- Color selected from the color palette.
- Width of 0 to 20 pixels.
- Style of solid, dashed, dotted, dash-dot, and dash-dot-dot

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

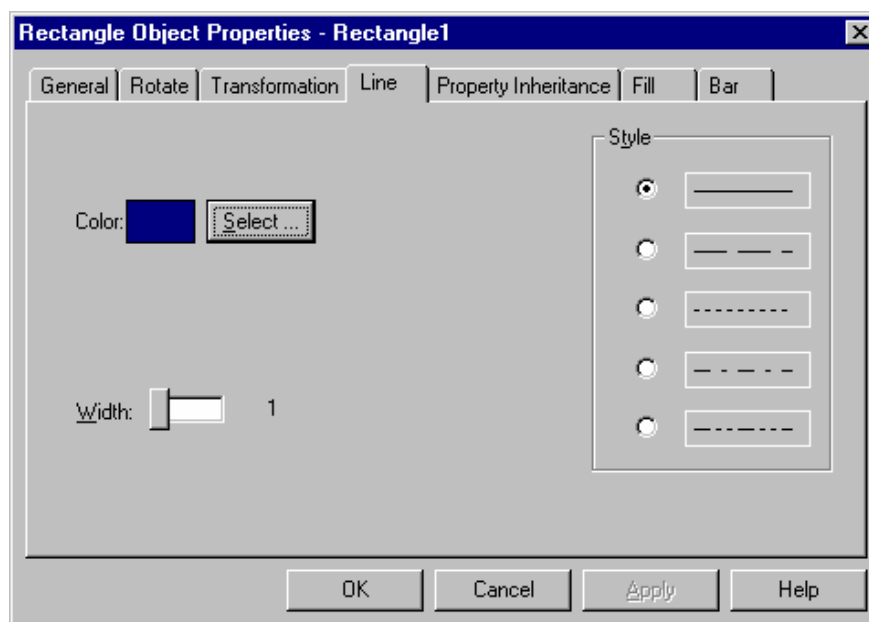
Changing a line on an object with its line property page

- 1** Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears.

8. Properties of Graphic Objects

8.23. Line Properties

- 2 Click the Line tab. The Line Property Page is displayed.



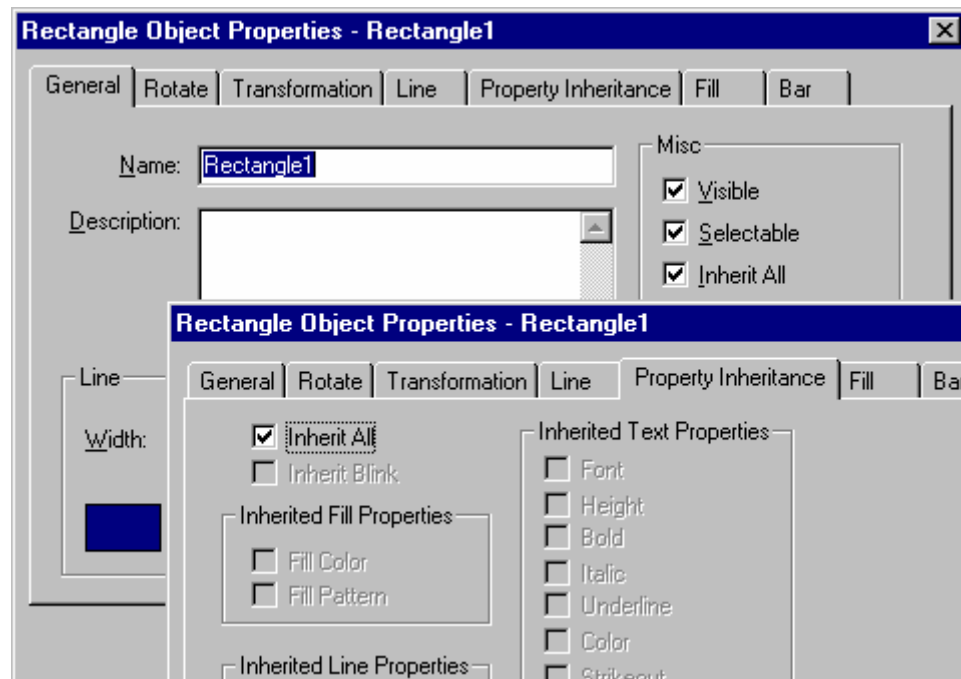
- 3 Accept default Line Color and Width or change as required.
- 4 Accept default Style or change to one of five styles (solid, dashed, dotted, dash-dot, and dash-dot-dot).
- 5 Click OK or Apply to accept attributes. OK also closes the Property Sheet.

8.24 Property Inheritance

Property Inheritance Page

The Property Inheritance page of the Object Properties dialog displays detailed inheritance properties that can be selected individually or in combination, providing a finer granularity of control over property inheritance.

In prior releases, scripts were required to allow the child to inherit some of the parent's properties but not others. Property Inheritance page selections now provide a consistent and easy-to-manage method of child and group inheritance.



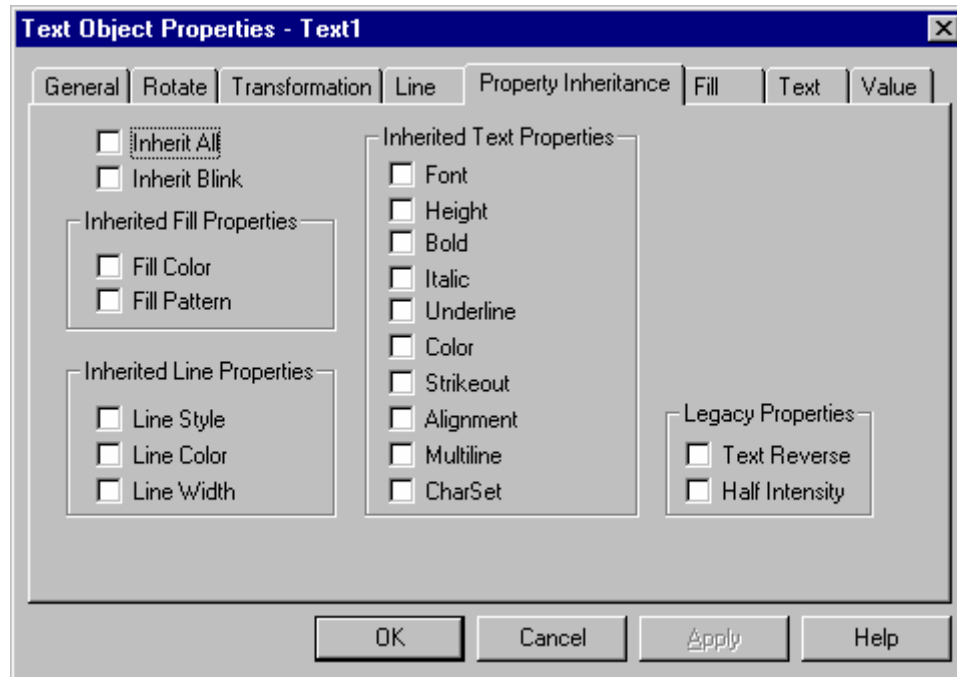
When the Inherit Enabled checkbox is selected on the General page of the Object Properties dialog box, the Inherit Enabled checkbox on the Property Inheritance page is selected automatically. The other checkboxes on the Property Inheritance page are disabled.

When Inherit Enabled is selected, all properties are inherited from the parent.

8. Properties of Graphic Objects

8.24. Property Inheritance

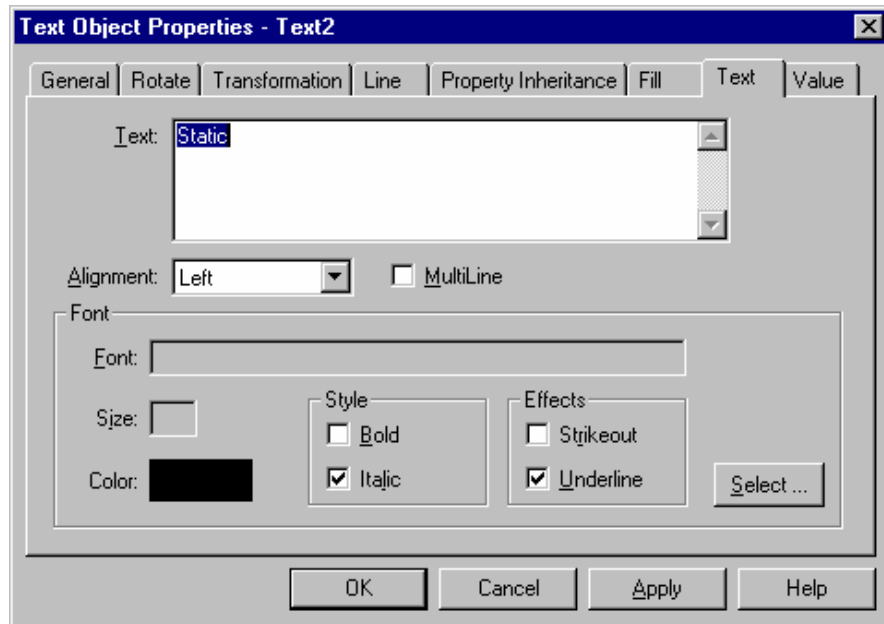
When the Inherit Enabled checkbox is deselected, the Property Inheritance checkboxes are enabled and the user can pick and choose which inherited properties will propagate to the child or group.



Propagation of some text properties from the parent but not others

- 1 Set the Display Builder text defaults (parent properties) to Bold, Italic, and Underline.
- 2 Select New from the File menu.
- 3 Select the text tool and click in the display window. The text that appears is bold, italicized, and underlined.
- 4 Right-click the text object and select Properties... from the menu. The General page of the Properties... dialog appears. Make sure the Inherit Enabled checkbox is deselected.
- 5 Select the Property Inheritance tab. The Property Inheritance page appears.
- 6 Select the Bold checkbox.

- 7 Select the Text tab. The Bold checkbox is selected and disabled. The bold text style will be inherited from the parent and cannot be changed at the child level unless the Bold checkbox on the child's Property Inheritance page is deselected.



- 8 On the Text page, deselect the Italic and Underline checkboxes. Click Apply and OK. The child text object changed independently from the parent default and no longer displays the italic and underline properties. The bold style, inherited from the parent, remains. If the parent style changes to non-bold, so will the child.

Property inheritance for groups

The Property Inheritance page functions in the same way for groups as for child objects.

8.25 Fill Properties

Fill Page

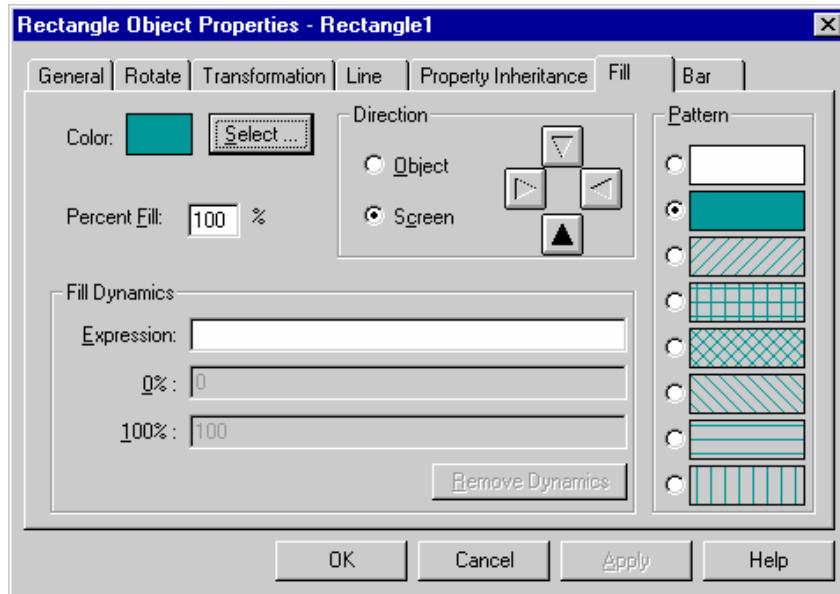
There are two different Fill Property Pages:

- Fill Property Page for a Graphic Object. It looks like the illustration in Step 2 below.
- Fill Property Page for the Builder Defaults, a Display, or an Embedded Display. It is identical to the illustration below but contains only Color and Pattern attributes (no Dynamics).

Default properties are determined from the object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Changing the fill with its Fill property page

- 1 Execute either step 1a, 1b, or 1c.
 - a) Change the fill properties of a Graphic Object or an Embedded Display, make sure the object is selected (has handles), then select Edit/Object Properties (or double-click the object or press the Alt + Enter key combination) or use the Right-mouse Button or the Toolbar.
 - b) Change only the Color and Pattern fill properties of a Display, select Edit/Display Properties (or double-click the display background) or use the Right-mouse Button.
 - c) Change only the Color and Pattern fill properties of the Builder Defaults, select Edit/Defaults.
- 2 Click the Fill tab. The Fill Property Page is displayed.
 - a) It is as shown below if the object has dynamic attributes.
 - b) It shows only Color and Pattern fill properties if there are no dynamic attributes.



- 3 Accept the current Color and Pattern attributes or change them as required.
- 4 Set Percent Fill to the upper limit the object should be allowed to reach. This is normally 100%, but could be less if, for example, there is an uncontrolled drain installed at the 90% mark.
- 5 If the object has no dynamic attributes, skip to step 8.
- 6 Accept Direction chosen or change as required.
 - a) Select an arrow to determine direction the object will fill.

The up arrow fills from bottom to top.

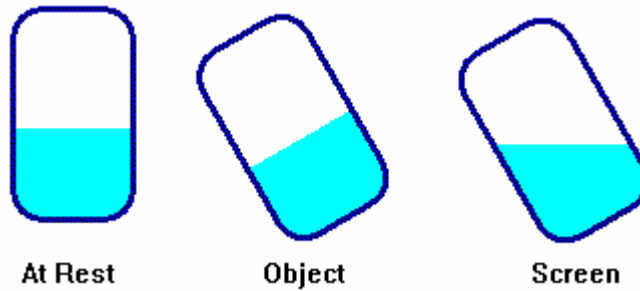
The arrow pointing to the right fills from left to right.

The arrow pointing to the left fills from right to left.

The down arrow fills from top to bottom.
 - b) Use Object direction to fill the object in relation to itself.
 - c) Use Screen direction to fill the object in relation to the screen.

8. Properties of Graphic Objects

8.25. Fill Properties



- 7 The Fill Dynamics entries must be complete. Be sure to include an LCN or DDB reference as specified in the note below.
 - a) Enter any valid Expression or Data Reference. Scrolling is supported.
 - b) Enter 0% and 100% of Range to accept any valid expression, data reference, literal, or constant. Scrolling is supported.
 - c) Click Remove Dynamics to erase dynamics entries.

- 8 Click OK or Apply to accept attributes. OK also closes the Property Sheet.

NOTE: The controlling variable or expression for the basic object dynamics (for example, fill, rotate, value, and bar) may not contain a public scripting variable. It must include an LCN or DDB reference. Literals and constants should never be used alone in these fields.

8.26 Text Properties

Text Page

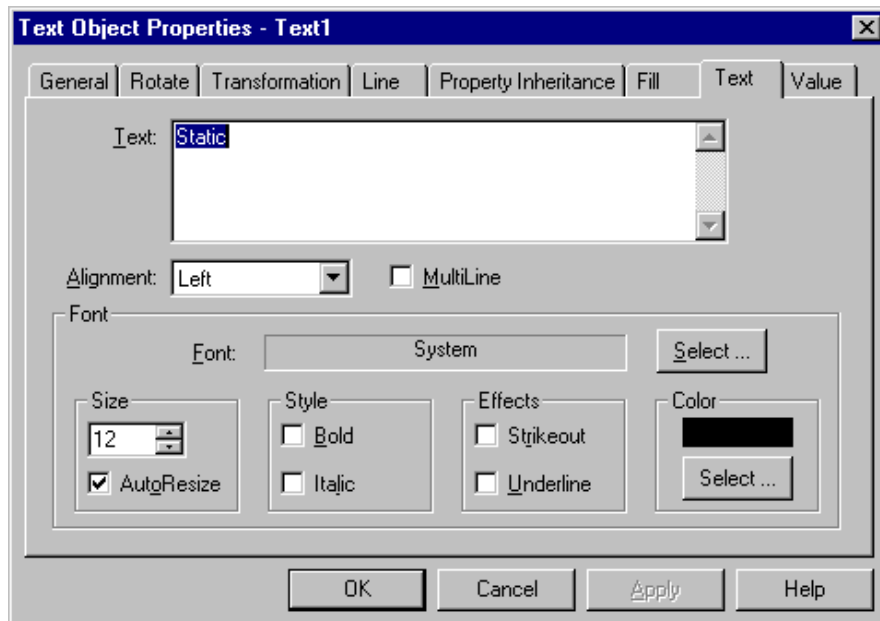
This property page defines various text attributes

- text content,
- text alignment,
- font size, style and effect.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting Text properties with its text property page

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (you may also double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Object Property Sheet appears. Click the Text tab to display the Text Property page.



8. Properties of Graphic Objects

8.27. Properties Unique to Object Types

- 2 Enter or modify text content (default text is "Static"). If more than one line is required, check Multiline.

NOTE: For word-wrap to work properly in Multiline mode, each line of multiline text must contain one or more space characters on which the lines can break.
- 3 Choose the Font size. If Multiline is Not checked, checking AutoResize will cause the font size to automatically change if the text object when the object is resized vertically. Note: If the object is configured to inherit the font size, the inherited font size will disable the AutoResize option.
- 4 Chose text alignment (Left, Center, and Right).
- 5 Click the Font Select button to alter the Font Type, Size, Style, or Effect.
- 6 Click OK or Apply to accept the properties. OK also closes the Property Sheet.

8.27 Properties Unique to Object Types

This section contains the properties unique to individual objects:

- Rectangle-Bar Properties
- Value Properties (Value Page)
- ListBox Properties (List Page)
- ComboBox Properties (List Page)

8.28 Rectangle-Bar Properties

This property page determines how a bar-shaped rectangular object fills under dynamic conditions.

NOTE: Use this Bar Property Page only to set the Dynamic Direction (Vertical or Horizontal) and the Range (Top, Bottom, or Origin) of the bar.

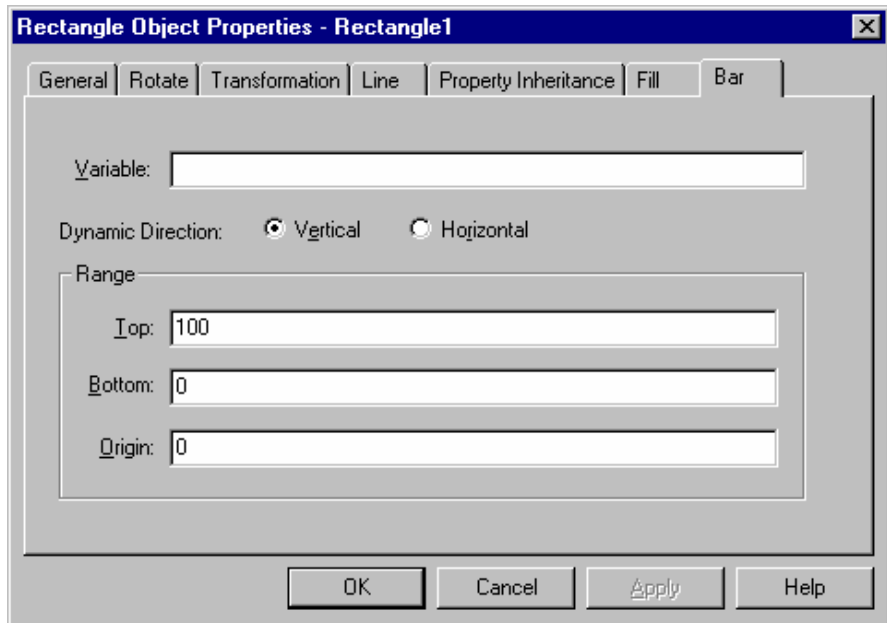
You must use the Fill Properties (Fill Page) to set the bar's

- color
- direction (Reference to Object or Screen, to the Left, Right, Up or Down)
- pattern
- fill dynamics

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting Bar properties with its bar property page

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (or double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Rectangle Object Property Sheet appears.
- 2 Click the Bar tab. The Bar Property Page is displayed.



Rectangle Object Properties - Rectangle1

General | Rotate | Transformation | Line | Property Inheritance | Fill | **Bar**

Variable:

Dynamic Direction: ☒ Vertical ☐ Horizontal

Range

Top:

Bottom:

Origin:

OK Cancel Apply Help

- 3 Enter a Variable as required. Be sure to include an LCN or DDB reference as specified in the note below.
- 4 Select Vertical or Horizontal to define orientation of bar.
- 5 Enter constant, expression, or data reference range value for Top or Right end of bar.

8. Properties of Graphic Objects

8.28. Rectangle-Bar Properties

- 6 Enter constant, expression, or data reference range value for Bottom or Left end of bar.
- 7 Enter constant, expression, or data reference value for bar's dynamic origin.
- 8 Click OK or Apply to accept command.

NOTE: The controlling variable or expression for the basic object dynamics (for example, fill, rotate, value, and bar) may not contain a public scripting variable. It must include an LCN or DDB reference. Literals and constants should never be used alone in these fields.

8.29 Value Properties

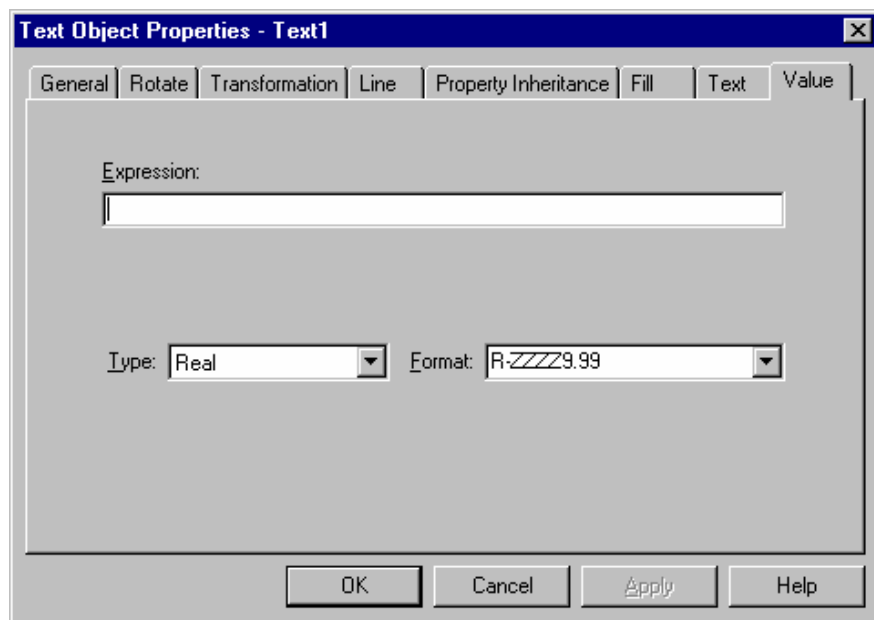
Value Page

This property page defines dynamic values of LCN Data Types.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Setting Value properties with its value property page

- 1 Make sure the object to be modified is selected (has handles), then select Edit/Object Properties (or double-click the object, or press Alt + Enter, or use the Right-mouse Button, or click the Property Sheet button  on the Standard toolbar). The Text Object Property Sheet appears. Click the Value tab to display the Value Property page.



8. Properties of Graphic Objects

8.29. Value Properties

- 2 Enter the Expression required. Be sure to include an LCN or DDB reference as specified in the note below. It can be any expression that evaluates into a valid TPS GUS value format. As an example, in its simplest form this could be a point value reference such as LCN.Valve1.PV.

NOTES:

(A) The controlling variable or expression for the basic object dynamics (for example, fill, rotate, value, and bar) may not contain a public scripting variable. It must include an LCN or DDB reference. Literals and constants should never be used alone in these fields.

(B) The Display Builder will allow a non-existent function to be entered in this expression field. No error is reported when the text sheet is saved or the display is validated. At runtime, the error is not reported; the text object displays "#####" for an uninitialized value.

(C) The concatenation operator, &, does not behave at runtime in the same manner in a script. For example, `me.text = lcn.a100.pv & " gpm"` in a script works, but the same expression, `lcn.a100.pv & " gpm"`, entered on the value page of a text object does not work. The correct way to do this is to use two text objects to display the required results. For example, 1575 gpm (data value and the units). The concatenation operator in a value field should only be used when both referenced items are elements of the same data collection group.

- 3 Enter the required LCN Data Type (Real, Boolean, String, Enumeration, SD Enumeration, DateTime, or Unknown) and its associated Format. Consult the related Parameter Reference Dictionary for a specific parameter's Type.

NOTES: (A) Enumerations across the Network Gateway must be handled as strings, not enumerations.

(B) DateTime fields (DisplDB, public or LCN CDS variables) are defined to be Doubles, but when they are formatted as Reals (eg. R99999.9999) in a VTO, they are displayed rounded to 6 significant digits, which is only tenths of a day. To get the proper degree of rounding/display, use OnDataChange:

```
public_double = dispdb.datetime1  
me.text = public_double
```

- 4 Click OK or Apply to accept the properties. OK also closes the Property Sheet.

8.30 ListBox Properties

List Page

For the ListBox Object primitive, the list page includes tools for management of list items.

- 1** From the ListBox Object Properties dialog, select the List tab. The List page appears.
- 2** Click the Text field in the Item area and type in the name of the first list item.
- 3** In the Data field, enter the integer value that may be associated with the particular text entry. At runtime, a script may set and retrieve the value of the Data field and take the appropriate action without retrieving the item's text.
- 4** Click the Add button to add the text entry to the list. The entry appears at the end of the list, or in alphabetical order if the Sorted checkbox is selected.
- 5** Click the Insert button to add an entry in front of the currently selected item in the list.
- 6** Click the Replace button to replace the currently selected list item with the Text field entry.
- 7** Click Delete to remove the currently selected item from the list.
- 8** Click OK when the list is complete. The listbox appears with the new entries.

8.31 ComboBox Properties

List Page

The List page contains the tools you need to create and manage list items for the ComboBox Object primitive.

Setting up a ComboBox list

- 1 From the ComboBox Object Properties sheet, select the List tab. The List page appears.
- 2 Click the Text field in the Item area and type in the name of the first list item.
- 3 In the Data field, enter the integer value that may be associated with the text entry just made. At runtime, a script may set and retrieve the value of the Data field and take the appropriate action without retrieving the item's text.
- 4 Click the Add button to add the text entry to the list as a new item. This new item will appear at the end of the list, or in alphabetical order if the Sorted check box is selected.
- 5 Click the Insert button to add the text entry to the list as a new item, in front of the one currently selected.
- 6 Click the Replace button to replace the currently selected list item with the Text field entry.
- 7 Click Remove to remove the currently selected item from the list.
- 8 Select the Force List Items check box to force the user to choose an item from the list at runtime.
- 9 Select the Sorted check box to alphabetize the list at runtime.
- 10 Click OK. The values and selections made are saved, the List page disappears, and the ComboBox header appears.
- 11 To verify your changes, select Display/Run on the menu bar or the Run Display button on the Standard taskbar. The ComboBox object with the new entries (items) appears in the display.

8.32 Line Object Properties

This is the list of Property Pages found on the Line Object Property Sheet.

- Line Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)

8.33 Rectangle Object Properties

This is the list of Property Pages found on the Rectangle Object Property Sheet.

- Rectangle Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)
- Rectangle-Bar Properties

8.34 Rounded Rectangle Object Properties

This is the list of Property Pages found on the Rounded Rectangle Object Property Sheet.

- Rounded-Rectangle Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)

8.35 Ellipse Object Properties

This is the list of Property Pages found on the Ellipse Object Property Sheet.

- Ellipse Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)

8.36 Open Polygon Object Properties

This is the list of Property Pages found on the Open polygon Object Property Sheet.

- Open Polygon Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)

8.37 Closed Polygon Object Properties

This is the list of Property Pages found on the Closed Polygon Object Property Sheet.

- Closed Polygon Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)

8.38 Open Bezier Curve Object Properties

This is the list of Property Pages found on the Open Bezier Curve Object Property Sheet.

- Open Bezier Curve Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)

8.39 Closed Bezier Curve Object Properties

This is the list of Property Pages found on the Closed Bezier Curve Object Property Sheet.

- Closed Bezier Curve Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)

8.40 Arc Object Properties

This is the list of Property Pages found on the Arc Object Property Sheet.

- Arc Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)

8. Properties of Graphic Objects

8.41. Text Object Properties

8.41 Text Object Properties

This is the list of Property Pages found on the Text Object Property Sheet.

- Text Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Line Properties (Line Page)
- Property Inheritance (Property Inheritance Page)
- Fill Properties (Fill Page)
- Text Properties (Text Page)
- Value Properties (Value Page)

8.42 Bitmap Object Properties

This is the list of Property Pages found on the Bitmap Object Property Sheet.

- Bitmap Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)

8.43 Button Object Properties

This is the list of Property Pages found on the Button Object Property Sheet.

- Button Object General Properties
- Line Properties (Line Page)
- Fill Properties (Fill Page)
- Property Inheritance (Property Inheritance Page)
- Text Properties (Text Page)
- Button Label/Color Properties. (Labels/Colors Page)

8.44 Data Entry Object Properties

This is the list of Property Pages found on the Data Entry Object Property Sheet.

- Data Entry Object General Properties
- Property Inheritance (Property Inheritance Page)
- Text Properties (Text Page)

8.45 ListBox Object Properties

This is the list of Property Pages found on the ListBox Object Property Sheet.

- ListBox Object General Properties
- Property Inheritance (Property Inheritance Page)
- Text Properties (Text Page)
- ListBox Properties (List Page)

8.46 ComboBox Object Properties

This is the list of Property Pages found on the ComboBox Object Property Sheet.

- ComboBox Object General Properties
- Property Inheritance (Property Inheritance Page)
- Text Properties (Text Page)
- ComboBox Properties (List Page)

8.47 Group Object Properties

This is the list of Property Pages found on the Group Object Property Sheet.

- Group Object General Properties
- Rotate Properties (Rotate Page)
- Transformation Properties (Transformation Page)
- Property Inheritance (Property Inheritance Page)

8. Properties of Graphic Objects

8.47. Group Object Properties

9. Accessing Properties of Graphic Objects

9.1 Overview

NOTE: These commands are identical when accessing Embedded Display Properties.

Topics in this section are:

- Accessing the Properties of Graphic Objects
- Accessing Properties of Graphic Object Scripts
- Accessing a Graphic Object Script (Edit/Object Script menu)

9.2 Accessing the Properties of Graphic Objects

There are several ways to access the Property Sheet of a graphic object

- from the Menu Bar (or by a keystroke)
- by using the right-mouse button
- by double-clicking the Graphic Object (default action).

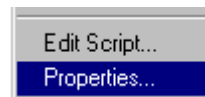
Accessing a Graphic Object Property Sheet from the Menu (Edit/Object Properties)

- 1 If the Graphic Object is not active, select it by clicking once on it (handles show).
- 2 Select Edit/Object Properties (not Edit/Display Properties) from the menu or press the <Alt + Enter> keys or use the Toolbar.

RESULT: The Properties Sheet for that Graphic Object appears.

Accessing a Graphic Object Property Sheet using right-mouse button (Properties)

- 1 With the mouse pointer within the required object (it doesn't need to be selected), click and hold the right-mouse button. A drop-down menu appears.



- 2 Move to the Properties item and release the mouse button.

RESULT: The Properties Sheet for that Graphic Object appears.

9. Accessing Properties of Graphic Objects

9.3. Accessing Properties of Graphic Object Scripts

Accessing a Graphic Object Property Sheet as the default action

With the mouse pointer within the required object (it doesn't need to be selected), double-click the mouse button.

RESULT: The Properties Sheet for that Graphic Object appears.

9.3 Accessing Properties of Graphic Object Scripts

There are two ways to access the Property Sheet of a graphic object

- from the Menu Bar
- by Using the Right-mouse button

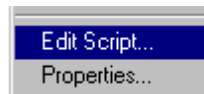
Accessing a Graphic Object Script from the Menu (Edit/Object Script)

- 1 If the Graphic Object is not active, select it by clicking once on it (handles show).
- 2 Select Edit/Object Script from the menu.

RESULT: The Script ListBox for that Graphic Object appears.

Accessing a graphic object script using the right-mouse button

- 1 With the mouse pointer within the required object (it doesn't need to be selected), click and hold the right-mouse button. A drop-down menu appears.



- 2 Move to the Edit Script item and release the mouse button

RESULT: The Script ListBox for that Graphic Object appears

10. Properties of Honeywell OLE Controls

10.1 Overview

The Honeywell GUS Display Builder is designed to be an OLE container and is aware of OLE controls. Because the Display Builder is an OLE container, you can put both Honeywell OLE Controls and Third-Party OLE Controls (non-Honeywell controls) in a display.

An OLE Control makes its properties and methods visible so that you can write to its properties and call its methods. You can access an OLE Control's properties via its Property Sheet and via scripting. You can script what functionality is invoked when a recognized event occurs on the OLE Control.

Honeywell OLE Controls have additional properties and methods that are unique to process control.

Topics covered in this section are:

- Properties Common to Honeywell OLE Controls
- Command Button OLE Control Properties
- ButtonPlus OLE Control Properties
- CheckBox OLE Control Properties
- Data Entry OLE Control Properties
- ListBox OLE Control Properties
- Trend OLE Control Properties
- Versioning Honeywell OLE Controls

NOTE: You will find Third-Party OLE Controls (non-Honeywell controls) available in the marketplace. Honeywell, however, will not be responsible for their proper operation with Honeywell equipment. You must test and verify Third Party OLE controls for suitability in your application.

10. Properties of Honeywell OLE Controls

10.2. Properties Common to Honeywell OLE Controls

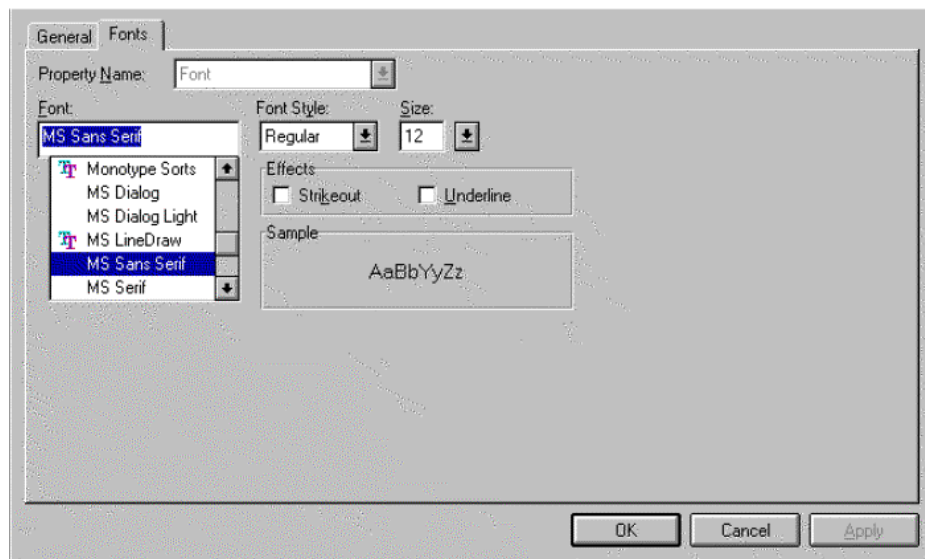
10.2 Properties Common to Honeywell OLE Controls

All Honeywell OLE Controls that display text have a Fonts page in their Property Sheet.

Honeywell OLE Controls that use an input timer have an Input page in their Property Sheet.

Setting fonts properties with the Fonts property page

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed.
- 2 Click the Fonts tab. The Fonts page is displayed.



- 3 Accept the Font and its properties (size, style, etc.) or change the font and its properties as required.
- 4 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

10.3 Command Button OLE Control Properties

A command button is used to invoke a command. When the button is clicked, the command associated with the button is invoked.



The Property Sheet of the Command Button OLE Control has two pages, a General page and a Fonts page. The Fonts page is common to all Honeywell OLE Controls. Refer to Properties of Honeywell OLE Controls Overview

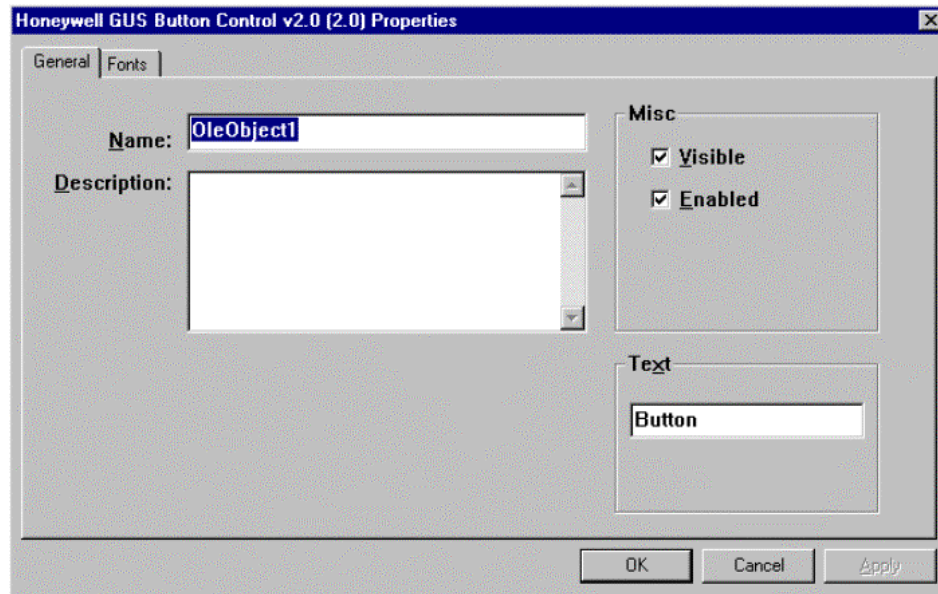
10.4 Command button General properties page

Setting general properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed. The General page is displayed by default (because it is the first tab).

10. Properties of Honeywell OLE Controls

10.4. Command button General properties page



- 2 Change the default name if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.

NOTE: An entry is required in the Name field.
- 3 Enter a description if required.
- 4 Accept the miscellaneous options as defaulted or change each one as required.
- 5 Change the label on the button by editing the text in the control labeled "Text."
- 6 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

Property Definitions for properties found on Command Button Properties Page

Property Name	Definition	Syntax	Required Entry
Name	Object name that must be unique in the display. This is the name used in referencing the object in a script. Default name is supplied by the Display Builder.	Refer to Display Object names in the <i>Display Scripting User's Guide</i> , in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.	Yes
Description	User description There is no default.	None	No
Visible	Determines the visibility of the object at buildtime and at display invocation. Default is "Visible" on.	NA	NA
Enabled	Determines the state (enabled or not) of the object at buildtime and at display invocation. Default is "Enabled" true (check on).	NA	NA
Text	Label displayed on the button. Default is "Button."	None	No

10.5 ButtonPlus OLE Control Properties

The ButtonPlus OLE control allows the use of custom labeled buttons to do a single action (as shown in the **Turn On Pump** button below) or to select a single action from a list of valid choices (as in the **Heat/Cool/Auto** group of buttons below). In either example, only one button is active at a time.



The Property Sheet of the ButtonPlus OLE Control has three pages:

- **General Page**—Refer to ButtonPlus General Properties Page.
- **Fonts Page**—The Fonts page is common to Honeywell OLE Controls. Refer to Properties Common to Honeywell OLE Controls.
- **Labels/Colors Page**—Refer to ButtonPlus General Properties Page.

10.6 ButtonPlus General Properties Page

Setting general properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed. The General page is displayed by default (because it is the first tab).

Honeywell GUS Button Plus Control v2.0 [2.0] Properties

General | Fonts | Labels/Colors

Name:

Description:

Type:

Orientation:

Misc

☒ Visible

☒ Enable

☐ Border

Number Buttons:

Horizontal Spacing:

Vertical Spacing:

OK Cancel Apply

- 2 Change the default name if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.

NOTE: An entry is required in the Name field.
- 3 Enter a description if required.
- 4 Accept the miscellaneous options (Visible, Enable, and Border) as defaulted or change each one as required.
- 5 Choose the required button switch Type (Momentary, Interlocking, or Latching).
- 6 Decide the Orientation of the buttons (Vertical or Horizontal).
- 7 Choose the Number of buttons (1 through 8).
- 8 Choose the Horizontal and Vertical Spacing of the buttons.
- 9 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.



TIP

If required, be sure to change the Fonts and Labels/Colors property pages.

10. Properties of Honeywell OLE Controls

10.6. ButtonPlus General Properties Page

Property Definitions for properties found on ButtonPlus General Properties Page

Property Name	Definition	Syntax	Required Entry
Name	Object name that must be unique in the display. This is the name used in referencing the object in a script. Default name is supplied by the Display Builder.	Refer to Display Object names in the <i>Display Scripting User's Guide</i> , in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.	Yes
Description	User description There is no default.	None	No
Visible	Determines the visibility of the object at buildtime and at display invocation. Default is "Visible" on.	NA	NA
Enable	Determines the state (enabled or not) of the object at buildtime and at display invocation. Default is "Enabled" true (check on).	NA	NA

10. Properties of Honeywell OLE Controls

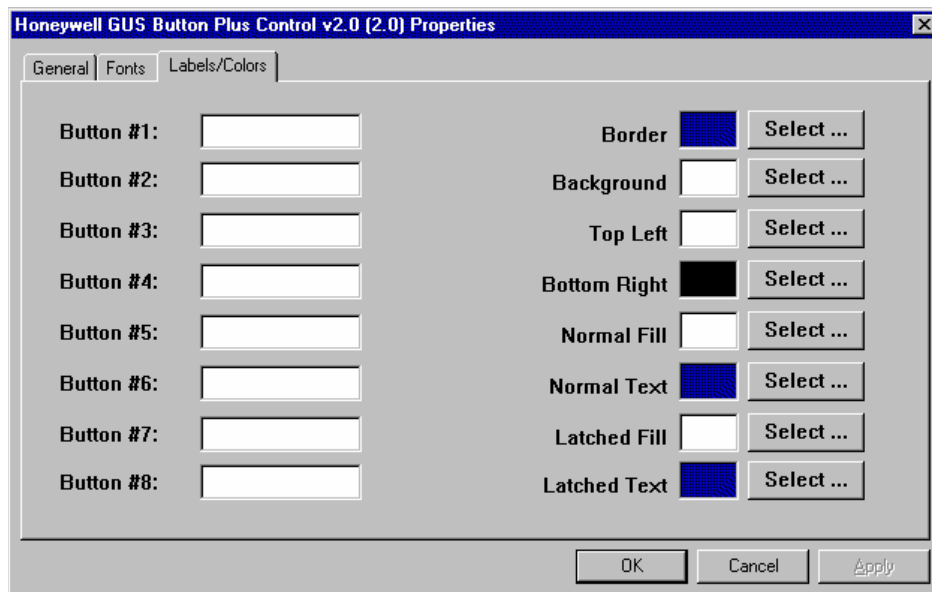
10.6. ButtonPlus General Properties Page

Property Name	Definition	Syntax	Required Entry
Border	Determines whether the button box has a border (the outline around the buttons) or not. A border is displayed if the check is on, and a border is not displayed if the check is off. Default is "Border" false (check off).	NA	NA
Type	Momentary—Active while button is down Interlocking—Button that is clicked (button down) is active. All other buttons unlatch and are inactive. Latching—Toggles in active (button down) or inactive (button up) state each time button is clicked.	NA	Yes
Orientation	Vertical—Buttons line up top to bottom. Horizontal—Buttons line up from left to right.	NA	Yes
Number Buttons	Number of total buttons.	NA	Yes
Horizontal Spacing	Left/right space between buttons.	NA	No
Vertical Spacing	Top/bottom space between buttons.	NA	No

10.7 ButtonPlus Labels/Colors Properties Page

Setting Labels/Colors properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed.
- 2 Click the Labels/Colors tab. The Labels/Colors page is displayed.



- 3 For each button used (Button #1 through #8) label with the required text.
- 4 Border color is the outline around the entire ButtonPlus control.
- 5 Background color is the color of the backplate behind the buttons.
- 6 Top Left is the shading color on the top and left side of all buttons.
- 7 Bottom Right is the shading color on the bottom and right side of all buttons.
- 8 Normal Fill is the color of the buttons when in normal (unlatched) mode.
- 9 Normal Text is the color of the button text when in normal (unlatched) mode.

- 10** Latched Fill is the color of the buttons when in latched mode.
- 11** Latched Text is the color of the button text when in latched mode.
- 12** Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

Property Definitions for properties found on ButtonPlus Labels/Colors Properties Page

Property Name	Definition	Syntax	Required Entry
Button #n	Text that is to be shown on the button face. Currently, use up to 8 buttons, so n= 1 through 8.	NA	No
Border	The outline around the entire ButtonPlus control.	NA	No
Background	Background color of the control.	NA	No
Top Left	The outline shading on the top and left side of all buttons.	NA	No
Bottom Right	The outline shading on the bottom and right side of all buttons.	NA	No
Normal Fill	Fill color of a button under normal or unlatched conditions.	NA	No
Normal Text	The color of the text on a normal button.	NA	No
Latched Fill	Fill color of a button under latched conditions.	NA	No
Latched Text	The color of the text on a latched button.	NA	No

10.8 CheckBox OLE Control Properties

Checkboxes support options that are either on or off. Use checkboxes for independent or nonexclusive choices.

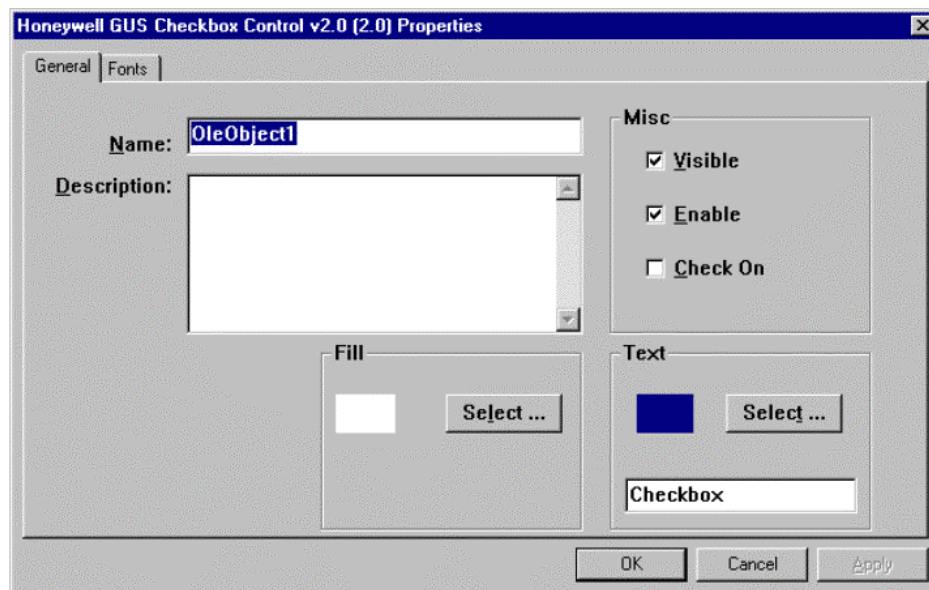


The Property Sheet of the CheckBox OLE Control has two pages, a General page and a Fonts page. The Fonts page is common to Honeywell OLE Controls. Refer to Properties Common to Honeywell OLE Controls.

10.9 CheckBox General Properties page

Setting general properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed. The General page is displayed by default.



- 2 Change the default name if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.
NOTE: An entry is required in the Name field.
- 3 Enter a description if required.
- 4 Accept the miscellaneous options as defaulted or change each one as required.
 - Visible—Make the object visible or invisible.
 - Enable—Determines the state of the object at buildtime and at display invocation.
 - Check On—The checkbox is checked in the default state.
- 5 Click Select in the Fill group to display the color palette from which you can select a fill color for the checkbox.
- 6 Change the label on the button by editing the text in the text control found in the "Text" group.
- 7 Click Select in the Text group to display the color palette from which you can select a text color for the text of the checkbox.
- 8 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

10. Properties of Honeywell OLE Controls

10.9. CheckBox General Properties page

Property Definitions for properties found on CheckBox General Properties Page

Property Name	Definition	Syntax	Required Entry
Name	Object name that must be unique in the display. This is the name used in referencing the object in a script. Default name is supplied by the Display Builder.	Refer to Display Object names in the <i>Display Scripting User's Guide</i> , in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.	Yes
Description	User description There is no default.	None	No
Visible	Determines the visibility of the object at buildtime and at display invocation. Default is "Visible" true (check on).	NA	NA
Enabled	Determines the state (enabled or not) of the object at buildtime and at display invocation. Default is "Enabled" true (check on).	NA	NA
Check on	Determines the state (checked or not checked) of the checkbox at buildtime and at display invocation. Default is "check on" false (not checked).	NA	NA

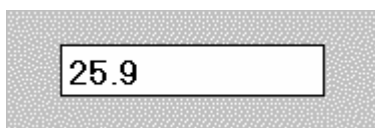
10. Properties of Honeywell OLE Controls

10.10. Data Entry OLE Control Properties

Property Name	Definition	Syntax	Required Entry
Fill Color	The fill color of the button. Another way of thinking of fill color is that it is the background color of the checkbox. Default is "light gray" (index 116 on the color palette).	NA	NA
Text Color	Color of the checkbox and its label. Default is "black."	NA	NA
Text	Label displayed on the button. Default is "Button."	Text only	No

10.10 Data Entry OLE Control Properties

Data entry control supports entry of data from the keyboard.



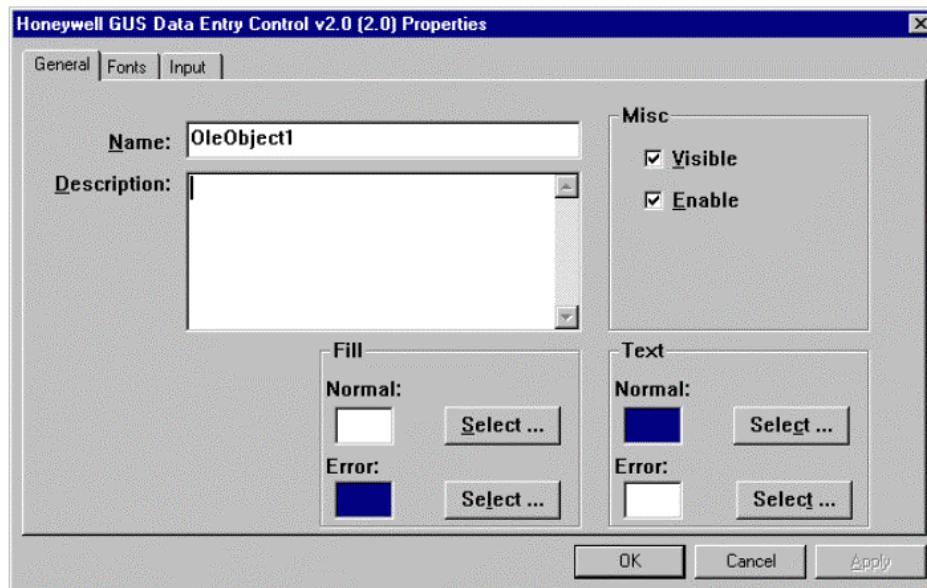
The Property Sheet of the Data Entry OLE Control has three pages:

- A General page. Refer to Data Entry General Properties Page.
- A Fonts page. Refer to Properties Common to Honeywell OLE Controls.
- An Input page. Refer to Data Entry Input Properties Page.

10.11 Data Entry General Properties Page

Setting general properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed. The General page is displayed by default (because it is the first tab).



- 2 Change the default name if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.

NOTE: An entry is required in the Name field.
- 3 Enter a description if required.
- 4 Accept the miscellaneous options as defaulted or change each one as required.

10. Properties of Honeywell OLE Controls

10.11. Data Entry General Properties Page

- 5 Click Select adjacent to the "Normal" color rectangle in the Fill group to display the color palette from which you can select a fill color to be used for the data entry control in normal state.
- 6 Click Select adjacent to the "Error" color rectangle in the Fill group to display the color palette from which you can select a fill color to be used for the data entry control in error state.
- 7 Click Select adjacent to the "Normal" color rectangle in the Text group to display the color palette from which you can select a text color for the text of the data entry control in normal state.
- 8 Click Select adjacent to the "Error" color rectangle in the Text group to display the color palette from which you can select a text color for the text of the data entry control in error state.
- 9 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

Property Definitions for properties found on Data Entry General Properties Page

Property Name	Definition	Syntax	Required Entry
Name	Object name that must be unique in the display. This is the name used in referencing the object in a script. Default name is supplied by the Display Builder.	Refer to Display Object names in the <i>Display Scripting User's Guide</i> , in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax.	Yes
Description	User description There is no default.	None	No
Visible	Determines the visibility of the object at buildtime and at display invocation. Default is "Visible" true (check on).	NA	NA
Enable	Determines the state (enabled or not) of the object at buildtime and at display invocation. Default is "Enabled" true (check on).	NA	NA

10. Properties of Honeywell OLE Controls

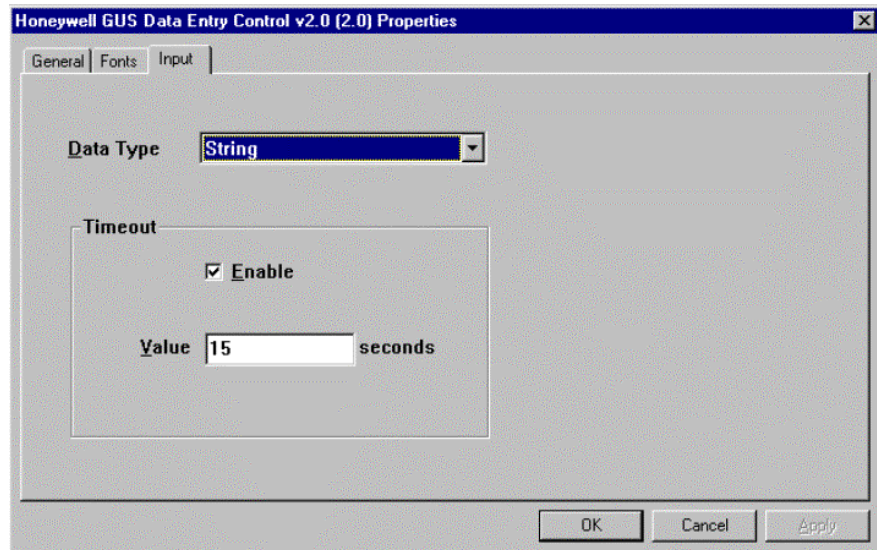
10.11. Data Entry General Properties Page

Property Name	Definition	Syntax	Required Entry
Fill Color/ Normal	The fill color of the data entry control at buildtime and at display invocation. NOTE: The "Normal Fill Color" remains the color during runtime unless it is changed by a script. Default is "light gray" (index 116 on the color palette).	NA	NA
Fill Color/ Error	The fill color of the data entry control when the data displayed is in error. NOTE: The user must use a script to change the fill color to the "error fill color." There is no built-in error handling routine that changes the fill color of a data entry control. Default is "black."	NA	NA
Text/ Normal Color	The text color of the data entry control at buildtime and at display invocation. NOTE: The "Normal Text Color" remains the color during runtime unless it is changed by a script. Default is "black."	NA	NA
Text/ Error Color	The text color of the data entry control when the data displayed is in error. NOTE: The user must use a script to change the text color to the "error text color." There is no built-in error handling routine that changes the text color of a data entry control. Default is "light gray" (index 116 on the color palette).	NA	NA

10.12 Data Entry Input Properties Page

Setting input properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed.
- 2 Click the Input tab. The Input page is displayed.



- 3 Select the TPS GUS data type that will be entered in the data entry control.
- 4 Accept the timeout-enabled option as defaulted or change it as required.
- 5 Enter the number of seconds at which a timeout event is generated for this control.
- 6 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

10. Properties of Honeywell OLE Controls

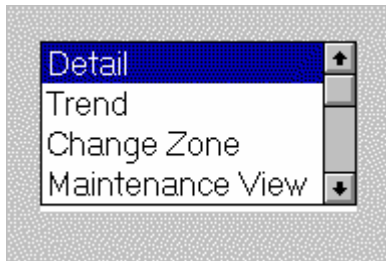
10.13. ListBox OLE Control Properties

Property Definitions for properties found on Data Entry Input Properties Page

Property Name	Definition	Syntax	Required Entry
Data Type	The TPS GUS data type of the data that will be entered in this control. Default is "String."	Choose from the list only	Yes
Timeout Enable	Timeout is enabled (checked) means that timeout events for the control will be generated every specified number of seconds. Timeout is disabled (not checked) means that timeout events for the control will not be generated every specified number of seconds.	NA	NA
Timeout Value	The number of seconds at which a timeout event is generated for this control. Timeout event is generated every n seconds if the control is enabled and not in the process of receiving keyboard entry. Default is 15 seconds.	Integer	Yes

10.13 ListBox OLE Control Properties

A ListBox is a control for displaying a list of textual choices for the user. The purpose of a ListBox is to display a collection of items and, in most cases, support selection of a choice of an item in the list. This ListBox control supports single selection only.



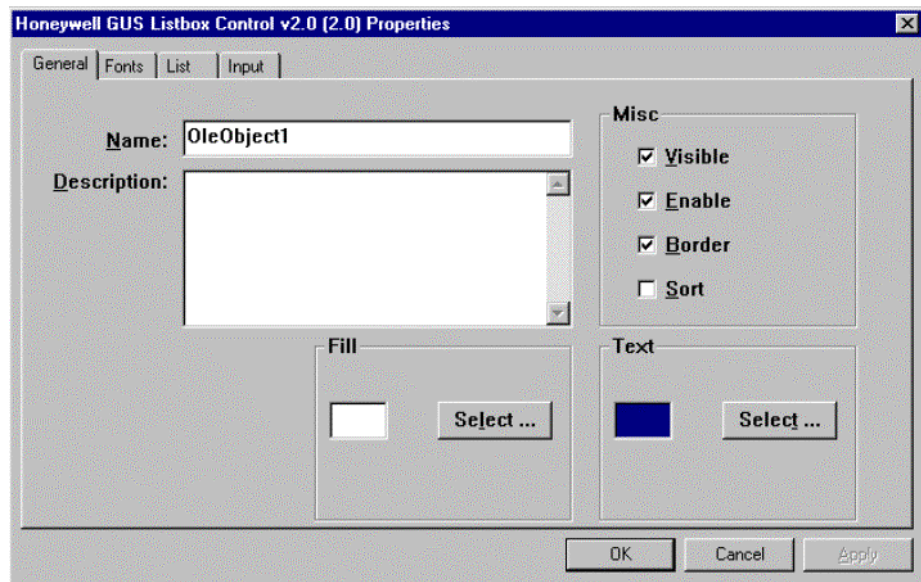
The Property Sheet of the ListBox OLE Control has the following four pages:

- **General Page**—Refer to ListBox General Properties Page.
- **Fonts Page**—The Fonts page is common to Honeywell OLE Controls. Refer to Properties Common to Honeywell OLE Controls.
- **List Page**—Refer to ListBox List Properties Page.
- **Input Page**—Refer to ListBox Input Properties Page.

10.14 ListBox General Properties Page

Setting general properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the Right-mouse Button menu of the control). The OLE Control's Property Sheet is displayed. The General page is displayed by default (because it is the first tab).



- 2 Change the default name if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.

NOTE: An entry is required in the Name field.

- 3 Enter a description if required. A description is not required.

10. Properties of Honeywell OLE Controls

10.14. ListBox General Properties Page

- 4 Accept the miscellaneous options as defaulted or change as explained in table below.
- 5 Click Select in the Fill group to display the color palette from which you can select a fill color for the ListBox.
- 6 Click Select in the Text group to display the color palette from which you can select a text color for the ListBox items.
- 7 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

Property Definitions for properties found on Data Entry General Properties Page

Property Name	Definition	Syntax	Required Entry
Name	Object name that must be unique in the display. This is the name used in referencing the object in a script. Default name is supplied by the Display Builder.	Refer to <i>Display Scripting User's Guide</i> , GUS Scripting Introduction, Data Entry Primitive section, Display Object names	Yes
Description	User description There is no default.	None	No
Visible	Determines the visibility of the object at buildtime and at display invocation. Default is "Visible" true (check on).	NA	NA
Enabled	Determines the state (enabled or not) of the object at buildtime and at display invocation. Default is "Enabled" true (check on).	NA	NA
Border	Determines whether the ListBox has a border (the outline around the list) or not. A border is displayed if the check is on, and a border is not displayed if the check is off. Default is "Border" true (check on).	NA	NA

10. Properties of Honeywell OLE Controls

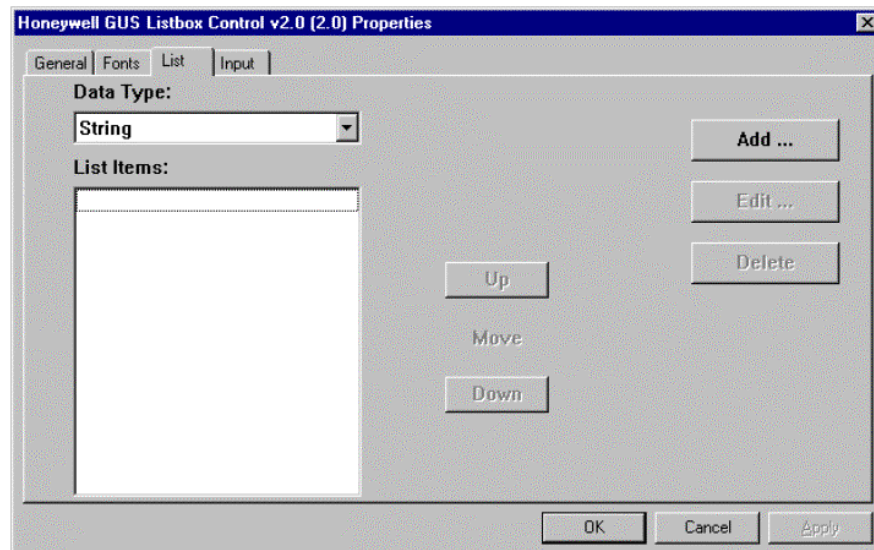
10.14. ListBox General Properties Page

Property Name	Definition	Syntax	Required Entry
Sort	Determines if the list items are alphabetically sorted (check on) or not (check off). The sort is done at buildtime and runtime. Default is "Sort" false (not checked).	NA	NA
Fill Color	The fill color of the ListBox. Another way of thinking of fill color is that it is the background color of the ListBox. Default is "light gray" (index 116 on the color palette).	NA	NA
Text Color	Color of the checkbox and its label. Default is "black."	NA	NA

10.15 ListBox List Properties Page

Setting list properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the right-mouse button menu of the control). The OLE Control's Property Sheet is displayed.
- 2 Click the List tab. The List page is displayed.



- 3 Select the Data Type of the list. Choose from String or Enumeration type.

NOTE 1: Enumerations across the Network Gateway must be handled as strings, not enumerations.

NOTE 2: The list must be empty to select a different data type.

Deleting list items from the list

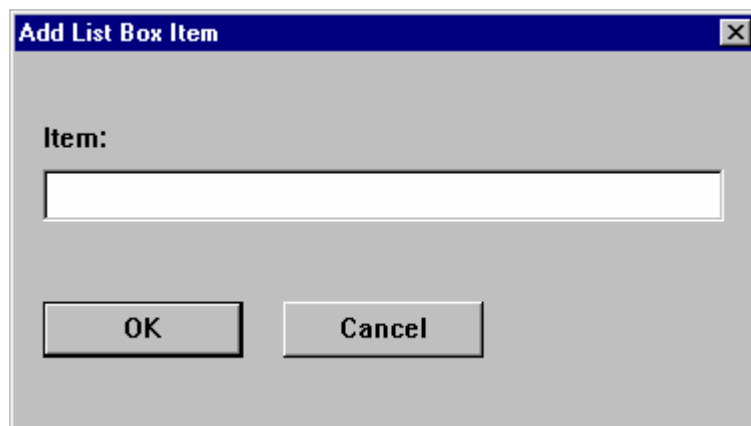
- 1 On the list page, select one or more items in the list that you want to delete. To select more than one item in the list, use Shift + Click to select contiguous items, and use Ctrl + Click to select noncontiguous items.
- 2 Click Delete on the list page. The list item(s) are deleted from the list.

Moving list items

To move a list item, select the list item to be moved. Then click Up or Down to move the item up or down one position in the list. NOTE: If the sort option is checked, the sort will override manual positioning when OK is clicked.

Adding list items of type string to the list

- 1 On the list page, select the item in the list after that you want to insert the new item.
- 2 Click Add... on the list page. The "Add ListBox Item" dialog is displayed.



- 3 In the "Add ListBox Items" dialog, type the string. Click OK. The item appears in the list on the list page after the selected item.

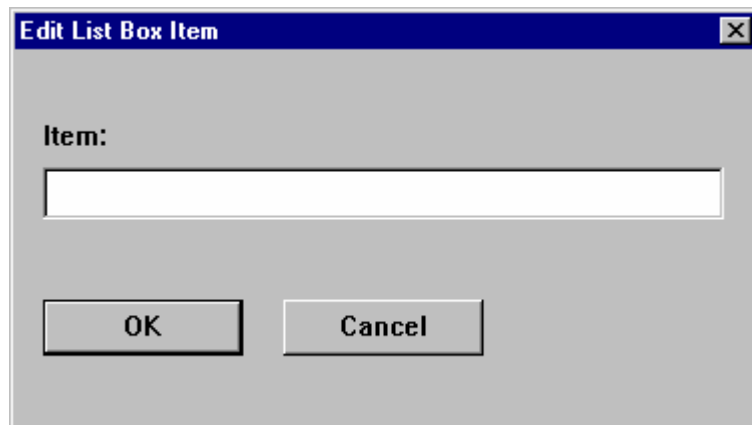
NOTE: Duplicate list items are allowed.

10. Properties of Honeywell OLE Controls

10.15. ListBox List Properties Page

Editing list items of type string to the list

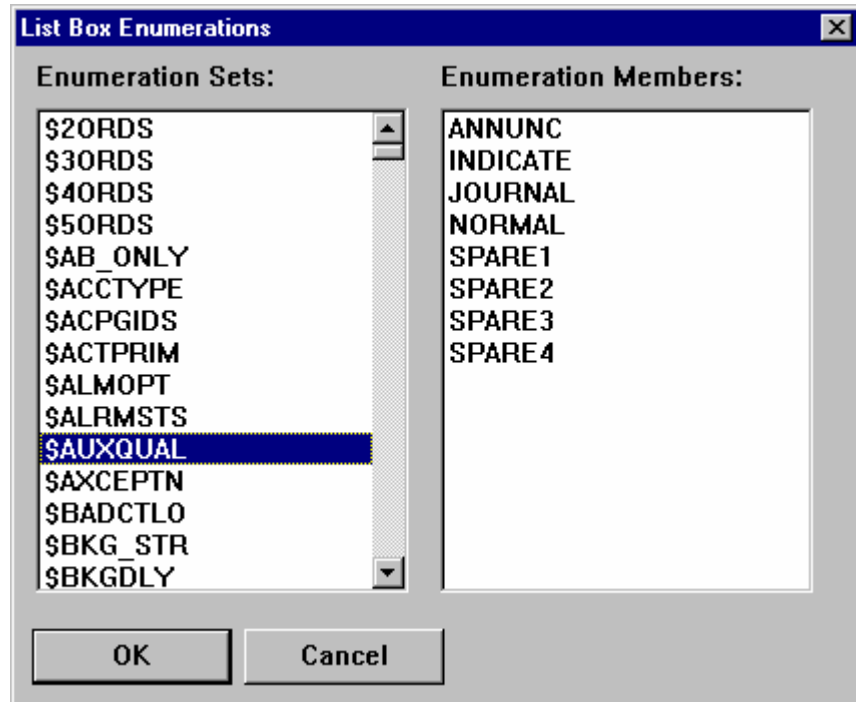
- 1 On the list page, select the item in the list that you want to edit.
- 2 Click Edit... on the list page. The "Edit ListBox Item" dialog is displayed.



- 3 In the "Edit ListBox Item" dialog, edit the string. Click OK. The edited item appears in the list on the list page. NOTE: Duplicate list items are allowed.

Adding list items of type enumeration to the list

- 1 On the list page, select the item in the list after which you want to insert the new item.
NOTE: "Add" is disabled if more than one item is selected in the list.
- 2 Click Add... on the list page. The "ListBox Enumerations" dialog is displayed.



- 3 In the "ListBox Enumerations" dialog, select the enumeration set. The members for the selected enumeration set appear in the enumeration members list.
- 4 Select the enumeration members to be displayed in the list. To select more than one item in the list, use Shift + Click to select contiguous items, and use Ctrl + Click to select noncontiguous items.
- 5 Click OK in the "ListBox Enumerations" dialog box. The item(s) appears in the list on the list page after the selected item.

NOTE: Duplicate list items are allowed.

Editing list items of type enumeration

NOTE: You cannot edit list items of type enumeration.

Enumerations across the Network Gateway must be handled as strings, not enumerations.

10. Properties of Honeywell OLE Controls

10.16. ListBox Input Properties Page

Property Definitions for properties found on ListBox List Properties Page

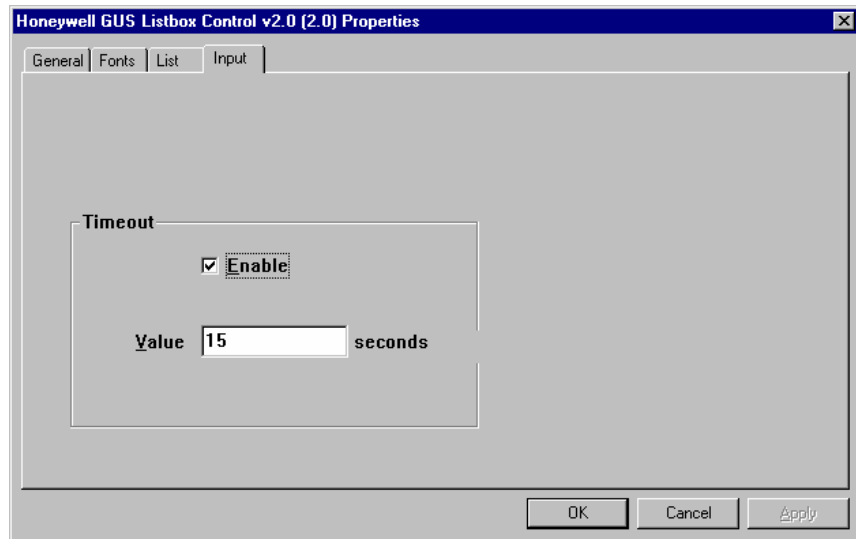
Property Name	Definition	Syntax	Required Entry
Data type	Determines the data type of all items in the list. NOTE: Data types cannot be mixed.		Yes
List items	List of items to be displayed at runtime.	None for strings. NOTE: 1) Enumeration set members cannot be edited. 1) Duplicate list items are allowed.	No

10.16 ListBox Input Properties Page

Setting input properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the right-mouse button menu of the control). The OLE Control's Property Sheet is displayed.

- 2** Click the Input tab. The Input page is displayed.



- 3** Accept the timeout enabled option as defaulted or change it as required.
- 4** Enter the number of seconds at which a timeout event is generated for this control.
- 5** Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

10. Properties of Honeywell OLE Controls

10.16. ListBox Input Properties Page

Property Definitions for properties on ListBox Input Properties Page

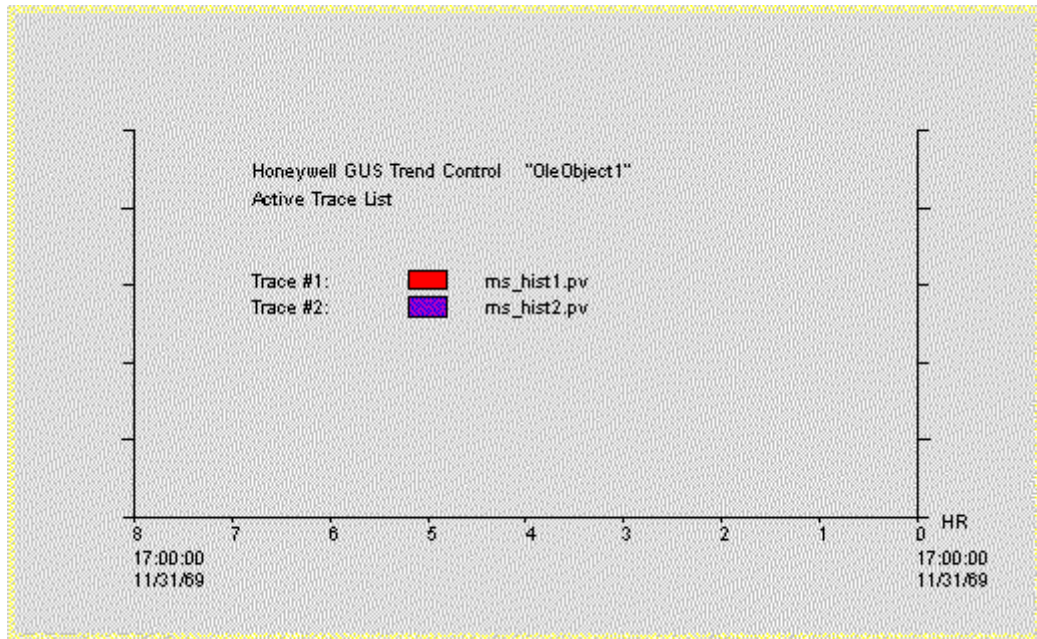
Property Name	Definition	Syntax	Required Entry
Timeout/ Enable	Timeout is enabled (checked) means that timeout events for the control will be generated every specified number of seconds. Timeout is disabled (not checked) means that timeout events for the control will not be generated every specified number of seconds. Default is timeout "enabled."	NA	NA
Value/ seconds	The number of seconds at which a timeout event is generated for this control. The default is 15 seconds. Timeout event is generated every n seconds if the control is enabled and not receiving user actions (mouse clicks or keyboard selection).	Number entry only	Yes

10.17 Trend OLE Control Properties

The Honeywell Trend OLE Control is a control for displaying up to eight traces of TPS Network process control values at runtime. Up to seven TPS Network process control variables are plotted against a base variable in an xy-plot.

Example of Trend OLE Control at buildtime

Here is an example of the Honeywell Trend OLE Control at buildtime, in the Plot Plus Labels trend style.

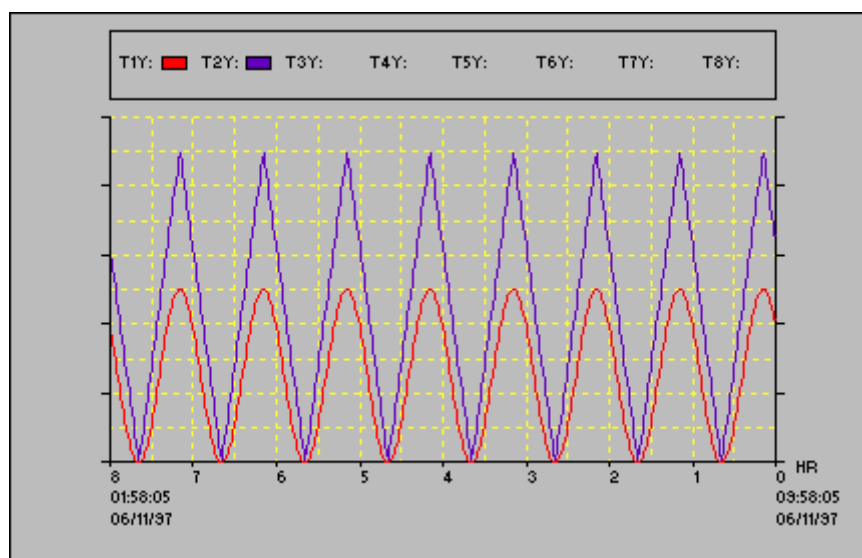


10. Properties of Honeywell OLE Controls

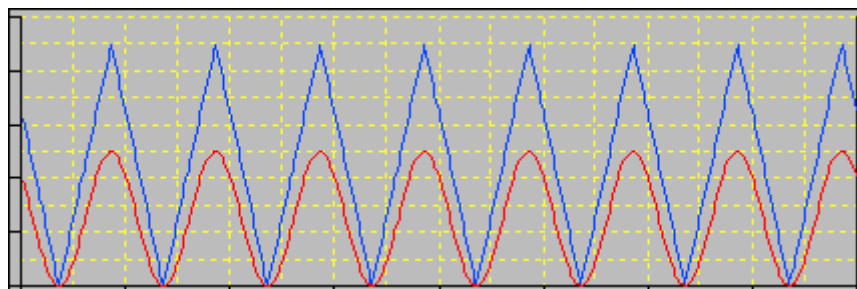
10.17. Trend OLE Control Properties

Example of Trend OLE Control at runtime

This is an example of the Honeywell Trend OLE Control at runtime, in the Plot Plus Labels trend style.



This is an example of the Honeywell Trend OLE Control at runtime, in the Plot Area Only trend style.



You can provide standard xy-plot capability for Trend OLE Control. The xy-plot allows trending of one trace against the other. The xy-plot allows you to plot 1 to 7 traces against one trace.

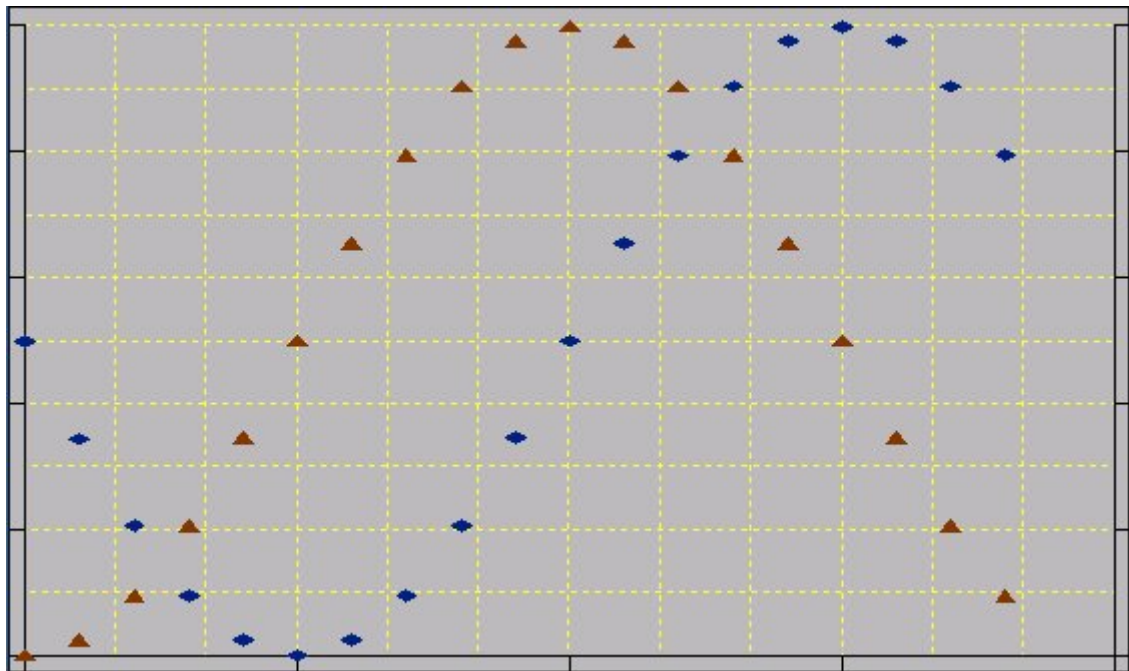
The xy-plots can be plotted as:

- Normal scatter plot
- Operation point plot

NOTE: By default, trace 1 corresponds to the X-axis and traces 2 through 8 are plotted on the Y-axis against the X-axis value with the matching timestamp. If the X-axis trace is deleted then the X-axis trace is set to zero and no graph is plotted on the xy-plot.

Example of normal scatter plot

Here is an example of the standard scatter plot without lines where three datasets are plotted against a fourth (X) dataset. The xy-plots can be distinguished by their shape and color. The xy-plots are plotted as rectangles, diamonds, Xs, bowties, and triangles pointing in different directions.

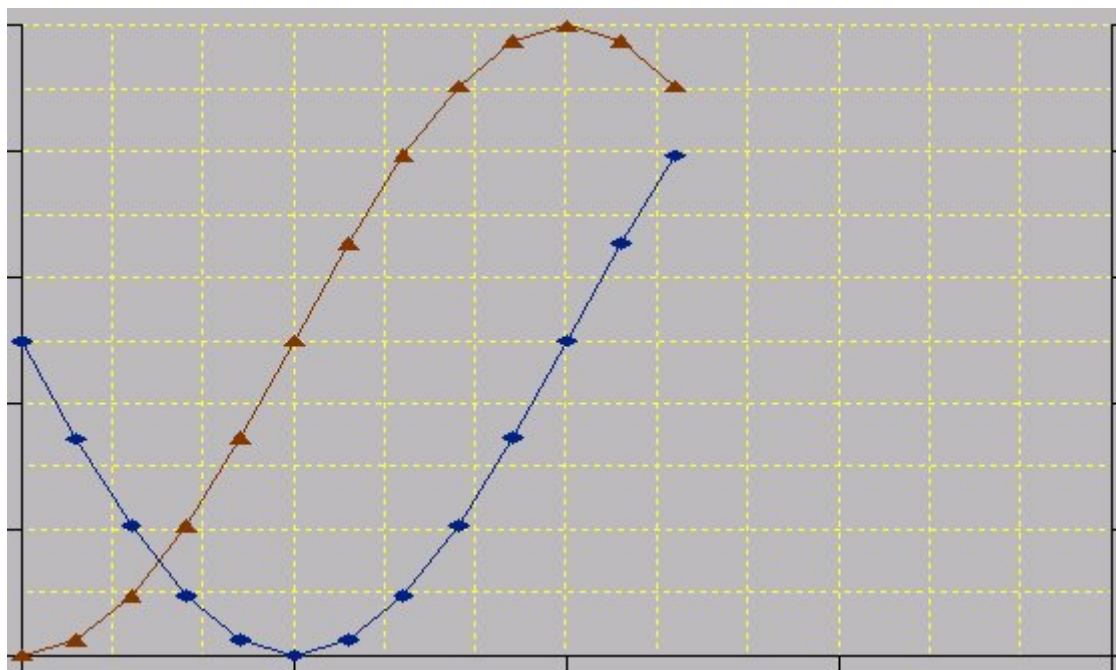


10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Example of scatter plot with lines

Here is an example of the scatter plot with lines where three datasets are plotted against a fourth (X) dataset and are joined by lines.



Example of operation point plot series with limits

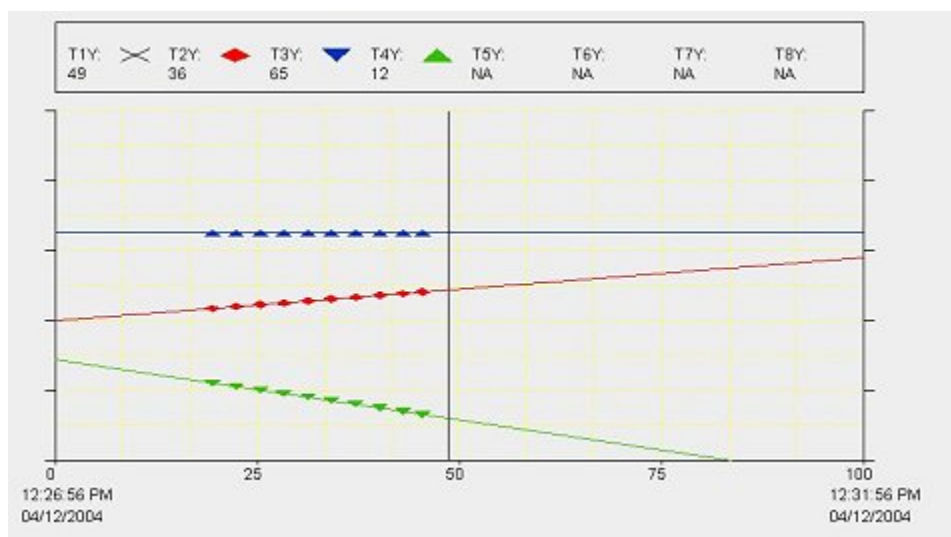
Here is an example of the operation point plot series with limits where three datasets are plotted against a fourth (X) dataset. Only the latest value of xy-plot is displayed on the screen.



10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

This is an example of the Honeywell Trend OLE Control at runtime. In the Plot Plus Labels xy-plot style, the interpolated values are displayed at a hairline cursor.

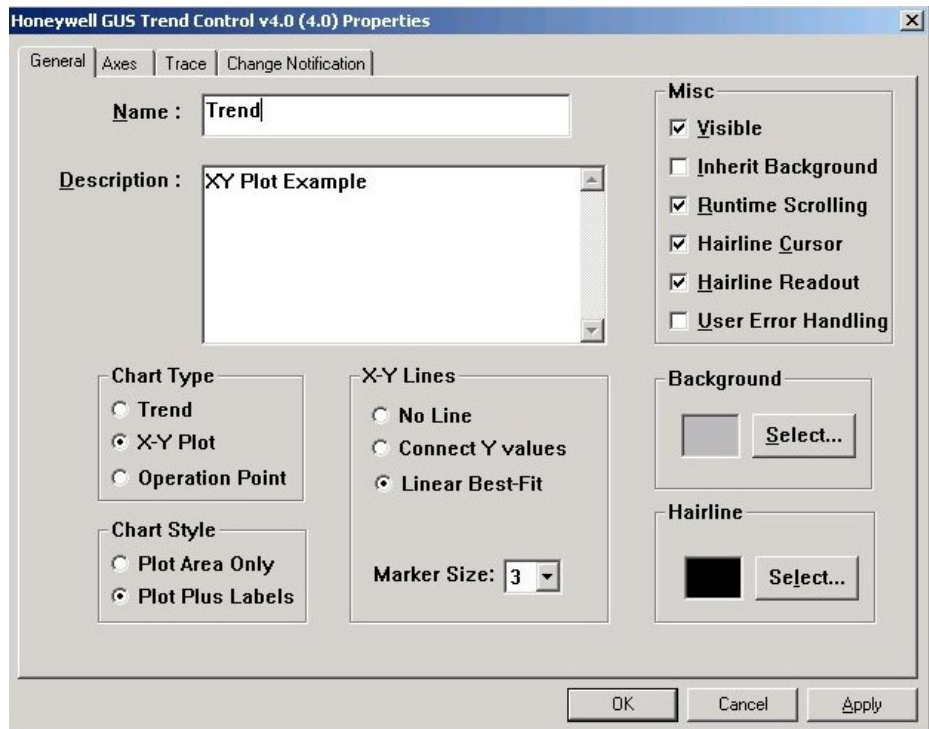


Setting general properties

The Property Sheet of the Honeywell Trend OLE Control has four pages: a General page, an Axes page, a Trace page, and a Change Notification page.

NOTE: No trend property supports parameterized input. For example, you cannot insert a display parameter into any trend property.

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the right-mouse button menu of the control). The OLE Control's Property Sheet is displayed. The General page is displayed by default (because it is the first tab).



- 2 Change the default name if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name. An entry is required in the Name field.
- 3 Enter a description if required. A description is not required.

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

- 4 Accept the miscellaneous options as defaulted or change each one as required. Refer to the table in [Property Definitions for Trend General Page](#).
- 5 Accept the default for the "Chart Type" option, Trend, or change it to either X-Y Plot or Operation Point if required.
- 6 Accept the default for the "Chart Style" option, Plot Area Only, or change it to Plot Plus Labels if required.
- 7 Accept the default for the "X-Y Lines" option, No Line, or change it to either Connect Y values or Linear Best-Fit.
- 8 Accept the default for the marker size or select a different size of the marking from the dropdown selection list.
- 9 Click Select in the Background group to display the color palette from which you can select a background color for the trend.
- 10 Click Select in the Hairline group to display the color palette from which you can select a color for the hairline cursor.
- 11 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Property Definitions for Trend General Page

Property Name	Definition	Syntax	Required Entry
Name	Object name that must be unique in the display. This is the name used in referencing the object in a script. Modified at buildtime, read-only at runtime. Default name is supplied by the Display Builder.	Refer to Display Object names in the <i>Display Scripting User's Guide</i> .	Yes
Description	User description of the Trend Control object. Available at both buildtime and runtime. There is no default.	None	No
Background Color	Color of the background of the trend. Available at both buildtime and runtime. Default is gray.	NA	NA
Show Grid	Determines whether a grid is displayed behind the traces. Available at both buildtime and runtime. Default is "Show Grid" true (check on).	NA	NA
Hairline Color	Color of the hairline cursor. Available at both buildtime and runtime. Default is black.	NA	NA
Grid Color	Color of the grid. Available at both buildtime and runtime. Default is yellow.	NA	NA
Visible	Determines the visibility of the object at buildtime and at display invocation. Available at both buildtime and runtime. Default is "Visible" true (check on).	NA	NA
Inherit Background (Inherit Properties)	Determines whether the object is going to inherit properties from the display's properties. The only property that the Trend Control inherits is background color if "Inherit Properties" is TRUE. Available at both buildtime and runtime. NOTE: The "Inherit Background" setting is accessible in scripting by the property "Inherit Properties." Default is "Inherit Background" false (no check).	NA	NA

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Property Name	Definition	Syntax	Required Entry
Runtime Scrolling	Determines whether runtime scrolling of the trend is enabled or disabled. When the RuntimeScrolling property is set to TRUE, the user can scroll backward and forward through the trace data using the ScrollBackTime and ScrollForwardTime properties. The RuntimeScrolling property is available at both buildtime and runtime. Default is "Runtime Scrolling" true (check on).	NA	NA
Hairline Cursor	Determines if the hairline cursor functionality is available (enabled). When the HairlineCursor property is set to TRUE, the user may turn the hairline cursor on and off at runtime by use of the mouse. If the HairlineCursor property is set to FALSE, the hairline cursor will not be activated at runtime when the user selects the left mouse button. Available at both buildtime and runtime. Default is "Hairline Cursor" true (check on). NOTE: The hairline cursor is not automatically displayed when this property is set. The user can only activate the hairline cursor by using the mouse.	NA	NA
Hairline Readout	Determines if the Y-values and the time at the hairline cursor are displayed on the Trend Control at runtime. These values are automatically displayed for the Plot Plus Labels trend style when the HairlineCursor property is set to TRUE. For the Plot Area Only trend style, the user must create text objects and scripts to display the properties equivalent to the hairline readout information. Available at both buildtime and runtime. Default is "Hairline Readout" true (Check on).	NA	NA
User Error Handling	When UserErrorHandling is set to TRUE, any error that is detected within trend will raise an event instead of displaying an error dialog box. The response to this event is determined by the user written OnError script for this object. Default is false (check off).	NA	NA

Chart Type

The three types of charts that can be selected at buildtime are:

1 Trend

This default chart type shows the trend during runtime. The X-axis shows time and the other traces show trend lines.

2 X-Y Plot

This chart type shows the xy-plot during runtime. You can see the xy-plots in Custom Graphic displays. The xy-plot allows you to plot up to 7 traces against one trace. The configured value of trace 1 is the X-axis trace and all other configured values of the trace (trace 2 through 8) are plotted against the X-axis trace.

3 Operation Point Plot

This chart type displays the operation point plot with the most recent value of each trace. The operation point plot operates similar to xy-plot. In xy-plot, configured value of trace 1 is the X-axis trace, whereas in operation point plot, the most recent value of each trace is plotted.

To display the required chart type you must select one of the options from the General page of the Trend Control Properties dialog box. This option is set at buildtime.

Chart style

There are two styles of trends from which the user may select at buildtime.

1 Plot Area Only

This chart style displays only the plot area in the Trend Control. None of the ancillary information (such as hairline cursor values or time of the last collected value) is displayed. For this chart style, the plot area is equal to the size of the Trend Control. When using the Plot Area Only trend style, it is the responsibility of the user, via display objects and scripting, to display the required ancillary information. Properties are available to the user to display all ancillary information.

The OnPropertyChange() event is needed by scripting to display accurate values for the ancillary information. The Plot Area Only chart style is the default chart style.

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

2 Plot Plus Labels

This chart style displays the plot area and automatically displays all the ancillary information associated with the trend. This information includes, but is not limited to, axes labels, legends, hairline cursor values and time when the last value was collected.

To display the Plot Plus Labels trend the user must select the Plot Plus Labels option from the General page of the Trend Control Properties dialog. This option can only be set at buildtime. There is no property available to change the trend style at runtime.

On the Trend Control Property Sheets, all Plot Plus Labels user-configurable options can also be configured for the Plot Area Only chart style with the exception of the HairlineCursorReadout property. This property is only valid when the Plot Plus Labels chart style is selected. The "Display Tic Percentages" property is applicable only when the Plot Plus Labels trend style is selected.

At runtime, with the Plot Plus Labels Trend Control, the user activates the hairline cursor by pressing the left mouse button inside the plot area. To disable the hairline cursor, the user presses the left mouse button outside the plot area, yet still inside the Trend Control. Since the Plot Area Only chart style does not contain this area for the user to disable the hairline cursor, a second mechanism is provided. For both chart styles, the user can select the right-mouse button to disable the hairline cursor.

X-Y Lines

Select one of the X-Y line type options from the General page of the Trend Control Properties dialog to connect points on the xy-plot. This option can be set only at buildtime.

The three types of X-Y lines are:

- No line - XY-plot is plotted without the connecting lines.
- Connect Y values - XY-plot is plotted with connecting lines from point to point.
- Linear Best-Fit -XY-plot is plotted with connecting lines by linear regression.

NOTE: The hairline cursor is available only if X-Y line is selected as Linear Best-Fit. 'No line' is the default X-Y trace line.

NOTE: This feature is not available for Trends and Operation Point Plots.

Setting axes properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the right-mouse button menu of the control). The OLE Control's Property Sheet is displayed.
- 2 Click the Axes tab. The Axes page is displayed.

The screenshot shows the 'Honeywell GUS Trend Control v4.0 (4.0) Properties' dialog box with the 'Axes' tab selected. The dialog has four tabs: 'General', 'Axes', 'Trace', and 'Change Notification'. The 'Axes' tab is active, showing settings for the 'Vertical Axis' and 'Horizontal Axis'. The 'Vertical Axis' section includes 'Y Scale High' (100 %), 'Y Scale Low' (0 %), '# of Tic Marks' (5), '# Minor Grid Lines' (1), and 'Show Tic Percents' (None). The 'Horizontal Axis' section includes 'XY Plot', 'X Scale High' (100 %), 'X Scale Low' (0 %), '# of Tic Marks' (4), and '# Minor Grid Lines' (2). Below these are 'Limit Lines' for the vertical axis with 'Upper Limit Line' (80 %) and 'Lower Limit Line' (-1 %), and a 'Color' selection (black). The 'Display Labels' section has a checked box for 'Relative/Percent (vs. Absolute/E.U.)'. The 'Axes' section has a checked box for 'Show Axes' and a 'Select...' button. The 'Grid' section has a checked box for 'Show Grid' and a 'Select...' button. The 'Time Base' is set to '20 minutes'. At the bottom are 'OK', 'Cancel', and 'Apply' buttons.

- 3 For the Vertical Axis, accept the defaults for Y Scale High and Y Scale Low or change one or both of them.
- 4 For the Number of Tic Marks on the Vertical Axis, accept the default or change it to the required number (valid range is 1 to 20).
- 5 For the Number of Minor Grid Lines on the Vertical Axis, accept the default or change it to the required number (valid range is 0 to 20).

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

- 6 For Show Tic Percents, accept the default or change it to the required option from the dropdown selection list. This is applicable only to the Plot Plus Labels trend style.
- 7 For the Horizontal Axis in xy-plot, accept the defaults for X Scale High and X Scale Low or change one or both of them.
- 8 For the Number of Tic Marks on the Horizontal Axis in xy-plot, accept the default or change it to the required number (valid range is 1 to 20).
- 9 For the Number of Minor Grid Lines on the Horizontal Axis in the xy-plot, accept the default or change it to the required number (valid range is 0 to 20).
- 10 Accept or change the option for the display labels of the trend ("Relative/Percent (vs. Absolute/E.U)").
- 11 For the Upper Limit Lines, accept the default or change it to draw the xy-plot.
- 12 For the Lower Limit Lines, accept the default or change it to draw the xy-plot. Note that the line is drawn only if the value is between "Y scale Low" and "Y scale High." If the lower limit is -1% then no lines are drawn.
- 13 Accept or change the option for showing the Limit Lines color ("Select") button.
- 14 Accept or change the option for showing the axes of the trend ("Show Axes").
- 15 Accept or change the option for showing the axes color ("Select") button.
- 16 Accept or change the option for showing the grid of the trend ("Show Grid").
- 17 Accept or change the option for showing the grid color ("Select") button.
- 18 Accept or change the time base or select a different time base of the trend or operation point plot from the dropdown selection list.

Property Definitions for Trend / XY-Plot Axes Page

Property Name	Definition	Syntax	Required Entry
Vertical Axis/ Y Scale High	Percent of full-scale high limit on Y-axis. The maximum is 100%. Available at both buildtime and runtime. Default is 100%.	Float	Yes
Vertical Axis/ Y Scale Low	Percent of full-scale low limit on Y-axis. The minimum is 0%. Available at both buildtime and runtime. Default is 0%.	Float	Yes

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Property Name	Definition	Syntax	Required Entry
Vertical Axis/ Number of Tic Marks	Determines the number of Y-axis tic marks to be displayed evenly across the Y scale range. Available at both build time and run time. The default value is 5 and the valid range is 1 to 20. NOTE: The top most tic mark specifies the high value of the Y-scale. The horizontal axis (the base) is not considered as a tic mark for the total number of tic marks.	Short	Yes
Vertical Axis/ Number of Minor Grid Lines	Determines the number of "minor" grid lines to be displayed between the tic marks. Available at both build time and run time. The default value is 1 and the valid range is 0 to 20.	Short	Yes
Vertical Axis/ Display Tic Percentages	Determines how the Y-axis (vertical) tic percentages are to be displayed. Available at both build time and runtime. The default is "None." The other valid entries are: None (do not display percentages) Both Sides (display on both vertical axes) Left Side (display on left axis) Right Side (display on right axis) NOTE: Fractional percentage values are rounded to the nearest 1/10%.	NA	Yes
Horizontal Axis – XY Plot / X Scale High	Percent of full-scale high limit on X-axis. The maximum is 100%. Available at both buildtime and runtime. The default is 100%. NOTE: This property has no effect on Trend plots.	Float	Yes
Horizontal Axis– XY Plot / X Scale Low	Percent of full-scale low limit on X-axis. The minimum is 0%. Available at both buildtime and runtime. The default is 0%. NOTE: This property has no effect on Trend plots.	Float	Yes

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Property Name	Definition	Syntax	Required Entry
Horizontal Axis– XY Plot / Number of Tic Marks	Determines the number of X-axis tic marks to be displayed evenly across the X scale range. Available at both buildtime and runtime. The default is 5 and the valid range is 1 to 20. NOTE: The extreme right tic mark specifies the high value of the X-scale. The vertical axis (the base) is not considered as a tic mark for the total number of tic marks. NOTE: This property has no effect on Trend plots.	Short	Yes
Horizontal Axis– XY Plot / Number of Minor Grid Lines	Determines the number of “minor” grid lines to be displayed between the tic marks. Available at both buildtime and runtime. The default is 1 and the valid range is 0 to 20. NOTE: This property has no effect on Trend plots.	Short	Yes
Horizontal Axis/ Display Relative Time (vs. Absolute Time)	Determines whether the time is relative or absolute. When displaying relative time, “time now” is the most current value and is the rightmost value on the X-axis. Available at both buildtime and runtime. Default is “Relative Time” true (check on). NOTE: This property only affects Trend charts.	NA	NA
Upper Limit Line	Determines the percentage of the full-scale high limit drawn on the horizontal axis (Y-axis). Available at both buildtime and runtime. The maximum limit is 100%. The default is –1.0% (no line).	Float	Yes
Lower Limit Line	Determines the percentage of the full-scale lower limit line drawn on the horizontal axis (Y-axis). Available at both buildtime and runtime. The minimum limit is -1.0%. Default is -1.0% (no line).	Float	Yes

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

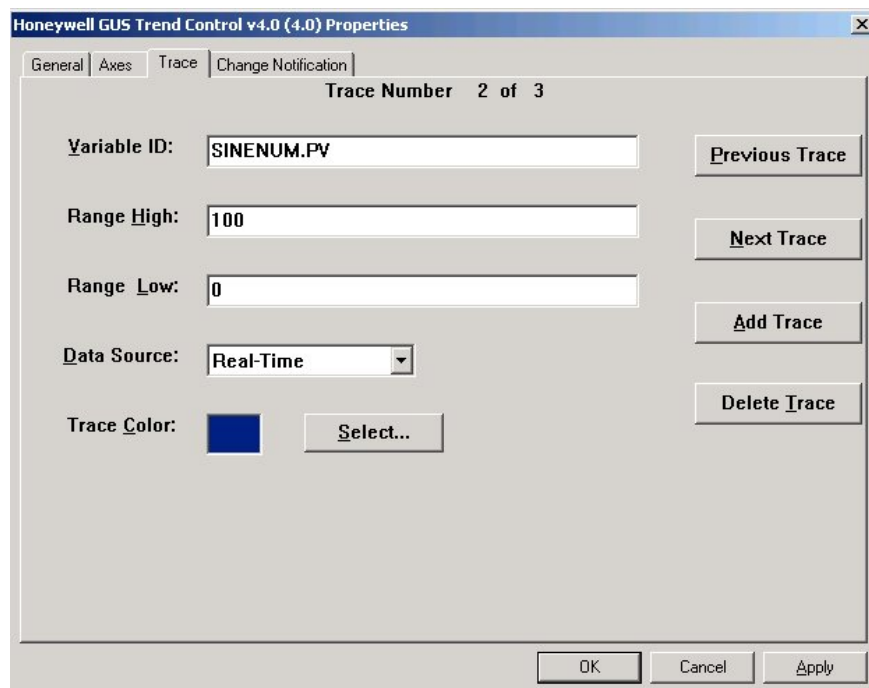
Property Name	Definition	Syntax	Required Entry
Limit Lines Color	Shows the color of the limit lines. Available at both buildtime and runtime. Default is Black.	NA	Yes
Axes Color	Shows the color of the X and Y-axes. Available at both buildtime and runtime. Default is Black.	NA	Yes
Show Axes	Determines whether the axes are displayed. Available at both buildtime and runtime. Default is "Show Axes" true (check on).	NA	NA
Grid Color	Shows the color of the grid. Available at both buildtime and runtime. Default is yellow.	NA	NA
Show Grid	Determines whether a grid is displayed behind the traces. Available at both buildtime and runtime. Default is "Show Grid" true (check on).	NA	NA
Horizontal Axis/ Time Base	Determines the Timebase used for the X-axis. Available at both buildtime and runtime. NOTE: This property only affects Trend charts.	NA	Yes

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Setting trace properties

- 1 Make sure the OLE Control to be modified is selected (has handles), then select Edit/Object Properties (or double-click the control or select Properties from the right-mouse button menu of the control). The OLE Control's Property Sheet is displayed.
- 2 Click the Trace tab. The Trace page is displayed.



Navigating from one trace to another

- 1 Click "Previous Trace" to move back one trace in the trace list.
- 2 Click "Next Trace" to move forward one trace in the trace list.

NOTE: The "Previous Trace" button will be disabled when you have reached the beginning of the trace list. The "Next Trace" button will be disabled when you have reached the end of the trace list.

Adding a Trace

- 1 On the trend trace properties page, navigate to the trace after which you want to add the new trace.
- 2 Click "Add Trace" to get into add mode. A new trace page is displayed. The number of the new trace is displayed, and the total number of traces is incremented and displayed.
- 3 Enter an LCN variable ID in the Variable ID text box. This should be the point.parameter for a numeric or enumeration variable. Note: For enumeration sets the values are scaled zero to 100 based on the number of possible states. For a 2-state flag, OFF plots as 0 and ON plots as 100; For a 4-member enumeration, the values are plotted as 0, 33.3, 66.7 and 100.
- 4 Accept the default for Range High and Range Low or change one or both of the values.
- 5 Select a data source from the dropdown list.
- 6 Click Select to display the color palette from which you can select a color for the trace.
- 7 Click Apply to save the new trace and continue.

Canceling an Add Trace

If you selected "Add Trace" and do not want to follow through with adding a trace, click "Delete Trace."

Deleting a Trace

- 1 Navigate to the trace that you want to delete.
- 2 Click "Delete Trace."

NOTE: A trace number uniquely defines a Variable ID and its selected properties. The trace number is bound to a given trace at runtime, therefore, if you delete a trace at runtime, none of the other trace numbers are affected. For example, deleting Trace #2 at runtime frees up this trace number for subsequent re-use. Then, assuming that Trace #1 is active, invoking an "AddTrace" will return the first available trace number, which is "2."

10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Property Definitions for Trend Trace Page

Property Name	Definition	Syntax	Required Entry
Variable ID	Point name (i.e., Variable ID) for trace NOTE: Each character in Point/Parameter Names must be alpha, numeric, !, \$, or underscore. Names cannot begin or end with underscore and cannot have two sequential underscores. Dot is not allowed. Read-only property at runtime.	LCN Variable ID Syntax The format is 1) Optional 2-char (char or digit) LCN ID plus backslash separator; 1) Required point name (16 char max.) plus dot separator; 1) Required parameter name (8 char max.). NOTE: Do not use the GUS Display Scripting syntax for variable specification.	Yes
Range High	High range on X and Y-axes for trace in engineering units. Maximum value is FLT_MAX. Available at both buildtime and runtime. Default is 100.	Float	Yes
Range Low	Low range on X and Y-axes for trace in engineering units. Minimum value is (-1)*FLT_MAX. Available at both buildtime and runtime. Default is 0.	Float	Yes

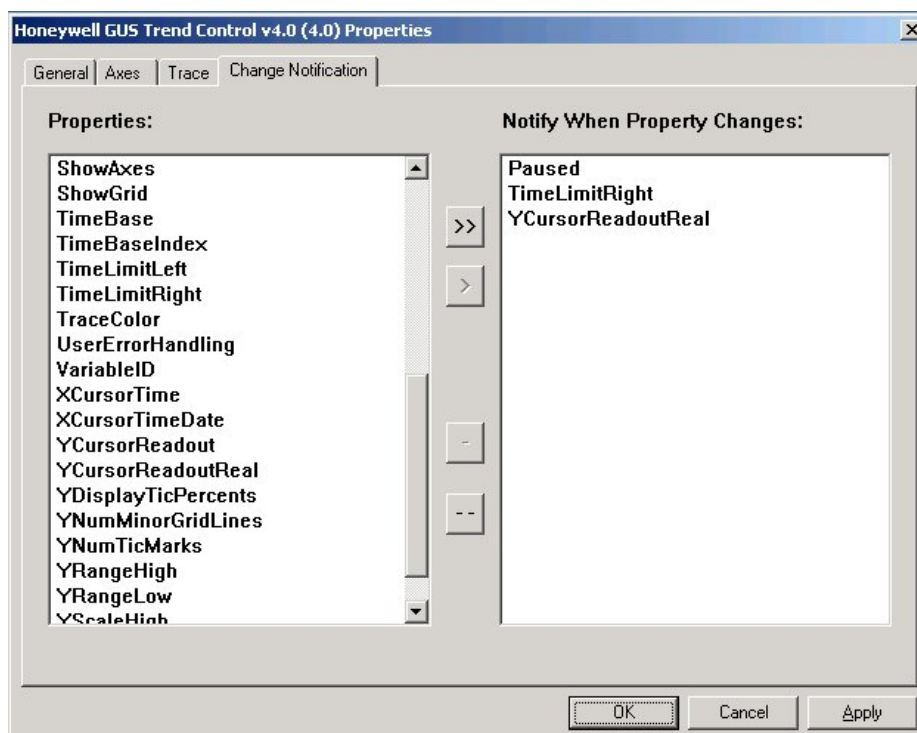
10. Properties of Honeywell OLE Controls

10.17. Trend OLE Control Properties

Property Name	Definition	Syntax	Required Entry
Data Source	Data source associated with each trace. Available at both buildtime and runtime.	NA	Yes
Trace Color	Color of the trace. Available at both buildtime and runtime.	NA	Yes

Change Notification properties page

The Change Notification build time properties page is provided to allow the user to select which properties will activate events. The default for all events is off (no activation of events). The user cannot change the notification events at runtime.



10. Properties of Honeywell OLE Controls

10.18. Versioning Honeywell OLE Controls

This property page is closely coupled with the `OnPropertyChange()` event. This event is described in detail in the *Display Scripting User's Guide* topic, **OnPropertyChange() Event**. The `OnPropertyChange()` event is executed each time the value of one of the selected properties on the Change Notification property page changes.

The left side of the display is the “Properties” listbox that contains a list of all the Trend Control runtime properties that will provide notification when the property value changes. The listbox on the right side of the display, titled “Notify When Property Changes,” contains a list of the properties the user has chosen to provide notification when the property value changes. A scrollbar will be activated in either listbox when more items are in the list than can be displayed at one time.

The user can request notification of all property changes by selecting the “>>” pushbutton, which places all the properties in the “Notify When Property Changes” listbox. Selecting “- -” removes all properties from the user-requested list. Selecting the “>” pushbutton when a property is selected in the “Properties” listbox causes the property to be added to the user-requested list. Selecting the “-” pushbutton deletes the selected property from the user-requested list.

The user may also select multiple items in either listbox by using the SHIFT key and the mouse to select a contiguous number of properties. Using the CTRL key and the mouse allows multiple selections of non-contiguous properties. Either of these selection methods can be used in coordination with the scroll bar, when activated.

Notification of all properties that are trace specific (for example `TraceColor(nTraceID)`) applies to all traces. The user cannot specify property change notification for a subset of active traces.

10.18 Versioning Honeywell OLE Controls

All Honeywell GUS OLE Controls support versioning. Each Honeywell GUS OLE Control is registered with a version number. The version number consists of an external number and an internal number. For example, the registered control “Honeywell GUS Trend Control v2.0 (2.3)” identifies “2.0” as the external number and “2.3” as the internal number.

Internal numbering is used to track additions and deletions of properties. The external version number is derived from the internal version number, and is used to identify significant updates in product functionality.

11. Embedded Displays

11.1 Overview

Topics covered in this section are:

- Making a Parameterized Display
- Defining Display Parameters (Display/Define Parameters and Help)
- Adding and Editing Display Parameters
- Saving Parameterized Displays
- Embedded Display Properties
- Accessing Embedded Display Properties
- Accessing Embedded Display Script
- Entering Display Parameters (Edit/Enter Parameters)



ATTENTION

Embedding a display with variables does not maintain any different collection set rates for references in the embedded display. Therefore, any data collection rates and groups that had been set for a .pct will NOT be included when that display is embedded in another display.

11.2 Making a Parameterized Display

A parameterized display has only one limitation: a display with an inline parameter that does not have a default value can neither be validated nor run.

Parameters

Parameter Definition—The creator of the parameterized display can define any number of parameters, using the Define Parameters Sheet shown in Defining Display Parameters (Display/Define Parameters and Help) (that follows on the next page).

11. Embedded Displays

11.2. Making a Parameterized Display

For each parameter, you can define

- parameter identifier (refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name).
- parameter type
- prompt (string used to aid the person who reuses this display)

Each parameter name in a given display must be unique within the scope of that display. Parameter names must not conflict with other properties of the display.

Parameter Types—Parameters can be any of the following types.

integer (value type)	date (value type)	object (value type)
long (value type)	currency (value type)	entity (reference type)
single (value type)	string (value type)	variable (reference type)
double (value type)	Boolean (value type)	Inline (inline type)

Referencing Parameters in the Display—You can reference parameters in Basic Dynamic expressions and scripts within the display, in place of a variable, or anywhere a variable reference is allowed. Display parameters are referenced as properties of the display, such as “display.params.param1.”

Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax.

Inline (substitution) Parameters in the Display—Inline parameters cause a substitution at compile time.

Authoring Rules for an Embedded Display

Embedding a display with variables does not maintain any different collection set rates for references in the embedded display. Therefore any data collection rates and groups that had been set for a .pct will NOT be included when that display is embedded in another display.

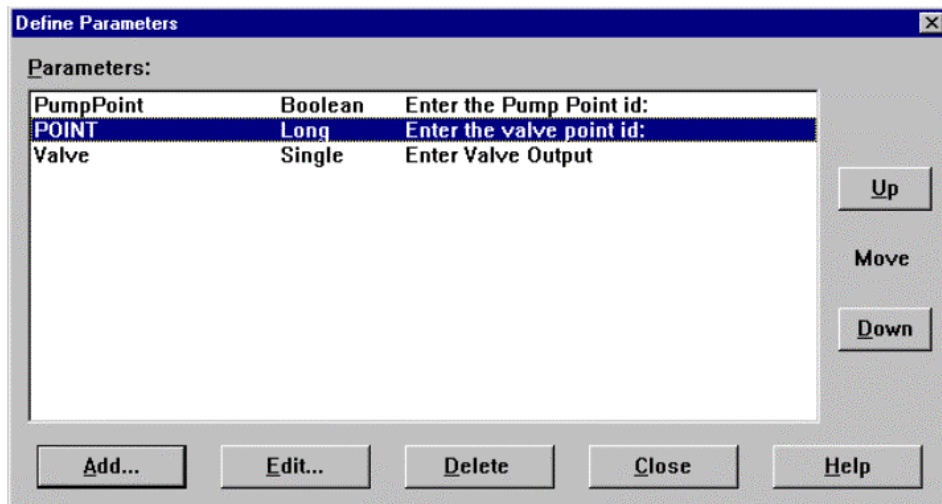
Don't reference display parameters in a script handling the "OnDisplayStartup" event.

For further information, refer to General Guidelines for Value Type Display Parameter Usage in the *Display Scripting User's Guide*, in the GUS Scripting Introduction section.

11.3 Defining Display Parameters

Display/Define Parameters and Help

Access the Define Parameters dialog box shown here by selecting Define Parameters from the Display menu.



The buttons at the bottom are used as follows:

Up and Down Buttons—Use these buttons to change the order of parameters in the list. Select a parameter, then click either button to move that parameter relative to others in the list.

Add Button—Use this button to open the Add Parameter dialog box. Refer to Adding Display Parameters.

Edit Button—Use this button to open the Edit Parameter dialog box. Refer to Editing Display Parameters.

Delete Button—Use this button to delete a parameter. Select the parameter and then click the Delete button.

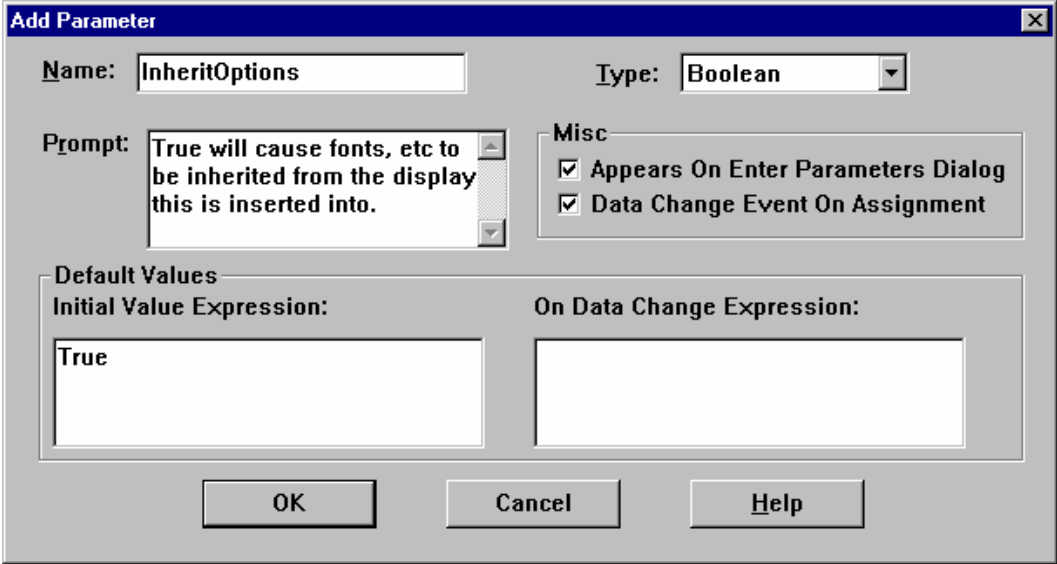
Close Button—Use this button to close the dialog box.

Help Button—This button selects Help information.

11.4 Adding Display Parameters

Add parameters by using the Add Parameter dialog box shown below.

You can access this dialog box by using the Add button  on the Define Parameters dialog box. Refer to Defining Display Parameters for a picture of this dialog box.



The image shows the 'Add Parameter' dialog box. It has a title bar with 'Add Parameter' and a close button. The dialog is divided into several sections. The 'Name' section has a text box containing 'InheritOptions'. The 'Type' section has a dropdown menu set to 'Boolean'. The 'Prompt' section has a text box containing 'True will cause fonts, etc to be inherited from the display this is inserted into.' The 'Misc' section has two checked checkboxes: 'Appears On Enter Parameters Dialog' and 'Data Change Event On Assignment'. The 'Default Values' section has two text boxes: 'Initial Value Expression' containing 'True' and 'On Data Change Expression' which is empty. At the bottom are three buttons: 'OK', 'Cancel', and 'Help'.

“Name” edit box—Type the required parameter name into this box. The maximum size of the parameter name is 24 characters. The parameter name follows GUS R220 embedded display parameter name syntax rules. Parameter names must be unique across the entire display.

“Type” dropdown list—The list displays the parameter types. The default parameter is Long.

“Prompt” edit box—Type the required prompt into this box. There is no limit to the size of the prompt. The edit box is multiline with word-wrap. A prompt is optional.

“Misc”

“Appears On Enter Parameters Dialog” checkbox—If checked, this parameter will appear on the “Enter Parameters” dialog box. If not checked, it won’t. The default is “checked.” The intent is to allow the user to prevent the “Enter Parameters” dialog box from being cluttered with parameters that are really only being used to hold global data within the embedded display, and are not intended for exposure to the containing picture. This check box is grayed out when the type of the parameter is Variable, Entity or Inline.

“Data Change Event On Assignment” checkbox—If checked, an assignment to this parameter from within a script will cause dependent scripts to be executed. Dependent scripts are “OnChange” subroutines that contain a reference to this display parameter. If not checked, this parameter does not cause data change events when it is modified. The default is “checked.” When this item is not checked, parameters are treated as simple data items. It may also be used to eliminate race conditions when data change behavior is not needed. This check box is grayed out when the type of the parameter is Variable, Entity or Inline.

“Default Values”

“Initial Value Expression” edit box— This is the expression that determines the initial value of this parameter when the display containing this parameter is invoked.

“On Data Change Expression” edit box— This is the default expression that causes an OnDataChange event to be triggered for this parameter.

“OK” button—The “OK” button performs the following validations.

- There must be a parameter entered. The parameter is unique across the entire display. The parameters follow GUS R210 embedded display parameter name syntax rules.
- A type is selected. (This happens by nature of a dropdown list; no checking is required.)

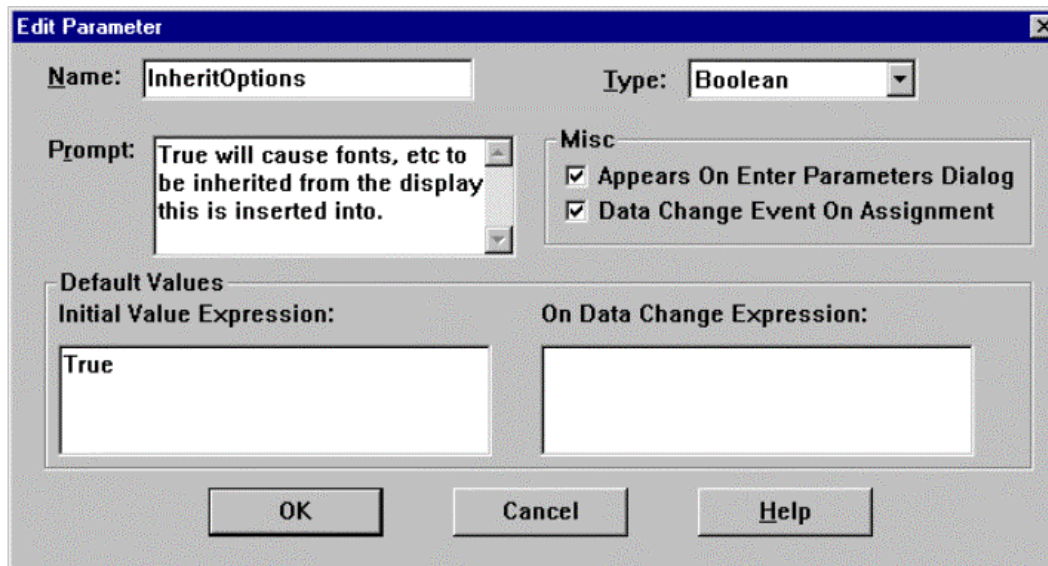
RESULT—The dialog box is closed. The parameter and its properties are saved in the internal definition of the display. The parameter is appended to the Parameters list of the “Define Parameters” sheet. **NOTE:** The addition of the parameter cannot be undone.

“Cancel” button—The “Cancel” button cancels the add operation and then closes this dialog box.

“Help” button—The “Help” button displays help on the dialog box.

11.5 Editing Display Parameters

You can edit existing parameters by selecting the “Edit...” button on the Define Parameters dialog box, which invokes the Edit Parameter dialog box.



When the “Edit Parameter” dialog box is invoked, the controls contain the values of the parameter being edited.

The “edited” parameter definition replaces the definition that existed internally and in the Parameters list of the “Define Parameters” dialog box.

The contents of the Add Parameter and Edit Parameter dialog boxes are identical. Refer to Adding Display Parameters for a description of the fields in this display.

The Default values include both the Initial Value Expression and the On Data Change Expression. The Initial Value Expression is an expression that determines the initial value of the parameter when the display is invoked. The On Data Change Expression is an expression that causes an OnDataChange event to be triggered for the parameter.

Refer to Adding Display Parameters for a definition of the Appears on Enter Parameters Dialog and the Data Change Event on Assignment check boxes that appear in the Misc. section.

11.6 Saving Parameterized Displays

Embedded displays are saved as display files, using the same “save” dialog as for displays.

Because they are displays, they appear in the file system as display files, with no characteristics, distinguishing them as embedded displays. The file extension is defaulted to “.pct,” but the user can change the file extension to indicate that it is intended to be used as an embedded display.

If the file extension is not “.pct,” it cannot be executed at runtime; however, it can be executed for testing within the Display Builder.

Saving unchanged parameterized display changes its revision

A display can have only one version of a parameterized display inserted in it because Display Builder uses the Save Date and Time of the parameterized display to determine its “version.”

For example, you perform these steps:

- 1** You have a display EXAMPLE1.PCT that has two instances of the parameterized display, PARDISP.PCT, inserted in it.
- 2** You inadvertently save PARDISP.PCT when no changes were made to it.

RESULT: You cannot add a third instance of PARDISP.PCT to display EXAMPLE1.PCT.

REASON: The Display Builder thinks PARDISP.PCT is a newer version because the Save Date and Time have changed.

11. Embedded Displays

11.7. Replacing an Embedded Display

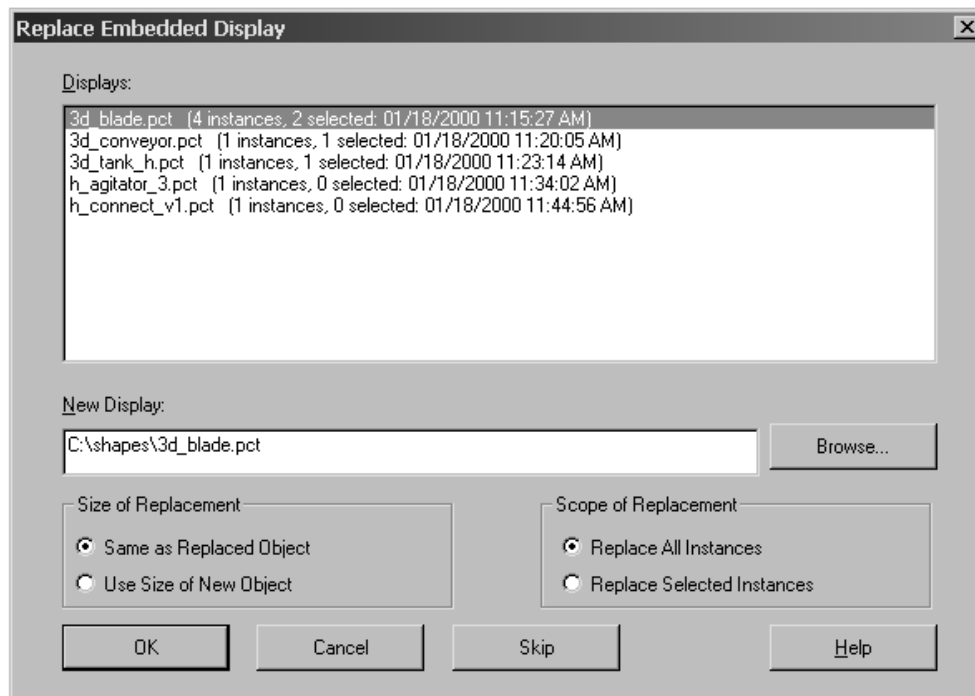
11.7 Replacing an Embedded Display

Edit/Replace Embedded menu and Help

Prior to R640, you can replace an embedded display with a new version by using the “Replace Embedded Display” command from the “Edit” menu. With R640, you can also replace selected instances of an embedded display in a GUS display with another embedded display by using the “Replace Embedded Display” command from the “Edit” menu.

Follow these steps:

- 1 If you want to replace selected instances, select the instances of the embedded displays you wish to replace.
- 2 From the “Edit” menu, select “Replace Embedded Display”. A dialog box appears identifying the display to be replaced.



- 3 Select the display to be replaced.

- 4 Type in the name of the new display or find the file by using the Browse button.
- 5 Select the required "Size of Replacement" option. "Same as Replaced Object" retains the original size of the object. "Use Size of New Object" uses the size of the new object.
- 6 Select the required "Scope of Replacement" option.
- 7 Replace All Instances: Replaces all the instances of the embedded display.
- 8 Replace Selected Instances: The selected instances of the embedded display selected from the list box are replaced.

NOTE: The replacement is executed with the following rules:

If all the instances of the embedded display are selected, all the instances will be replaced.

If there are no instances of the embedded display selected, all the instances will be replaced.

If a subset of the instances is selected, the name of the new embedded display must be unique to the display. The new embedded display is read from the media and the selected instances are replaced.

TIP

To replace selected instances when the objects are grouped, the user must first ungroup the group object containing the embedded display. Then, the user can select and replace the embedded display instance using the replace selected instances functionality. Finally, the user can then group the new embedded display instance and the other group objects back into a group object.

NOTE: If a group object containing an embedded display is selected, and the user executes the replace selected instances functionality for the grouped embedded display, the embedded display in the group object is ignored. It is not replaced with the new embedded display and no error message is displayed.

- 9 Click "OK" to execute or "Cancel" to cancel the operation.

Determining the Default Type

When you replace an embedded display, parameters can either assume the default values of the new embedded display or maintain their original values.

Original parameter values will be maintained only if

- (a) parameter name and type are the same in both versions, or
- (b) parameter name is the same in both versions and the parameter type is “inline” in the new version.

If neither (a) nor (b) is true, the default values of the new display will be used.

11.8 Bulk Replacement Tool

The Bulk Replacement Tool was introduced in GUS R330. The tool allows you to replace all the instances of an embedded display in multiple GUS displays with a different embedded display, without having to manually read the displays into the GUS Display Builder.

With GUS R350, the Bulk Replacement Tool has been enhanced to allow you to view and change all instances of display parameter values in the embedded displays in multiple GUS displays. This can be done without having to manually read the displays into the GUS Display Builder

With the Bulk Replacement Tool you can:

- Specify one or more GUS displays to search and provide the directory for the modified displays to be written.
- Replace embedded displays. The replacement is executed with the following settings:
 - Specify the current embedded display names (from) and the new name (to) to replace in each of the GUS displays.
 - Specify the dimensional size option of each new embedded display using its original size, or resize the new embedded display to the size of the existing one.
- Replace display parameters.
 - Create a list of display parameters of the top-level embedded displays with current values, for the selected GUS displays.
 - Save the display parameters list in a CSV file, modify the file, and read the updated file back into the display parameters list.

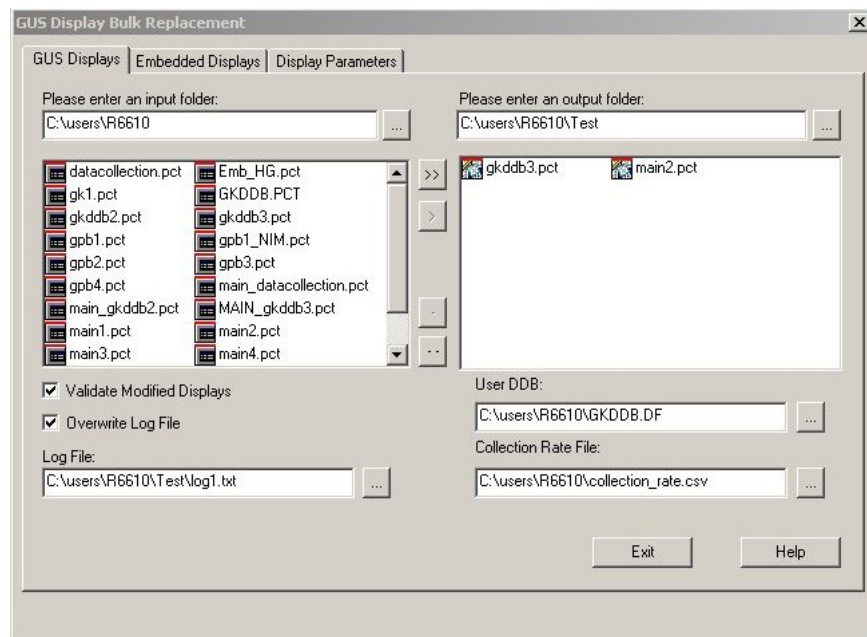
- Specify the current display parameter values and the new values that you want to replace in all instances, in each of the top-level embedded displays in the selected GUS displays.
- Validate the modified GUS displays after the replacements have been made. The DDB file (.df) and the Parameter Collection Rate file (.csv) can be specified for validation. This option is disabled by default.
- Specify a log file name to save the results of each operation.

This can be done without reading the displays manually into the GUS Display Builder. To invoke the tool from the Start menu, select Start > Programs > Honeywell TPS > GUS Display Bulk Replacement.

Adding GUS Displays to the Bulk Replacement Tool List

To add GUS Displays to the Bulk Replacement tool list, follow these steps:

- 1 In the “GUS Display Bulk Replacement” dialog box, select the GUS Displays tab.



- 2 Enter the input and output folder either by typing the directory names in the edit box or click [...] and browse to select the required directory.

11. Embedded Displays

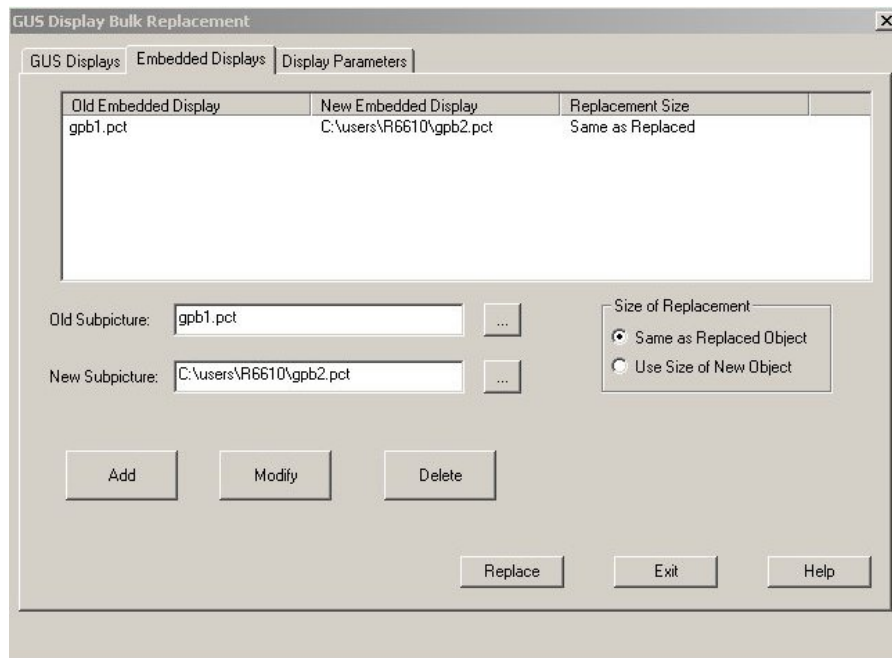
11.8. Bulk Replacement Tool

- 3 Select the files to be replaced and click ">" button. To select all the files to be replaced click ">>" button. The selected files appear in the output folder.
- 4 To remove a file from the output folder so that the file is not replaced, select the file and click the "-" button. To remove all the files click the "- -" button. The selected files are removed from the output folder.
- 5 If required, check the "Validate Modified Displays" box to validate the modified GUS displays when the replace operation is finished. Checking this option enables the "User DDB" and the "Collection Rate File".
- 6 In the edit box of the "User DDB" and the "Collection Rate File", type the file name or click [...] and browse to select the required files.
- 7 If required, specify the name of a Log File where the results are stored. Type the file name in the "Log File" edit box or click [...] and browse to select the required file. If a file name is not specified, the log file name is defaulted to replacelog.txt in the output folder.
- 8 Select the "Overwrite Log File" option to overwrite the existing log file or leave the box unchecked to append the results.
- 9 Click "Help" to invoke help information.
- 10 Click "Exit" to close the Bulk Replacement tool without executing the tool.

Adding and Replacing Embedded Displays using the Bulk Replacement Tool List

To add and replace embedded displays using the Bulk Replacement tool, follow these steps:

- 1 In the “GUS Display Bulk Replacement” dialog box, select the “Embedded Displays” tab.



- 2 Type the name of the current Embedded Displays in the “Old Subpicture” edit box or click [...] to browse and select the required displays.
- 3 Type the name of the New Embedded Display in the “New Subpicture” edit box or click [...] to browse and select the required displays.

NOTE: Circular or interwoven replacement of embedded displays is not allowed. If a Circular replacement is found in the list box, an error message is displayed with the name of the encountered Circular embedded display.

Circular replacement involves replacing one embedded display with a second, and further in the list box, replacing the second embedded display with a third.

Example: Replace A with B, and then later, replace B with C.

11. Embedded Displays

11.8. Bulk Replacement Tool

- 4 Select "Size of Replacement" option to define the required dimension size for the new embedded display.

NOTE: The option "Same as Replaced Object" resizes the embedded display to the size of the existing display. The option "Use Size of New Object" indicates to use the size of the new display.
- 5 Click the "Add" button to add the old and new embedded displays, and replacement size into the embedded displays list box.

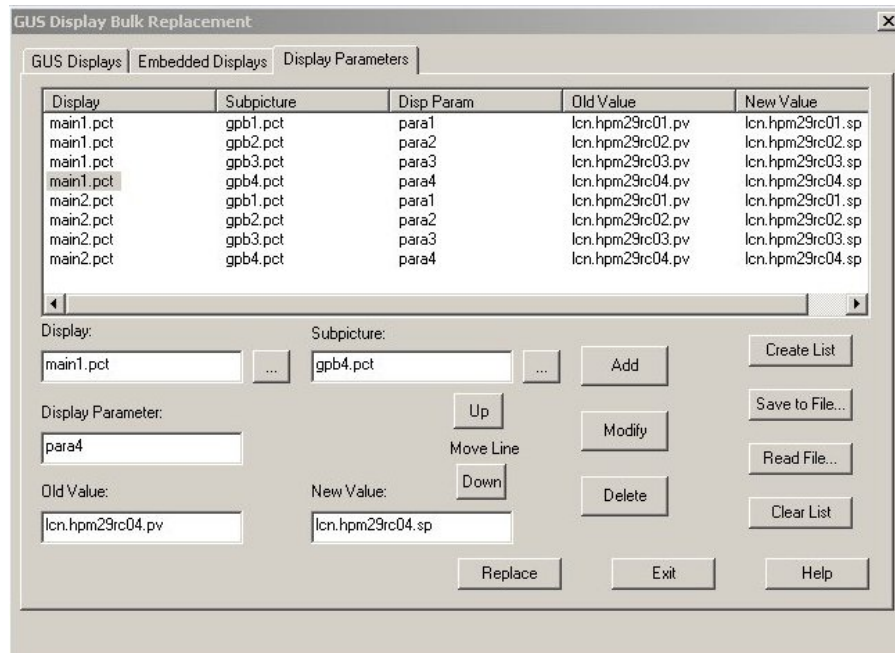
NOTE: The "Add" button is enabled when the user has specified both an old and new embedded display name.
- 6 To modify a relationship from the Embedded Display list box, select the old embedded display name in the list box. The information from the selected line is copied to the input controls. Make the changes in the input controls and click the "Modify" button.
- 7 Once modified, if required, the user can add the embedded display name. To add the embedded display name, click the "Add" button. If the embedded display names in the edit boxes are identical to the corresponding names in the original selected line, then the original selected line in the embedded displays list box is modified. If the names are not identical, a new line is added to the end of the list box with the modified controls. Thus, allowing the relationships with similar embedded display names to be added with minimal data entry.

Example: Replace valve1.pct with C:\subs\valve1.pct, then replace valve2.pct with C:\subs\valve2.pct, and so on.
- 8 To delete a relationship, select the relationship to be removed from the "Embedded Displays" list box and click the "Delete" button.
- 9 To execute the Bulk replacement, click the "Replace" button.
- 10 Click "Help" to invoke help information.
- 11 Click "Exit" to close the Bulk Replacement tool without executing the tool.
- 12 To cancel the execution process, click "Cancel" that appears in the "Replacing Embedded Displays" dialog box. The Bulk replacement tool will abort after completing the display that is being processed.

Viewing and Replacing Display Parameter Values using the Bulk Replacement Tool List

To view and replace the display parameters using the Bulk Replacement tool, follow these steps:

- 1 In the “GUS Display Bulk Replacement” dialog box, select the “Display Parameters” tab.



- 2 Type the name of the required GUS Display in the “Display” edit box or click [...] to browse and select the required displays.
- 3 Type the name of the required Embedded Display in the “Subpicture” edit box or click [...] to browse and select the required displays.
- 4 Type the name of the required Display Parameter in the “Display Parameter” edit box.

11. Embedded Displays

11.8. Bulk Replacement Tool

- 5 Type the required value of the Display parameter in the “Old Value” edit box and type the new value (value to be replaced) in the “New Value” edit box. Note that the old parameter value refers to both the Initial value field (that is used for value parameters, inline parameters, and reference parameters), and the On data change expression field.

NOTE 1: You can also use wildcard characters in the edit boxes. The regular expression pattern characters are similar to those in SafeView. Refer to the SafeView Users' Guide “Regular Expression Pattern Syntax” section for detailed information about wildcard characters.

Typical wildcard characters are as follows.

“?” – match any single character except new line.

“*” – Closure character (match 0 or more occurrences of the previous pattern or character)

In the old value field, the only wildcard characters allowed are “?” and “*”. In the new value field, the only wildcard character allowed is “*”. The wildcard character “*” indicates to substitute the same characters found in the old display parameter values field. The number of wildcard characters in the old and new display parameter value fields should be the same. A “?” combination in the old display parameter value field counts as 1 wildcard for the number count (implies 0 or more characters).

Example: Replace old value ‘nmalm34’ with ‘nmalm43’. The new value is changed to ‘nmalm43’. To replace all instances of old parameter values starting with ‘nmalm’ with ‘amalm’ use the wildcard character and enter the old value as ‘nmalm?’ and the new value as ‘amalm*’. Click “Replace”. The old value ‘nmalm43’ is replaced with ‘amalm43.’

NOTE 2: If the relationship between any of the old and new display parameter values in the list box results either in chain* or cyclic** replacements, the replacement is executed sequentially up to the last line. The last line of the cycle or chain represents the final replacement result.

* Example for Chain replacement: Line 1 – X replaced with Y; line 3 – Y replaced with Z.

** Example for Cyclic replacement: Line 1 – X replaced with Y; line 3 – Y replaced with Z; line 5 – Z replaced with X.

- 6 Click “Up” or “Down” buttons to change the order of parameters in the list. Select a parameter and click either of the buttons to move the selected parameter up or down. These buttons are enabled only when you select a line in the list box.

NOTE: If the top line of the list is selected, or if a line is moved to the top of the list, the “Up” button is disabled. Similarly, if the bottom line of the list is selected, or if a line is moved to the bottom of the list, the “Down” button is disabled.

- 7** Click “Add”, “Modify”, or “Delete” button to alter the display parameter list.

To add the display information from the input controls into the display parameters list box, click the “Add” button.

NOTE: The “Add” button is enabled when you enter data in all the input controls edit boxes.

To modify a relationship from the Display Parameters list box, select the subpicture name in the list box. The information from the selected line is copied to the input controls. Make the changes in the input controls and click the “Modify” button.

Once modified, if required, the user can add the display parameter relationship. To add the modified relationship, click the “Add” button. This allows the relationships with similar names to be added with minimal data entry.

Example: Replace parameter TankValve1 old value A100.PV with B100.PV, then replace parameter TankValve2 old value A100.SP with B100.SP, and so on.

To delete a relationship, select that relationship from the “Display Parameter” list box and click “Delete”.

NOTE: After “Add”, “Modify”, or “Delete” operations, the edit boxes are cleared and the respective buttons are disabled in the display.

- 8** Click “Create List” to populate the display parameter list box with all the display relationships found in the selected GUS displays. Because it deletes the existing display relationships listed in the display parameter list box, a message box appears requesting you to confirm the delete operation. Click “Yes” to delete the existing display parameter list and create a new list, or click “No” to cancel the command.

NOTE: The “Create List” button is enabled only when you select at least one GUS display from the GUS displays page.

- 9** Click “Clear List” to delete all the display relationships. Since it deletes the existing display relationships listed in the display parameter list box, a message box appears to confirm the operation. Click “Yes” to delete the existing display parameter list or click “No” to cancel the command.

- 10** Click “Save to File” to save display relationships listed in the display parameters list box to a .csv file.

NOTE: If the file exists, the user is prompted to overwrite the file. If the user clicks OK to the overwrite prompt, the current file information is overwritten. The .csv files can be modified using Excel, Word, and Note Pad. Any field that is entered after the “New Value” is considered to be comments.

11. Embedded Displays

11.8. Bulk Replacement Tool

- 11** Click "Read File" to read the .csv file into the Bulk Replacement Tool to populate the display parameters list box. Because it deletes the existing display relationships in the display parameter list box, a message box appears requesting you to confirm the delete operation. Click "Yes" to delete the existing display parameter list and to read the file, or click "No" to cancel the command.

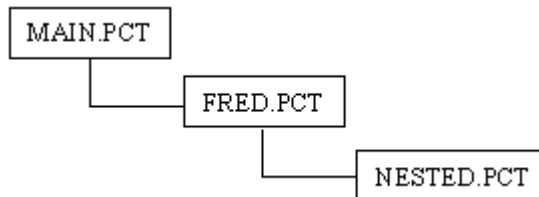
NOTE: Any errors encountered during the read are displayed identifying the error lines. Lines with errors are not read into the display parameter list.

- 12** Click "Replace" to execute bulk replacement. The old display parameter value is replaced with the new display parameter value for all instances of the specified embedded display in the selected GUS display.

GUS display is validated against the DDB and the Collection Rate file, if the appropriate option is selected in the GUS displays page. Select at least one GUS display and one display parameter relationship for processing.

NOTE: When using the Bulk Replacement tool it is not possible to modify parameter values of nested embedded displays. To search and replace parameter values of nested embedded display, refer to the following example.

The top-level GUS display, MAIN.PCT contains embedded display, FRED.PCT; FRED.PCT contains embedded display, NESTED.PCT.



To search for instances of an embedded display in NESTED.PCT, select FRED.PCT as a top-level GUS display. If you select MAIN.PCT as GUS display, the tool finds and replaces the parameters for FRED.PCT; but the parameters for NESTED.PCT are not modified. To modify the parameters for NESTED.PCT, FRED.PCT has to be selected as a top-level GUS display.

- 13** Click "Exit" to exit the Bulk Replacement tool.
- 14** Click "Help" to invoke online help information.

- 15** To cancel the execution process, click "Cancel" that appears in the "Replacing Display Parameters" dialog box. When selected, the Bulk Replacement tool replaces all display parameter requests in the current file. If the validate option is selected, the current file is validated. After the validation is complete, the session is aborted with a message indicating that the operator has canceled the operation and the same is logged in the log file.
- 16** If both the embedded displays and the display parameter values are to be replaced, do the following:
- a) Go to the embedded displays page. Perform the replace embedded display procedure.
 - b) Go to the display parameter page. Click "Create List" to load the list of display parameters in the page. Select and modify the display parameters and click "Replace" to replace the display parameters.

11. Embedded Displays

11.9. Rebinding Embedded Display Parameters

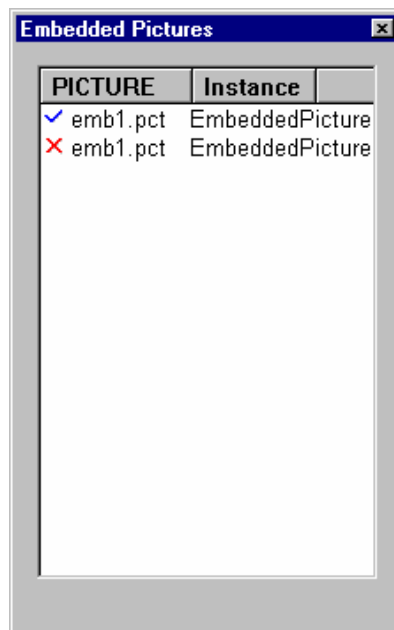
11.9 Rebinding Embedded Display Parameters

For each embedded display that has been replaced, any previously defined parameter is automatically rebound if the parameter names and types in the new version are the same as in the previous version.

If a parameter name or type has changed, or if a new parameter has been added, those parameter entries are shown invalidated. You will be required to invoke the “Enter Parameter” command and enter the values of the new or changed parameters.

Use this procedure:

- 1 From the View menu, select Embedded Displays. The Embedded Pictures ListBox is displayed.



Note that in this example, display **emb1.pct** is listed twice.

- a) The first listing has a blue check mark in front of it, indicating that display has been validated.
- b) The second listing has a red “x” in front of it, indicating it has NOT been validated.

- 2 Notice the Instance name of the unvalidated embedded display. In this example, it is named "EmbeddedPicture" You can use the left-right scroll bar to read the full name.
- 3 Now, double-click the filename that isn't validated (has a red "x" beside it).

RESULT: The Enter Parameters dialog box opens.
- 4 Enter the expressions for the changed parameters as directed in Entering Display Parameters (Edit/Enter Parameters).

After the parameters are defined, they will be shown as Validated.

11.10 Embedded Display Properties

The properties of an embedded display include the display properties that are applicable when a display is embedded, plus properties that are applicable because the embedded display is an object in a main display.

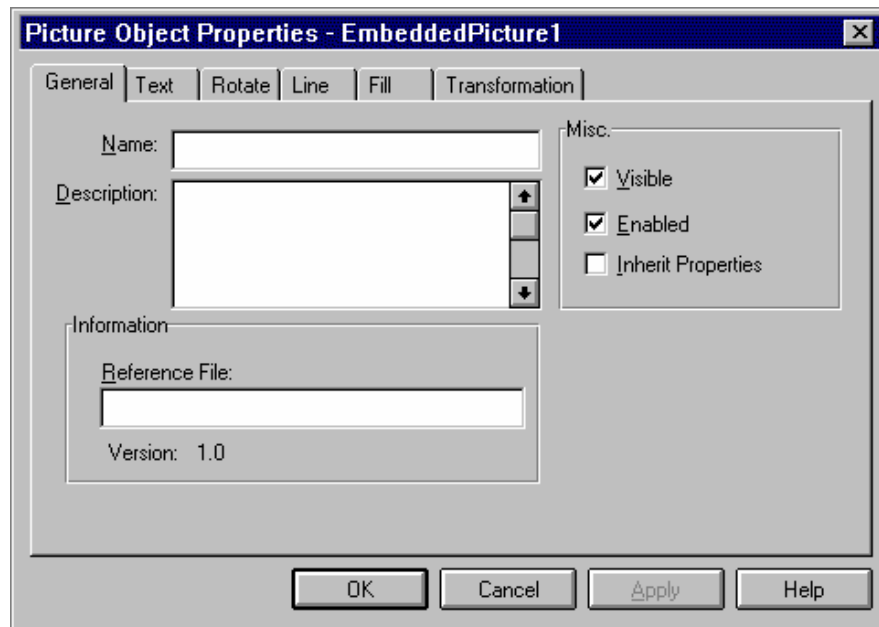
These properties are accessible from Property Sheets at buildtime and from scripts at runtime.

Setting General Properties

- 1 Make sure the embedded display to be modified is selected (has handles), then select Edit/Object Properties (or double-click the object or select Properties from the right-mouse button menu of the object). The embedded display's Display Object Property sheet is displayed. If the General Page is not already displayed, click the General page tab.

11. Embedded Displays

11.10. Embedded Display Properties



- 2 Change the default name if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name.

NOTE: An entry is required in the Name field.
- 3 Accept the description inherited from the reference file or edit the description if required.
- 4 Accept the miscellaneous options as defaulted or change each one as required.

NOTE: The reference file pathname is informational only and cannot be changed.
- 5 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

Property Definitions for properties found on Embedded Display General Properties Page

Property Name	Definition	Syntax	Required Entry
Name	Object name that must be unique in the display. This is the name used in referencing the object in a script. Default name is supplied by the Display Builder.	Refer to Display Object names in the <i>Display Scripting User's Guide</i> .	Yes
Description	User description. The initial description is inherited from the reference file.	None	No
Visible	Determines the visibility of the embedded display at buildtime and at display invocation. Default is "Visible" true (check on).	NA	NA
Enabled	Determines whether the embedded display is selectable at runtime. Default is "Enabled" true (check on).	NA	NA
Inherit Properties	Determines whether the embedded display inherits property values from its container. Default is "Inherit Properties" false (check off).	NA	NA
Reference File	The pathname of the reference file for the embedded display. NOTE: This is read-only.	NA	NA

11. Embedded Displays

11.11. Accessing Embedded Display Properties

Embedded display—property pages

- Embedded Display Properties
- Text Properties (Text Page)
- Rotate Properties (Rotate Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)
- Transformation Properties (Transformation Page)

11.11 Accessing Embedded Display Properties

There are several ways to access the Property Sheet of an Embedded Display:

- from the Menu Bar (or by a keystroke)
- by Using the Right-mouse Button
- by double-clicking the Embedded Display (default action).

These procedures are the same as for Graphic Objects, and are repeated below.

Accessing embedded display properties from the Menu

- 1 If the embedded display is not active, select it by clicking once on it (handles show).
- 2 Select Edit/Object Properties (not Edit/Display Properties) from the menu or press the keystroke Alt + Enter.

RESULT: The Properties Sheet for that embedded display appears.

Accessing embedded display properties from right-mouse button menu

- 1 With the mouse pointer within the required embedded display (it doesn't need to be selected), click and hold the right-mouse button. A drop-down menu appears.



- 2 Move to the Properties item and release the mouse button.

RESULT: The Properties Sheet for that embedded display appears.

Accessing embedded display properties as the default action

With the mouse pointer within the required embedded display (it doesn't need to be selected), double-click the mouse button.

RESULT: The Properties Sheet for that embedded display appears.

11.12 Accessing Embedded Display Script

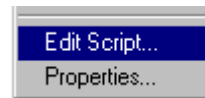
You access embedded display scripts in the same way you access Graphic Object scripts:

- 1 If the embedded display is not active, select it by clicking once on it (handles show).
- 2 Select Edit/Object Script from the menu (Don't select Edit/Display Script).

RESULT: The Script ListBox for that embedded display appears.

Accessing an embedded display script using the right-mouse button

- 1 With the mouse pointer within the embedded display (it doesn't need to be selected), click and hold the right-mouse button. A drop-down menu appears.



- 2 Move to the Edit Script item and release the mouse button.

RESULT: The Script ListBox for that embedded display appears.

11.13 Entering Display Parameters

Binding formal parameters to actual parameters

When a parameterized display is added to a display (by using the Insert/Display function or by copying it from another display), you may then invoke the embedded display's Enter Parameters dialog box so parameters can be entered.

You can either enter the parameters or defer their specification. The parameter values can be TPS GUS variables, DDB variables, constants, display parameter IDs, and any expression containing any combination of these.

Deferring parameter binding

You can defer entering any or all of the parameterized display's parameters. If any parameter is not entered, the parameterized display will appear as invalidated in the containing display, and an error will be reported when the containing display is validated.

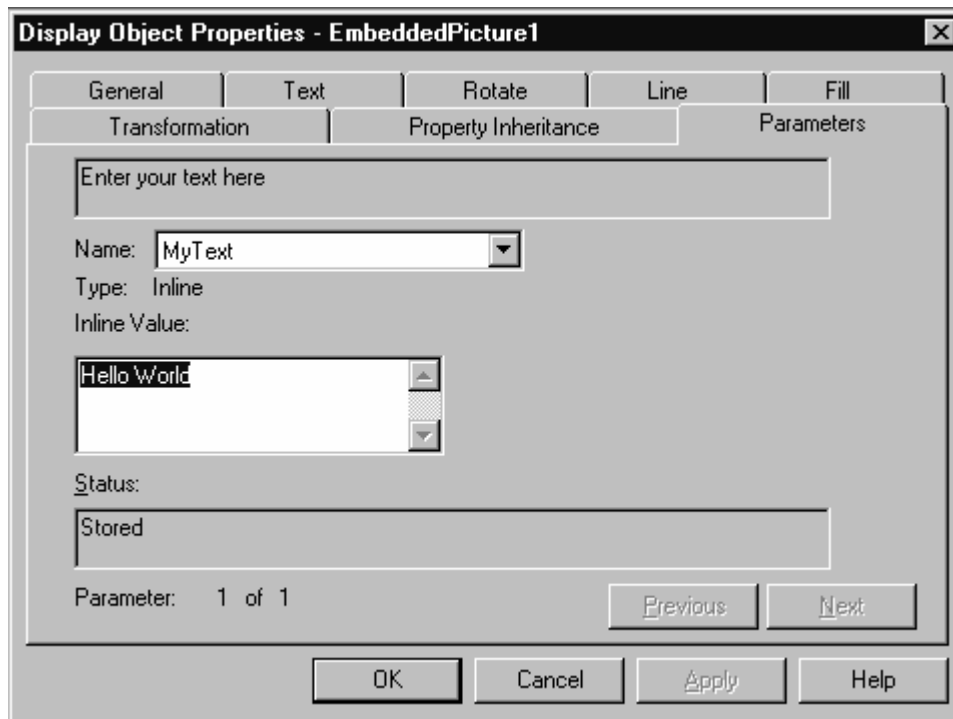
11. Embedded Displays

11.13. Entering Display Parameters

Ways to access parameters

Access the Enter Parameters dialog box in the following ways.

- Automatically opened by the application when an embedded display is initially inserted in a display (only if the “Auto Param. Prompts” option has been checked on the display’s General page).
- From the Edit/Enter Parameters command on the menu.



Description and rules of information

Follow these guidelines when putting data in the Enter Parameters dialog box.

Prompt (read-only)—the user's description of the parameter

Name (read-only)—the name of the parameter. This name is also used in scripting.

Type (read-only)—The parameter's Type

Initial Value expression/**On Data Change** expression

- Enter data in ONLY ONE of these text boxes.
- Enter data in the *Initial Value Expression* box if it contains only literals and constants. Remember, if you are entering a string literal, be sure it is enclosed in quotes.
- Enter data in the *On Data Change Expression* box if it contains one or more variables or references to display parameters. If the expression entered contains LCN and/or DDB variables that are known to be static, such as configuration data, still enter it in the On Data Change dialog box. For further information, refer to General Guidelines for Value Type Display Parameter Usage in the *Display Scripting User's Guide*, in the GUS Scripting Introduction section.

Status—the status control displays the syntax and semantic error(s) for the entered expression. If the message is too long, the message can be scrolled.

Missing quotes around a single-word string are not identified at validation

If you omit the quotes around a single-word string entry for an embedded display parameter, you will not get an error during the validation of the display.

The error will be detected at runtime with a misleading error message. The message is "Error 35 running script on EmbeddedPicture1 Line ##--Sub or Function not defined."

Navigating through the list of parameters

Only one parameter can be displayed at a time in the Enter Parameters dialog box. If the display has more than one parameter, scroll bars appear at the right side of the dialog box.

Use the scroll bars to "page" from one parameter to the next. As you move the scroll bar up and down, each parameter is displayed in the order of definition entry.

11. Embedded Displays

11.13. Entering Display Parameters

Default values of parameters

If a parameter value is undefined, this table lists the default display.param values that will be entered automatically.

Parameter	Default
Long	0
Single	0
Date	12:00:00 AM
Currency	0
String	""
Boolean	False
Object	No Default (creates a runtime error)
Entity	No Default (creates a runtime error)
Variable	No Default (creates a runtime error)
Inline	Empty String

12. Properties of the Builder and Displays

12.1 Overview

Topics in this section are:

- Defining collection rates and groups for display variables (Display/Data Collection and Data Collection Help)
- Default Collection Rate
- Collection Rate File
- General —ges--Default and Display
- Accessing Builder Default Properties (Edit/Defaults)
- Display Properties
- Accessing Display Properties (Edit/Display Properties)
- Accessing the Display Script (Edit/Display Script)

12.2 Defining collection rates and groups for display variables

Display/Data Collection and Data Collection Help

Naming Examples

The Display Builder will display a variable's syntax exactly as the user entered the variable name in a script or a basic dynamic. The variable name is the final data reference, not an intermediate reference.

Examples of single variable names are as follows:

DISDB.ENT01.PV (not DDB.ENT01)

LCN.A100.X.Y (not A100.X)

An example of an indexed variable is

A100.X(INT01 + INT02).

12. Properties of the Builder and Displays

12.2. Defining collection rates and groups for display variables

Data References that are displayed in the Data Collection Dialog:

The Data Collection dialog box shows information on how the system polls for the data used by the display. Not all displays and script data reference variables will be present in this list—only data that is polled periodically.

The following system variables will show in the list

- LCN Data references in scripts for these events: OnPeriodicUpdate onDataChange
- Embedded Display Parameters
- Object Dynamics
- Display Database references Definition of Variables

There is no predefinition of variables in the GUS Display Builder.

The Display Builder identifies the display variables from the individual scripts and basic dynamics, therefore, the list of variables displayed in the Set Collection function is the one the application built from the scripts and basic dynamics.

The Set Collection displays the list of variables built from the last validation.

Each time a display is validated, a new symbol table is built. The Data Collection dialog uses this symbol table to get its variable information. In this way, variable information is as current as the last validation.

Collection Rates

Rate in Seconds—The collection rate for each group is specified in whole seconds.

Default Rate—The default rate for each group is 4 sec.

Maximum Rate—The maximum rate is 215 second.

Minimum Rate—The minimum rate is 0 second.

Initial Rate—A rate of “0” indicates that the variable is updated only on initial display invocation.

Group IDs

Default—The default group ID for variables is “1.”

Range—The range of Group IDs is from 0 to 245.

Reserved ID—Group IDs—246 - 255 are reserved.

Fast ID—Group ID “FST” is used for the Fast Data Collection group. Items in this group have their values updated once per second while the display is being run in Fast Update mode.

Change Zone ID—Group ID “CZE” is used for the Change Zone entity group.

PMK ID—Group ID “PMK” is used for the PMK share group.

Demand Update Function

Collection on Demand is implemented by scripting:

LCN.Update n

Where n is the required Group ID.

Full Display update—If group “0” is specified in the demand update function, then all groups will be collected on demand.

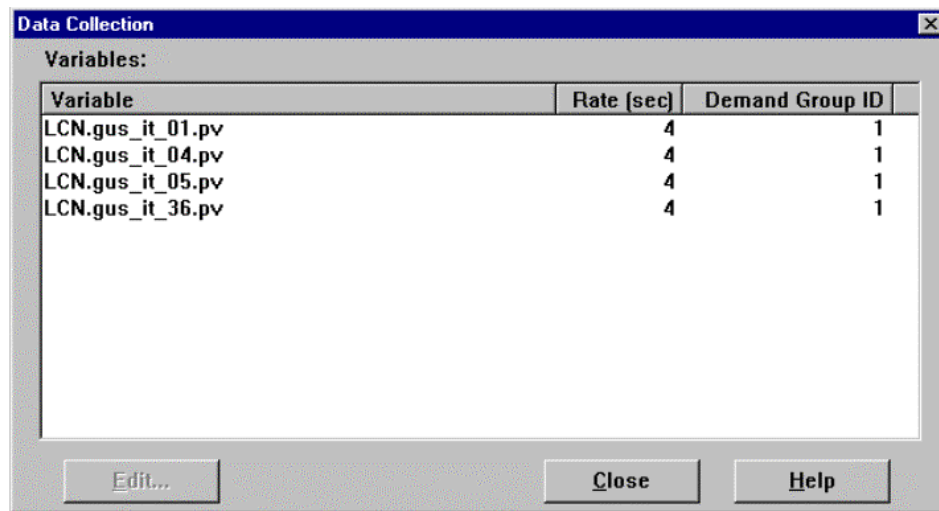
Individual Group —If an individual group IDs (1 to 255) is specified in the demand update function, then only that groups will be collected on demand.

12. Properties of the Builder and Displays

12.2. Defining collection rates and groups for display variables

Access and edit data collection variables (Display/Data Collection and Help)

When you select Data Collection from the Display menu, the Data Collection dialog box opens with current variables displayed:



Headers at the top of the list identify each variable's Name, Rate, and Demand Group ID. Initially, the list is sorted alphabetically by variable name.

Setting Column Widths—You can drag the dividers between the Headers to adjust column widths.

Sorting Variable Data—When you click one of the Header's buttons, the selected column is sorted alphabetically.

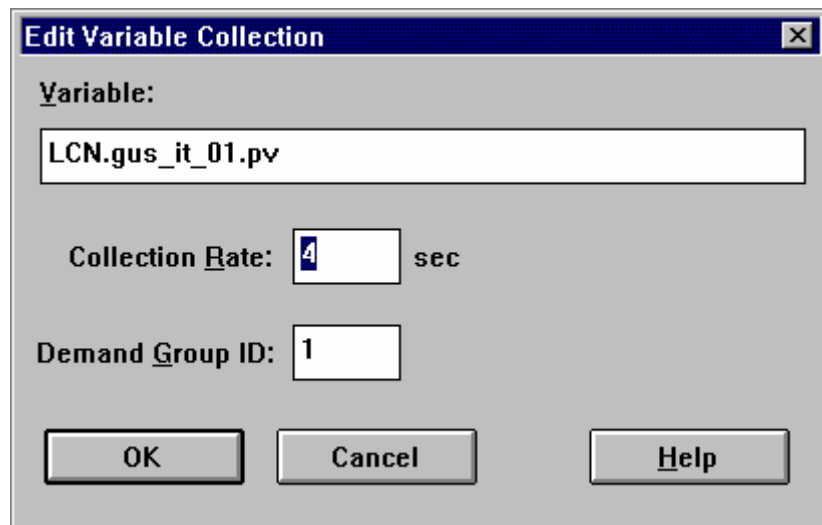
Edit—this button opens the Edit Variable Collection dialog box.

Close—this button closes the dialog box.

Help—The Help button displays this Help information.

Editing the Variable Collection List

When you click the Edit button on the Variable Collection dialog box, the Edit Variable Collection edit box appears:



Variable—The variable name you selected from the variable list is shown.

Collection Rate/Group ID—Each variable in a group can have a unique collection rate. You can accept or edit the Collection Rate and the Demand Group ID. All variables in the same group are collected at the same rate when that group is put in Demand Update.

OK—When the OK button is pressed the Group ID is validated and the variable is checked to see that

- a) it does not violate the maximum size of group, and
- b) the collection rate is within the valid range.

The variable's data shown here and in the Data Collection dialog are updated and the dialog box closes.

Cancel—this button closes the dialog box without updating the collection rate or group ID.

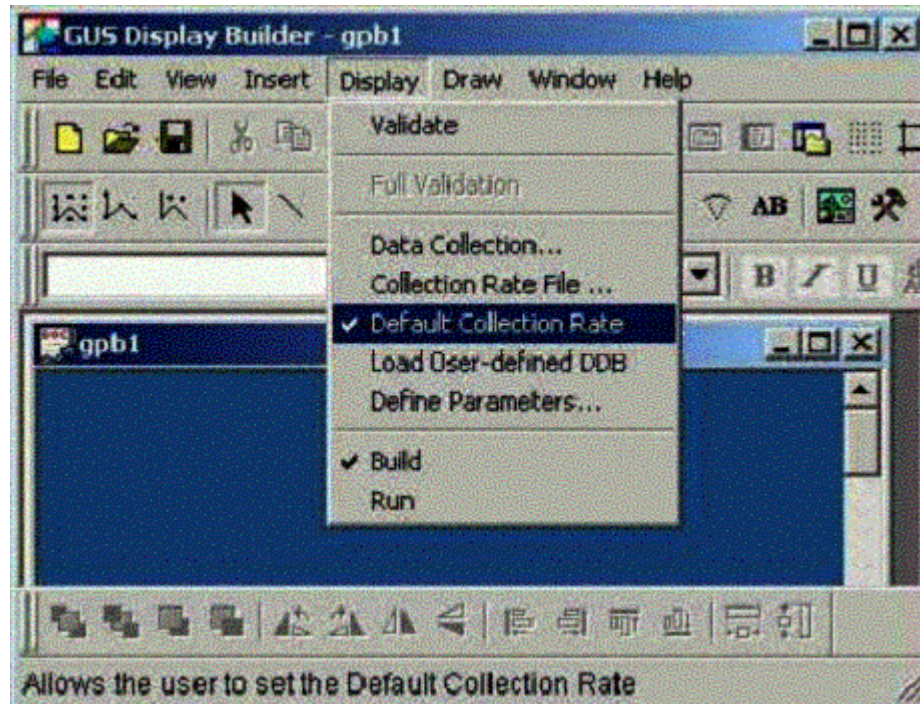
Help—The Help button displays this Help information.

12. Properties of the Builder and Displays

12.3. Default Collection Rate

12.3 Default Collection Rate

When the GUS Display Builder is started for the first time after installation, by default, the Default Collection Rate is checked in the Display menu, as shown below.



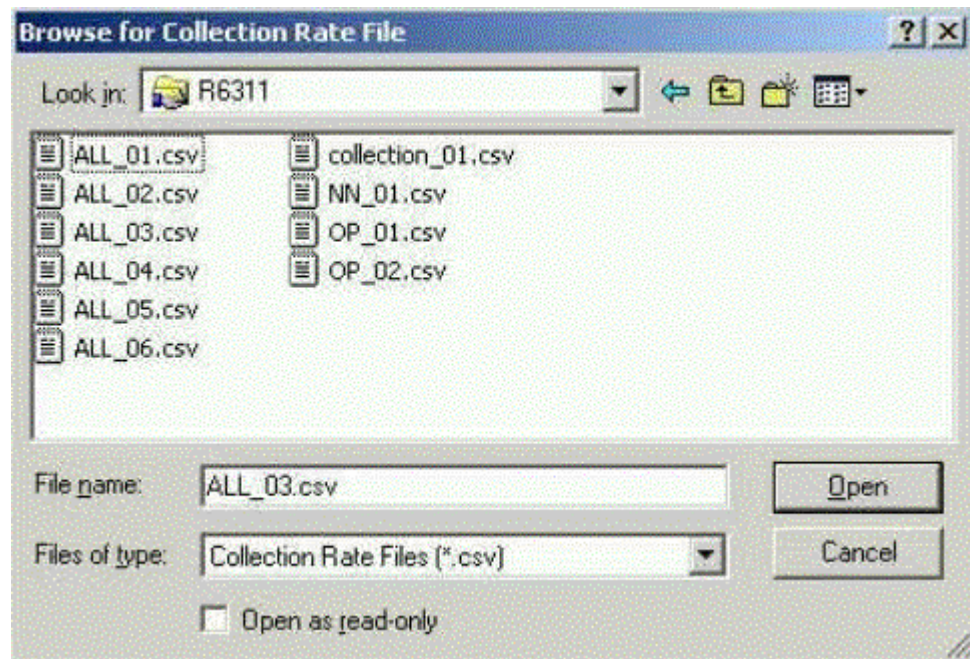
The first time GUS Display Builder is invoked after it is installed, an initial collection rate of 4 seconds and a group ID of 1 are specified for the display.

The display should be saved before validating to see the updated display name. The changes made to the Collection rate and Group ID will be reflected in the display when the display is saved.

12.4 Collection Rate File

This option allows you to specify the default collection rates and Group IDs (on a per-parameter-name basis) that are stored in a Collection Rate (.csv) file. A Collection Rate template file (sample_collectionrate_file_xxx_x.csv) containing some basic set of display names, point.parameters, associated default collection rates, and Group IDs are available in the location \Program Files\Honeywell\TPS\RAC\templates. You can generate your own Collection Rate files by making copies of the Collection Rate template file and then editing the copies using either Notepad or MS Excel. A collection rate of 4 seconds and a Group ID of 1 are used if the collection rate is not specified for a parameter name in the file.

When you select Collection Rate File from the Display menu, a file selection dialog box for selecting a Collection Rate file appears, showing the .csv files. An example follows:



The file path of the previously selected Collection Rate File is always shown in the Look-in box. If you edit and save the Collection Rate file, you must select the file through the Collection Rate File menu item to make the changes visible in the display. The collection rate of 4 seconds is assumed if the collection rate is not specified for a parameter name in the file.

12. Properties of the Builder and Displays

12.4. Collection Rate File

NOTE: In order to incorporate the collection rates and the Group IDs specified in the selected collection rate file, you must save the display before you validate it.

If there are invalid entries in the Collection Rate file selected by the user, error messages are written in the error log text file (CSErrorLog.txt) with the date and time. The error log file is stored in the same directory where the selected Collection Rate file is present. If an error occurs, when the collection rate file is selected, appropriate error messages are displayed.

When a Collection Rate File is selected, the Collection Rate File menu item is checked and the Default Collection Rate menu is unchecked.

From the Display menu, you can select the Default Collection Rate when it is unchecked to set the default collection rate of 4 seconds, and the Group ID of 1 for subsequently added parameters. Prior to selecting this menu, you must validate any parameter that has been added with the currently active collection rate file. Once you select this menu item, the Display Builder stores the status of the Default Collection Rate menu item, inactivates the Collection Rate file and checkmarks the Default Collection Rate. The status of the Default Collection Rate is retained even if the application is closed and reopened again. If you select Default Collection Rate when it is already checked, then there will not be any action.

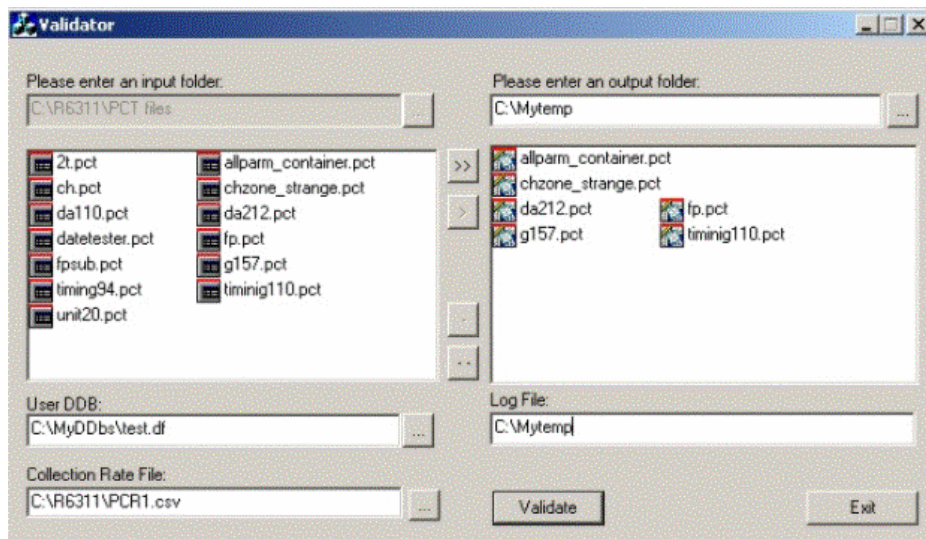
However, you can continue to change the collection rates and Group ID for any given Point.Parameter on an individual basis using the GUS Display Builder Data Collection dialog box. You can use the Data Collection dialog box if you want to specify a collection rate value and Group ID other than what is specified in the Collection Rate File, or by the default collection rate of 4 seconds. When you change the collection rates and Group ID through the Data Collection dialog box, the collection rates and Group ID specified in the Collection Rate File or the selected default collection rate of 4 seconds will not have any effect.

Batch Processing Using the Display Validator

Batch processing enables you to apply a specified Collection Rate File to a set of selected display files. This function is available in the GUS Display Validator.

You can perform batch processing as follows:

- 1 Select an input folder in the GUS Display Validator as shown in the following display. All the displays (.pct files) available in the folder are shown.



- 2 Select the displays to which the collection rates are to be applied.
- 3 Select the output folder in which the selected displays with the specified collection rates are to be stored.
- 4 Click the file selection button in the Collection Rate File edit box. A file selection dialog appears, showing the .csv files. Select the required Collection Rate File.
- 5 Select the folder in which the validation log file is to be stored. This log file is different from the error log text file (CRErrorLog.txt) created by the batch processing function. While parsing the Collection Rate File, the error log text file (CRErrorLog.txt) is created in the same directory where the Collection Rate File is located. If an error log file is already available in the directory, the new error log information is appended to the existing list with the new time and date.
- 6 Click the Validate button to apply the collection rates to all the selected displays.

12. Properties of the Builder and Displays

12.4. Collection Rate File

The collection rates and Group IDs in the selected displays are overwritten by the revised collection rates listed in the selected Collection Rate File. If a particular parameter is not specified in the Collection Rate File, then the collection rates and Group IDs in the selected displays remain the same. If a wildcard (*) is entered for a Collection Rate and/or for a Group in the Collection Rate File, then the respective values in the display remain unchanged. Therefore, you have the option of modifying the collection rates and Group IDs of a few parameters and leaving the collection rates and Group IDs of other parameters unchanged.

NOTE: If you want only the Validator functionality to be used without the set collection batch processing functionality, then the Collection Rate File edit box should be empty.

Format of the Collection Rate file

You can open Collection Rate file and edit it using Notepad or MS Excel.

If you open the file using MS-Excel, here is an example of the file that will appear.

Display	Parameter	Rate	Group	
BOILER1_U1.pct	Lcn.tag1.pv	8	1	
BOILER1_U1.pct	Lcn.tag1.sp	8	1	
BOILER2_U1.pct	Hci.server.tag1.pv	12	2	
BOILER3_U1.pct	Dispdb.ent01.pv	12	2	
BOILER1_U1.pct	Pv	8	1	
BOILER1_U1.pct	*	8	1	
*	PV	4	*	Comment 1
*	SP	*	*	
*	PVHITP	8	*	Comment 2
*	PVLOTP	8	*	
*	OP	8	*	
*	*	8	*	

Validation does not take place in the **Display** column. If the **Display** column is blank, the display name is considered as *.pct. The display name can have the displayname.pct or the displayname format. If the Group column is blank or contains an "*", the Group ID of 1 is assigned. The fifth (blank) column is optional and can be used for entering comments, as shown by the 2 entries Comment 1 and Comment 2.

When the Collection Rate file is parsed, only the first four columns are read. All the other columns are ignored.

The collection rate and Group ID are assigned based on a series of searches. If there is a match in the Data Collection file, the corresponding collection rate and Group ID are assigned by the GPB.

For example, to assign the collection rate for parameter LCN.tag1.param1 in display BOILER1_U1.pct, the following steps are executed to search for the right match.

1. Look for "BOILER1_U1.pct, LCN.tag1.param1"
2. If not found, look for "BOILER1_U1.pct, param1"
3. If not found, look for "*.pct, LCN.tag1.param1"
4. If not found, look for "*.pct, param1" (original format ",v,4,*")
5. If not found, look for "BOILER1_U1pct,*"
6. If not found, look for "*,*"

NOTE 1: After editing, when you save or close a .csv file in MS Excel, you are prompted to save it as a .xls file. You can ignore the prompt and save the Collection Rate file with a .csv extension.

NOTE 2: The first line in the file generated by the GUS tool is a comment, which specifies the format of the data to be entered. If there is a line in the .csv file containing the string "Display,Parameter,Rate,Group", then it is considered as a comment and is ignored.

NOTE 3: The parameter name has to be in full form (such as "LCN.tag1.pv"). The names that are in fragments (such as "tag1.pv") are not supported.

NOTE 4: In order to use the correct display name for the match, you must save the display before you validate it. A display name of "*" will be used for the match if you validate an unsaved file.

12. Properties of the Builder and Displays

12.4. Collection Rate File

NOTE 5: Collection Rate functionality will not override the special group handling of \$CZ_ENTY and PMK entity variables.

NOTE 6: The parameter name can be the name of a collector, such as "\$ACKSTAT".

If you open the same file using Notepad, the file appears as shown below. Each value is separated by a comma. Only one set of records (Parameter, Collection Rate) is allowed per line.

```
Display,Parameter,Rate,Group,
BOILER1_U1.pct,Lcn.tag1.pv,8,1,
BOILER1_U1.pct,Lcn.tag1.sp,8,1,
BOILER2_U1.pct,Hci.server.tag1.pv,12,2,
BOILER3_U1.pct,Dispdb.ent01.pv,12,2,
BOILER1_U1.pct,Pv,8,1,
BOILER1_U1.pct,*,8,1,
*,PV,4,*,Comment 1
*,SP,*,*,
*,PVHITP,8,*,Comment 2
*,PVLOTP,8,*,
*,OP,8,*,
*,*,8,*,
```

It is not mandatory to keep “*,8,*,” as the last line in the file. You can, however, keep it as the last line for the purpose of consistency. A single line “*,8,*,” in the file is similar to setting the default to 8 seconds. If this line is not present in the file, the following rules are applied for all the other parameters for which the collection rate is not explicitly mentioned in the file.

- The collection rate is set to 4 seconds while editing the display using GUS Display Builder.
- The existing collection rates are not changed during batch processing.

If the same parameter has different collection rates, the last entry is taken in to account. A comma is the only separator allowed between values. Spaces are not allowed. By using the above file format, you can use the following combinations to set the collection rates.

- To set the collection rates for some parameters individually (x - *,v,3,*,)
- To set the collection rates for other parameters through default (ex - *,*,6,*,)

NOTE 1: Any subscripted parameter name in the Collection Rate File is invalid (Example: PV(3)).

NOTE 2: An unsubscripted parameter name in the Collection Rate File applies the specified collection rate to all the elements of the parameter name. For example, an entry “*,pv,6,*” in the collection rate file, applies the collection rate of 6 to all the elements of PV (Pv, Pv(1), Pv(2)).

Applying Multiple Collection Rate Files for the same display

You can apply collection rates from different Collection Rate Files while building a new display as follows:

1. In the GUS Display Builder, proceed as follows:
 - for a new display, open it and then save it before validating the display to match the correct display name from the Collection Rate file, or
 - for an existing display, just open it.
2. Select a Collection Rate File (for example, pcr1.csv).
3. Add the required parameters and click the Validate Menu item.
4. The collection rates selected in step 2 will be applied to the parameters added in step 3.
5. Select a different Collection Rate File (for example, pcr2.csv).
6. Add the required parameters and click the Validate menu item.
7. The collection rates selected in step 5 (pcr2.csv) will be applied to the parameters added in step 6 only. The collection rates of parameters added in step 3 are not affected.

12. Properties of the Builder and Displays

12.5. General Pages—Default and Display

Applying the same Collection Rate Files to multiple displays

The selected Collection Rate File will be used to set the collection rate for any parameter added to displays that are open at any given time.

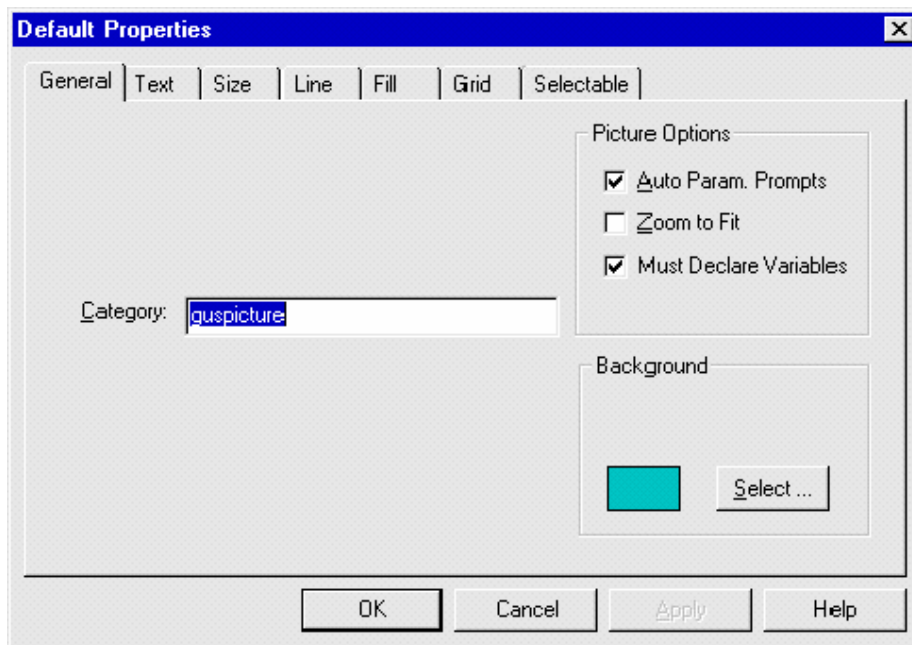
12.5 General Pages—Default and Display

Two General Property Pages use the same Help button. For information on the general properties of each display, refer to the following list for the required property page:

- General Page-Display Builder Default Property Sheet
- General Page-Display Object Property Sheet

12.6 General Page-Display Builder Default Property Sheet

- 1 From the Edit menu, select Defaults. The Default Properties sheet opens with the General Page showing.



- 2 Enter a Category if required. The default Category is "guspicture."

12. Properties of the Builder and Displays

12.6. General Page-Display Builder Default Property Sheet

NOTE: If the category is empty, the display does not register a category and is not managed by SafeView. However, if the category is "NOPRIOR," the display is managed by SafeView without appearing in the history or being invoked by the Prior Display.

- 3 In the Picture Options section, check Auto Param. Prompts if you intend to embed this display into another display and want to be able to insert new parameters before you position the display. If Auto Param. Prompts is checked, the Parameters page of the Display Object Properties window for the embedded display will be displayed before the crosshair cursor, used for display positioning, appears. If Auto Param. Prompts is unchecked, the Parameters page will not be displayed and you will not be able to enter new parameters before you position the embedded display.
- 4 In the Picture Options section, check Zoom to Fit if you want the size of your display to be adjusted to fit its window. This is a Runtime property only. Use caution when designing displays to use this feature because, in some cases, text may be unreadable. Refer to Considerations when using Zoom to Fit displays at Runtime for more information.
- 5 In the Picture Options section, check Must Declare Variables if you want the script compiler to behave as if you had added the "option explicit" statement to all scripts. The result is that you must declare all variables and functions prior to using them. Use of this option is recommended because it increases the level of error checking performed by the GUS Basic compiler.
- 6 Change the Background default color if required. Here, you change the color of ALL new displays. If you want to change the color of ONLY the active display, refer to

General Page-Display Object Property Sheet.

NOTE: Do not use these gray background colors:

- Index=120 with RGB = 136, 136, 136
- Index=121 with RGB = 125, 125, 125

If used, these colors may make the Highlight Rectangle appear invisible at Runtime.

- 7 Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.

12.7 General Page-Display Object Property Sheet

- 1 Select Display Properties from the Edit menu. The Display Object Properties-Display sheet opens with the General Page on display.

Display Object Properties - Display

General | Text | Size | Line | Fill | Grid | Selectable

Name:

Description:

Category:

Information

Reference File:

Validation State: NOT VALIDATED

Picture Options

☒ Auto Param. Prompts

☐ Zoom to Fit (runtime)

☐ Must Declare Variables

Background

Password

- 2 Enter or change the Name of your default property window if required. Refer to Display Object names in the *Display Scripting User's Guide*, in the GUS Scripting Introduction, Data Entry Primitive section for the correct syntax for the name. NOTE: An entry is NOT required in the Name field.
- 3 Enter a Description and Category if required.

NOTE: You cannot enter the following characters in the Category field: ? * \ "
- 4 Under Picture Options, check Auto Param. Prompts if you want the Enter Parameters dialog box to be automatically displayed after the Add Parameters dialog box is closed. If this option is disabled, you must manually invoke the Edit/Enter Parameters command after adding a parameter. NOTE: THIS FEATURE IS NOT CURRENTLY AVAILABLE.

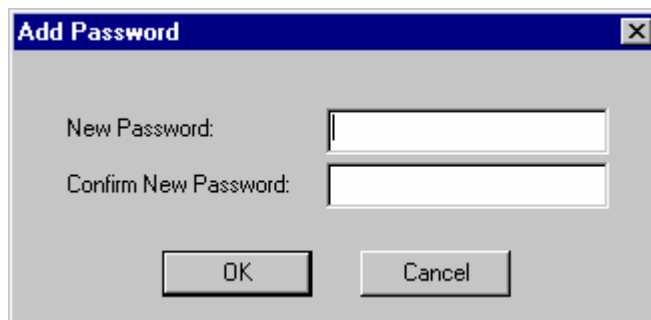
- 5 Under Picture Options, check Zoom to Fit if you want the size of your display to be adjusted to fit its window. This is a Runtime property only. Use caution when designing displays to use this feature because, in some cases, text may be unreadable. Refer to Considerations when using Zoom to Fit displays at Runtime for more information.
- 6 Under Picture Options, check Must Declare Variables if you want the script compiler to behave as if you had added the "option explicit" statement to all scripts. The result is that you must declare all variables and functions prior to using them. Use of this option is recommended because it increases the level of error checking performed by the GUS Basic compiler.
- 7 Change the Background default color if required. Here, you can change the background color for ONLY the currently ACTIVE display. If you want to change the color of all future NEW displays, refer to General Page-Display Builder Default Property Sheet.

NOTE: Do not use these gray background colors:

- Index=120 with RGB = 136, 136, 136
- Index=121 with RGB = 125, 125, 125

If used, these colors may make the Highlight Rectangle appear invisible at Runtime.

- 8 If required, add a Password by executing the following steps:
Select the Add button in the Password group box. The Add Password dialog box appears.



For security reasons, the characters you type in this dialog box will be displayed as asterisks.

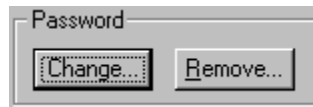
12. Properties of the Builder and Displays

12.7. General Page-Display Object Property Sheet

Enter the password you want to add in the New Password data port.

Enter the same password in the Confirm New Password data port.

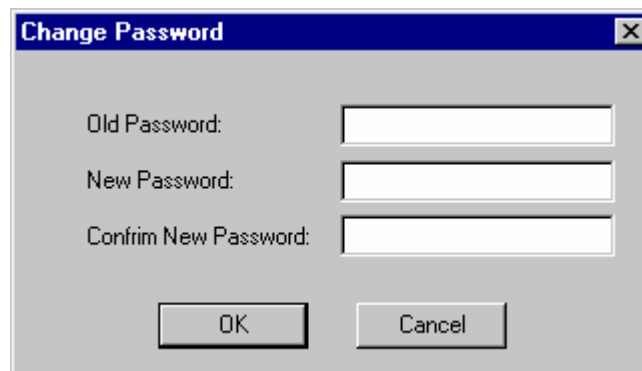
Click OK. The Add Password dialog box disappears and the new password is saved. The Add button in the Password group box disappears and is replaced by the Change and Remove buttons.



NOTE: The display is not password-protected until you save it.

- 9 If required, change the current Password by executing the following steps:

Select the Change button in the Password group box. The Change Password dialog box appears.



For security reasons, the characters you type in this dialog box will be displayed as asterisks.

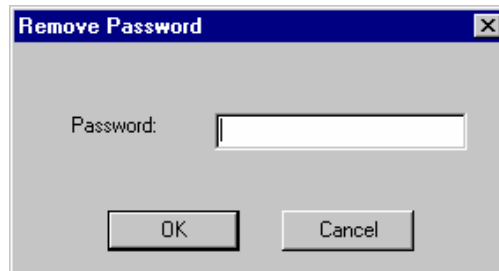
Enter the current password in the Old Password data port.

Enter the new password in the New Password data port.

Enter the same password in the Confirm New Password data port.

Click OK. The Change Password dialog box disappears and the new password is saved.

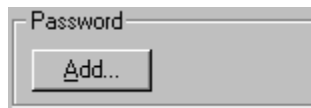
- 10** Remove the current Password if required by executing the following steps:
Select the Remove button in the Password group box. The Remove Password dialog box appears.



For security, the characters you type in this dialog box will appear as asterisks.

Enter the current password in the Password data port.

Click OK. The Remove Password dialog box disappears and the password is removed (deleted). The Change and Remove buttons in the Password group box disappear and are replaced by the Add button.



- 11** Click Apply to accept the changes and go to another property page (tab), or click OK to accept the changes and close the Property Sheet.



ATTENTION

When editing display objects, selecting "Apply," followed by "Cancel" can cause the changes **not** to be applied. Be sure to select "OK" after applying changes, not "Cancel."

12. Properties of the Builder and Displays

12.8. Display Builder Default Property Sheet

12.8 Display Builder Default Property Sheet

Display Builder Default Property Pages are:

- General Pages—Default and Display
- Text Properties (Text Page)
- Size Properties (Size Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)
- Grid Properties (Grid Page)
- Selectable Property (Selectable Page)

12.9 Size Properties

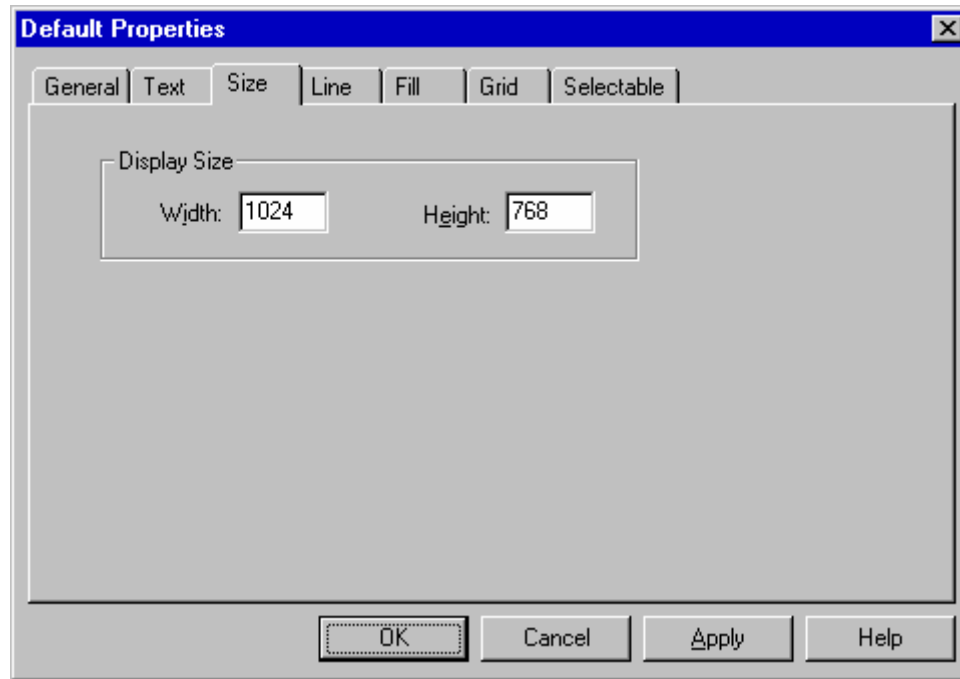
Size Page

This property page defines the Display Size of an object. The size is described by the number of pixels wide and high for the object.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to [How Default Properties Propagate](#).

Changing the size of an object

- 1 With the required Property Sheet opened, click the Size tab. The Size Property Page is displayed.



- 2 Observe the current Width and Height. Accept the defaults or change as required.
- 3 Click OK or Apply to accept attributes. OK also closes the Property Sheet.

12. Properties of the Builder and Displays

12.10. Grid Properties

12.10 Grid Properties

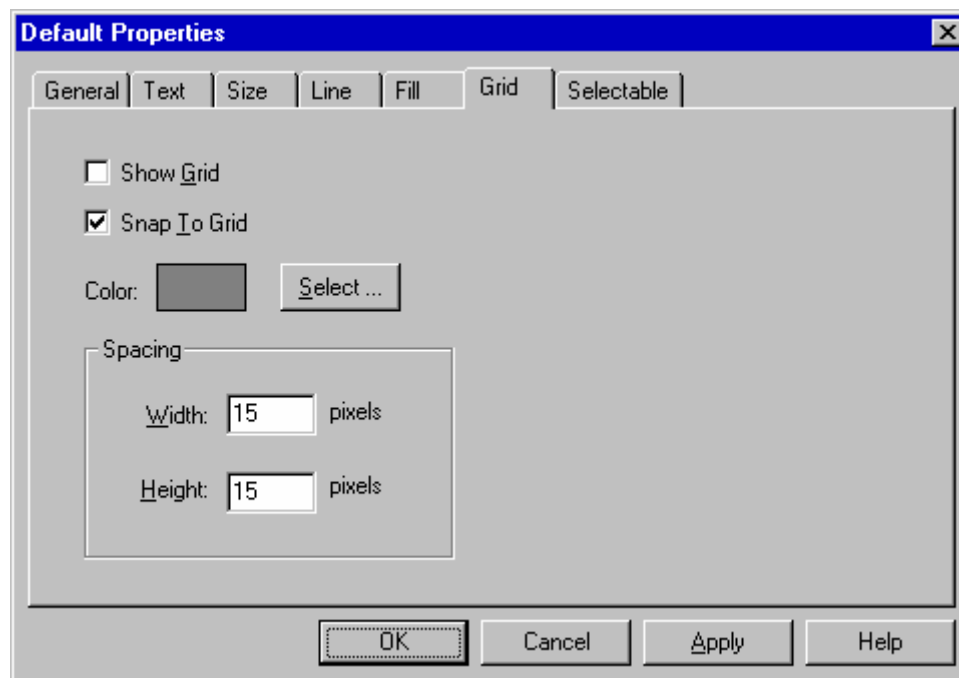
Grid Page

This property page defines a grid that can be put on your display object. This grid can help you to better align and size small objects in your display.

Default properties are determined from an object's parent. To review the propagation of display properties, refer to How Default Properties Propagate.

Changing the grid for an object

- 1 With the required Property Sheet open, click the Grid tab. The Grid Property Page is shown.



- 2 If Show Grid is checked, the grid will show. Accept the default or change as required.
- 3 If Snap to Grid is checked, objects you draw or move in the future will snap to the grid. Accept the default or change as required.

- 4 Accept the Color or change as required.
- 5 Observe the grid's current Width and Height in pixels. Accept the setting or change as required.
- 6 Click OK or Apply to accept attributes. OK also closes the Property Sheet.

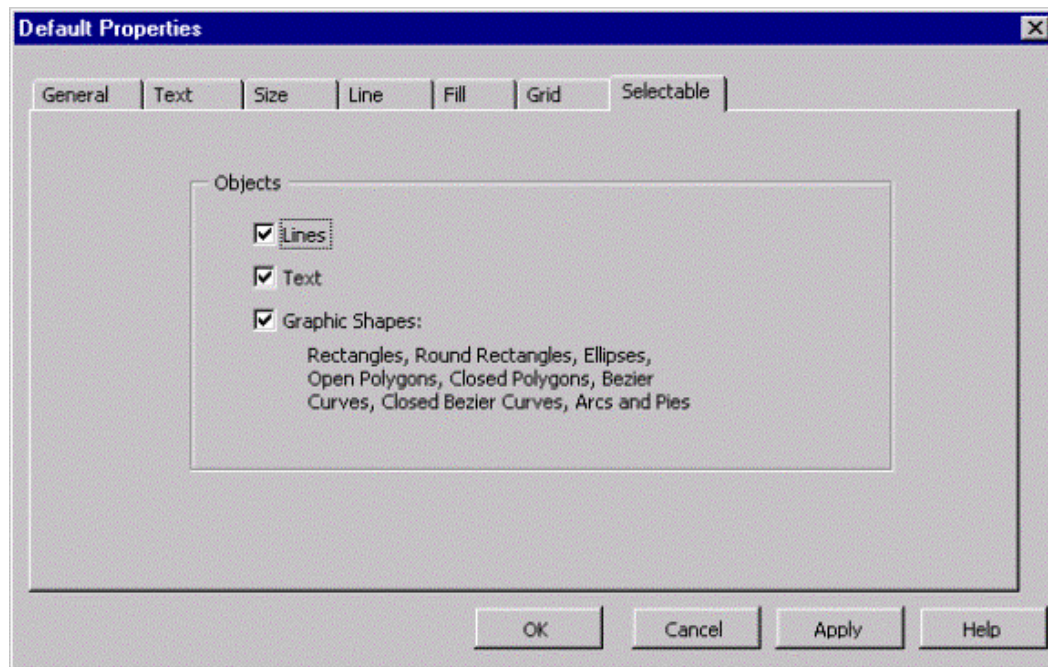
12.11 Selectable Properties

Selectable Page

This property page allows the user to determine the default selectable property of objects. The default selectable property determines whether objects such as lines, text, and graphic shapes are selectable or non-selectable.

Setting the Default for Selectable property

Select “Defaults” from the Edit menu. When the following Default Properties display appears, select the “Selectable” Page.



12. Properties of the Builder and Displays

12.11. Selectable Properties

This page has check boxes for setting the default selectable property for lines, text, and graphic shapes. A checked box indicates that the object category (lines, text, or graphic shapes) is selectable; an unchecked box indicates that the object is not selectable.

To change the selectable property for an object that has been added to the picture, go to the object's Display Properties page and select the Selectable Page.

The selectable default property is not applicable to objects such as embedded displays, embedded pictures, controls, buttons, edit/combo boxes, etc.

13. Accessing Builder Default Properties (Edit/Defaults)

13.1 Overview

The Default properties are applied when a new Display is opened (changing these defaults will not affect objects added to an existing display).

From the Edit menu, select Defaults. The Default Properties Sheet opens.

Property Pages are:

- General Pages—Default and Display
- Text Properties (Text Page)
- Size Properties (Size Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)
- Grid Properties (Grid Page)
- Selectable Property (Selectable Page)

13.2 Display Properties

Display Property Pages are:

- General Page-Display Property Sheet
- Size Properties (Size Page)
- Line Properties (Line Page)
- Fill Properties (Fill Page)
- Grid Properties (Grid Page)

13. Accessing Builder Default Properties (Edit/Defaults)

13.3. Accessing Display Properties

13.3 Accessing Display Properties

There are several ways to access the Display Property Sheet:

- from the Menu Bar (or by a keystroke)
- by Using the Right-mouse Button
- by double-clicking the Display (default action).

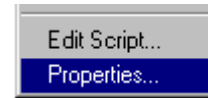
Accessing embedded display properties from the menu

Select Edit/Display Properties from the menu or press the keystroke Alt + Backspace.

RESULT: The Properties Sheet for the Display Object Properties-Display appears.

Accessing embedded display properties from right-mouse button menu

Place the mouse pointer over the display background (not over any object in the display), click and hold the right-mouse button. A drop-down menu appears.



Move to the Properties item and release the mouse button.

RESULT: The Properties Sheet for the Display Object Properties-Display appears.

Accessing embedded display properties as the default action

Place the mouse pointer over the display background (not over any object in the display) and double-click the mouse button.

RESULT: The Properties Sheet for the Display Object Properties-Display appears.

13.4 Accessing the Display Script

There are two ways to access the Display Script:

- from the Menu Bar
- by Using the right-mouse button

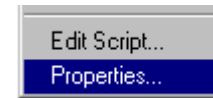
Accessing the display script from the menu

Select Edit/Display Script from the menu.

RESULT: The Script Edit ComboBox for the display appears.

Accessing the display script from right-mouse button menu

Place the mouse pointer over the required object. Click and hold the right-mouse button. A drop-down menu appears.



Move to the Edit Script item and release the mouse button.

RESULT: The Script Edit ComboBox for the display appears.

13. Accessing Builder Default Properties (Edit/Defaults)

13.4. Accessing the Display Script

14. OLE Objects

14.1 Overview

The topics covered in this section are:

- Inserting OLE Object (Insert/OLE Object menu and Help)
- Missing OLE Objects and ActiveX Controls
- Change Zones
- Loading a User-Defined DDB

14.2 Inserting OLE Object

Insert/OLE Object menu and Help



ATTENTION

Attempting to use the Insert/OLE Object menu to insert a GUS display will not work. Instead, always use the Insert/Display menu. For more information, refer to [Inserting an Embedded Display](#).

NOTE: Do not insert 16-bit applications in a display. Only 32-bit applications can be inserted.

The Insert/OLE Object command inserts an object, such as a spreadsheet or document, into a display window.

When the object is inserted, the external application in which the object was created becomes active in the display window, as well as its menus and toolbars.

While the application is active, the entire document can be displayed using the application's scrolling and paging functions. While the display window is active, only the object's first page is displayed.

Objects can be linked in a display window.

- If an object is linked, changes can be made to the object while the display window is closed. These changes will be displayed the next time the display window is opened.
- If an object is embedded, changes to the object must be made in the active application while the display window is opened.

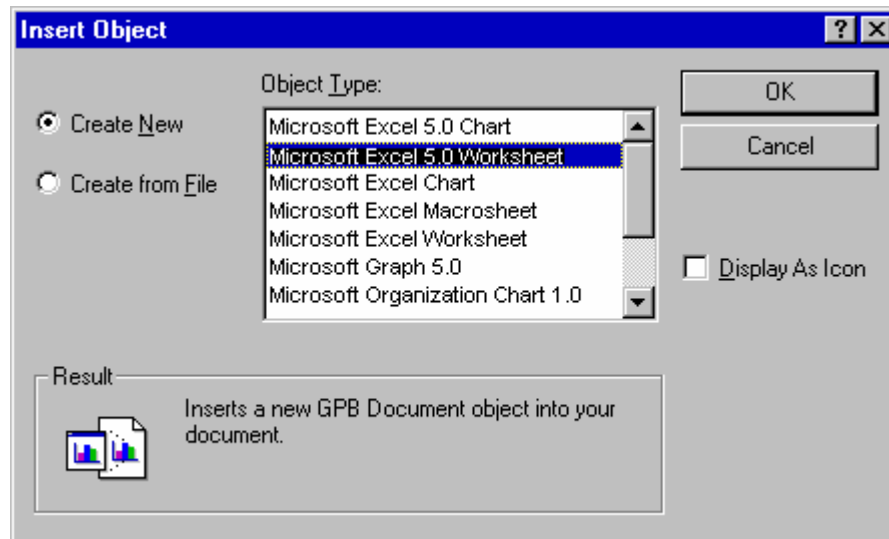
The object to be inserted can be a new object or an existing object.

14. OLE Objects

14.2. Inserting OLE Object

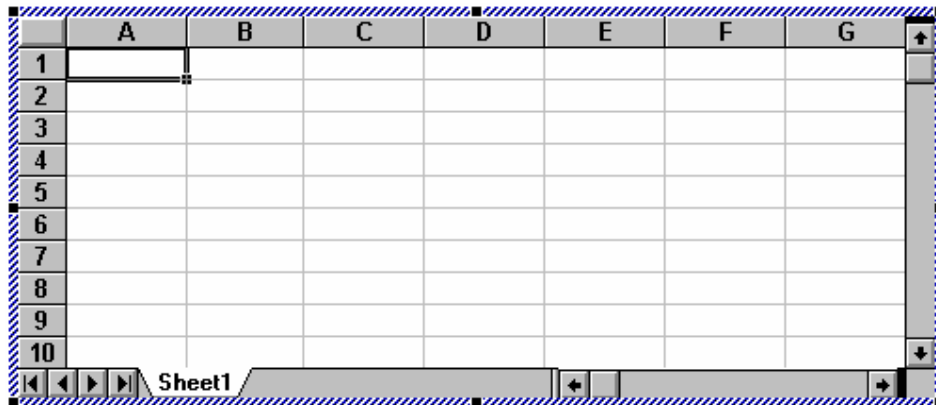
Inserting a new OLE Object

- 1 Select the display window.
- 2 Select Insert/OLE Object from the menu. The Insert Object dialog box appears.



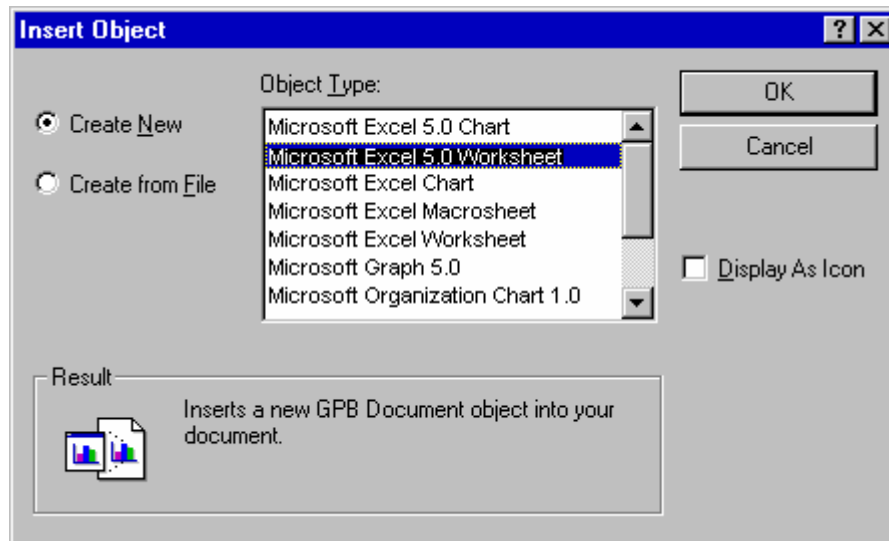
- 3 Select the Create New button.
- 4 Select the type of object from the Object Type list.
- 5 If the object is to appear as an icon, select the Display As Icon box. The icon can be changed by selecting the Change Icon button. To learn about each feature of the Insert Object dialog, click the  in the upper right corner of the box; then click the required feature for a short explanation.
- 6 Click OK.

RESULT: A new, active object of the type selected is inserted in the display window. This illustrates an Excel spreadsheet inserted as a new OLE object.



Inserting an existing OLE object

- 1 Select the display window.
- 2 Select Insert/Object from the menu. The Insert Object dialog box appears.



- 3 Select the Create From File button.

14. OLE Objects

14.2. Inserting OLE Object

- 4** A File box is displayed. Type the name of the file or use the Browse button to select from a list of file types in a specific drive and directory.
- 5** If the object is to be linked to the existing file, select the Link box.
- 5** If the object is to appear as an icon, select the Display As Icon box. The icon can be changed by selecting the Change Icon button.
- 7** Click OK.

RESULT: An existing object of the type selected is inserted in the display window. This object is now active.

Editing an OLE Object and its properties

Although OLE objects do not have property pages, their properties can be changed through their associated applications.

- 1** Select the object.
- 2** Open the Right Mouse menu or Object Browser menu.
- 3** Select Properties from the menu. The object's application is activated.
- 4** Select the appropriate property from the list.
- 5** Make the necessary change.
- 6** Do these steps again until all changes have been made.
- 7** Click OK.

RESULT: The changes are confirmed.

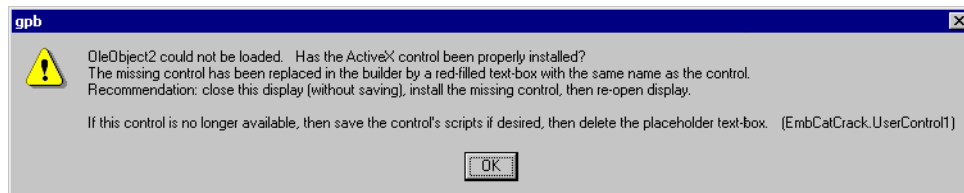
14.3 Missing OLE Objects and ActiveX Controls

In an existing GUS picture, when a missing OLE or ActiveX control object is detected, two things happen at buildtime, runtime, and during a display migration:

- 1 The missing control is replaced on the display by a “red text box,” and
- 2 An appropriate error message is displayed. The actual message displayed depends on the operating environment in which the error is detected:

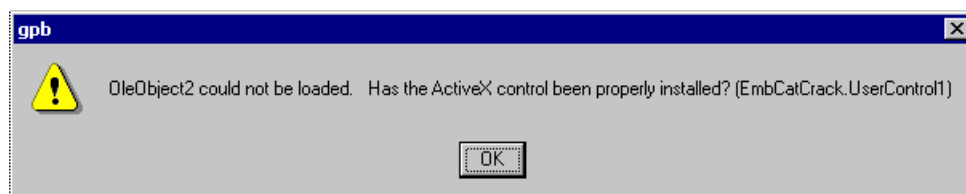
At build time

A missing OLE object or ActiveX control will initiate display of the following error message:



At run time

A missing OLE object or ActiveX control will initiate display of the following error message:



At display migration time

When a missing OLE object or ActiveX control is detected, messages will be entered in the validation and migration log files. Here are examples of the messages that would be entered if one missing control were detected. NOTE: The following log file messages appear only when you are migrating forward from GUS 210.

14. OLE Objects

14.3. Missing OLE Objects and ActiveX Controls

1. Example of a validation log file message:

```
-----  
Migration warnings for c:\test\migr_out\cracker.pct  
-----  
File c:\test\bundle3a\cracker.pct open error.  
File may be in use by another application, or it may have an  
incompatible file version. Check the Event Log for detailed  
information.
```

2. Example of a migration log file message:

```
-----  
Migration warnings for c:\test\migr_out\cracker.pct  
-----  
OleObject2 could not be loaded. Has the ActiveX control been  
properly installed?  
The missing control has been replaced by a red-filled text-box  
with the same name as the missing control.  
Recommendation: install the missing control and re-migrate this  
display.  
If this control is no longer available, then edit the migrated  
display, saving the control's scripts if required, and delete the  
red placeholder text-box.  
(EmbCatCrack.UserControl1)  
  
Migration of the directory c:\test\bundle3a into directory  
c:\test\migr_out has been completed.  
Total Files      = 1  
Failed           = 1  
Log File         = c:\test\migr_out\validate.txt
```


14.4 Change Zones

Displays

The Change Zone displays a subset of a point's parameters and enables the operator to change values for those parameters that can be changed by an operator. The subset of parameters displayed in a Change Zone depends on the point type and other variables such as algorithm type.



ATTENTION

When a point is displayed in a Change Zone and the user invokes a Change Zone for a different point, the new pointname is the first value to be updated. It may take a few seconds to update the rest of the Change Zone data for the new point.

Until all the data for the new point is displayed, the user will be blocked from changing any values via the Change Zone. Selecting an object, such as PV, SP, OP, or PTXST, on the Change Zone before the new point's data is completely displayed will result only in an audible beep.

Selecting the pointname in a Change Zone will invoke the Detail Display for that point in the Native Window.

Change Zone Limited Inheritance

Change Zone displays support limited inheritance, and the following properties can be set on the top level display and will propagate to the objects contained in the display:

- TextColor (default to black) -- If ChangeZone is inserted over a black background, we suggest changing to bright Blue
- Font and Charset (default to Arial)

Note: Text items that require a non-proportional font for alignment use a courier new typeface and are NOT affected by inheritance.

- FillColor -- applied to DataEntry & List boxes -- defaulted to dull blue-green

Setting up a Change Zone

- 1 Insert the correct Change Zone .pct file for your system into a display.
For TPN R5xx releases, use the "changezone_R5xx.pct" file.
For TPN R6xx releases, use the "changezone.pct" file.
- 2 Assign a pointname to dispdb.[`$CZ_ENTY`].
- 3 Optionally, change the data collection update rate.

Inserting changezone.pct into a display



ATTENTION

This documentation uses generic filenames. For both the Change Zone and the associated Ramp Key display, a version identifier may be embedded in the filename. For example, the file referred to as changezone.pct in this documentation may actually be named changezone_mr3.pct, or something similar, on your system.

The Change Zone is implemented as a .pct file that can be inserted into a display. While in the Display Builder, select Insert, then select Display, and then select \\Honeywell\TPS\rac\chg_zone\changezone.pct to add the Change Zone to a display.

When changezone.pct is inserted into a display, the user will be prompted to enter the values for three parameters:

- **rampkey_used**—indicates whether rampkey.pct is to be used in the display. Enter TRUE if rampkey.pct (see following section) will also be inserted into the display. Otherwise, enter FALSE.
- **group_for_guscz01**—the collection group assigned to \$GUSCZ01. This parameter has meaning only if rampkey_used = TRUE. For optimum ramp key performance, \$GUSCZ01 should be assigned its own collection group.
- **share_PMK**—indicates whether or not you want to share the functionality of the global PMK object. Enter TRUE if you want to share the PMK functionality with an object outside of the Change Zone. You will also have to add the necessary PMK scripting to the non-Change Zone object to enable the sharing. Enter FALSE if you don't want to share the functionality of the global PMK object. Even if you enter FALSE, the Point Manipulation keys in the Change Zone will still work.

If a display already contains changezone.pct and the “Replace Embedded Display” command is used to put a new version of changezone.pct in the display, the user is not prompted to enter values for the three parameters. In that case, the “Enter Parameters” command should be used to enter values for these parameters. For optimum ramp key performance, values should be entered for the three parameters.

NOTE: When inserting a Change Zone in a display that has a black background, position a light-colored rectangle behind the Change Zone to ensure visibility of the Change Zone's black text.

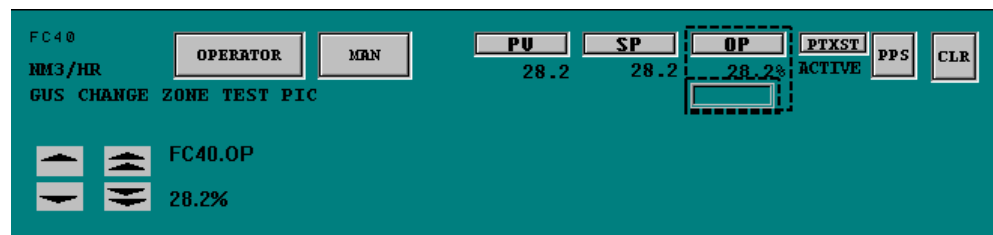
For more information, refer to Inserting an Embedded Display.

Inserting rampkey.pct into a Change Zone display

The rampkey.pct file can be inserted anywhere in a GUS display containing a changezone.pct. The rampkey.pct file displays Raise, Lower, Fast Raise, and Fast Lower keys similar in function to the ramp keys found on the operator keyboard.

Once the rampkey file is embedded (Insert/Display), whenever a SP or OP button is selected from the Change Zone the ramp keys appear in the display. Although the SP or OP input ports have a timeout, the ramp keys remain active until the user changes the parameter-of-interest.

The ramp key area pictured here includes the point name (FC40) followed by the input type (OP) and the OP value.



Error checking functions the same as when keyboard ramp keys are used.

For more information, refer to Inserting an Embedded Display.

Data Entry into a GUS Change Zone Display



ATTENTION

You need to select the **Numeric Keypad * Handling** option to enable this functionality.

1. Choose **Start> All Programs> Honeywell TPS > Configuration Utility**.
2. Click **Configure**, and then select **GUS Display Runtime Settings**.
3. Select **Numeric Keypad * Handling**.
4. Click **OK**.

With GUS R350, when you select the SP or OP button from the GUS Change Zone, an input port opens for data entry. If the data entered in the port is incorrect and you have not pressed ENTER, you can clear the entry by pressing the corresponding SP, OP, or CLR ENTR / numeric-keypad * key of the IKB or OEP and enter the correct data.

Examples of correcting invalid GUS Change Zone SP or OP entries are as follows:

14. OLE Objects

14.4. Change Zones

After selecting the GUS Change Zone /Group SP or OP target, if the operator enters a wrong value such as 12A when 12 is the intended value, the error can be corrected by using one of the following methods:

Press the CLR ENTR key of the OEP keyboard, and then type in 12 and press ENTER.

OR

If the **Treat numeric keypad * as CLR ENTR** option is selected, press * of IKB/engineer's numeric- keypad key, and then type in 12 and press ENTER.

OR

Press the same SP or OP target or button again, and then type in 12 and press ENTER.

OR

Press IKB/engineer's keyboard backspace key once and press ENTER

The result of any of these operations is that the invalid 12A value is erased and the correct value of 12 is entered.

Point Manipulation Key functionality

The following PMK functions can be scripted into the Change Zone:

- MAN
- AUTO
- NORM
- SP
- OP
- RAISE/LOWER
- FAST RAISE/FAST LOWER

For more information on PMK functions, refer to PMK (Point Manipulation Key) Object.

Runtime behavior of a Change Zone

Using the embedded point manipulation keys that can be inserted into a Change Zone, you can perform both incremental and continuous point ramping.



ATTENTION

There is an important difference between incremental and continuous ramping.

If you click a ramp key at a rate faster than the GUS software can process, the clicks will be queued (one ramp increment for each click). The ramped and displayed values may continue to change even after you stop clicking. If you hold down the ramp key, the value will ramp continuously until you release the key. The value on display when the key is released is the final ramped value.

Assigning a point name to \$CZ_ENTY

Change Zone uses dispdb.[\$CZ_ENTY] to determine the point to be displayed. After changezone.pct has been inserted into the display, a point name needs to be assigned to dispdb.[\$CZ_ENTY].

Below are examples of how to use dispdb.[\$CZ_ENTY] in a display in which changezone.pct has been inserted:

- Assign a point name in a subroutine script such as OnDisplayStartup() or OnLButtonClick().
- Call up a dialog box using a dialog box request.
- Assign a point name from a DDB entity ID.
- Assign a point name from \$al_enty.

OnDisplayStartup script

Set dispdb.[\$CZ_ENTY] to a valid point name in the imbedded changezone subpicture's OnDisplayStartup script.

The following is an example of using OnDisplayStartup script to display the Change Zone of a point named "a100":

```
Sub OnDisplayStartup()  
    dispdb.[$CZ_ENTY] = "a100"  
End Sub
```

Dialog box request

Create a dialog request to ask the user for the point name and store it into dispdb.[\$CZ_ENTY].

14. OLE Objects

14.4. Change Zones

The following is an example of a dialog box request:

```
Sub OnLButtonClick()  
    dispdb.[%CZ_ENTY] = askbox$("Enter point name")  
End Sub
```

DDB Entity ID assignment

Set dispdb.[%CZ_ENTY] to a valid point name in a script using a DDB entity ID.

```
Set dispdb.[%CZ_ENTY] = dispdb.ent01  
    dispdb.[%al_enty] assignment
```

Use the following assignment to display the Change Zone of a point selected on the Alarm Summary display:

```
set dispdb.[%CZ_ENTY] = dispdb.[%al_enty]
```

Validating a display containing changezone.pct

In the validation process, all the Change Zone variables (for example, dispdb.[%cz_enty].sp, dispdb.[%cz_enty].mode) are assigned to the CZE collection group for optimum Change Zone performance. The update rate of all variables in the CZE group is set to one second. Adding other variables to the CZE group may be detrimental to Change Zone performance.

Changing the data collection update rate for associated variables

You can set the collection rate to 0 for \$almclr, \$ealmclr, \$halmclr, \$lalmclr, \$lng_tag, and \$redyel, since these are relatively static variables. An NCF change is required to change the value of any of those variables.

Consult the TPS Network Bookset *Actors Manual* for related information on TPS Network Picture Editor Change Zones.

Standard Change Zone Displays

PM/APM/HPM Analog Input point

56FI7001
T/H
CHARGE C4 -> C701

PU SP

PTXST
ACTIVE

CLR

Fieldbus Analog Input point

AC\2APM_63
_A1_01_01
A0_03_01
TAIL TO MOUTH SIMULATION

AUTO MAN
AUTO CAS

PV SP
50.000

PTXST
ACTIVE

CLR

PM/APM/HPM Analog Output point

NIMAO
bspermin
ED_S GOOD DEAL

OPERATOR MAN

OP
-6.9% ACTIVE

PTXST

CLR

Fieldbus Analog Output point

AC\2APM_63
_A1_01_01
A0_03_01
TAIL TO MOUTH SIMULATION

AUTO MAN
AUTO CAS

PV SP
50.000

PTXST
ACTIVE

CLR

PM/APM/HPM Regulatory Control point

TC100
DEG F
TEMP CONTRLER/ASM TEST

OPERATOR MAN

PU SP OP
77.0 33.0% ACTIVE

PTXST

CLR

14. OLE Objects

14.4. Change Zones

Fieldbus Regulatory Control point



PM/APM/HPM Device Control point



PM/APM/HPM Digital Input point



PM/APM/HPM Digital Output point



PM/APM/HPM Digital Composite point



PM/APM/HPM Regulatory PV point

56MI7001
0-1000
PM PV DATAACQ ALGORITHM

PU

SP

-----1.0

PTXST

ACTIVE

CLR

PM/APM/HPM Logic point

M5LS01

LOGIC SLOT TEST POINT 1

PTXST

ACTIVE

CLR

PM/APM/HPM Timer point

FS\L63TM01
0.01SEC
LSTD, LM TIMER POINT

START

STOP

RESET

DISABLE

PV

SV

231424.1

CLR

PM/APM/HPM Process Module point

MMIL T
ГМВБСНОД

SEQUENCE

OLE

IF

UNIO

ОБЕБ

EXEC

MODE

DMVZE

ЗЛЕБ

СГБ

PM/APM/HPM Numeric point

56C703DECL

ETAT DECL./INDIS. C703

PU

0

CLR

14. OLE Objects

14.4. Change Zones

PM/APM/HPM Flag point

MSG_FLAG	OPERATOR	MAN	PV	PTXST	CLR
ALARM MESSAGE TESTNG/ASM			ALARM OK	!!!!!!	

HG Flag point

FS\MC_FLAG	PV	CLR
	UPPER LOWER	

HG Numeric point

HGNUMBER	PV	CLR
NUMERIC	40	

HG Process Module point

MSHGTIME	SEQUENCE	OPER	EXEC	MODE	PHASE	STEP	CLR
UNIT 1		OFF	NL	AUTO			

HG Analog Composite point

FS\B0PC701	PROGRAM	MAN	PV	SP	OP	CLR
ANALOG COMP			-7	-6.9	40.0%	

HG Analog Input point

FS\B1PI1

PIU AI POINT TYPE

PU

65.0

SP

65.0

CLR

HG Analog Output point

FS\B1PO1

PIU AO POINT TYPE

OPERATOR

MAN

OP

RRRRRRR%

CLR

HG Logic Block point

FS\MC_LOG_BLK

LOGIC BLOCK TEST

PV

UPPER

LOWER

CLR

HG Control Counter point

FS\MC_CCNT

CCCCCCCCCCCCCCCCCCCCCCCCCCCC

CLR

HG Digital Composite point

FS\B1DC1

PIU DIGITAL COMPOSITE

OPERATOR

MAN

PV

LOWER

OP

LOWER

CLR

14. OLE Objects

14.4. Change Zones

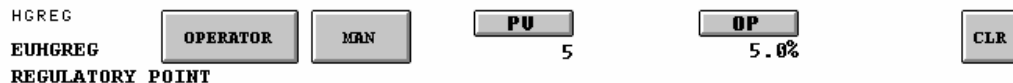
HG Digital Output point



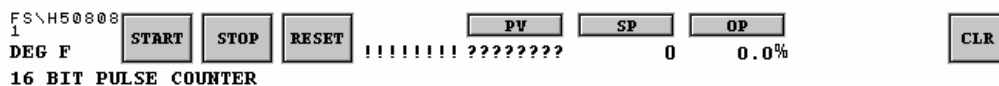
HG Digital Input point



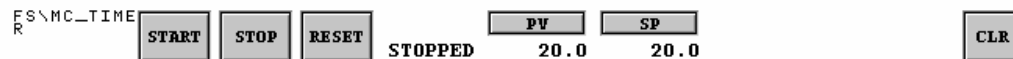
HG Regulatory Control point



HG Counter point



HG Timer point



AM Timer point

MSTIMER

START

STOP

RESET

STOPPED

PV

0.0

SP

25.0

PTXST

INACTI

CLR

AM Counter point

AMCOUNT1

START

STOP

RESET

STOPPED

PV

0.0

SP

35.0

OP

0.0%

PTXST

INACTI

PPS

CLR

eudesc

GUS CHANGE ZONE TEST PIC

AM Switch point

MSSWITCH

PHASE 1

PTXST

ACTIVE

PPS

CLR

GUS CHANGE ZONE TEST PIC

AM Custom point

THE_AM

PTXST

ACTIVE

PPS

CLR

LINKS CL CODE

AM Regulatory Control point

FC40

OPERATOR

MAN

PU

SP

77.0

OP

77.0%

PTXST

INACTI

PPS

CLR

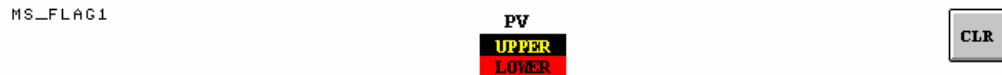
NM3/HR

GUS CHANGE ZONE TEST PIC

14. OLE Objects

14.4. Change Zones

AM Flag point



AM Numeric point



AM switch point Change Zone: Native Window vs. GUS display

An AM switch point contains up to five Self-Defined Enumeration states.

Native Window Change Zone:

All the possible states for an AM switch point are drawn as targets. The current state (S1CURSTS) is drawn with a yellow background. The requested state (S1REQSTS) is drawn in cyan. Any other states are drawn in green.



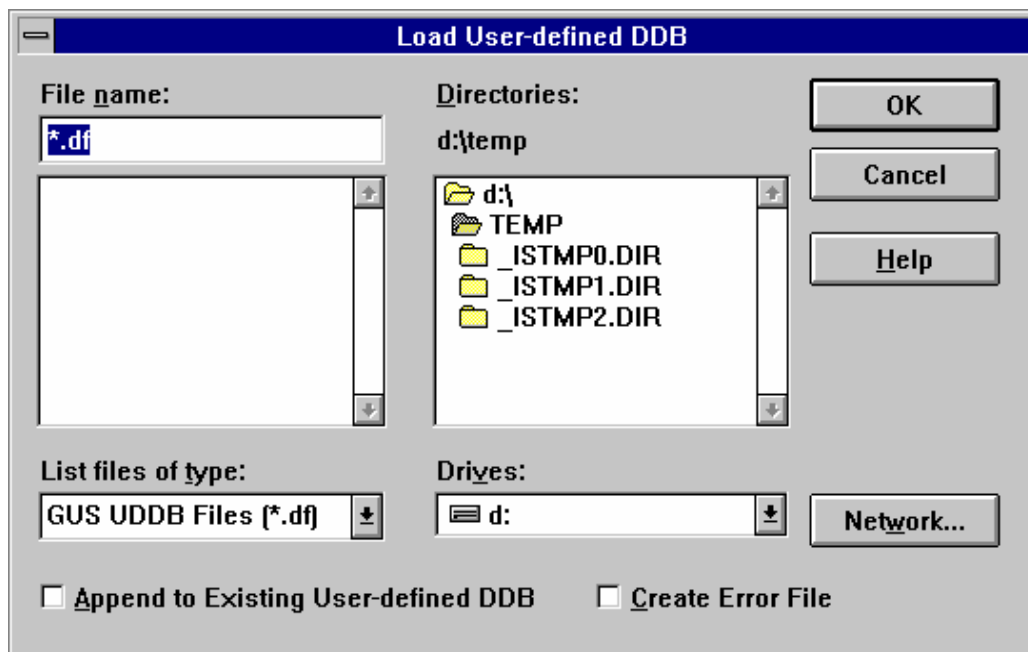
To change the requested state, select any of the states drawn in green and press ENTER.

14. OLE Objects

14.5. Loading a User-Defined DDB

14.5 Loading a User-Defined DDB

You can load a user-defined DDB from the **Display/Load User-defined DDB** menu item. The following dialog box appears:



Use the dialog box to navigate and find your user-defined DDB containing a ".df" extension.

All of the Syntax Rules found in the *LCN Actors Manual*, Section A3 *User-Defined DDB Variables*, are applicable with the following enhancements.

- There is an additional overwrite option for modifying a previous DDB declaration statement that contains name duplication. The default value of this option is “False” or “F.”

NOTE: This overwrite option cannot be used to duplicate either names in the DDB file or standard DDB names.

- There is no “maximum number” of DDB variable names per file.
- There is no “memory limitation” for User-Defined global DDB Variables.

At runtime, all of these user-defined DDB elements are accessible the same way as standard DDBs, except they are not automatically initialized. For this reason, users must first initialize a user-defined DDB before reading a value from it.

Create Error File: If this checkbox is selected, an error file with the same path and filename as the “.df” file, but with file extension “.err” will be created. Information about errors encountered during processing of the “.df” file will be placed in the error file.

14. OLE Objects

14.5. Loading a User-Defined DDB

15. Displaying System Data in Displays

15.1 Overview

GUS System displays are used to monitor and control the customer's process or environment. GUS enables customers to monitor their systems by allowing real-time and historical data to be shown in GUS displays.

Real-time data in displays

Real-time data can be represented in GUS displays in many ways. These techniques, discussed below, include the following:

- Showing Real-time System Data in Textual Form
- Showing Real-time System Data in Graphical Form
- Showing Historical Data in Displays

15.2 Showing Real-time System Data in Textual Form

The primary method for showing textual system data in displays involves the use of text objects. Text objects can be added to displays by using the text object button on the drawing toolbar. Once added, the text can be sized and properties such as font, color, etc. can be set. Creation of the link between a text drawing object and system data can be achieved in the following ways:

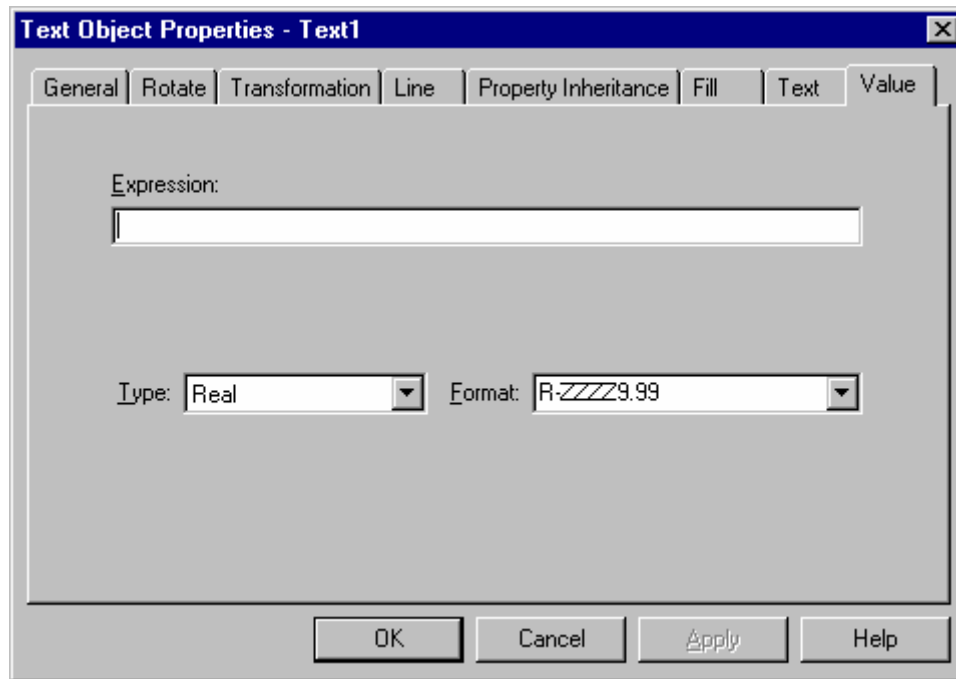
- Using the "Value" page in the Text Object Property Sheet
- Writing a Script

15. Displaying System Data in Displays

15.2. Showing Real-time System Data in Textual Form

Value Page

Dynamic value is expressed with the Value Page in the Text Object Property Sheet, shown here.



This page consists of an expression that includes the system data reference and formatting information that governs how the data is displayed.

The expression can be any expression that evaluates into a valid TPS GUS value format. In its simplest form this would simply be a point value reference such as LCN.Valve1.PV.

The formatting selections include a data type and display format used to convert and format the data for display. These format options are discussed in Appendix A of the *TPS GUS Picture Editor Reference Manual*.

Script method

Another method for displaying textual data involves writing a script. For more complex expressions and computations, you can create a script that will output the system data directly to the “Text” property of a text object in a display.

This script can be executed based on a script event such as a mouse button click or when the data referenced changes. In its simplest form the script might look like

```
Sub OnDataChange()  
  
    Text1.Text = LCN.TEMP1.PV  
  
End Sub
```

15.3 Showing Real-time System Data in Graphical Form

There are many ways to represent system data graphically in a GUS display. Some of these methods, discussed below, include:

- Changing object properties through scripts
- Using built in object dynamics
- Showing data by using graphical third-party OLE controls

There is an order for graphical transformations. The fixed order for applying graphical transformations to objects is as follows:

- 1 Scale
- 2 Rotate
- 3 Translate

Be aware of this order when applying multiple transformations, or dynamics, to objects.

Changing object properties through scripts

GUS displays contain both simple and complex objects (that is, groups or embedded displays). All objects have properties that can be modified by scripts at runtime based on system data.

Some system data values are measurements, conditions, and equipment states. Examples of object properties that can be used to reflect system values include color and blink.

15. Displaying System Data in Displays

15.3. Showing Real-time System Data in Graphical Form

A complete list of properties for a given object can be obtained by using the script “Properties” window. This is a simple example that uses color to represent the status of a valve.

```
Sub OnDataChange()  
    If LCN.Valve1.PV = "OPEN" = Then  
        me.FillColor = TDC_GREEN  
    Else  
        me.FillColor = TDC_RED  
    End If  
End Sub
```

NOTE: In this example TDC_GREEN AND TDC_RED are built-in constants.

Current rotation and direction

Current rotation and direction attributes are present in both fixed and dynamic states.

When No Dynamics are Present—Direction refers to the angle from the object’s creation point.

When Dynamics are Present—Direction refers to the direction imposed by the dynamic expression.

Using built-in object dynamics

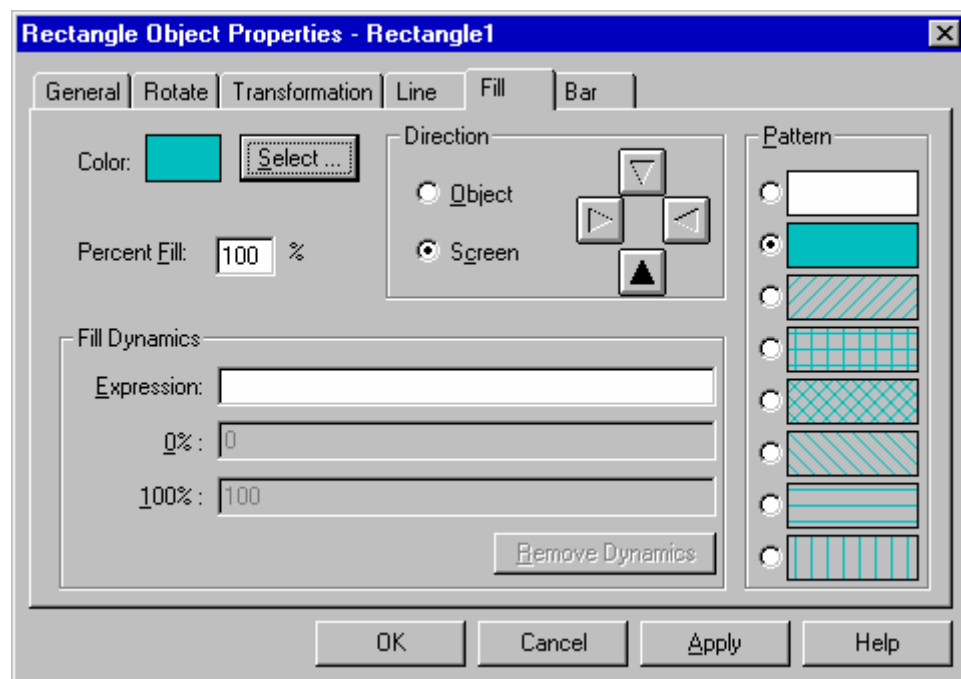
GUS objects provide some built-in dynamics options to allow simple graphical data representation without the need for scripting. These options include

- Dynamic Fill
- Dynamic Rotation
- Dynamic Bar
- Showing Data by using graphical third-party OLE controls

Dynamic Fill

You can use an object's fill level dynamic property to reflect the direct or manipulated value of system data in displays. A typical application can be to graphically represent the level of liquid in a container.

This dynamic capability is available for most drawing objects. The Fill property page for setting up this relationship is shown here:



To create a dynamic relationship, enter the fields described below.

Expression—This is the specification of the system data or controlling variable represented by the fill level. This expression must contain a system data reference. In the simplest form it might just be a direct point reference such as LCN.TANKLEVL.PV.

0%—This can be a constant or expression that represents the value of the controlling variable for which the fill level will be 0% filled. Any value less than this will also result in 0% filled.

15. Displaying System Data in Displays

15.3. Showing Real-time System Data in Graphical Form

100%—This can be a constant or an expression that represents the value of the controlling variable for which the fill level will be 100% filled. Any value greater than this will also result in 100% filled.

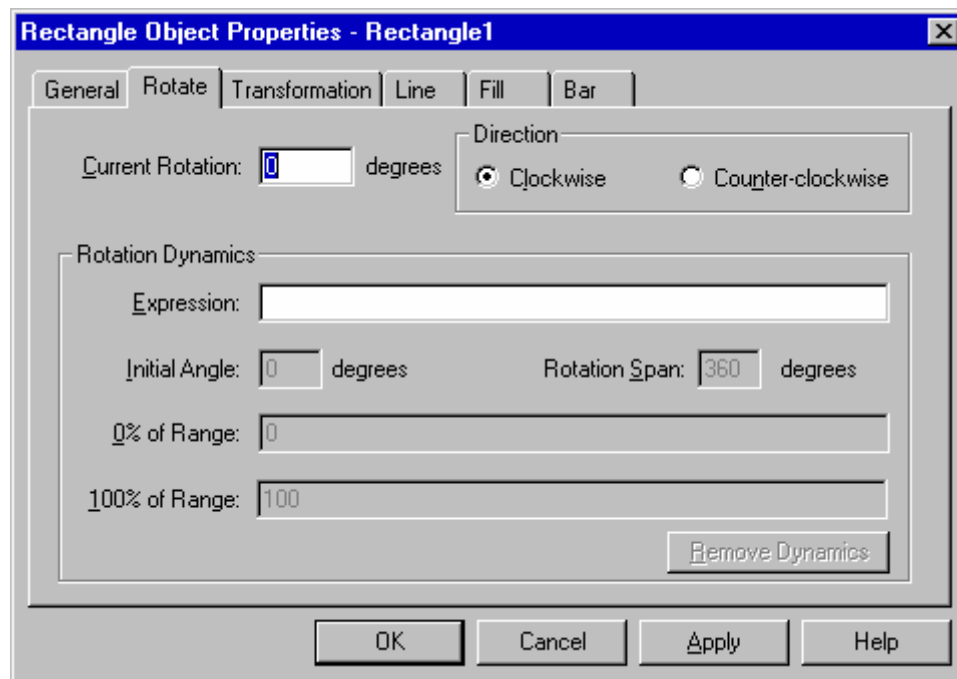
Note that, at runtime, as the controlling variable travels between the 0% and 100%, the value of the fill level is adjusted proportionally.

In addition to this dynamic relationship, you can also make appropriate selections for fill color, fill pattern, fill direction, and fill orientation.

Dynamic Rotation

You can use object rotation to reflect the direct or manipulated value of system data in displays. A typical application for this is to graphically represent the position of a valve or a damper.

This dynamic capability is available for all display-object types. The Rotate property page for setting up this relationship is shown here:



To create a dynamic relationship, enter the fields described below.

Expression—This is the specification of the system data or controlling variable represented by the rotation angle. This expression must contain a system data reference. In the simplest form it might just be a direct point reference such as LCN.VALVE1.PV.

Initial Angle—This is the angle that the object has when the controlling variable indicates minimum rotation. Typically this would be equal to the Current Rotation, i.e., the angle drawn.

Rotation Span—This is the degrees offset from the initial angle when the controlling variable indicates maximum rotation.

0% of Range—This can be a constant or expression that represents the value of the controlling variable for which the rotation will be 0 degrees from the initial angle or in other words at the initial angle specified. Any value less than this will also have the same result.

100% of Range—This can be a constant or an expression that represents the value of the controlling variable for which the rotation will be 0 at the maximum rotation angle. Any value greater than this will also have the same result. The maximum rotation angle is computed by rotating the object in the specified direction of rotation the degrees specified as Rotation Span.

Note that at runtime, as the controlling variable travels between 0% and 100%, the value of the rotation angle is adjusted proportionally.

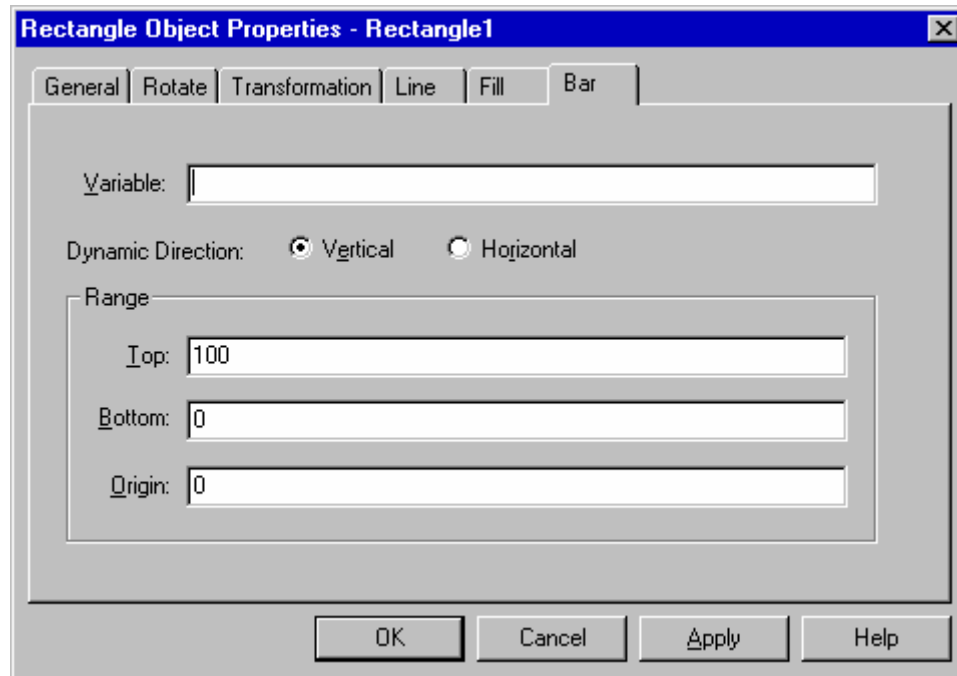
Dynamic Bar

You can use the length of a bar to reflect the direct or manipulated value of system data in displays. A typical application for this is to show the relationship of multiple variables in a bar graph format.

15. Displaying System Data in Displays

15.3. Showing Real-time System Data in Graphical Form

This capability is available by using the rectangle object. The property page for setting up this relationship is shown here:



The image shows a dialog box titled "Rectangle Object Properties - Rectangle1". It has a tabbed interface with tabs for "General", "Rotate", "Transformation", "Line", "Fill", and "Bar". The "Bar" tab is currently selected. Inside the dialog, there is a "Variable:" text box. Below it, the "Dynamic Direction:" section has two radio buttons: "Vertical" (which is selected) and "Horizontal". A "Range" section is also present, containing three text boxes: "Top:" with the value "100", "Bottom:" with the value "0", and "Origin:" with the value "0". At the bottom of the dialog are four buttons: "OK", "Cancel", "Apply", and "Help".

To create a dynamic relationship, enter the fields described below.

Variable—This is the specification of the system data or controlling variable represented by the bar. This expression must contain a system data reference. In the simplest form it might just be a direct point reference such as LCN.TEMP25.PV.

Dynamic Direction—This field indicates whether the bar expands vertically or horizontally.

Top/Right—This can be a constant or an expression that represents the value of the controlling variable for which the bar will extend from the origin position to the Top/Right maximum position. Any value greater than this will also have the same result.

Bottom/Left—This can be a constant or an expression that represents the value of the controlling variable for which the bar will extend from the origin position to the Bottom/Left position. Any value less than this will also have the same result. In the case of a simple bar (that is, one that grows in only one direction as the value of the variable increases) this would be the value of the controlling variable for which the bar has zero length.

Origin—This is the value of the controlling variable for which the bar has zero length. Typically it would be equal to the Bottom/Left value but it can also be equal to the Top/Right (for a bar that grows down or to the left) or somewhere in between (the case for a bar that could grow in either direction).

Notice that, at runtime, as the controlling variable travels between the Bottom/Left and Top/Right values the bar length is adjusted proportionally.

The maximum length of the bar is the length of the rectangle that was initially drawn on the display.

Showing Data by using graphical third-party OLE controls

Because GUS is an OLE container, you can embed a variety of third-party OLE controls into displays. These controls offer several graphical options for displaying system data.

OLE controls available through third-party vendors include such things as gauges and graphs. The relationship between system data and these OLE controls require the creation of a script.

Typically, these controls have properties such as “Value” that can be set to represent the required system data item. An example of what this script might look like is

```
Sub OnDataChange( )  
    GAUGE1.Value = LCN.TEMP1.PV  
End Sub
```

15. Displaying System Data in Displays

15.4. Update Display

15.4 Update Display

This function causes the display to be refreshed immediately. All objects that have changed in appearance since the display was last updated will be redrawn.

This function is particularly useful in subroutines responding to UI events that modify display properties prior to showing a message box or prompt. A call to UpdateDisplay prior to the message box or prompt function will cause the modified properties to be drawn.

For example:

```
Sub OnLButtonDown()  
    me.textcolor = TDC_RED  
    'cause the color of me to be updated now  
    UpdateDisplay  
    msgbox "Display color has been updated."  
End Sub
```

The runtime drawing model of GUS displays is to refresh the display during idle time or immediately after a script handling a user interface event runs to completion. Idle time occurs when no scripts are running. If no idle time has occurred within a configurable period of time (the default is 3 seconds), a display refresh is forced.

UpdateDisplay is used to override the runtime drawing model, to force the display to update during non-idle time.



ATTENTION

UpdateDisplay should be used sparingly because it can negatively impact performance of the display.

15.5 Showing Historical Data in Displays

GUS displays can contain historical data as well as real-time data. This data is available through the use of the Trend OLE control provided by Honeywell. This control shows data collected and saved over time in a graphical plot format. This historical data is available from several sources. These sources include

- Real-time Data
- History Module
- Hiway Gateway

Configuring a trend trace

Several properties are key to defining the layout of the trend trace data within its graph. These are YscaleHigh, YscaleLow, RangeHigh, RangeLow and the TimeBase. YscaleHigh and YscaleLow apply globally to all the trend traces within a Trend OLE Control. RangeHigh and RangeLow can be specified on a per-trace basis. The RangeHigh specifies the maximum value in engineering units that corresponds to the top position of the Y-Axis. The RangeLow specifies the minimum value in engineering units that corresponds to the bottom position of the Y-Axis. Since each trace can be defined with a unique set of RangeLow and RangeHigh limits, each trace can logically be viewed as having its own Y-Axis. Using these range limits, you can “zoom” the trace data in and out at runtime via scripting.

For all traces, the YscaleHigh and YscaleLow limits define the region of the graph that should be depicted on the Trend OLE Control display at any given time. The default values are full scale at YscaleHigh of 100 and YscaleLow as 0. Varying these limits enables the user to examine different portions of the graph. For example, assume Trace #1 has been defined with a RangeHigh of 60 and a RangeLow of 20, and the default scaling for YscaleHigh and YscaleLow. Since a YscaleHigh of 100 and a YscaleLow of 0 specifies full-scale or 100% of the graph, these defaults would assign 60 to the top position of the Y-Axis and 20 to the bottom position of the Y-Axis, with a delta of 40. If instead, the user wants to focus on just the top half of the graph, he would adjust the YscaleLow to 50. A YscaleHigh of 100 and a YscaleLow of 50 denotes the top 50% of the graph. Fifty percent of the delta of 40 is 20. Therefore, the new limits redefine 40 to be the bottom position of the Y-Axis, which is 60 minus the new delta of 20. The top position 60 remains unchanged. This new scaling would result in a display, which shows just the top half of the graph, but stretched into the entire display.

15. Displaying System Data in Displays

15.5. Showing Historical Data in Displays

Configuring an xy-plot trace

XY-plot trace configuration is similar to the Trend trace. In xy-plot, Trace #1 is used for plotting the X-axis trace. Note that in xy-plot if Trace #1 is deleted, no trace is plotted. When a trace is deleted in xy-plot, the trace remains empty until you configure a new value.

These properties are configured using the Trend Control Property Sheet. The Trace property page for setting up the system data relationship is:

The screenshot shows the 'Honeywell GUS Trend Control v4.0 (4.0) Properties' dialog box with the 'Trace' tab selected. The 'Trace Number' is 2 of 3. The 'Variable ID' is 'SINENUM.PV'. The 'Range High' is '100' and the 'Range Low' is '0'. The 'Data Source' is 'Real-Time'. The 'Trace Color' is a blue square with a 'Select...' button. On the right side, there are buttons for 'Previous Trace', 'Next Trace', 'Add Trace', and 'Delete Trace'. At the bottom are 'OK', 'Cancel', and 'Apply' buttons.

Trace Number 2 of 3	
Variable ID:	SINENUM.PV
Range High:	100
Range Low:	0
Data Source:	Real-Time
Trace Color:	Blue

To create a dynamic relationship, enter the fields described below.

Variable ID—This is the specification of the system data or controlling variable represented as a trace in the Trend display. This expression must contain a system data reference. In the simplest form it might be just a direct point reference such as TEMP25.PV. For the trend control, the data references use the TPS GUS variable ID format.

Range High and Range Low—For all values associated with a given Variable ID, these fields indicate the maximum and minimum values in engineering units.

Data Source—This field specifies the data source.

Trace Color—This field specifies the color of the trace.

In addition to the trace setup, the amount of data displayed can be set either at runtime or buildtime by using the Axes page. Other display aspects can be controlled by using the General page. Refer to Trend OLE Control Properties for more details.

15.6 Trend OLE Control Display Runtime

This section contains some runtime capabilities and restrictions for the Trend Control at Display Runtime.

When Trend Control initialization errors occur at display startup, the Trend Control's background will change to red, and a large black X will be displayed in place of trace data. Additionally, the event log will contain a description of possible causes for the initialization errors.

The maximum number of trends that can run at any one time on a GUS node is 16, and closing a running display that contains many trends can take several seconds. Therefore, if a user closes a display which contains many trends, while simultaneously opening a display which contains trends, there is a small window of opportunity for trend initialization errors to occur for the new display, and some of the trends will not be operational. The user is advised to close the display containing uninitialized trends and reopen the display.

The hairline cursor values displayed for real time data on the Trend Control may vary slightly for the same point.parameter. This variation is due to each trace being collected from the LCN data owner separately, regardless of whether the point.parameter is the same for two traces. In this case, there may be a slight difference in the value when it is collected independently from the data owner. History Module data will be the same for more than one of the same point.parameter on a Trend.

The user is not allowed to change the data source of any traces on a Trend while the Trend is scrolled back in time.

Trends will not run if the traces are automatic and HM collection is turned off (disabled).

Runtime plotting of trend trace data

Trace data is plotted against time on the X-Axis. Each value is plotted within the graph based on the specified scaling and ranges. As new data is received for a given trace, the existing data is scrolled from right to left so as to consistently maintain a maximum of one time base worth of data for all traces.

The difference between the newest value just received, and the TimeBase specified by the user is the current oldest value. The oldest value is always the leftmost point, and the newest value is always the rightmost point. Each receipt of new data for just one trace causes all active traces to be scrolled once the receipt is complete so that all traces have no data older in time than the currently-defined oldest value. In this way time consistency across all traces is guaranteed.

Trace values that exceed the limits of their Y-Axis appear to “leave” the graph. Trace values that are marked as invalid by the HOPCServer appear as “holes” along the trace line.

Runtime scrolling of Honeywell trend OLE controls

The Trend OLE Control provides a runtime scrolling capability. In GUS Release 100, you can request to scroll back in time by setting the “ScrollBackTime” property to a required number of seconds. After having scrolled backward in time, you can subsequently advance forward by setting “ScrollForwardTime” to a required number of seconds. The valid values for these two properties are not limited to the valid values for the time base. That is, it is legitimate to request to scroll back in time by three hours (that is, 10800 seconds), although 10800 is not a valid value for the “TimeBase” property. Positive values are expected for these two properties; a negative value for either property has no significance. Note that the scrolling capability does not apply to a trace with a data source of Real-time.

Subsequent to your request, you can verify the scrolling amounts provided by the HOPCServer by reading back the properties “ScrollBackTime” and/or “ScrollForwardTime.”

In a future release of GUS, selecting “Runtime Scrolling” from the General Property Page will result in the creation and management of scroll buttons to facilitate runtime scrolling.

Hairline cursor in Honeywell trend OLE controls

Since each active trace can be defined with a different set of YRangeHigh and YRangeLow limits, there can be up to eight unique logical Y-Axes within a given Trend OLE Control. (In contrast to the range limits, recall that the YscaleHigh and YscaleLow limits globally apply to all traces.) Therefore, in order to discern the values of Y at any given point in time, the Trend OLE Control provides you with a Hairline Cursor capability in runtime mode. To activate the hairline cursor, you must check “Hairline Cursor” on the General Property Page at build-time (or via scripting in runtime mode) and you must subsequently click the left mouse button in the trace plotting area of the Trend OLE Control at runtime. Whenever you select the hairline cursor, a vertical line is dropped at the selected point. At any time, you can re-select the hairline cursor to a different position on the Trend OLE Control. Any selection outside of the valid display of data will de-activate the hairline cursor.

If the value of the “HairlineReadout” property is TRUE (=1), the Y-values at the points of intersection for all traces will be displayed in a readout box at the top portion of the Trend OLE Control display. The Absolute Time of the intersection of the cursor with the X-Axis is also displayed. Optionally, you can read each distinct Y-value at the hairline cursor via the “YcursorReadout” property for any active trace.

- An “NA” value for any Y-value indicates that trace is currently not active.
- An “ERR” value indicates a problem in the internal calculation of a Y-value (such as, clicking at a position where real-time data has not yet arrived).
- A “STAT” value indicates a Y-value that has arrived from the HOPCServer has invalid status, and therefore cannot be plotted.

Note that, until you re-click, the hairline cursor remains fixed at the point where it was initially selected as its associated Y-values change in time. That is, the hairline cursor does not scroll from right to left with the data.

NOTE: In xy-plots, the hairline cursor appears only if “Linear Best-fit” in the xy-line is selected.

15.7 Change GUS Cursor Over Target Object

At GUS runtime, the shape of the cursor changes from an arrow to a cross-hair when the cursor is positioned over a target object. When the cursor is moved away from the target, the cursor changes to an arrow. If the cursor is stationary on an object, and if the target object changes to a non-target object or vice versa, the shape of the cursor changes accordingly. If the cursor is stationary and the object moves, the cursor changes accordingly. That is, the cursor shape should track the available mouse button events.

NOTE: The cursor shape does not change when it is placed over target objects such as Active X, DataEntry, ListBox, and ComboBox.

The following table lists the Button events affected by this function.

Item or Button Event	Cursor Shape	Notes
Non-selectable object or group		
Selectable object without script		
Selectable object or group that scripts any of the following events: OnLButtonDown, OnLButtonUp, OnLButtonClick, OnLButtonDoubleClick, OnRButtonDown, OnRButtonUp, OnRButtonDoubleClick.		If the object is grouped or is part of the embedded display, all parent objects must be selectable.

16. Issuing System Commands from Displays

16.1 Overview

GUS displays are used to monitor and control the customer's process or environment. GUS enables customers to control their systems by allowing them to modify system parameters directly from GUS displays. This can be done in a variety of ways.

- Using scripts written for any display object
- Using Honeywell OLE input controls and scripting
- Using third-party OLE controls
- Using Parameterized Displays

16.2 Using scripts written for any display object

All of the methods for modifying system parameters include a script line to assign a value to a system parameter. In the case of embedded displays this script can be hidden from the user. Examples of this script line might look like:

```
LCN.TEMP25.PV = 130
```

```
LCN.VALVE.PV = OPEN
```

NOTE: OPEN is a predefined constant

Scripts can be written for any screen object and the events that those objects support. An example of an event that can be scripted would be a mouse click.

16.3 Using Honeywell OLE input controls and scripting

Honeywell has provided a set of OLE controls designed to facilitate user input and selection of system parameter values. These controls, along with some simple scripting, provide users with several options for system control. These options include

- Typing in values for system parameters
- Selecting from lists of choices
- Using buttons to invoke actions

16. Issuing System Commands from Displays

16.3. Using Honeywell OLE input controls and scripting

The Honeywell OLE controls provided are:

- Data Entry Control
- Listbox Control
- ButtonPlus Control

Data entry control

The Data Entry control is provided for inputting typed values to control the system processes or parameters. This control looks like the following:



Like all GUS objects it is configured by using the object Property Sheet. The control is configured for the type of data to be input along with other options such as data entry timeout. Also in some cases this object can be made visible only based on some system conditions or user action. If this were the case, the visible property would not be selected.

At runtime this control accepts user input when the user presses the “Enter” key. If the user entry is successfully validated, the value the user entered is passed as the “sData” parameter to the “GoodEntry” event script. This value is also contained in the “Text” property of the data entry control. To transfer values to system parameters, a script would be written for the “GoodEntry” event. This script would look like:

```
Sub GoodEntry(sData As String)
    LCN.PARAM1.PV = sData
End Sub
```

Setting focus to a Data Entry Control

Let’s assume when a button on your display is clicked, you want keyboard focus moved to a Data Entry control named **OleEntry1** so the user can enter text.

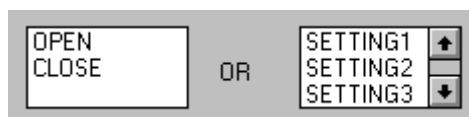
Place a script in the button object something like this:

```
Sub OnLButtonClick
    OleEntry1.SetFocus
End Sub
```

When the left mouse button is clicked, keyboard focus will be moved to the Data Entry control and an I-beam cursor will be inserted in the control box to indicate text can be entered.

Listbox control

The Listbox control is provided to allow the user to select from a list of valid choices. For discreet value cases where there are a number of selections, this is a good choice because it eliminates typing and the need to remember the selections. It also takes up a relatively small amount of space in the display. If there is not space in the display for the entire list the Listbox can be drawn smaller and vertical scrolling will be provided automatically. This control appears like this:



The Listbox is configured by using the object Property Sheet. The control is configured for list selections along with other options such as user entry timeout.

At runtime this control processes user input when the user clicks on a choice in the Listbox. The string the user selected is passed as the “sName” parameter to the SelectionChange event script and is also contained in the “SelectedString” property of the Listbox control. To transfer values to system parameters a script is written for the “SelectionChange” event that might look like

```
Sub SelectionChange(sName As String,nMemberId As Long)
    If sName = "OPEN" then
        LCN.PARAM1.PV = OPENVALUE
    else
        LCN.PARAM1.PV = CLOSEVALUE
    End if
End Sub
```

NOTE: OPENVALUE and CLOSEVALUE are predefined constants.

ButtonPlus control

The ButtonPlus control is provided to allow the use of custom labeled buttons to do a single action or to select from a list of valid choices. For discreet values with a small number of options this is a good choice because it eliminates typing, is easily selectable, and clearly shows the state of the process or equipment. An example of this control looks like this:



The ButtonPlus control is configured by using the object Property Sheet. The control is configured for number of buttons along with other options such as labels and whether the buttons are interlocked; that is, when one is selected all the others are not selected.

At runtime this control reacts when the user clicks on a button. The button that the user selected is accessed in the script as the variable “nIndex” where the 1st button has an nIndex value of 0 and the additional buttons in the case of multiple buttons would have nIndex values of 1, 2, 3, etc. To transfer values to system parameters when a button is pressed, a script would be written for the “ButtonOn” event. In the case of a single button that stays depressed (latched) when clicked, the user would most likely write scripts for both “ButtonOn” and “ButtonOff” events.

For the interlocked multibutton example shown above the user script might look like:

```
Sub ButtonOn(nIndex As Integer)
    If nIndex = 0 Then
        LCN.TEMP1.PV = HEAT
    Else
        If nIndex = 1 Then
            LCN.TEMP1.PV = COOL
        Else
            LCN.TEMP1.PV = AUTO
        End If
    End If
End Sub
```

In the above script HEAT, COOL, and AUTO are predefined constants.

16.4 Using third-party OLE controls

Because GUS is an OLE container the user can embed a variety of third-party OLE controls into displays. These controls offer a variety of graphical options for changing system data. OLE controls that are available through third-party vendors include such things as slider controls, custom shaped buttons, masked input controls, etc. The relationship between system data and these OLE controls would require the creation of a script. Typically these controls would have properties such as "Value" that could be read and assigned to a system data item when an event such as "OnUserChange" occurs. Consult the third-party documentation for details. This is an example of what a script using a third-party OLE control looks like:

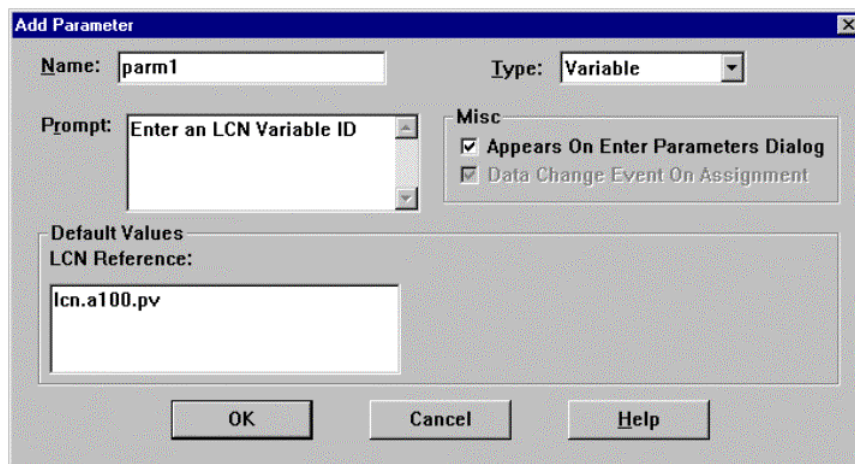
```
Sub OnUserChange( )
    LCN.TEMP1.PV = SLIDER1.Value
End Sub
```

16.5 Using Parameterized Displays

Parameterized Displays provide a way for users to create displays requiring user input without knowledge of the GUS scripting language. Using parameterized displays, a library of input options can be provided by combining the above OLE Controls with the scripts to do the actual system control. The system data item being controlled can be referenced as a parameter of the parameterized display. Thus when the parameterized display (for example a single data entry control and its associated script) is inserted into a display (becoming an embedded display) the user can simply add the system data reference.

The steps for creating this type of embedded display are:

- 1 Create a new display
- 2 Add the Data Entry OLE Control and configure it as described above.
- 3 Add a parameter to the new display. You can do this by using the following dialog box (Add Parameters), that you can invoke from the Define Parameters dialog box.



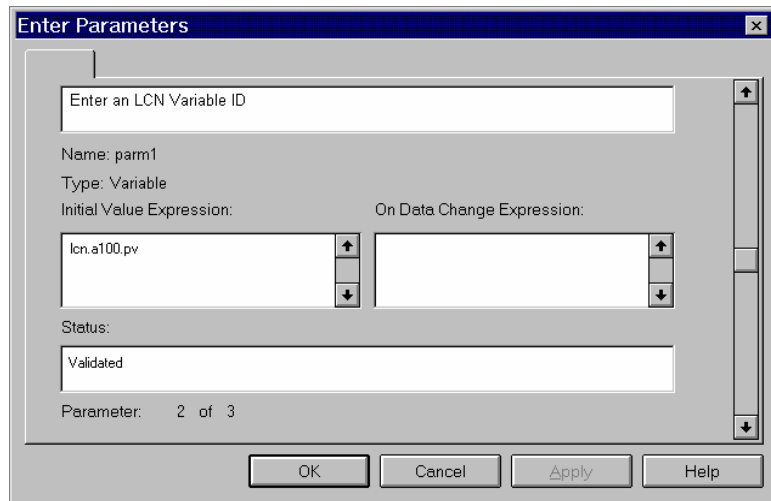
- 4 Refer to Adding Display Parameters for details.

- 5 Write the script for the control referencing the defined display parameter.
The script for a simple data entry control packaged as an embedded display would be as follows:

```
Sub GoodEntry(sData As String)
    display.params.parm1 = sData
```

```
End Sub
```

- 6 Save the display as one of a library of reusable displays.
- 7 Upon inserting the embedded display from the library, you can simply click the data entry control and, by using the "Enter Parameters" dialog box, enter the system data reference. This dialog box is shown below:



The "Enter Parameters" dialog box is a standard Windows-style window with a title bar and a close button. It contains a text input field at the top labeled "Enter an LCN Variable ID". Below this, the "Name" is set to "parm1" and the "Type" is "Variable". There are two main sections: "Initial Value Expression" and "On Data Change Expression". The "Initial Value Expression" field contains the text "lcn.a100.pv" and has up/down arrow buttons on its right. The "On Data Change Expression" field is empty and also has up/down arrow buttons on its right. Below these fields is a "Status:" label and a text box containing the word "Validated". At the bottom left, it says "Parameter: 2 of 3". At the bottom right, there are four buttons: "OK", "Cancel", "Apply", and "Help".

16.6 PMK (Point Manipulation Key) Object

The PMK Object provides users of GUS Basic the ability to ramp a point's SP or OP parameter up or down for analog data points based on discrete user-defined increments. In addition to ramping a point's parameter up or down, the PMK Object can send events to GUS display objects in response to method calls in scripting.

This description of the GUS scripting PMK Object includes:

- PMK Object properties
- PMK Object methods
- PMK Object scripting examples in GUS Basic
- IKB integration of point manipulation keys with GUS Basic

What it does

The PMK is a built-in GUS scripting object that implements the point manipulation keys found on the IKB and other Honeywell engineering keyboards. GUS scripts interact directly with the PMK Object through the invocation of methods (for example, PMK.Clear) or by setting properties.

Since the PMK Object is a built-in object in GUS displays it does not need to be declared (or DIMed in GUS Basic) by the user. There exists exactly one PMK object per GUS display, and it is global in scope. For example, an embedded display and the main display both use the same PMK object without modifications to scripts.

How it works

Once the PMK has a registered entity and a registered parameter, all point manipulation keys from the IKB will result in changes to the value of the registered entity if the AutoWrite property is set to TRUE. If the AutoWrite property is set to FALSE, changes will not be made to the registered data point. In either case, scriptable events will be fired when PMK keys are pressed.

Point manipulation keys can be emulated for PC keyboard users through GUS scripts that invoke methods on the PMK object (for example, PMK.Raise). Primitive objects in Display Builder can script responses to PMK events raised when point manipulation keys are pressed on the keyboard. TPS Network events such as OnRaise will still be generated for objects when point manipulation keys are pressed.

Keyboard Special Considerations

There are two areas where keyboard responses must be fully understood.

- **Differences in RAISE requests**—Holding down a pre-configured RAISE key on the IKB will send a continuous stream of RAISE messages to the PMK object. If the user scripted a configurable key on the IKB to cause a raise (through a QUE_KEY actor or some other means), the script would only be executed once when the key is released.
- **Differences between “Enter” keys**—The “Enter” keys on an Engineers Keyboard and an IKB are mapped to the Carriage_Return key code. No PMK manipulation is possible using these keys. The “ENTER” key on a Universal Station Operator Keyboard is mapped to OnPMKEnter, which, enables PMK manipulation.

Registering the PMK object

Once an entity has been registered with the PMK scripting object, the user can specify a parameter (SP or OP) to register and ramp the data point through the keyboard. The following script shows one method of registering the PMK object with the SP parameter.

```
Sub OnGotFocus()  
    Set PMK.Entity = LCN.A100  
    PMK.Key PMK_SP  
End Sub
```

Now that the PMK object has been linked to A100, future ramping will increment/decrement the entity A100. Since the PMK scripting object does not care about input focus, end users must take precautions not to ramp values for other PMK implementations in the same display.

Incorrect registration

Here is an example of a script that allows the user to inadvertently ramp the wrong value.

In this example, the user has two points to ramp, A100 and B100. The picture has two text objects: text1 registers the PMK object with A100 and text2 does nothing.

16. Issuing System Commands from Displays

16.6. PMK (Point Manipulation Key) Object

The script behind text1 is:

```
Sub OnLButtonClick()  
    'Register the A100 Entity  
    Set PMK.Entity = LCN.A100  
    'Tell the PMK to ramp the SP  
    PMK.Key PMK_SP  
End Sub
```

At first glance, this might seem like a good way to write a script for the PMK object. A possible problem comes into play with the following scenario:

The operator displays the picture. The operator left clicks on the first text object (text1) and starts pressing the raise key on the keyboard. The SP value for A100 begins to ramp as expected. Now the user clicks on the another text object (text2) and presses the raise key on the keyboard. The operator might expect the data point for B100.SP to begin increasing, however the A100.SP data point increases instead.

The problem is that the PMK object is global in scope. Although the operator left the scope of text1 one by setting focus to text2, the global PMK object still has A100 registered.

Correct registration

Here is an example of a script that protects the user from ramping the wrong value.

```
Sub OnGotFocus()  
    Set PMK.Entity = LCN.A100  
    PMK.Key PMK_SP  
End Sub  
Sub OnLostFocus()  
    PMK.Clear  
End Sub
```

Now the PMK unregisters the entity A100 when the object text1 loses focus. When the operator selects text2 and hits the Raise key nothing will happen, as A100 is no longer registered.

For more examples of PMK Object registration, refer to PMK Scenarios and Examples.

PMK events

Each time the operator hits a PMK on the keyboard, an equivalent PMK event can be raised. The event message is sent to the PMK Event Handler if the main display is the active window and when the key pressed was successful in operation.

For example, if the user hits the Raise key prior to setting a valid Entity, an error will occur and a Raise event will not be generated.

The Event Handler can script actions that occur when the operator presses a PMK.

For example, an engineer wanting to script a rectangle to rotate when the raise key was pressed on the keyboard could add the following script on a rectangle object.

```
Sub OnDisplayStartup()  
    Set PMK.EventHandler = me  
End Sub  
Sub OnPMKRaise()  
    'Rotate the rectangle when the raise key is hit  
    me.Angle = me.Angle + 10  
End Sub
```

The Event Handler is set to the rectangle object at startup. No matter what picture object has input focus when the raise key is pressed, the rectangle will always get the raise event for this display. Since the rectangle will get all PMK events regardless of what has input focus in the display, the Event Handler is considered global in scope for the active display.

PMK events can be emulated in scripting by the KEY method on the PMK object.

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16.6. PMK (Point Manipulation Key) Object

PMK error handling

PMK error handling is similar to PMK event handling. Like the Event Handler, the PMK object has an Error Handler. The engineer can create a script to act when a PMK error occurs at runtime. When an error occurs at runtime, an OnPMKError event is raised for the registered error handler object. The OnPMKError event passes in two parameters, the ErrCode and the ErrString. The ErrCode is a predefined numeric constant in GUS scripting. The ErrString contains information about the cause of the error. Values for ErrCode are:

ErrCode	Cause / Description
PMKERR_NOENTITY	A valid entity must be registered with the PMK before ramping or changing the mode.
PMKERR_NOPARAMETER	A parameter (SP or OUT) must be specified before ramping or changing the mode.
PMKERR_BADENTITY	An attempt to register an invalid entity to the PMK has occurred.
PMKERR_BADEVENTHANDLER	An attempt to register an invalid event handler to the PMK has occurred.
PMKERR_RAMPFALIED	Ramping the PMK object failed.
PMKERR_KEYFALIED	The PMK Key method failed.
PMKERR_MODEFALIED	The PMK object could not change the entities mode.
PMKERR_UNKNOWN	The PMK raised an unknown error.

What is raised/lowered

Not all data points can be raised/lowered by the PMK object. Only points that are of type real can be raised. In other words digital points, flag points, switches and logic points can not be raised/lowered by the PMK object. Regulatory, Analog, Process Module and Counter points can be raised/lowered by the PMK object if they have at least one of the following parameter types: OP, SP, PVTV and AVTV.

All real data points raise/lower the OP parameter when the OUT button is pressed.

16. Issuing System Commands from Displays

16.6. PMK (Point Manipulation Key) Object

Most real data points raise/lower the SP parameter when the SP button is pressed, with the following exceptions.

Entity Type	Parameter Raised/Lowered by SP button press
COUNTHG	None
COUNTAM	PVTV
ANINNIM	PVTV
REGPVNIM	AVTV if the pvalgid = "TOTALIZR", otherwise PVTV

Updating the display

The PMK object only raises events and writes new values to the TPS Network, it does not update the display. As a result, ramped values on the display will appear to be updated only on the next LCN scan cycle, not after the user presses a raise/lower key. The only exception is a direct reference to the PMK.Entity.param in a data change script or by setting the reference property for a text object to the PMK.Entity.param. Data change events are generated for each value change caused by the PMK.

An example of a data change script behind a text object that would have an immediate update for a PMK Raise/Lower is:

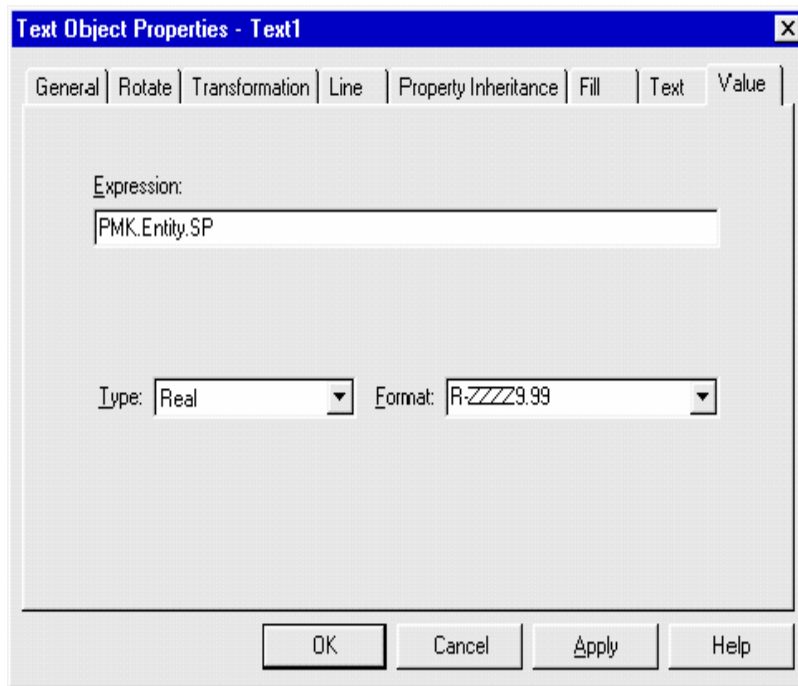
```
Sub OnDataChange()  
    me.text = PMK.Entity.SP  
End Sub
```

The data change event will only occur when the entity registered by the PMK object and the parameter registered by the PMK object change.

16. Issuing System Commands from Displays

16.6. PMK (Point Manipulation Key) Object

Alternatively, set the text object reference property as shown below:



Binding a reference on the PMK entity occurs when the entity is first set.

For example, suppose that the DispDB.Ent01 stores A100 as its entity. If the engineer sets the PMK Entity property to DispDB.Ent01 in a script, the PMK Entity will be bound to A100. If the engineer changes the value of DispDB.Ent01 to B100 later in the script, the PMK Entity will not change to B100, but will still reference A100.

16.7 PMK Object Properties

Property	Entity
Description	Registers an entity for future PMK ramping methods. A null entity will prevent any PMK ramping.
Syntax	[Set] PMK.Entity [=Entity] Entity is an object of type Entity Id.
Example	Set PMK.Entity = LCN.A100 Set PMK.Entity = DispDB.Ent01 Set PMK.Entity = GetEnt(askbox("enter entity ID")) Set PMK.Entity = LCN.A100.ent1 Set dispDB.Ent01 = PMK.Entity Text1.text = PMK.Entity.PV
Remarks	When the Entity is changed from a valid value to either a NULL value or another entity, an automatic journalization will occur if the point has been ramped and not previously journalized. Pressing a PMK on a NULL or invalid Entity will cause an OnPKMError event to occur on the Error Handler object.

Property	EventHandler
Description	Display object that will receive PMK events. A null value will block all PMK events. For more details on PMK events, refer to PMK Scriptable Events.
Syntax	[Set] PMK.EventHandler [=ObjectName] ObjectName is a GUS display object.
Example	Set PMK.EventHandler = text1 Set PMK.EventHandler = me PMK.EventHandler.FillColor = TDC_RED

16. Issuing System Commands from Displays

16.7. PMK Object Properties

Remarks	If the ObjectName passed into the EventHandler is not a valid object, the EventHandler will be NULL.
----------------	--

Property	ErrorHandler
-----------------	---------------------

Description	Display object that will receive OnPMKError events from the PMK object.
--------------------	---

For more details on PMK events, refer to

PMK Scriptable Events.

Syntax	[Set] PMK. ErrorHandler [=ObjectName]
---------------	---------------------------------------

ObjectName is a GUS display object.

Example	Set PMK. ErrorHandler = text1
----------------	-------------------------------

Set PMK. ErrorHandler = me

Remarks	If the ObjectName passed into the ErrorHandler is not a valid object, the ErrorHandler will be NULL. The ErrorHandler object can use the PMK.ErrCode and PMK.ErrString properties to determine why the OnPMKError event was fired.
----------------	--

Property	AutoUpdate
-----------------	-------------------

Description	When AutoUpdate is true, a relationship exists between the PMK entity property and the source entity name form from the assignment. The source entity name form is called the associated name form. During ramping, the PMK will use this relationship to do the following:
--------------------	---

- a) As ramp increments are completed, a data change event is fired for references to the ramped parameter via the associated name form. For more information, see the AutoDataChange property that follows.
- b) When a ramp operation is complete, the PMK will automatically perform an LCN.UPDATE for the group containing the ramped parameter reference through the associated name form.

Syntax	PMK. AutoUpdate [=Boolean]
---------------	----------------------------

16. Issuing System Commands from Displays

16.7. PMK Object Properties

Example PMK.AutoUpdate = TRUE

PMK.AutoDataChange = TRUE

Set PMK.entity = dispdb.[\$cz_enty]

Assume that the user is ramping the sp parameter. As ramping occurs, data change events for references to dispdb.[\$cz_enty].sp are fired by the PMK for each ramp increment. When the ramp operation is over, the PMK does an LCN.UPDATE method for the group containing dispdb.[\$cz_enty].sp.

Remarks The default value for AutoUpdate is True.

Property	AutoDataChange
----------	----------------

Description	When AutoDataChange is true, data change events are fired for references to the ramped parameter on the associated name form of the PMK entity as ramp increments are completed. Has effect only when AutoUpdate is TRUE.
--------------------	---

Syntax	PMK. AutoDataChange [=Boolean]
---------------	--------------------------------

Example	PMK.AutoUpdate = TRUE
	PMK.AutoDataChange = TRUE
	Set PMK.entity = dispdb.[\$cz_enty]
	Assume that the user is ramping the sp parameter. As ramping occurs, data change events for references to dispdb.[\$cz_enty].sp are fired by the PMK for each ramp increment. When the ramp operation is over, the PMK does an LCN.UPDATE method for the group containing dispdb.[\$cz_enty].sp.

Remarks	The default value for AutoDataChange is True.
----------------	---

16. Issuing System Commands from Displays

16.7. PMK Object Properties

Property	AutoWrite
Description	Allows the PMK to automatically change the value or mode for the registered entity. When TRUE, the PMK object can change the registered point's value or mode based on keyboard and scripting interaction. When FALSE, the PMK object cannot change point values or modes for the registered entity.
Syntax	[PMK. AutoWrite [=Boolean] Boolean. Determines if automatic method invocation will occur for the PMK object.
Example	PMK. AutoWrite = FALSE
Remarks	All PMK raise events are raised if AutoWrite is set to FALSE.

Property	EnableMode
Description	Allows the PMK to change the registered entity's mode when the MAN, AUTO or NORM keys are pressed.
Syntax	[PMK. AutoWrite [=Boolean] Boolean. Determines if mode changes will occur for the registered PMK entity.
Example	PMK. EnableMode = FALSE
Remarks	All PMK events are still raised regardless of the value. By default this value is set to TRUE.

Property	UserData
Description	Stores a Variant data type for use in scripting.
Syntax	[Set] [PMK. AutoWrite [=Variant] Variant. Data that can be used for scripting.
Example	PMK.UserData = "Hello World" Set PMK.UserData = Rectangle1
Remarks	This property stores data for later use in scripts.

16.8 PMK Object Functions and Methods

Method	Clear								
Parameters	None								
Description	Sets the following default properties on the PMK object: <table> <tr> <td>Entity</td><td>NULL</td></tr> <tr> <td>EventHandler</td><td>NULL</td></tr> <tr> <td>ObjectHandler</td><td>NULL</td></tr> <tr> <td>AutoWrite</td><td>TRUE</td></tr> </table>	Entity	NULL	EventHandler	NULL	ObjectHandler	NULL	AutoWrite	TRUE
Entity	NULL								
EventHandler	NULL								
ObjectHandler	NULL								
AutoWrite	TRUE								
Syntax	PMK.Clear								
Example	Calling the Clear method will implicitly deactivate the PMK object.								
Remarks									

Function	FastRaise/FastLower
Parameters	None
Return Type	Boolean
Description	Invokes a discrete (single) FastRaise/FastLower method on the PMK Object. This function will fail (return FALSE) if an invalid entity or point parameter exists or the PMK is deactivated.
Example	PMK.FastRaise

Function	Raise/Lower
Parameters	None
Return type	Boolean
Description	Invokes a discrete (single) Raise/Lower method on the PMK Object. This function will fail (return FALSE) if an invalid entity or point parameter exists or the PMK is deactivated.
Example	PMK.Raise

16. Issuing System Commands from Displays

16.8. PMK Object Functions and Methods

Method	Key
Parameters	Integer
Description	Emulates a point manipulation key keyboard press Pre defined scripting Key Constants are: PMK_OUT PMK_SP PMK_RAISE PMK_LOWER PMK_FASTRAISE PMK_FASTLOWER PMK_MAN PMK_AUTO PMK_NORM PMK_ENTER
Example	PMK.Key PMK_LOWER

16.9 PMK Scriptable Events

Event	OnPMKS
GUS Objects	All
Description	Fired when the SP key is pressed and the PMK object is activated.

Event	OnPMKError
Parameters	ErrCode – <i>Long</i> ErrString – <i>String</i> ErrCode can be one of the Error Handling constants. ErrString is a textual description of the error.
GUS Objects	All
Description	Fired when a PMK error occurs. Only the object registered as the ErrorObject will receive this event.

Event	OnPMKChange
Parameters	ObjectType– <i>Integer</i> ObjectType specifies what has changed in the PMK. Possible constants are: PMK_ENTITY – The Entity has changed. PMK_ERRORHANDLER – The Error handler has changed. PMK_EVENTHANDLER – The Event handler has changed. PMK_ALL – All of the above items have changed. Called when a Clear method is invoked
GUS Objects	All
Description	Fired when a PMK error occurs. Only the object registered as the ErrorObject will receive this event.

16. Issuing System Commands from Displays

16.9. PMK Scriptable Events

Event	OnPMKOUT
GUS Objects	All
Description	Fired when the OUT key is pressed and the PMK object is Activated.
Event	OnPMKRaise
Parameters	Param – <i>String</i> Value – <i>Variant</i> Param is the parameter that was raised. Value is the new value of the parameter that was raised.
GUS Objects	All
Description	Fired when the Raise key is successful and the PMK object is Activated. When the Raise key is held down, the Raise event will be repeated in a periodic manner based on the KeyBoard type.
Remarks	Value can be used to update text objects in a display. Text1.Text = Value
Event	OnPMKLower
Parameters	Param – <i>String</i> Value – <i>Variant</i> Param is the parameter that was lowered. Value is the new value of the parameter that was lowered.
GUS Objects	All
Description	Fired when the Raise key is successful and the PMK object is Activated. When the Lower key is held down, the Lower event will be repeated in a periodic manner, based on the KeyBoard type.
Remarks	Value can be used to update text objects in a display. Text1.Text = Value
Event	OnPMKFastRaise

16. Issuing System Commands from Displays

16.9. PMK Scriptable Events

Parameters	Param – <i>String</i> Value – <i>Variant</i> Param is the parameter that was raised. Value is the new value of the parameter that was raised.
GUS Objects	All
Description	Fired when the FastRaise key is successful and the PMK object is Activated. When the FastRaise key is held down, the FastRaise event will be repeated in a periodic manner based on the KeyBoard type.
Remarks	Value can be used to update text objects in a display. Text1.Text = Value

Event	OnPMKFastLower
--------------	-----------------------

Parameters	Param – <i>String</i> Value – <i>Variant</i> Param is the parameter that was lowered. Value is the new value of the parameter that was lowered.
GUS Objects	All
Description	Fired when the FastLower key is successful and the PMK object is Activated. When the FastLower key is held down, the FastLower event will be repeated in a periodic manner based on the KeyBoard type.
Remarks	Value can be used to update text objects in a display. Text1.Text = Value

16. Issuing System Commands from Displays

16.9. PMK Scriptable Events

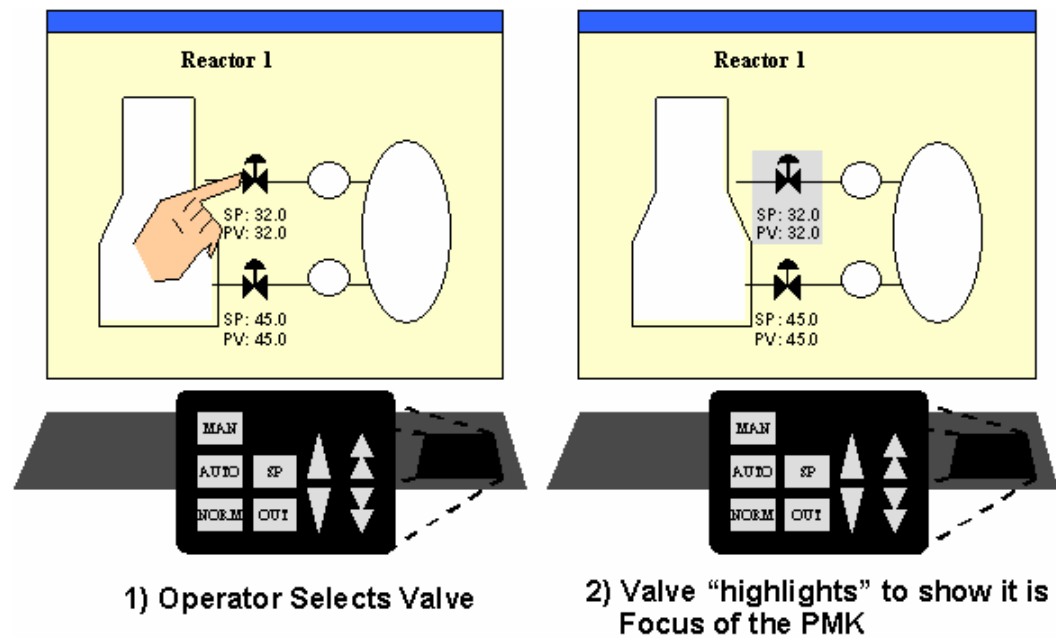
Event	OnPMKEnter
GUS Objects	All
Description	Fired when the Enter key is pressed and the PMK object is activated.
Prerequisites	The PMK object must be designated as an Event Handler. This is accomplished by executing the script <code>Set PMK.EventHandler = ObjectName</code> .
Event	OnPMKMAN
GUS Objects	All
Description	Fired when the MAN key is pressed and the PMK object is activated.
Event	OnPMKAUTO
GUS Objects	All
Description	Fired when the AUTO key is pressed and the PMK object is activated.
Event	OnPMKNORM
Parameters	NormMode – <i>String</i> Normal Mode specified the nMode parameter for the current entity.
GUS Objects	All
Description	Fired when the NORM key is pressed and the PMK object is activated.

16.10 PMK Scenarios and Examples

Example 1 – A Valve Embedded display

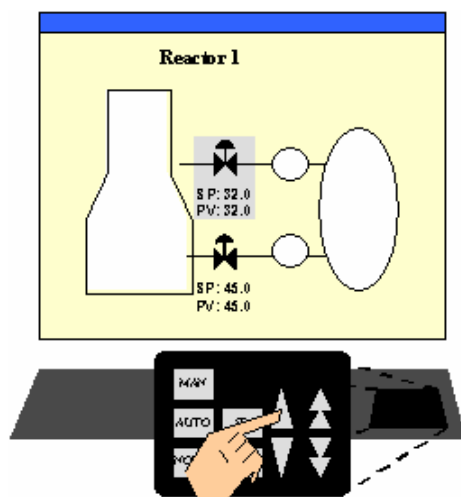
Behavior

The figures that follow illustrate the behavior of the Valve.PCT Embedded Display:

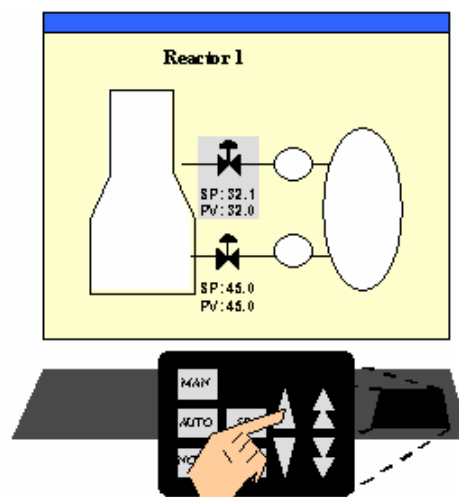


16. Issuing System Commands from Displays

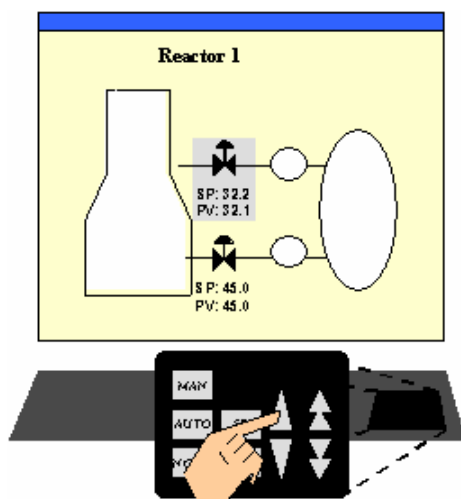
16.10. PMK Scenarios and Examples



3) Operator Ramps SP via PMK
(Valve was scripted to always
ramp SP)

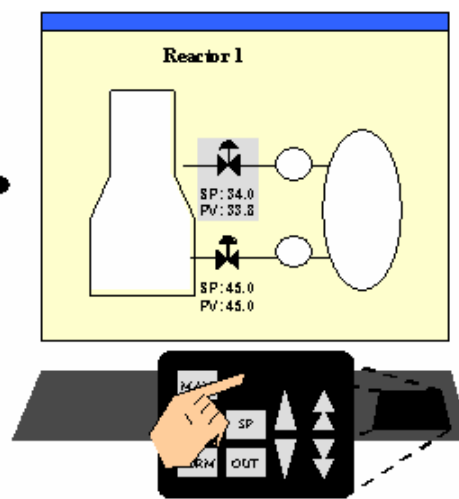


4) SP is ramped per the NCF configuration,
and at a rate of one increment per 0.7 secs

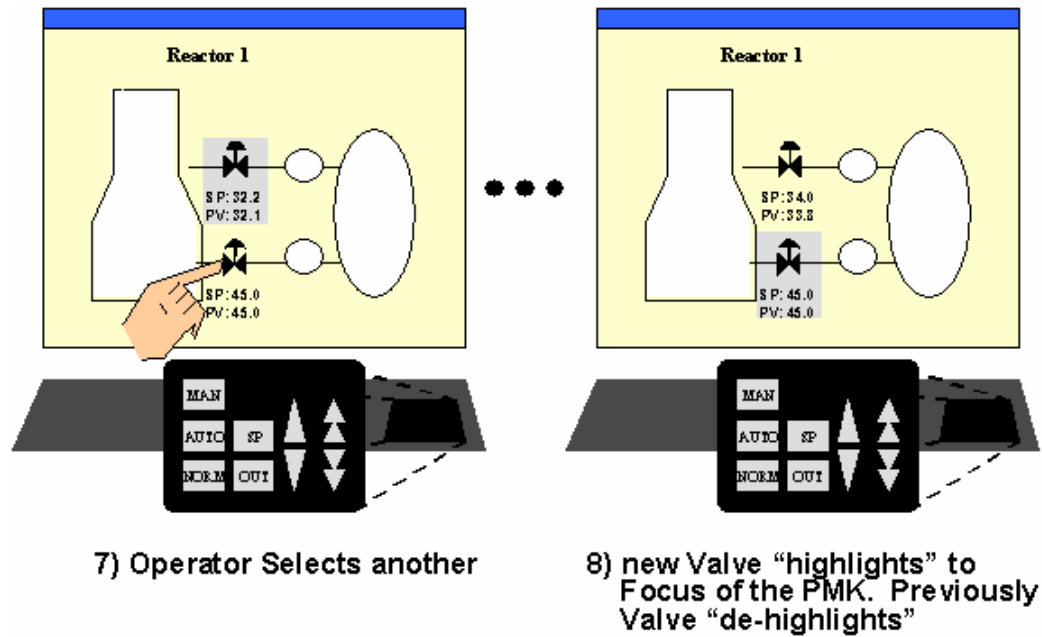


5) SP Continues to Ramp

...



6) When SP at desired value, the operator
releases the raise key, and ramping stops
There is no overshoot.



Using the Valve Embedded display

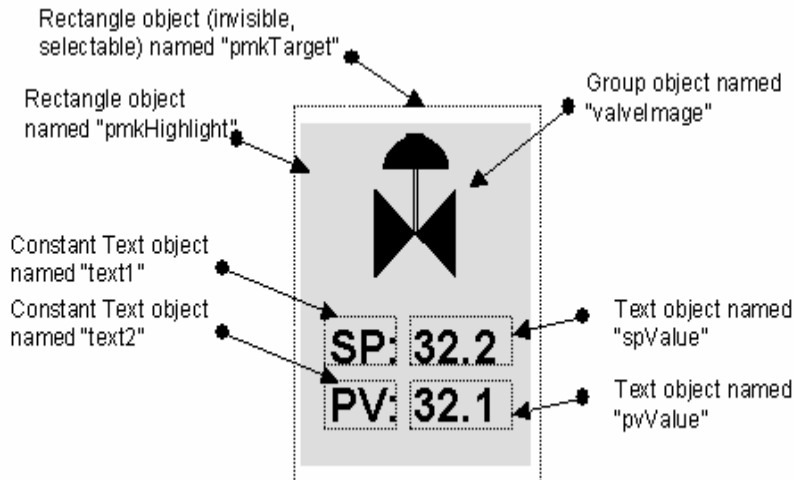
No scripting is required to use the Valve.PCT; it is merely inserted into the Display and the Parameter POINT bound to the point representing the valve.

Authoring the Valve Embedded display

The figure below illustrates the structure of the Valve Embedded display Valve.pct. Valve.pct has one parameter of type Entity, named Point.

16. Issuing System Commands from Displays

16.10. PMK Scenarios and Examples



Valve Embedded Display scripts

Rectangle object: PMKTarget

Sub OnLeftButtonClick

On Error Go To ErrorHandler

' set the PMK to the SP for the point representing this valve

Set PMK.Entity = display.params.point

' register the appropriate objects to handle PMK events

' (the text object spValue shows the SP Value)

Set PMK.EventHandler = spValue

'show this valve is the focus of the PMK

PMKHighlight.visible = True

Exit Sub

```
ErrorHandler:
    <statements TBD>
End Sub

Sub OnLostFocus
    ' Unhook from the PMK & unhighlight the focus rectangle
    PMK.clear
    PMKHighlight.visible = False
End Sub

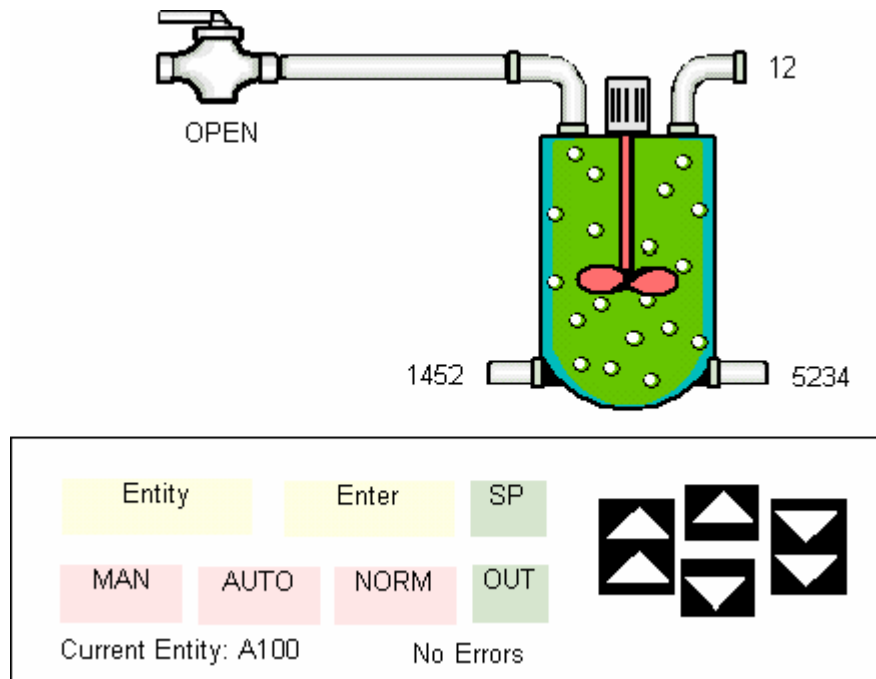
Text object: spValue
Sub OnDataChange
    ' changes to SP due to ramp result in normal OnDataChange
    ' event
    me.text = format (display.params.point.SP, ###.##)
End Sub

Text object: pvValue
Sub OnDataChange
    me.text = format (display.params.point.PV, ###.##)
End Sub
```

Example 2 – A Custom Change Zone

Behavior

The next example will create a custom change zone. The result will be a picture similar to the one that follows.



The above picture contains the following objects:

- 1 An embedded display named emdKeyPad, which represents the keypad object at the bottom of the display.
- 2 An embedded display named emdValve, which represents the valve in the upper-left corner of the display.
- 3 A bitmap object representing a boiler.
- 4 Three text objects named text1, text2, and text3 respectively. These text objects are around the bitmap and have values of 1452, 5234, and 12 respectively.

The embedded keypad acts like a virtual keyboard for operators who might not have an IKB keyboard. Hitting the Raise button on the embedded display will be equivalent to hitting the raise key on the IKB keyboard. Using the embedded point manipulation keys that can be inserted into a Change Zone, you can perform both incremental and continuous point ramping.



ATTENTION

There is an important difference between incremental and continuous ramping.

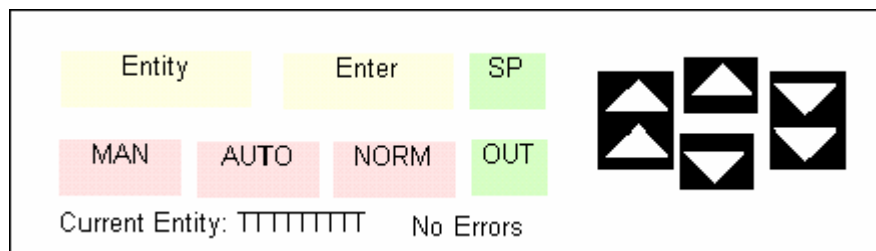
If you click a ramp key at a rate faster than the GUS software can process, the clicks will be queued (one ramp increment for each click), the ramped and displayed values may continue to change even after you stop clicking. If you hold down the ramp key, the value will ramp continuously until you release the key. The value on display when the key is released is the final ramped value.

When the user selects the embedded valve display and performs a raise/lower the valve will open or close. After selecting the text objects around the bitmap followed by an SP/OUT key, press the operator will be able to raise/lower the SP or OP parameter for the text object's data point.

16.11 PMK Scripts

Embedded display: embKeyPad

The embedded display consists of eleven buttons that perform various PMK keyboard manipulations. A text object named txtEntity displays the Entity that can be manipulated through the keypad. The txtError object will also display any PMK Errors that occur at runtime.



16. Issuing System Commands from Displays

16.11. PMK Scripts

```
Sub OnDisplayStartup()  
    Set PMK.EventHandler = txtEntity  
    Set PMK.ErrorHandler = txtError  
End Sub
```

```
Sub OnEnterFocus()  
    Set PMK.EventHandler = txtEntity  
    Set PMK.ErrorHandler = txtError  
End Sub
```

It should be noted that there is no OnExitFocus event to clear the PMK object. The embKeyPad display acts as a universal wrapper for the PMK object. In general this practice is not recommended (see the emdValve script for details).

Embedded display text objects

txtEntity

```
Sub OnDataChange()  
    me.Text = PMK.Entity.[name]  
End Sub
```

txtErrors

In addition to displaying the txtError object, it will also display any error messages raised by the PMK object at runtime.

```
Sub OnPMKError(ErrCode as long, ErrString as string)
    Select Case PMK.ErrCode
Case PMKERR_NOENTITY
    MsgBox "You forgot to register the PMK entity!"
Case PMKERR_NOEVENTHANDLER
    ' Do nothing for this error
Case Else
    MsgBox ErrString
End Sub

Sub OnLButtonClick()
    ' The user has acknowledged the error so clear it
    me.Text = ""
End Sub
```

16. Issuing System Commands from Displays

16.11. PMK Scripts

Embedded display buttons

The buttons behind the embedded display all follow the same format, just a call to the PMK.Key method. For simplicity, only the scripts behind the FastRaise and Auto button will be provided.



BtnFastRaise

```
Sub OnLButtonClick()  
    PMK.Key PMK_FASTRAISE  
End Sub
```

AUTO

btnAuto

```
Sub OnLButtonClick()  
    PMK.Key PMK_AUTO  
End Sub
```

Text objects on the display

The display has several text objects that can be ramped by the embedded change zone picture. The script behind one of the text objects follows:



```
Sub OnGotFocus()  
    Set PMK.Entity = LCN.A100  
End Sub  
Sub OnLostFocus()  
    Set PMK.Entity = ""  
End Sub
```

Note that the clear method is not called for the lost focus event. Calling clear would NULL the Error Handler and the Event Handler that was originally set in *embKeyPad*'s start up script. If clear was called after the lost focus event and the operator pressed the SP key on the keyboard the embedded change zone picture would not receive the OnPMKSP event. A good rule of thumb is the clear method should only be called if there is not a universal Event Handler or a universal error handler in the picture.

Embedded Display: *emdValve*



OPEN

The embedded display *embValve* represents a valve that can be opened or closed when the user hits the raise / lower key on the keyboard. When the valve receives input focus, it will set the Event Handler and Error handler to itself, allowing it to receive PMK events. This creates a problem with the custom change zone embedded display in the picture however. By setting the Event Handler and Error handler to itself, the valve steals and replaces the universal PMK Event Handler and Error handler for the custom change zone. To fix this, when the lost focus event is called for the embedded valve, the Event Handler and Error handler will be restored to the custom change zone.

```
Dim OldEventHandler as Object
```

```
Dim OldErrorHandler as Object
```

```
Sub OnEnterFocus()
```

```
    ' Save the old PMK handlers
```

```
    Set OldEventHandler = PMK.EventHandler
```

```
    Set OldErrorHandler = PMK.ErrorHandler
```

```
    PMK.Clear
```

```
    Set PMK.EventHandler = me
```

```
    Set PMK.ErrorHandler = me
```

```
End Sub
```

```
Sub OnExitFocus()
```

16. Issuing System Commands from Displays

16.11. PMK Scripts

```
    PMK.Clear
    ' Restore the old PMK handlers
        Set PMK.EventHandler = OldEventHandler
    Set PMK.ErrorHandler = OldErrorHandler
End Sub

Sub OnPMKError(ErrCode as long, ErrString as string)
    LCN.DigValve1 = OFF
End Sub

Sub OnPMKRaise(Param As String, Value As Variant)
    LCN.DigValve1 = ON
End Sub

Sub OnPMKLower(Param As String, Value As Variant)
    LCN.DigValve1 = OFF
End Sub
```

In general, it is not a recommended practice to use a universal PMK change zone in your embedded picture as shown above. End users may forget to save and restore the Event and Error Handler, causing unexpected behavior in the main display. Instead, it is recommended that engineers script OnEnterFocus events to register the PMK for their embedded picture and OnExitFocus to clear the PMK for their embedded pictures.

16.12 ButtonSvc Integration

The ButtonSvc handles all keyboard input from the IKB for the TPS system. This service will be responsible for relaying all the PMKs directly to the GUS display that has current input focus.

The GUS display, upon receiving PMK messages from the ButtonSvc, will internally invoke appropriate methods on the PMK Object. The following table lists the appropriate response by the GUS Display for a particular PMK key from the keyboard.

Point Manipulation Key	GUS Display Reaction
Raise	• If the AutoWrite property is set to TRUE, a PMK.Raise method will be invoked. • The OnPMKRaise event will be fired for the Event Handler. • The OnRaise event will be fired for the object of input focus.
Fast Raise	• If the AutoWrite property is set to TRUE, a PMK.FastRaise method will be invoked. • The OnPMKFastRaise event will be fired for the Event Handler. • The OnFastRaise event will be fired for the object of input focus.
Lower	• If the AutoWrite property is set to TRUE, a PMK.Lower method will be invoked. • The OnPMKLower event will be fired for the Event Handler. • The OnLower event will be fired for the object of input focus.
Fast Lower	• If the AutoWrite property is set to TRUE, a PMK.FastLower method will be invoked. • The OnPMKFastLower event will be fired for the Event Handler. • The OnFastLower event will be fired for the object of input focus.

16. Issuing System Commands from Displays

16.12. ButtonSvc Integration

Point Manipulation Key	GUS Display Reaction
MAN	PMK.Key PMK_MAN
AUTO	PMK.Key PMK_AUTO
NORM	PMK.Key PMK_NORM
SP	PMK.Key PMK_SP
OUT	PMK.Key PMK_OUT

17. GUS Display Builder Error List

17.1 Overview

The following is a table listing the errors that a user of the GUS Display Builder may encounter in building, testing, and running GUS displays. The errors are in alphabetical order.

The following information is available for each error.

- What user action(s) causes error checking
- Detailed explanation of the error message
- What actions you should take to resolve the error

You may also want to supply OnError handlers in your scripts to handle errors differently than in the default manner. For instance, entering an invalid point.parameter displays a default “1052 config error” error message. For further information, refer to Bad Status Examples in the *Display Scripting User's Guide*, in the GUS Scripting Introduction section.

NOTE: For Validation Errors, refer to Validation Error List (Error List Help).

17. GUS Display Builder Error List

17.2. Error Table

17.2 Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"<No message defined for this error.>"	Various.	The error string for this error could not be located in the Honeywell or system message table.	Contact Honeywell.
%1 is not a valid display parameter	During display validation.	A script is attempting to use a display parameter that is not defined.	Use the Script Editor to replace the display parameter reference, or use the Define Parameters dialog to define the referenced parameter.
"A parameter with this name already exists."	Click OK on add or edit display parameter dialog.	Display already contains a parameter with the entered name.	Change parameter name.
"AddTrace method failed. Possible causes include: (1) Eight traces already exist for this trend, (2) Input VariableID pointer is NULL or input VariableID has invalid syntax, (3) Input trace color did not validate."	Invoking the AddTrace method.		a) Call Delete Trace first if eight traces are already active for this OLE Trend Control. b) Ensure sVariableID pointer is non-NULL. c) Ensure clrTraceColor parameter is a valid "long."

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"An input value of zero was provided for either a ScrollBackTime or a ScrollForwardTime. No action will be taken."	Setting the ScrollBackTime or ScrollForwardTime properties.		Set the properties to a non-zero value.
"Attempt to connect to HOPC data server failed."	When opening a display at buildtime or runtime.	The display process could not connect to HOPC server. Prevents data access for display.	Try to use "On Line" command on file menu in builder. Ensure that the HOPC Server is running and connected to the Ethernet.
"Can't Create Script Object."	Translating TDC (TPS Network) file.	Can't translate TPS Network display parameter correctly.	Fix TDC (TPS Network) version and retranslate.
"Cannot locate validation error.\nDisplay object may have been deleted."	When invoking "Go to Error" function on Validation Error List dialog	The builder cannot find the object having the selected error.	Verify that the object has been deleted. You may want to validate the display again.
"Cannot open file: "	Click OK on file open dialog.	GUS could not open requested display: not found; read error; etc.	Check for existing display and reinvoke.
"Cannot save to already opened document:\n"	Click OK on file save dialog, or file save button/command.	GUS has 2 display windows open with the same file. Cannot save to 1 window while another window is displaying same file.	Close duplicate windows and reinvoke save command.
"Circular replace of embedded display not allowed - <display name>"	When a circular replacement is found in the embedded displays.	Circular replacement involves replacing one embedded display with a second display and replace the second with the third and so on.	Delete the circular replacement and enter the name of an embedded display.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Communication error"	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	The TPS Network point.parameter requested is unavailable. This indicates communication on the LCN is disrupted or a device has failed. This may be because either the data owner is not running, or the connection between the NT and the TPS Network is not operating.	Using the Native Window, verify the GUS personality is running. Verify the node or device on which the data resides is loaded and running.
"Configuration error"	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	The TPS Network point.parameter does not exist. The point.parameter has been misspelled, the point has been deleted from the TPS Network, or the value is equal to a "bad value" or "not a number."	Verify the point.parameter is spelled correctly and the point exists on the TPS Network. If it is an element of an array, verify the index is a valid index.
"Connection request exceeds maximum allowable."	When opening a display online.	User can only have four HOPC connections. All connections are being used; therefore, there is no data access.	Close a display and invoke the display again.
"Conversion From LCN Char Error."	Translating TDC (TPS Network) file.	Can't convert user-entered information from TPS Network to GUS (ASCII to UNICODE).	Fix TPS Network version and retranslate.
Creation_Fail "OCX Creation Failure"	Translating TDC (TPS Network) file.	Can't create OCX for translated TPS Network element.	

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"DeleteTrace method failed. Possible causes include: (1) Input TraceID is out of the valid range 1 through 8, or (2) Input TraceID is currently not active."	Invoking the DeleteTrace method.		a) Ensure nTraceID parameter is in the range 1 through 8. b) Call AddTrace first if the trace attempting to be deleted is not currently active.
"Display %1 contains more than 100 LCN scanned variables.\nDisplay 100 cannot run this display. Use Display Max or Multiple Displays."	Attempt to run a display on Display 100 with more than 100 points.	(message is explicit)	Install Display Max or Multiple Displays or remove some points from this display using the builder.
"Display %1 with different version exists in display.\nReplace not allowed."	Click OK on replace embedded display.	User attempted to replace 1 embedded display with an embedded display that already exists in the file, AND has a different time stamp than the currently existing one.	1) Replace older version of existing embedded display with newer version, and reinvoke. 2) Replace first requested embedded display with different embedded display.
"End of file not encountered."	Translating TDC (TPS Network) file.	TPS Network file missing some needed record.	NOTHING: Corrupt TPS Network file.
"Error in opening TDC file."	Click OK on open file dialog or when translating TDC (TPS Network) file.	User tried to open (had trouble translating) a TPS Network file.	Figure out where the translation went wrong, and fix it on the TPS Network. Then reopen.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Error in opening US display file."	Click OK on open file dialog.	User tried to open (had trouble translating) a TPS Network file.	Figure out where the translation went wrong, and fix it on the TPS Network. Then reopen.
"Error in processing US display file:\n"	Translating TDC (TPS Network) file.	Error occurred in translating: prior message gives more detail.	Fix TPS Network version and retranslate.
"Error Registering Constant Objects"	During program startup.	An internal software error.	Contact Honeywell.
"Error translating Target Expression."	Translating TDC (TPS Network) file.	Can't translate TPS Network target correctly.	Fix TPS Network version and retranslate.
"Error translating Value Expression."	Translating TDC (TPS Network) file.	Can't translate TPS Network value correctly.	Fix TPS Network version and retranslate.
"Exception caught in UpdateContinuousData."	Receiving new trace data from the HOPCServer.	Exception raised during data retrieval process with the HOPCServer.	Internal problem.
"Expecting a left parenthesis after the collector keyword"	During display validation.	A call to the collector function in a script is missing the closing left parenthesis.	
Expecting a left parenthesis after the evaluate keyword	During display validation.	A script syntax error was detected in an evaluate operator	Enter the closing parenthesis using the script editor.
"Expecting a string literal as argument for collector function"	During display validation.	The argument to the collector function must be a string literal.	

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
Expecting a string literal as argument for evaluate function	During display validation.	A script contains an evaluation operator in which the operand is not a string literal. The operand of the evaluate operator must be a string literal.	The operand of the evaluate operator must be a string literal.
“Failed to activate the Bitmap creation tool”	When Insert Bitmap is activated.	GPB will not attempt to insert a bitmap.	Save display and restart GPB.
“Failed to create the bitmap.InFile could be a bad format.”	When inserting or reading the bitmap file.	The file is either 1) not a bitmap, or 2) a bitmap with less than 256 colors, or 3) the bitmap file is corrupted.	Read in a correct bitmap file.
“Failed to end me table in OleObjScriptIn”	When adding an object to a display.	An internal software error.	Contact Honeywell.
“Failed to initialize Basic due to (%d)”	During program startup.	An internal software error.	Contact Honeywell.
“Failed to initialize Basic global objects table, may be out of memory”	During program startup.	An internal software error.	Contact Honeywell.
“Failed to initialize Basic prototype table, may be out of memory”	During program startup.	An internal software error.	Contact Honeywell.
“Failure to add trace with HOPCServer where TraceID and requested VariableID are as follows: “	Trend OLE Control's request to the HOPCServer via the History Client to add a trace.		Restart HOPCServer.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Failure to delete trace with HOPCServer where requested TraceID = "	Trend OLE Control's request to the HOPCServer via the History Client to delete a trace.		Restart HOPCServer.
"Failure to initialize history retrieval process with History Client for TraceID ="	Trend OLE Control's request to History Client to retrieve the new trace values following a notification of new data from the History Client.		Internal problem.
"Failure to initialize with History Client. Possible causes of failure are: (1) an attempt to load more than 16 Trend Controls per GUS at Runtime, (2) a failure of the HOPCSrvr, or (3) a mismatch in the count of traces between the Trend Control and the storage."	Trend OLE Control's initialization with the History Client.	The History Client is a static library that is linked with the OLE Trend Control. It interfaces with the HOPCServer on behalf of the OLE Trend Control. When the OLE Trend Control changes to runtime mode, an initialization process is initiated with the History Client. This process failed.	a) Restart HOPCServer b) Ensure that GUS does not have pictures loaded which collectively total more than 16 OLE Trend Controls.
"Failure to set data source with HOPCServer where TraceID and requested value are as follows: "	Trend OLE Control's request to the HOPCServer via the History Client to set the DataSource to the designated value.		Restart HOPCServer.
"Failure to set time base with HOPCServer where requested value = "	Trend OLE Control's request to the HOPCServer via the History Client to set the TimeBase to the designated value.		Restart HOPCServer.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Failure to set time offset with HOPCServer where requested value = "	Trend OLE Control's request to the HOPCServer via the History Client to set the TimeOffset to the designated value. (Reference ScrollBackTime and ScrollForwardTime.)		Restart HOPCServer.
"Failed to open file"	Reading of the file.	File could not be opened or a read error occurred, for example, File permissions. No room on the temporary storage directory defined by the Environment Variable "temp" or "tmp."	Change the file permissions. Purge files off the device that the temporary file is stored.
"Failed to read application configuration data from registry.\nRun GUS installation program to perform registry setup."	During builder or runtime startup.	Registry data is missing or corrupted.	Run setup to configure registry.
"Failed to register Basic prototype table for constants, may be out of memory"	During program startup.	An internal software error.	Contact Honeywell.
"Failed to register Basic prototype table, may be out of memory"	During program startup.	An internal software error.	Contact Honeywell.
"Failure to register trace Variable ID with Name History in On-line Mode."	Invoking the Data Access Client to register the Variable ID names at build-time when on-line.	A problem was experienced at build-time when on-line, in attempting to add the trace VariableID names.	Possible cause is a corrupt .pct file. Recreate file, and retry.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Failure to validate input trace Variable ID with LCN. Please re-enter known trace Variable ID."	Setting the trace Variable ID property.	The trace VariableID is not known by the TPS Network.	Set property to registered value for TPS Network.
"File type not GUS source file."	Click OK on file open dialog.	Selected display not GUS source file, and cannot be opened by GUS.	Select a new display to open.
"Header record not encountered."	Translating TDC (TPS Network) file.	Missing TPS Network header record for TDC file.	NOTHING: Corrupt TPS Network file.
"HOPC data server failed to read LCN data from display file."	When a display is invoked online under the following conditions. Picture Manager (picman) is running, and SafeView is not being used.	The maximum number of displays is running. The Picture Manager (picman) will not allow this display to be opened and displayed.	Close a display and invoke the display again.
"Incompatible US version number.\n Version number must be R400 or higher."	Translating TDC (TPS Network) file.	TPS Network file is not R500 or later.	User must translate to R500 version or later on the TPS Network, then retranslate on GUS.
"Input string for property is invalid and will not be set. A valid Name must satisfy the following two rules: (1) First character must be alpha; (2) All remaining characters must be alpha, numeric, or underscore."	Setting the Name property.		Set property to valid value.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Input string length for property is invalid and input will not be set. An invalid length of characters has been input. Valid length for "Name" Property is 1 to 24 characters, valid length for "Description" Property is 0 to 1000 characters and valid length for "VariableID" is 3 to 28 characters."	Setting a string parameter to an invalid length.	Either the Name, Description or VariableID property was set to a value with an invalid string length.	Set property to valid value.
"Input TraceID is out of the valid range 1 through 8, or is currently not active. Input value for property will not be set."	Calling a method or setting a property which requires the TraceID as a parameter.		Call method with TraceID between 1 and 8, or call AddTrace first if TraceID is not currently active.
"Input value for property is invalid and will not be set. Valid values for Boolean property are either TRUE(1) or FALSE(0)."	When attempting to set any property of type BOOL to a non-valid value.		Set property to valid BOOL value.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Input value for property is invalid and will not be set. Valid values for scrolling long integer are in the range 1 to LONG_MAX, where LONG_MAX = 2147483647. (NOTE: The absolute value of an input negative number will be taken.)"	When attempting to set any property of type "long" to an out-of-range value.		Set property to valid "long" value.
"Input value for property is invalid and will not be set. Valid values for YScaleHigh are 1 to 100 percent, and 0 to 99 for YScaleLow."	Setting either the YScaleHigh or YScaleLow property.		Set property to valid value.
"Input value for property was invalid and will not be set. The Time Base has been adjusted to the nearest valid value (seconds) which is "	Setting the TimeBase property.		Set property to alternate value if adjusted value is not acceptable.
"Input value for property was invalid and will not be set. Valid values for the Data Source are 0 through 3, where 0=Real-Time, 1=History Module, 2=Hiway Gateway and 3=Automatic selection."	Setting the DataSource property.		Set property to valid value.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Input value for property is invalid and will not be set. Value values for YRangeHigh and YRangeLow are (-1)*FLT_MAX to FLT_MAX, where FLT_MAX = 3.402823466e+38F."	Setting the YRangeHigh property.		Set property to valid value.
"Internal error in display builder during script compile"	During display validation.	An internal software error.	Contact Honeywell.
"Insert Display file %s not found"	Click OK on insert display dialog.	Display name entered by user not found in this directory; Explorer loses access to this directory.	Make sure name entered correctly, or directory exists. Correct as needed.
"Invalid character \"%c\" in name"	This occurs when the user enters an invalid object name in the Property Sheet for a display object.	(message explicit)	Consult the <i>Display Scripting User's Guide</i> for details on syntax rules for object naming.
"Invalid Data Type"	Translating TDC (TPS Network) file.	Translating TPS Network file.	Fix TPS Network version and retranslate.
Invalid display parameter type for binding to reference type display parameter	When using the "Enter Parameters" dialog.	A display parameter entered as the binding expression must be a reference type (either Variable or Entity type) when the parameter being bound is a reference type.	

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
Invalid object for binding to reference type display parameters	When using the "Enter Parameters" dialog.	The object entered as the binding expression must be a reference type when the parameter being bound is a reference type.	
"Invalid Record Type."	Translating TDC (TPS Network) file.	Wrong record type found; expected a different record.	Fix TPS Network version and retranslate.
"Maximum allowed number of scripted data access symbols exceeded"	During display validation.	There are too many process data object references in the display. The approximate limit is 3000 references.	Reduce the number of data object references.
"Maximum allowed number of scripted display objects exceeded"	When adding display objects to a display.	There are too many display objects in the display. The approximate limit is 6000 display objects.	Reduce the number of display objects.
"New embedded display exists in the picture. Cannot replace selected embedded display instances."	When the "Replace Selected Instances" option is checked and a subset of the instances is selected.	This error is reported when the name of the new embedded display exists in the GUS display.	Give a unique name to the new embedded display.
"Not yet supported"	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	This error is reported when a request is made to the Hopcsrvr, which it does not support.	Contact Honeywell.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"No error but no data"	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	Indicates an LCN indirect reference was added to the Hopcsrvr during picture validation. The indirection cannot be completely validated until runtime, therefore a warning of "No error but no data" is returned.	N/A
Picture Manager has not been initialized - run picmanc.exe first.	Attempting to run a GUS display.	The specified wait time (or 15 minutes if none was specified) has elapsed since a request was made to run a display but PICMANC initialization has not yet begun.	1. Ensure that the Native Window has completed the load with GUS personality. 2. Execute picmanc.exe, if login allows. 3. Run the display again.
Picture Manager initialization is not complete - please try invocation later.	Attempting to run a GUS display.	The specified wait time (or 15 minutes if none was specified) has elapsed since a request was made to run a display and PICMANC initialization has been started but has not yet been completed.	Wait 1 minute and then run the display again.
"Please enter a name that is not more than 24 characters"	This occurs when the user enters an invalid object name in the Property Sheet for a display object.	(message explicit)	Consult the <i>Display Scripting User's Guide</i> for details on syntax rules for object naming.
"Problem creating windows resources."	Attempting to dynamically create a Window's resource, such as a "CreatePen."	A create call by the OLE Trend Control for a Window's resource failed.	Check hard disk storage.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Range % not defined."	Click OK or Apply on Object Property Sheet	One or more of the range parameters for one of the object's basic dynamics is not defined.	Check the range parameter text entries on the basic dynamic page(s) to determine which one has not been defined. Define the missing range(s).
"Read Record Error."	Translating TDC (TPS Network) file.	OBSOLETE	OBSOLETE
"Resource Shortage in PC processor caused severe condition"	Runtime.	Resource shortage in memory, CPU, or disk bandwidth has prevented the workstation from servicing the TPS Network software in a timely fashion. In response, the TPS Network software has transitioned to the SEVERE state awaiting the availability of the necessary resources.	When the condition clears, the TPS Network node will automatically return to the OK state.
"Runtime scrolling is not currently activated. It can be activated by setting the RuntimeScrolling property."	Writing to either ScrollFowardTime or ScrollBackTime properties with the RuntimeScrolling property disabled.	The RuntimeScrolling property must first be set to TRUE before trying to activate scrolling.	Set the RuntimeScrolling property to TRUE, first.
"Requested method could not be invoked because History Client has not successfully initialized for this Trend Control."	Calling any method on the History Client after an unsuccessful initialization request.		Restart HOPCServer. Re-run picture.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Script shutdown sequence has not finished. Force termination?"	When user enters <ctrl-break> key combination while display is running the shutdown scripts.	The display is terminating slowly due to execution of OnDisplayShutdown scripts. Responding 'Yes' to this prompt will terminate these scripts, and exit the display immediately.	
"Store error"	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	The value which was supposed to be stored to the TPS Network was invalid for the specified TPS Network point.parameter.	Verify the point.parameter has write permissions. Verify the GUS keylock is in the appropriate position. Verify the data being stored matches the type of data of the parameter.
"Subpicture reference number not found"	Translating TDC (TPS Network) file.	TPS Network file contains reference to non-existing embedded display.	Fix TPS Network version and retranslate.
"Symbol Name Size : 0."	Translating TDC (TPS Network) file.	Variable in TPS Network picture found with a 0-length name (illegal).	Fix TPS Network version and retranslate.
"Symbol: bad record type %.1d"	Translating TDC (TPS Network) file.	Parsing TPS Network variables (symbol) table contains bad record.	Fix TPS Network version and retranslate.
"Syntax error"	During validation of an LCN reference entered in the "Enter Parameters" dialog for a reference type parameter.	The name entered in "Enter Parameters" dialog begins with an invalid symbol.	
"Text for Format is invalid"	Translating TDC (TPS Network) file.	Bad text in TPS Network text element.	Fix TPS Network version and retranslate.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
“The hairline cursor is not currently active. It can be activated by setting the HairlineCursor property. To display the Y-values at the hairline cursor points of intersection on the Trend OLE Control, also set the HairlineReadout property.”	Selecting the hairline cursor, when the HairlineCursor property is disabled.	The HairlineCursor property must first be set to TRUE before trying to select the hairline cursor while in runtime mode.	Set the HairlineCursor property to TRUE, first.
“The X Time-coordinate at the hairline cursor intersection is a read-only property.”	Attempting to write to the XCursorTime property.		Do not attempt to write to the XCursorTime property.
There is a limit of sixteen GUS OLE Trend Controls per GUS Station; i.e., sixteen represents the maximum number of Trend Controls across all loaded pictures at runtime. You are currently exceeding this limit, and will have no data access for those Trend Controls in runtime mode.	Whenever a new Trend Control is created in a GUS Pictures.	Not more than 16 OLE Trend Controls can exist in one picture, because at runtime, the maximum number of Trend Controls that can be handled by the HOPCServer is 16 across all loaded pictures.	Delete Trend Control objects such that the total number across all loaded pictures at runtime does not exceed 16.
“This operation will destroy the script on this group!”	During Ungroup.	Any script that is on the group will be lost if the action continues.	To save the script, if needed, copy the script before this action.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
“Trend YScaleHigh must be greater than Trend YScaleLow. Valid range for YScaleHigh is ≥ 0.01 and ≤ 100.0; valid range for YScaleLow is ≥ 0.0 and ≤ 99.99.”	Setting either the YScaleHigh or YScaleLow properties.		Set property to valid value.
“The number of traces is a read-only property.”	Attempting a write to the property NumTraces.		Do not attempt to set property “NumTraces.”
“The Y-values at the intersection of the hairline cursor are read-only properties.”	Attempting to write to the YCursorReadout property.		Do not attempt to write to any YCursorReadout property.
“This method is runtime only.”	Calling any runtime only method during build-time mode.		Call the given method only in build-time mode.
“Trace YRangeHigh must be greater than Trace YRangeLow. Valid float range is - 3.40E+38...-1.17E-38, 0, 1.17E-38...3.40E+38.”	Setting either the YRangeHigh or YRangeLow properties.		Set property to valid value.
“TrendName can only be updated while in Build Mode.”	Setting the Name property to a value while in runtime mode.		Do not attempt operation in runtime mode.
“Unable to complete Save operation.”	Loading/Storing OLE Trend Control properties from/to disk.	A problem was experienced when trying to save the OLE Trend Control's properties to persistent storage (.pct file).	Check hard disk storage.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Unable to complete Load operation."	Loading/Storing OLE Trend Control properties from/to disk.	A problem was experienced when trying to retrieve the OLE Trend Control's properties from persistent storage (.pct file) to internal variables.	Possible cause is a corrupt .pct file. Recreate file, and retry.
"Unable to properly initialize with History Client. No GPB storage available for input."	Requesting the storage name (.pct file) from the Display container.	A valid GPB storage name for the OLE Trend Control could not be located in order to proceed with the OLE Trend Control's initialization procedures.	Internal problem.
"Unable to register find/replace callback message"	During program startup.	An internal software error.	Contact Honeywell.
"Unexpected file format for TDC file."	Translating TDC (TPS Network) file.	File format for TPS Network is wrong.	NOTHING: Corrupt TPS Network file.
Unexpected text following expression	When using the "Enter Parameters" dialog.	There is a syntax error in the expression.	
"Unknown Data Access Error"	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	Internal error.	Contact Honeywell.
"Unknown object type for info rec"	Translating TDC (TPS Network) file.	Newer version of TPS Network file attempting to be translated.	
"Unknown record type for translation"	Translating TDC (TPS Network) file.	Newer version of TPS Network file attempting to be translated.	

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
“Unknown ref type reading subpicture description”	Translating TDC (TPS Network) file.	TPS Network embedded picture is not subpicture or phantom (Honeywell defined subpicture).	Fix TPS Network version and retranslate.
“Unsupported type or no type selected.”	Click OK on add or edit display parameter dialog.	Unsupported TPS Network parameter type selected.	Select supported parameter type.
“Use runpic.exe to invoke Display %1.”	Attempt to run a display not using PICMANC.	Always use the PICMANC command to invoke a display.	Execute picmanc.exe.
“Value: bad type. Location: %.3f, %.3f ln “	Translating TDC (TPS Network) file.	TPS Network value contains illegal type.	Fix TPS Network version and retranslate.
“Value error”	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	The TPS Network was unable to return a value for a given point.parameter.	Verify the point.parameter exists and has a valid value.
“Variable ID can only be updated while in Build Mode.”	Attempting to write to the Variable ID in test mode or runtime mode.		Do not attempt to set property in test mode or runtime mode.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
Variable ID must be formatted per LCN Variable ID syntax as follows: 1) Optional LCN ID of 2 alpha- numerics plus backslash separator. 2) Required 16-char max Point Name plus dot separator; 3) Required 8-char max Parameter Name. Each char in Point/Parameter Name must be alpha, numeric, ! , \$, or underscore. Names cannot begin or end in underscore, and cannot have two sequential underscores. Dot is not allowed. Please enter a valid name, or press "Cancel" to exit."	Setting the trace Variable ID property.	See syntax rules in error message.	Set property to valid value.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
"Write_error"	During display run mode, when a data object is referenced. This error message appears in a message box, in addition to information on the script and object in error.	The value that was attempted to be stored to the TPS Network was invalid for the specified TPS Network point.parameter. A string further identifying the result from the TPS Network is also given.	Verify the point.parameter has write permissions. Verify the GUS keylock is in the appropriate position. Verify the data being stored matches the type of data of the parameter.
"YRangeHigh is currently less than or equal to YRangeLow. Please ensure that YRangeHigh is greater than YRangeLow, or no plotting of trace values will occur."	At runtime, when the mathematical relationship between YRangeHigh and YRangeLow is checked every time one of the values is changed.	YRangeHigh is invalid with respect to YRangeLow. The initial default values for YRangeHigh and YRangeLow are 100.0 and 0.0, respectively. Although you can change these values, the order in which you change them is important because their mathematical relationship must be maintained to prevent an error. This order can change depending on whether the values are positive or negative. Thus, when setting new values, YRangeHigh must remain more positive/less negative than YRangeLow.	Set property to valid value, with these conditions: - Before setting a new YRangeHigh, ensure that its value will remain more positive/less negative than the current YRangeLow. - Before setting a new YRangeLow, ensure that its value will remain more negative/less positive than the current YRangeHigh.

17. GUS Display Builder Error List

17.2. Error Table

Error Message	When the Error Check is Done	Explanation	Resolution
“YRangeLow is currently greater than or equal to YRangeHigh. Please ensure that YRangeLow is less than YRangeHigh, or no plotting of trace values will occur.”	At runtime, when the mathematical relationship between YRangeHigh and YRangeLow is checked every time one of the values is changed.	YRangeLow is invalid with respect to YRangeHigh. The initial default values for YRangeHigh and YRangeLow are 100.0 and 0.0, respectively. Although you can change these values, the order in which you change them is important because their mathematical relationship must be maintained to prevent an error. This order can change depending on whether the values are positive or negative. Thus, when setting new values, YRangeLow must remain more negative/less positive than YRangeHigh.	Set property to valid value, with these conditions: - Before setting a new YRangeLow, ensure that its value will remain more negative/less positive than the current YRangeHigh. - Before setting a new YRangeHigh, ensure that its value will remain more positive/less negative than the current YRangeLow.
“YScaleHigh is currently less than or equal to YScaleLow. Please ensure that YScaleHigh is greater than YScaleLow, or no plotting of trace values will occur.”	Setting the YScaleHigh property.		Ensure YScaleHigh and YScaleLow have the correct relationship.
“YScaleLow is currently greater than or equal to YScaleHigh. Please ensure that YScaleLow is less than YScaleHigh, or no plotting of trace values will occur.”	Setting the YScaleLow property.		Ensure YScaleHigh and YScaleLow have the correct relationship.

18. Getting Help on the Display Builder

18.1 Overview

Online and context sensitive help are both available from the Help menu.



For help using these commands, refer to:

- Accessing the Help Topics browser
- Contents Tab
- Index Tab
- Find Tab
- Setting the Context-sensitive Help mode

18.2 Accessing the Help Topics browser

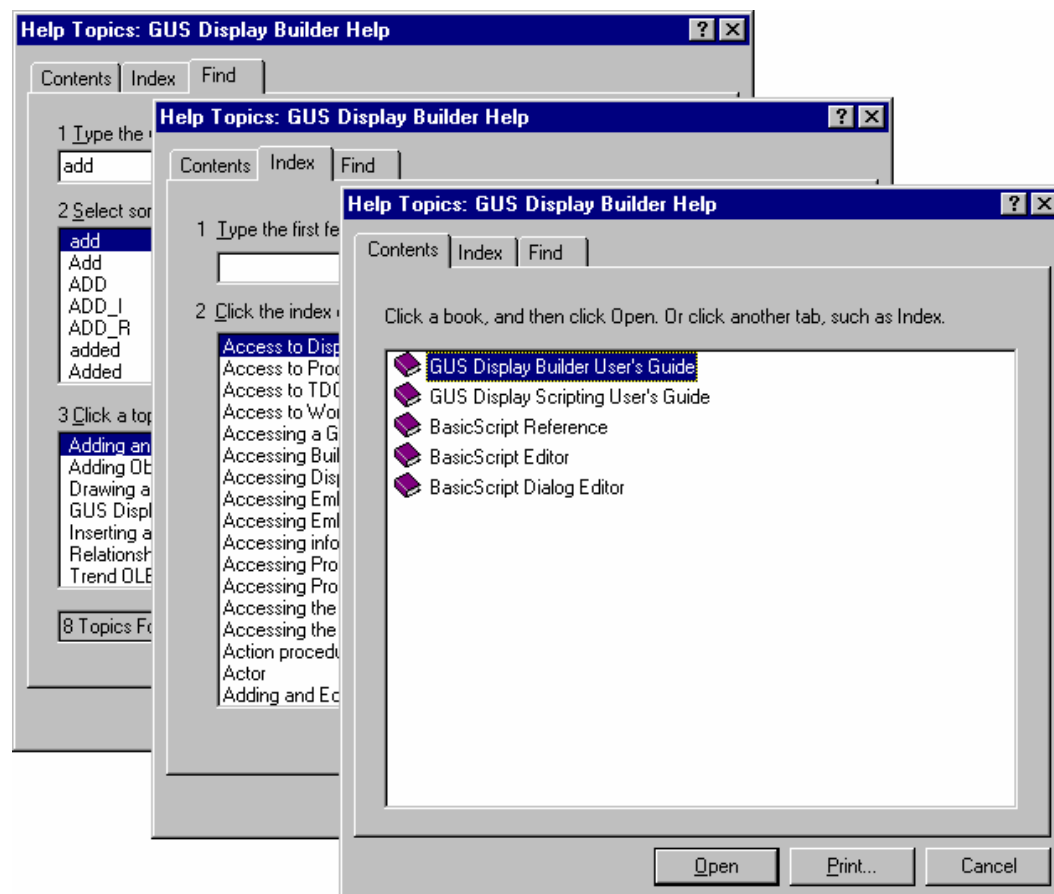
Help/Help Topics menu

Select Help Topics from the Help menu to open a tabbed window of help topics for the GUS Display Builder. The tabs are:

- A Contents Tab (shown at right)--A table of contents showing books and chapters
- An Index Tab of contents (shown in middle)--An alphabetized list of the table of contents
- A Find Tab (shown at left)--Lets you do a full-text search for any word in these help files.

18. Getting Help on the Display Builder

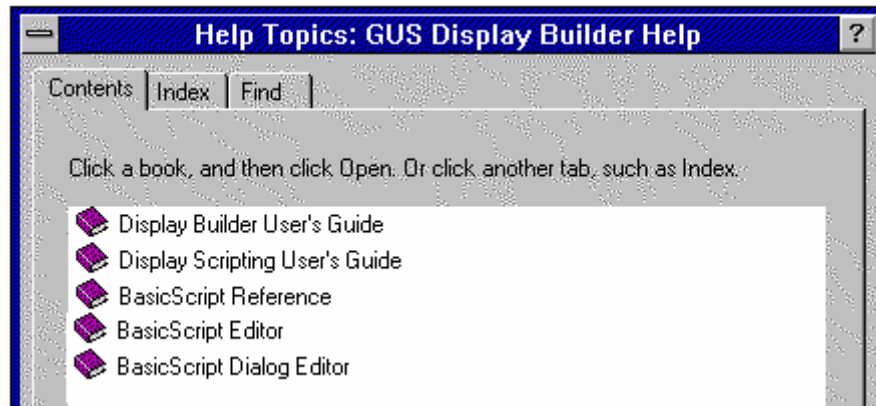
18.2. Accessing the Help Topics browser



18.3 Contents Tab

Lists the five “books” that make up this help system. Double-click a book to open it and find the topic pages within. To display a topic page, either (1) highlight its topic and click the Display button or (2) double-click its topic name.

- **Display Builder User’s Guide**—A guide to the use of the many functions of the Display Builder.
- **Display Scripting User’s Guide**—A guide to using the Scripting Language with the Display Builder as it is related to other Honeywell products. These guides are combined in a single Help file.
- **BasicScript Reference**—A reference to the elements of the Scripting Language, listed both alphabetically and by specific function.
- **BasicScript Editor**—A reference to the basic Script Editor. Honeywell’s modifications to this editor are contained in the Display Scripting User’s Guide listed above.
- **BasicScript Dialog Editor**—Use this editor to make Dialog Boxes for your project.



To learn about each feature of Contents Tab, click the  in the upper right corner of the window; then click the required feature for a short explanation.

18. Getting Help on the Display Builder

18.4. Index Tab

18.4 Index Tab

Lists all the topics under the Contents Tab in alphabetical order. Search for a required topic by typing the first few letters of the topic name.

As with the Contents Tab, you can display a topic page by either (1) highlighting its topic and clicking the Display button or (2) by double-clicking its topic name.

To learn about each feature of Index Tab, click the  in the upper right corner of the window; then click the required feature for a short explanation.

18.5 Find Tab

Does a full-text search simultaneously on all documents listed under the Contents Tab. Each word of the text is searched. To use, execute the following steps:

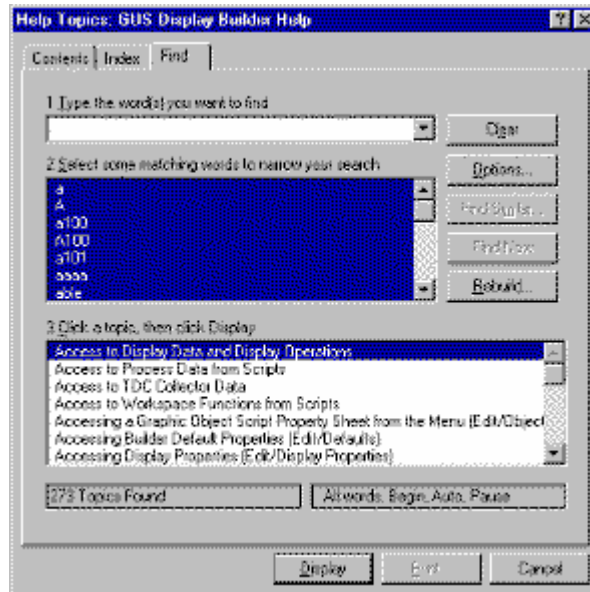
- 1 In ListBox #1, type the word or words you want to find.
If you type letters in lower case, Find Tab will search for words listed in both upper and lower case.
If you type words with leading capitals or all capitals, Find Tab will limit its search to words with that case.

RESULT: Words that match are displayed in ListBox #2.

NOTE: You can limit your full-text search to one or more of the help files listed in the Contents Tab by first clicking the Options button. Then, in the Options dialog box, click the Files button and select the required Help files.

18. Getting Help on the Display Builder

18.5. Find Tab



- 2 Click (highlight) a word in ListBox #2 to limit the search further.

RESULT: Topics containing the word(s) highlighted in ListBox #2 are displayed in ListBox #3.

- 3 Highlight a topic and click the Display button (or double-click the topic name).

RESULT: The topic page, containing the required word, is displayed by the help system.

- 4 Return to Find Tab for another search by clicking the Help Topics button at the top of the page.

To learn about each feature of Find Tab, click the  in the upper right corner of the window; then click the required feature for a short explanation.

NOTE: The first time the Find command is used in Help Topics, you must click first on the Next button and then on the Finish button so that Windows Help will build the search word list on your computer.

18.6 Setting the Context-sensitive Help mode

How to obtain Help on a Menu item or Toolbar icon

1. Select What's This? on the Help menu or select the Help button  on the Standard toolbar icon.
The normal cursor is changed to a Help cursor by adding a question mark 
2. Move the new Help cursor over a menu command or toolbar icon you want help information on and right-click. Help information for the item selected will be displayed.

How to obtain Help on a dialog box

If a dialog box displays a **Help** button, simply click the button to obtain help.

18.7 Accessing information about Display Builder

Help/About Honeywell menu

Select About Honeywell from the Help menu to find Version and Copyright information on this product.

Honeywell

Process Solutions
2500 West Union Hills Drive
Phoenix, AZ 85027